

(12) **United States Patent**
Ewer et al.

(10) **Patent No.:** **US 12,403,020 B2**
(45) **Date of Patent:** **Sep. 2, 2025**

(54) **METHODS AND INSTRUMENTATION FOR INTERVERTEBRAL CAGE EXPANSION**

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(73) Assignee: **Amplify Surgical, Inc.**, Laguna Hills, CA (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 69 days.

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(65) **Prior Publication Data**

US 2021/0315714 A1 Oct. 14, 2021

Related U.S. Application Data

(63) Continuation of application No. 15/793,950, filed on Oct. 25, 2017, now Pat. No. 10,993,815.

(60) Provisional application No. 62/412,781, filed on Oct. 25, 2016.

(51) **Int. Cl.**

A61F 2/46 (2006.01)
A61F 2/44 (2006.01)
A61B 17/17 (2006.01)
A61F 2/28 (2006.01)
A61F 2/30 (2006.01)

(52) **U.S. Cl.**

CPC **A61F 2/4611** (2013.01); **A61F 2/442** (2013.01); **A61F 2/4455** (2013.01); **A61B 17/1757** (2013.01); **A61F 2002/2835** (2013.01); **A61F 2002/30329** (2013.01); **A61F 2002/30331** (2013.01); **A61F 2002/30471** (2013.01); **A61F 2002/30545** (2013.01); **A61F 2002/3055** (2013.01); **A61F 2002/30556** (2013.01); **A61F 2002/30579** (2013.01); **A61F 2/4603** (2013.01); **A61F 2002/4625** (2013.01); **A61F 2002/4627** (2013.01); **A61F 2002/4629** (2013.01); **A61F 2002/4638** (2013.01)

(58) **Field of Classification Search**

CPC **A61F 2/4611**; **A61F 2/442**; **A61F 2/4455**
See application file for complete search history.

(56)

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* cited by examiner

Primary Examiner — Sameh R Boles

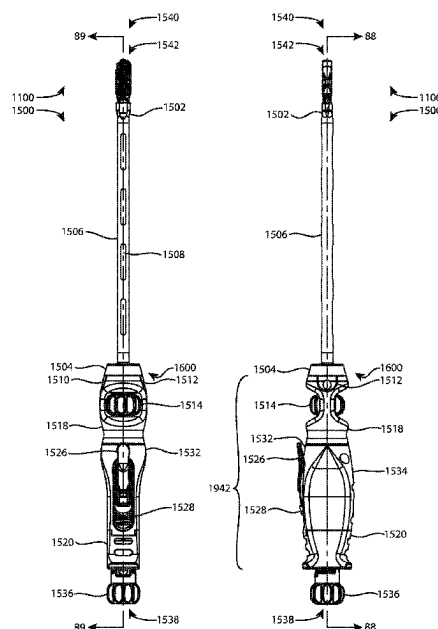
(74) *Attorney, Agent, or Firm* — David Meibos;
Maywood IP Law

(57)

ABSTRACT

An instrument may be coupled to a multi-axis expandable intervertebral cage so that the cage may be inserted into an intervertebral space, expanded along multiple different directions, filled with bone graft, and locked with a fastener. The instrument may be part of an instrument set that includes auxiliary instruments to determine implant size, insert bone graft into the cage, and deliver the fastener.

5 Claims, 72 Drawing Sheets



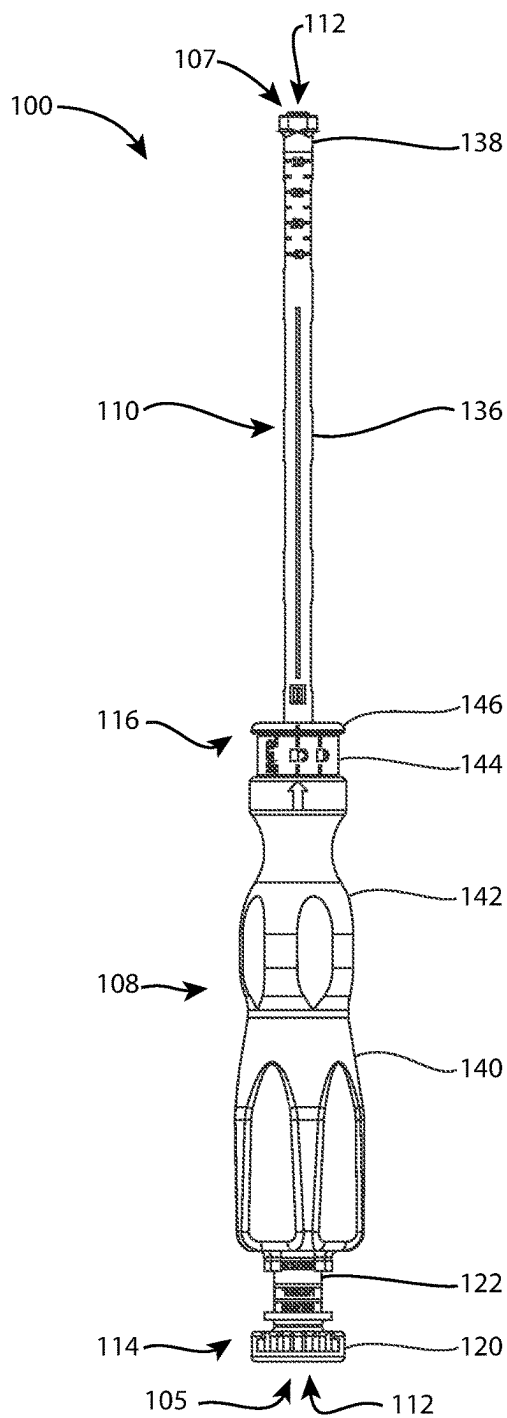


FIG. 1

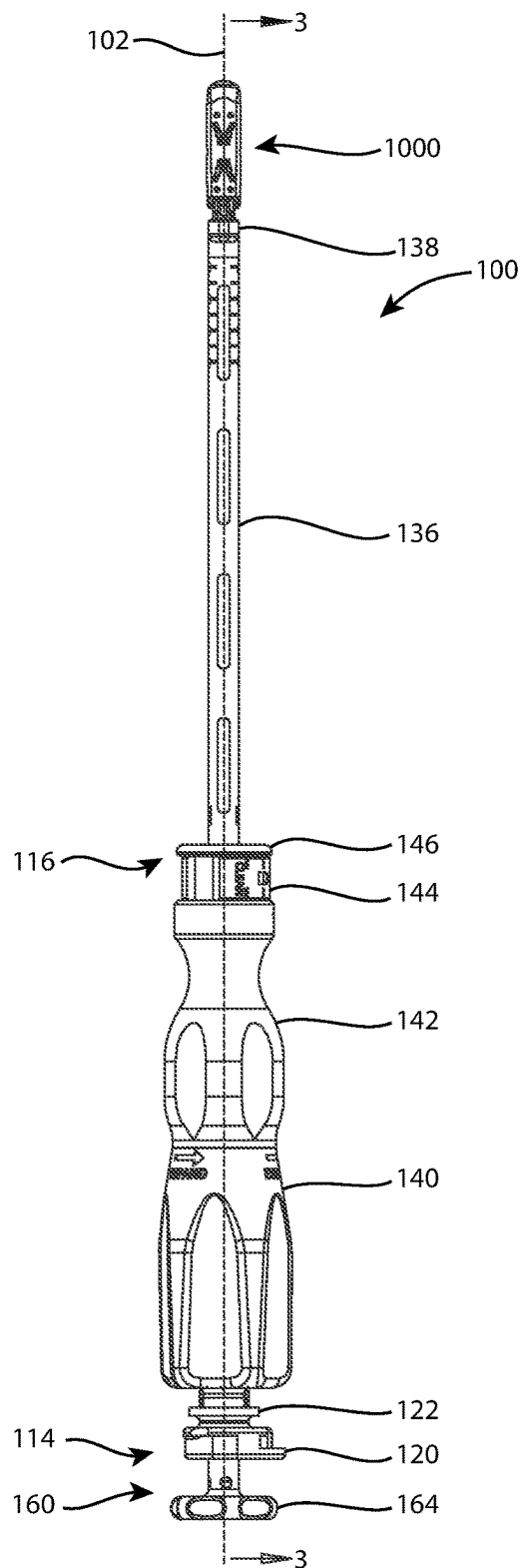


FIG. 2

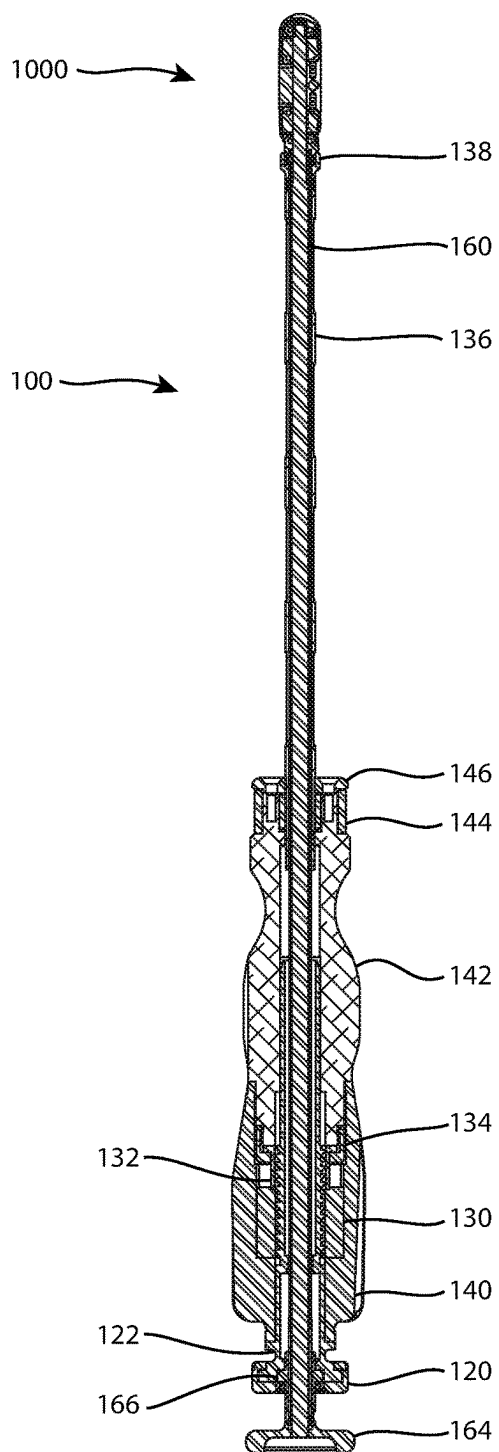


FIG. 3

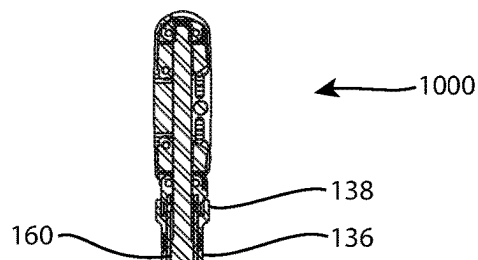


FIG. 4

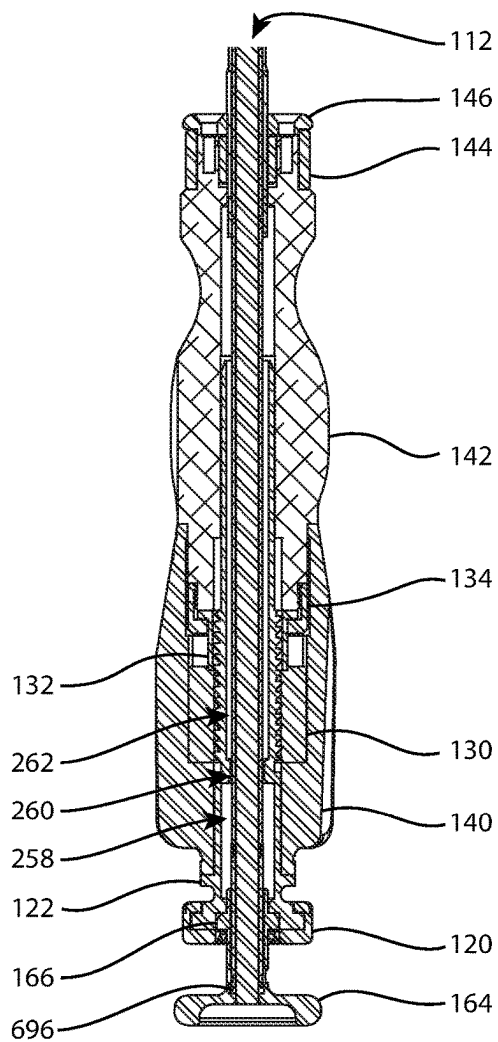


FIG. 5

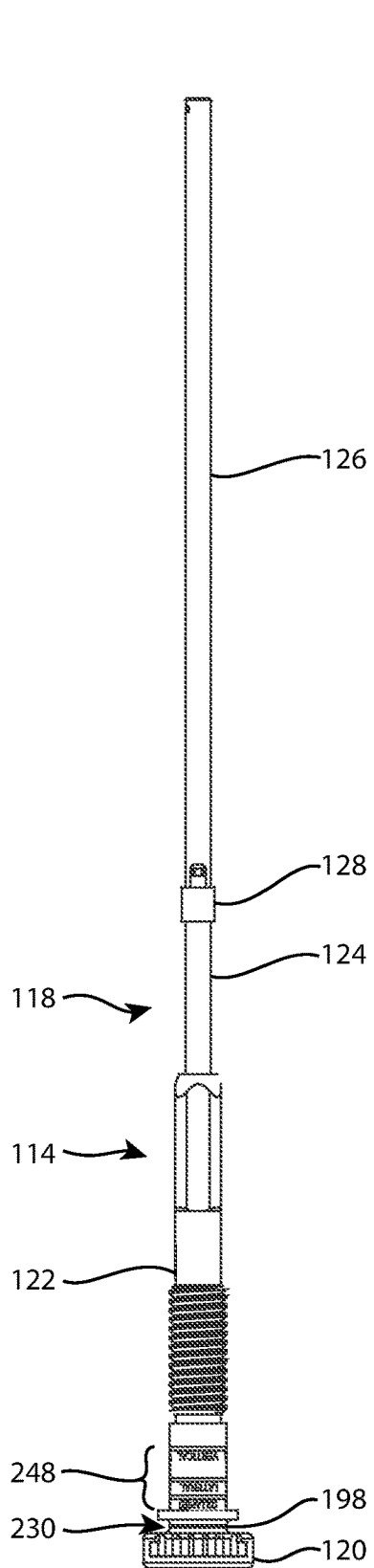


FIG. 6

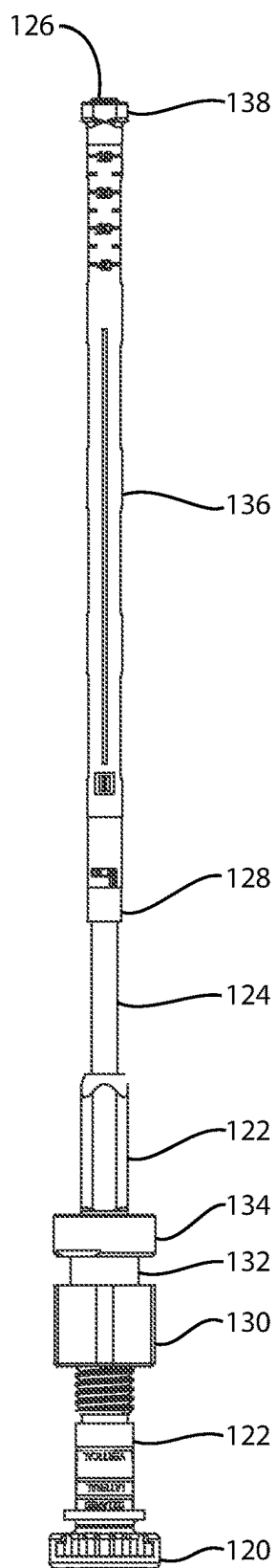


FIG. 7

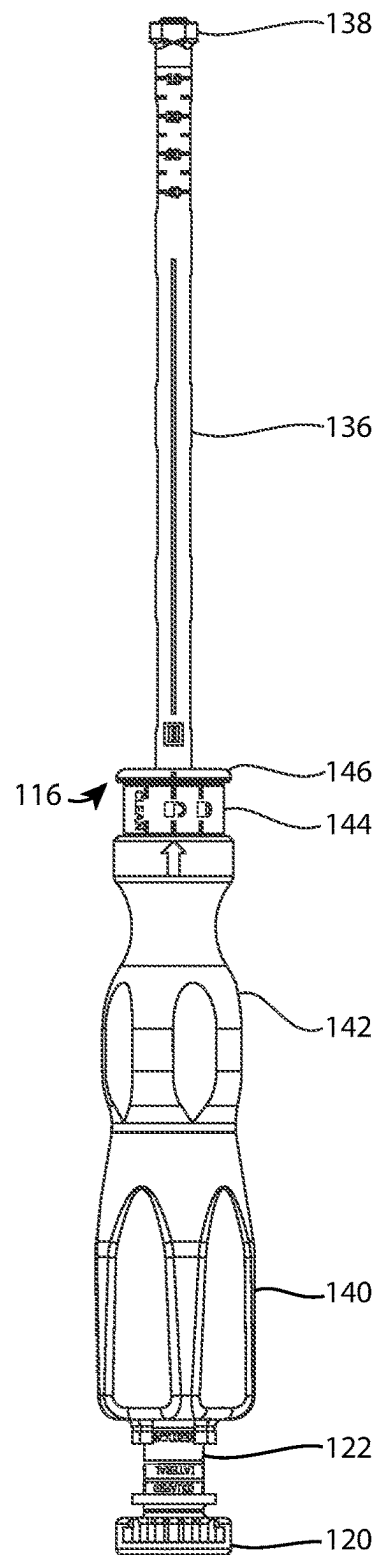
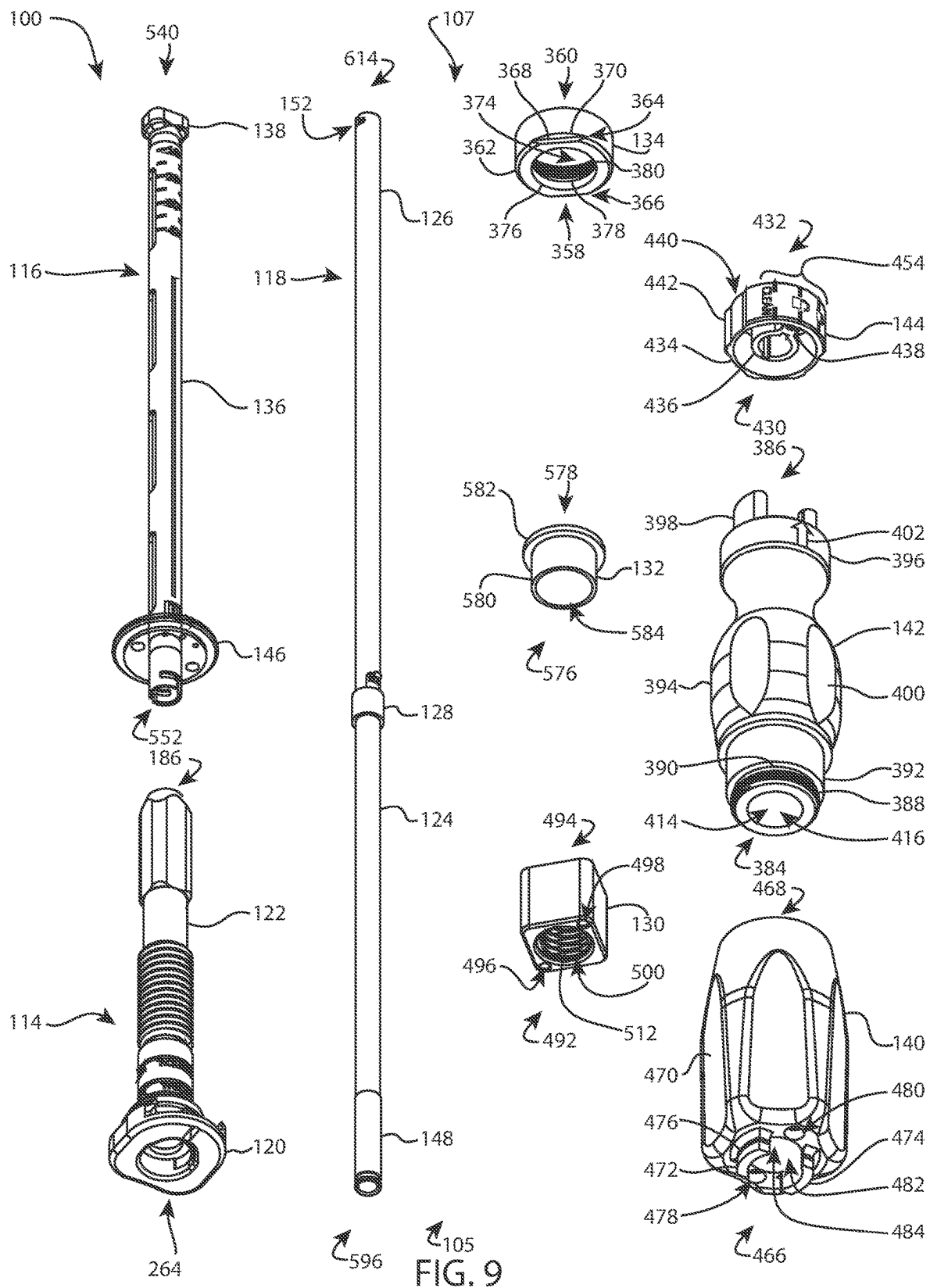
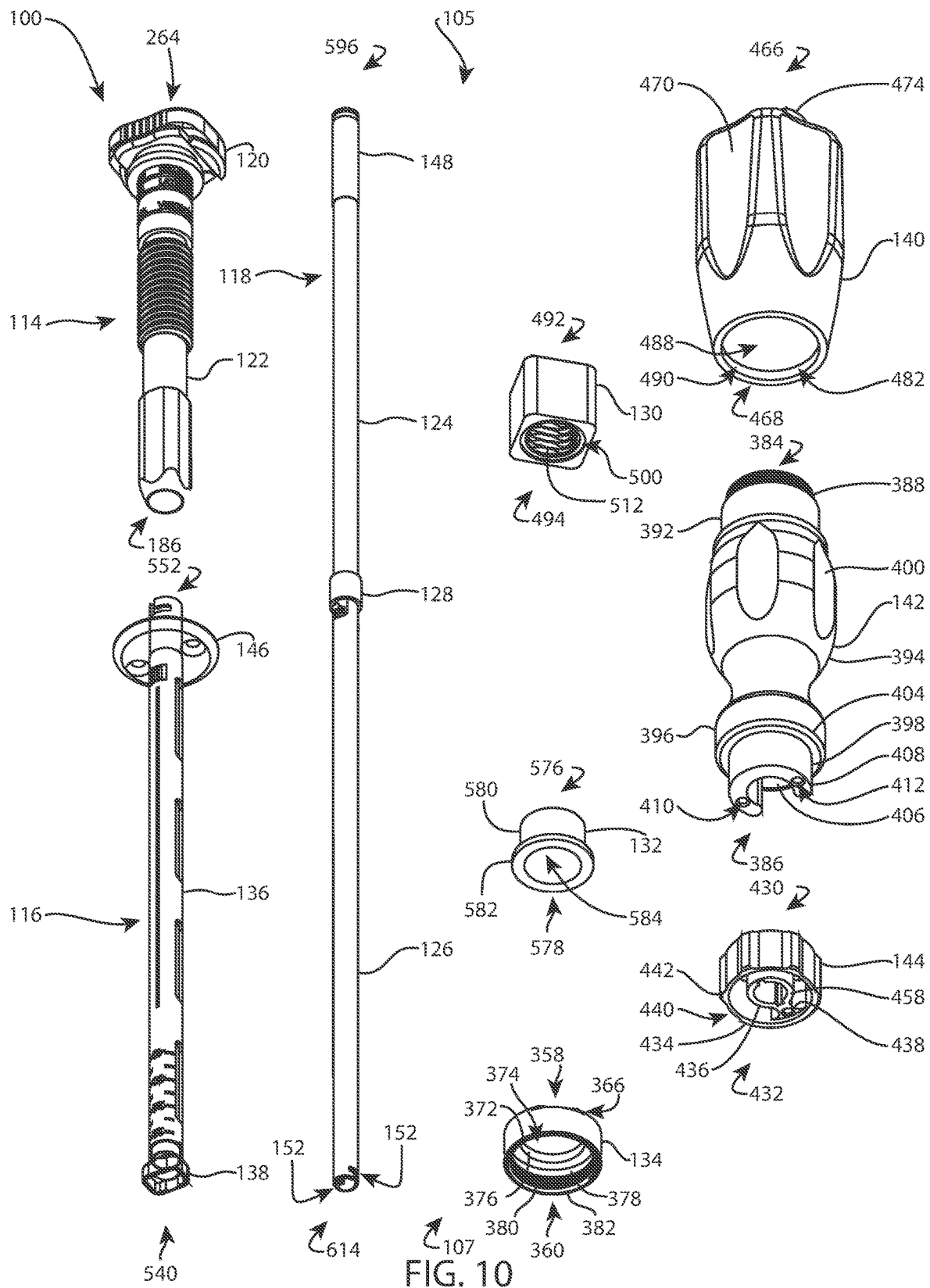


FIG. 8





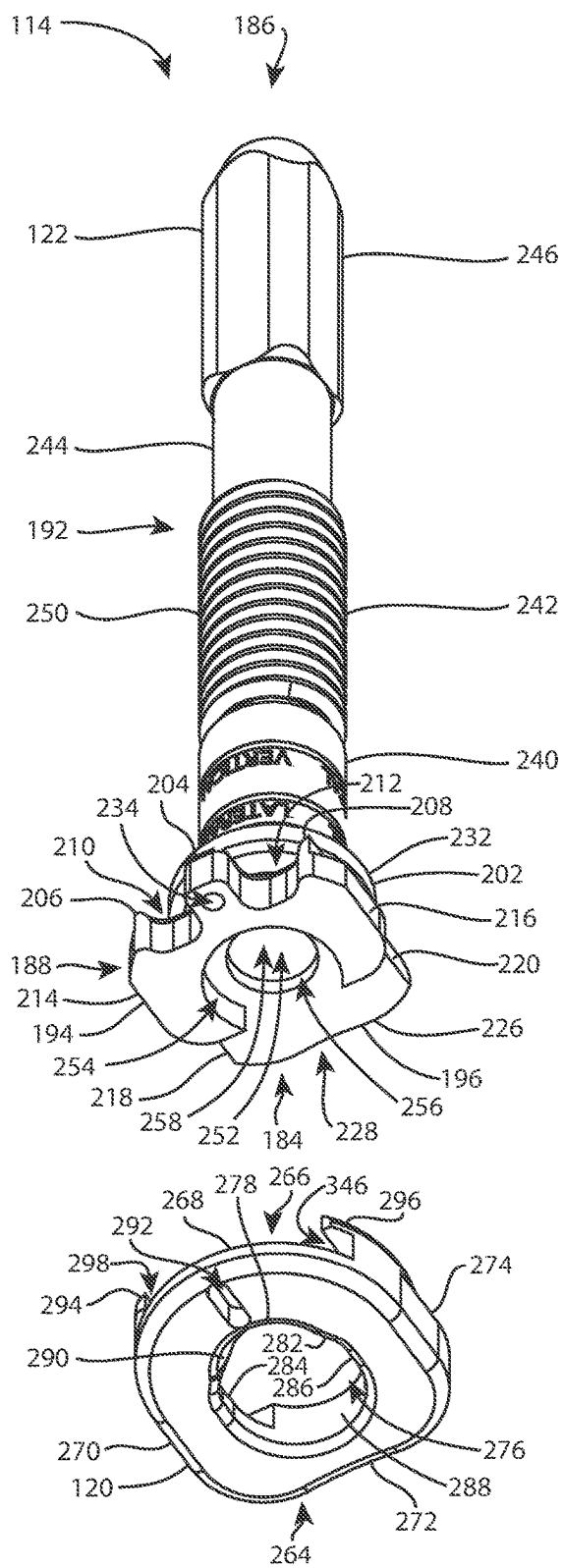


FIG. 11

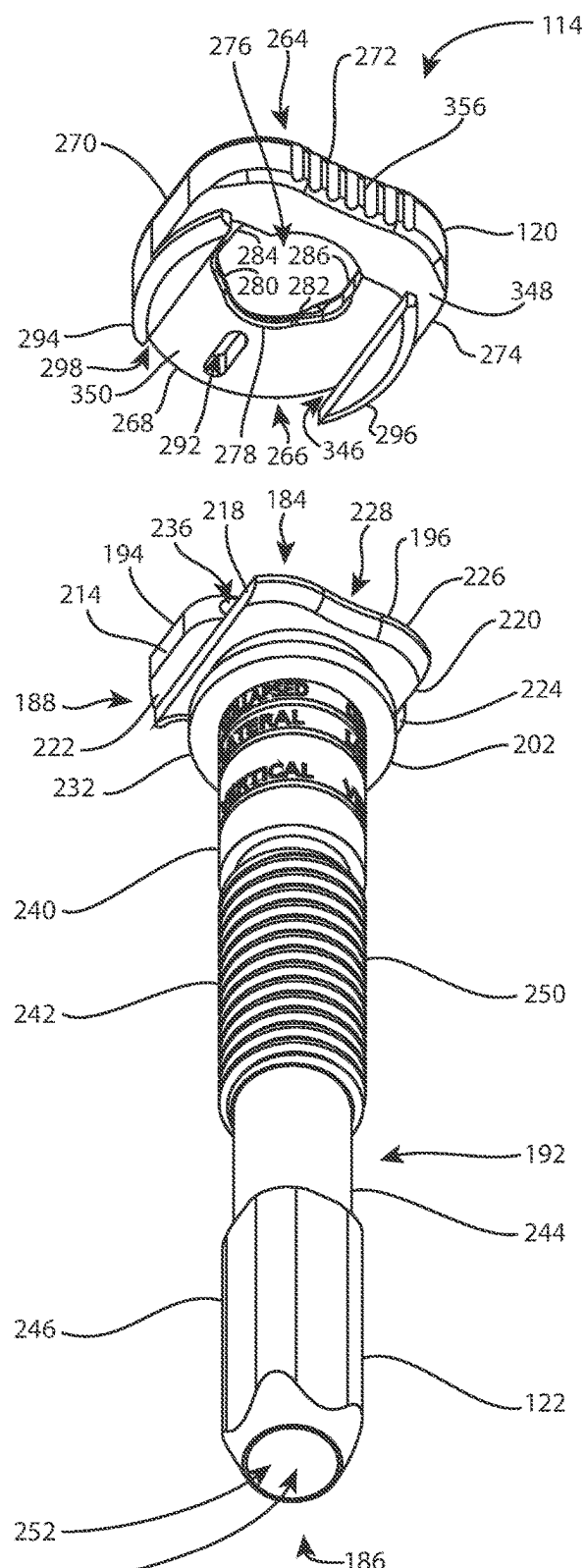


FIG. 12

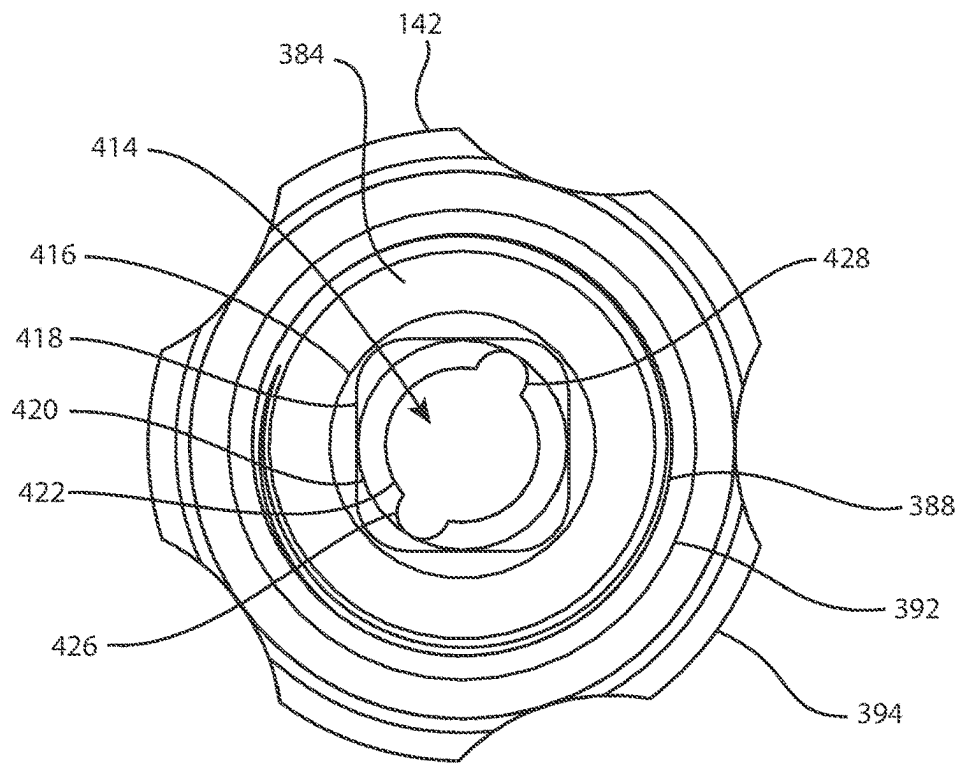


FIG. 13

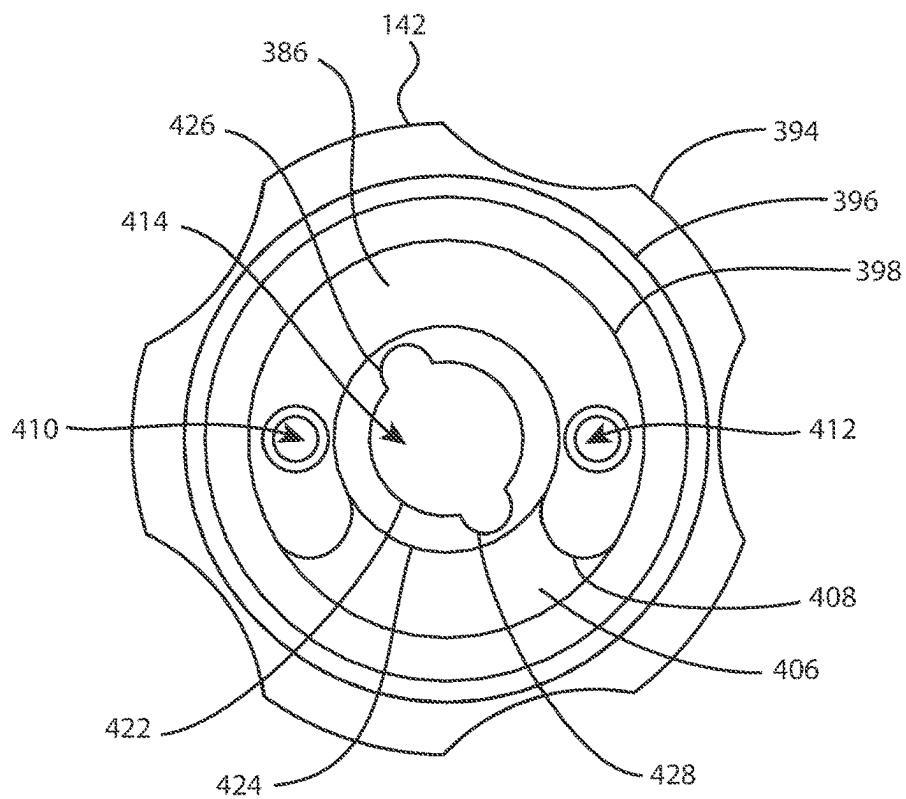


FIG. 14

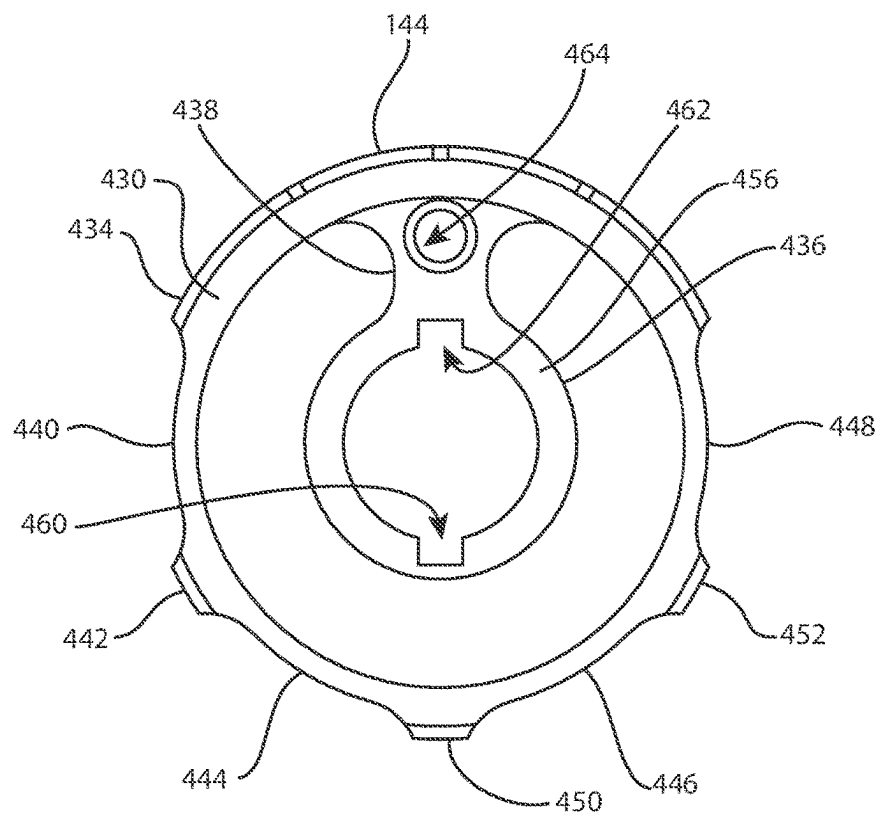


FIG. 15

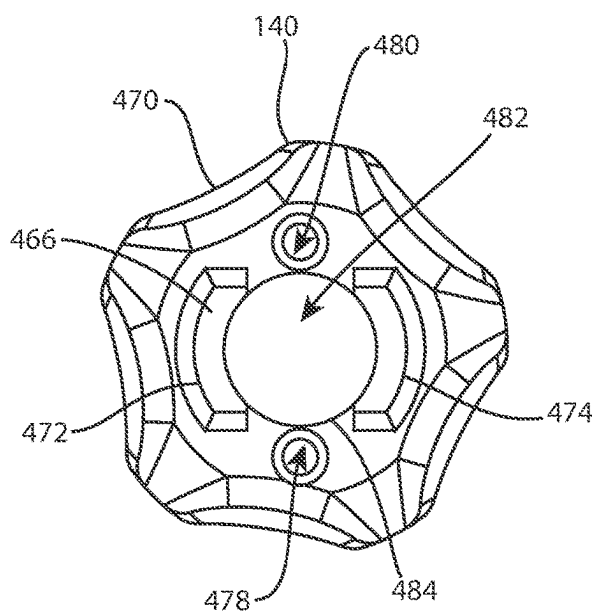


FIG. 16

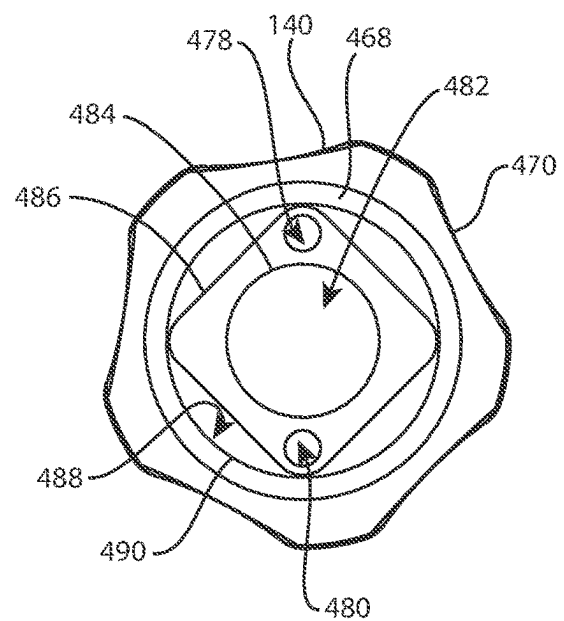


FIG. 17

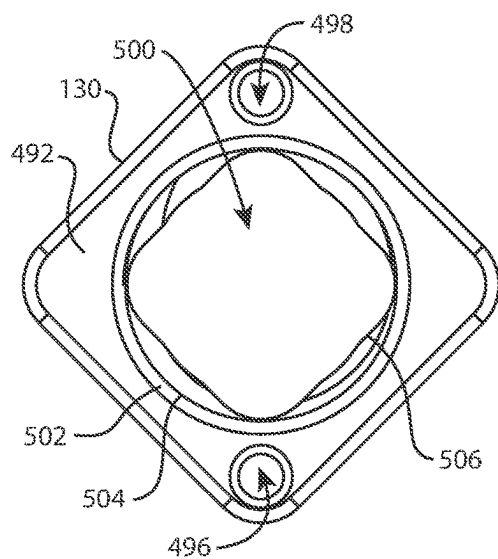


FIG. 18

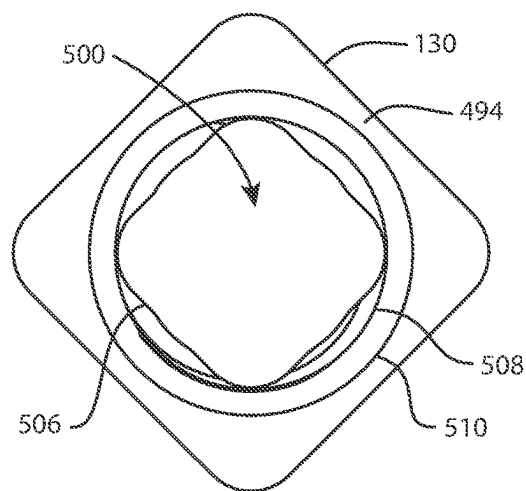


FIG. 19

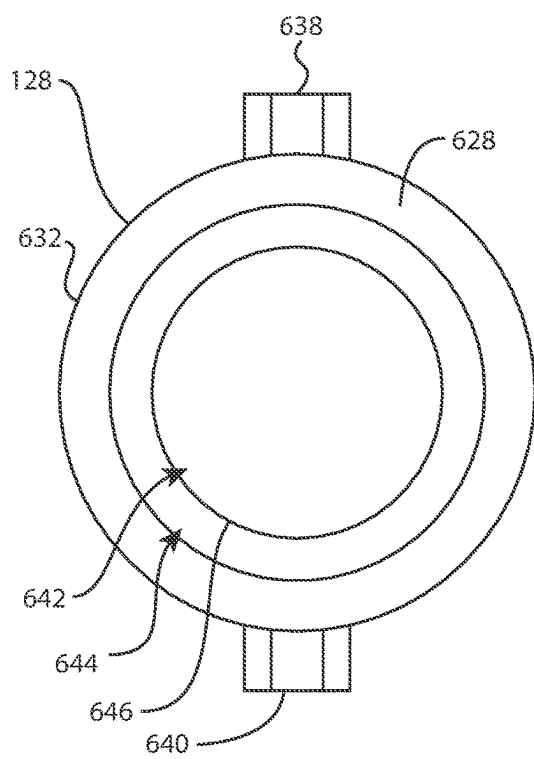


FIG. 20

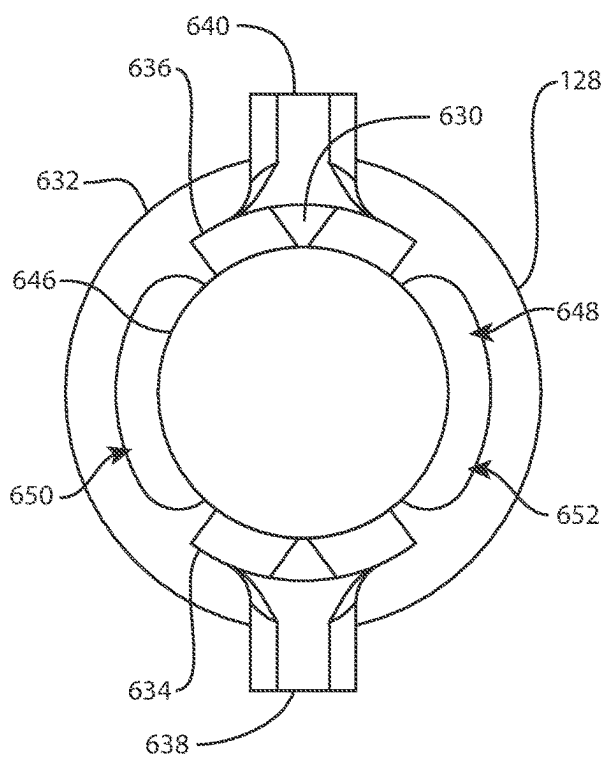
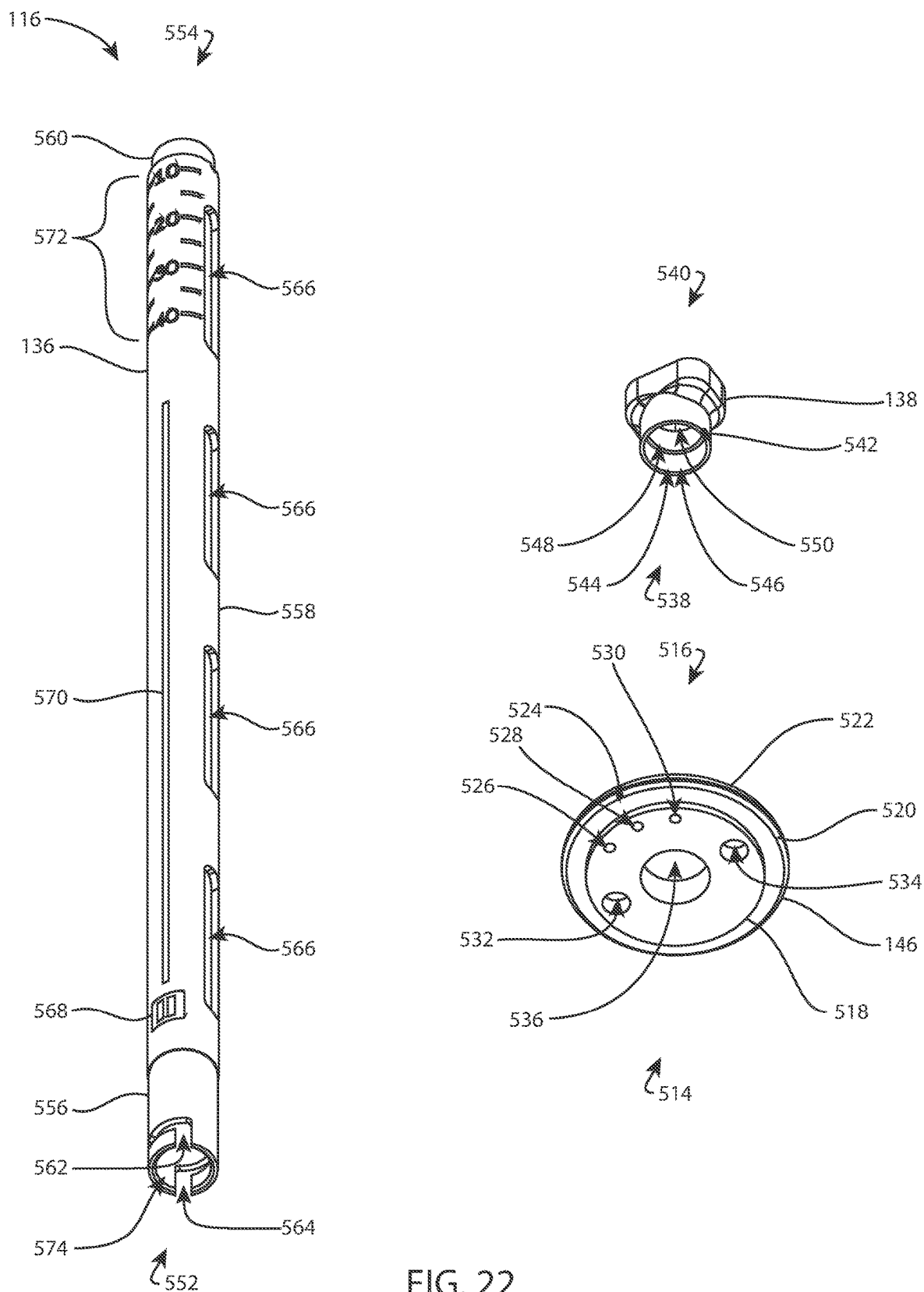


FIG. 21



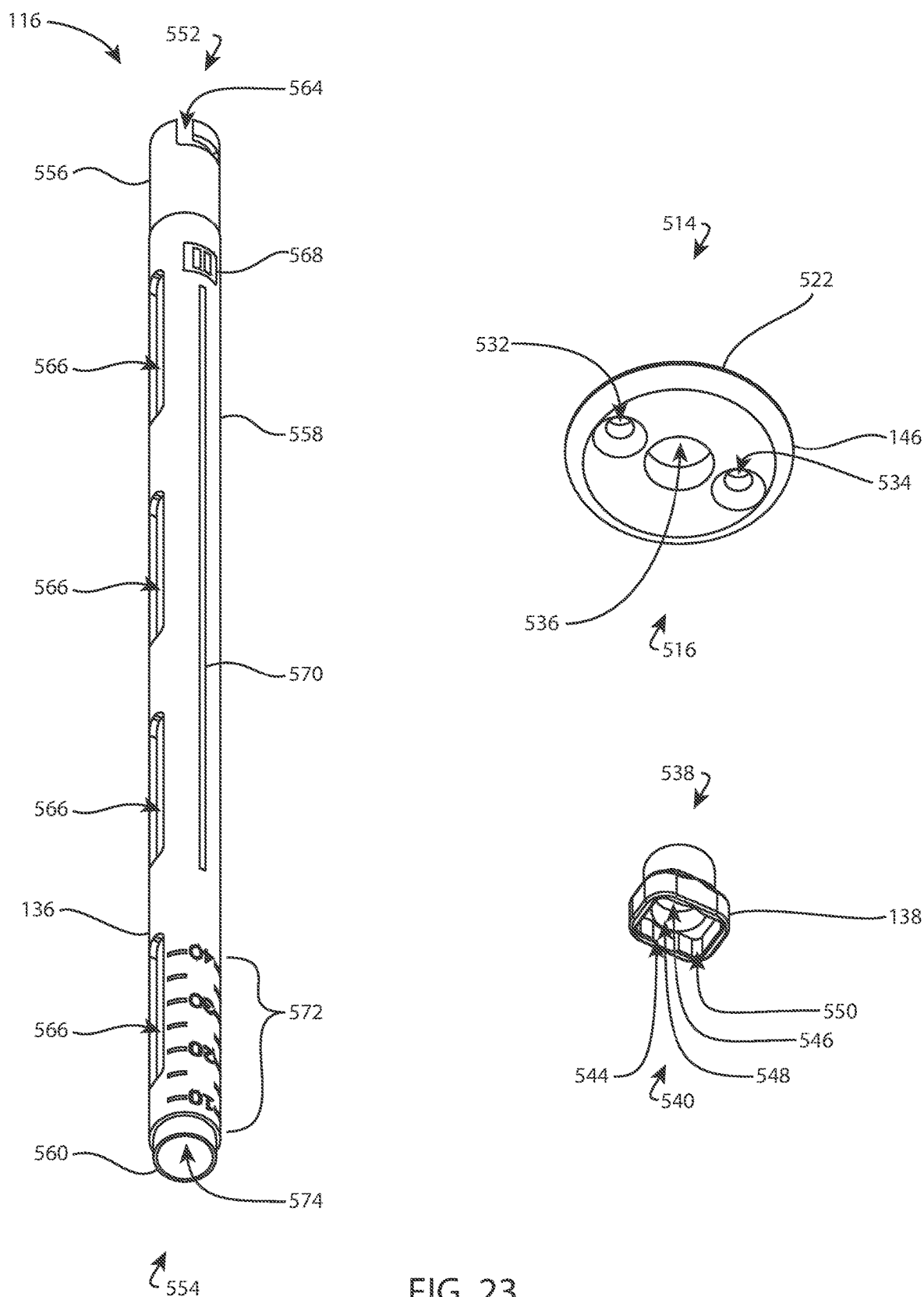


FIG. 23

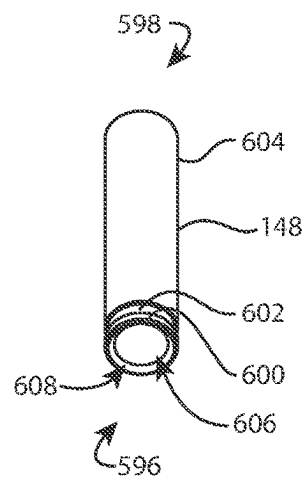
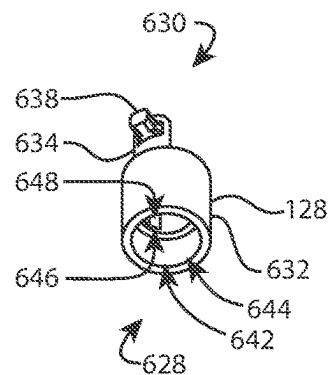
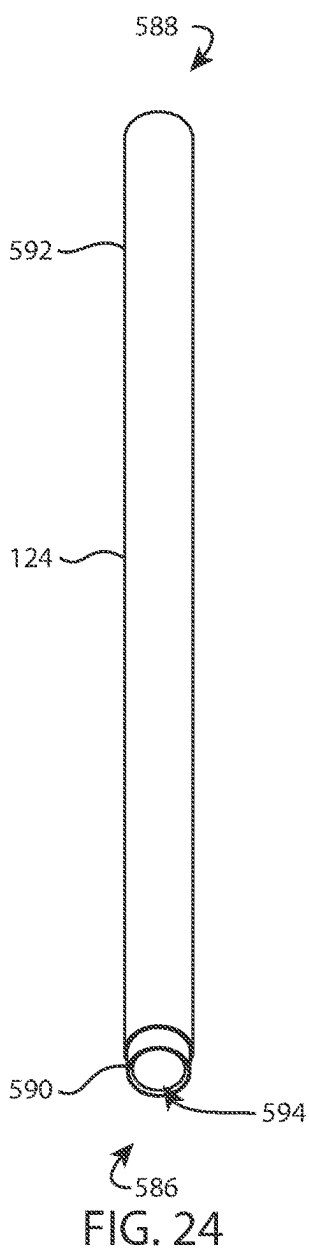
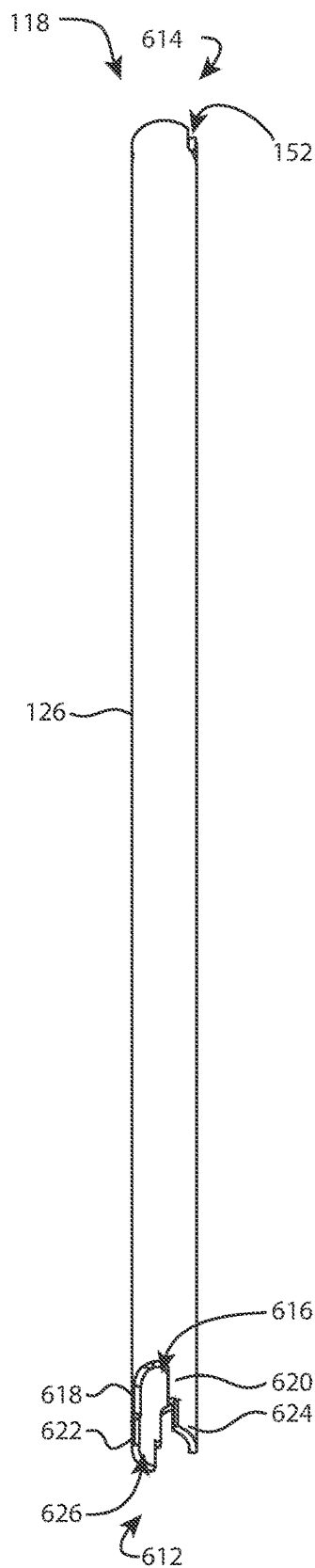
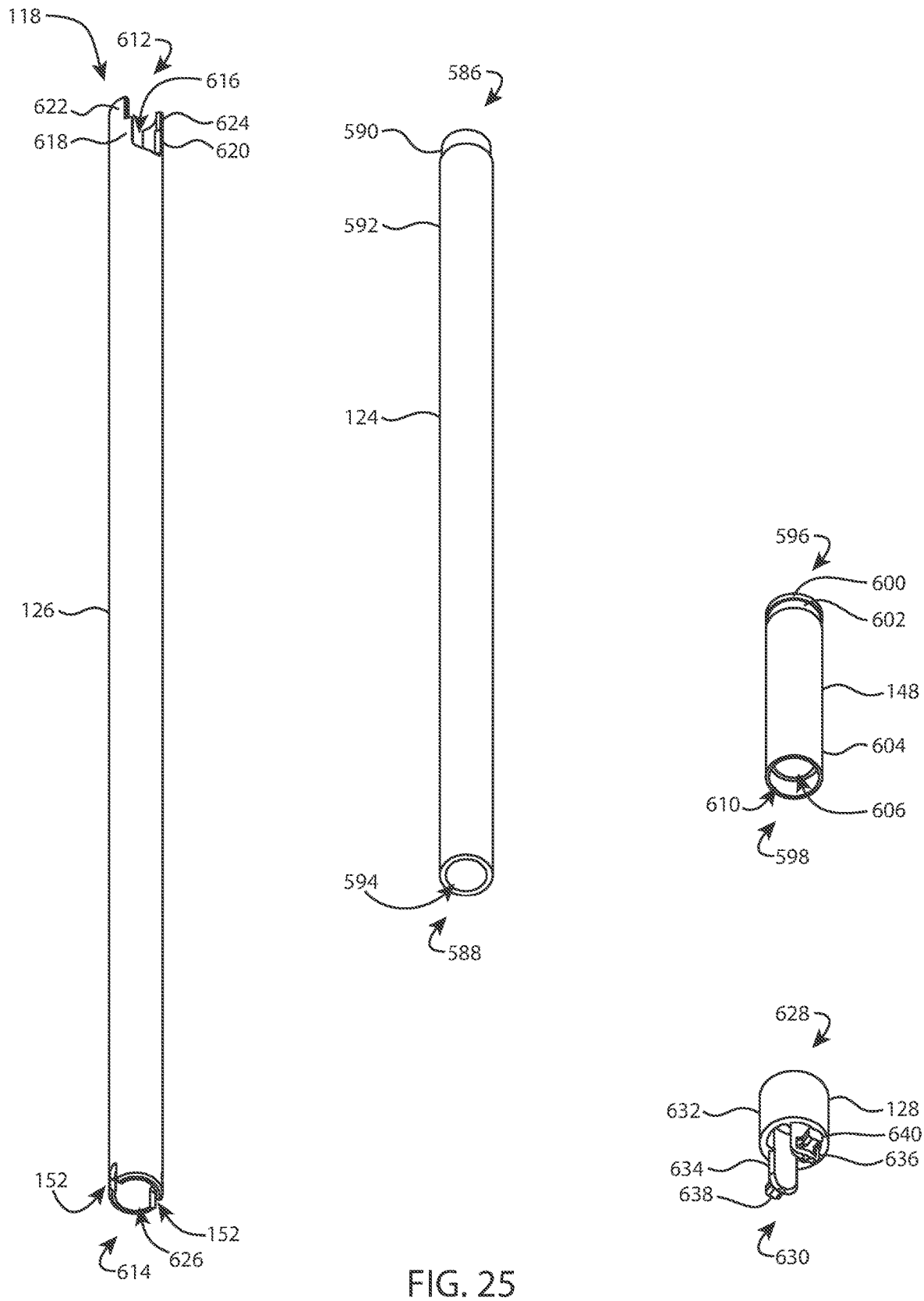


FIG. 24



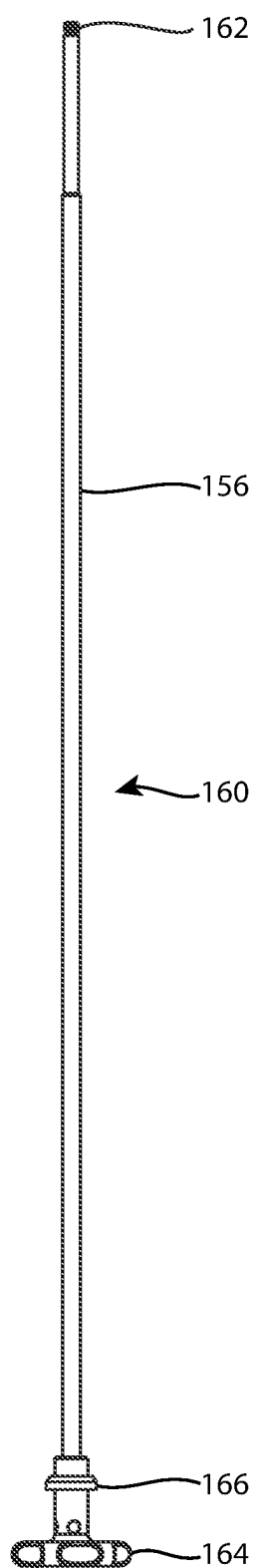


FIG. 26

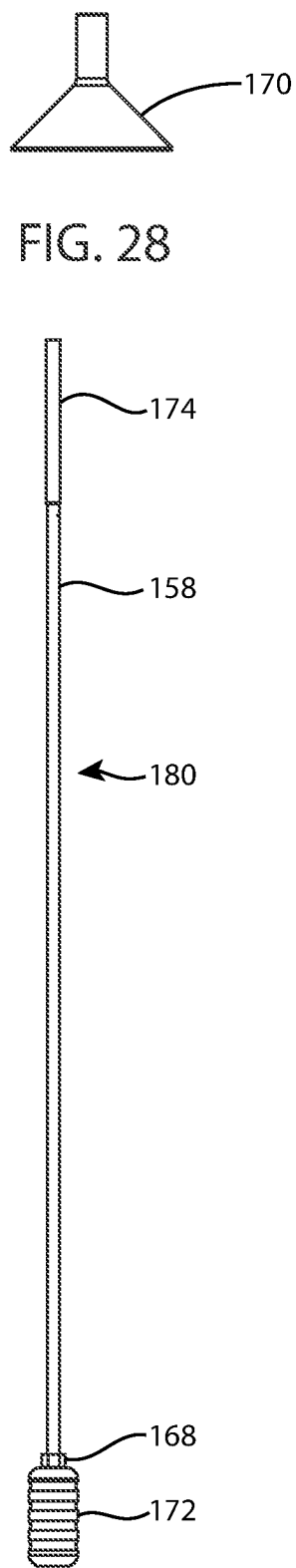


FIG. 27

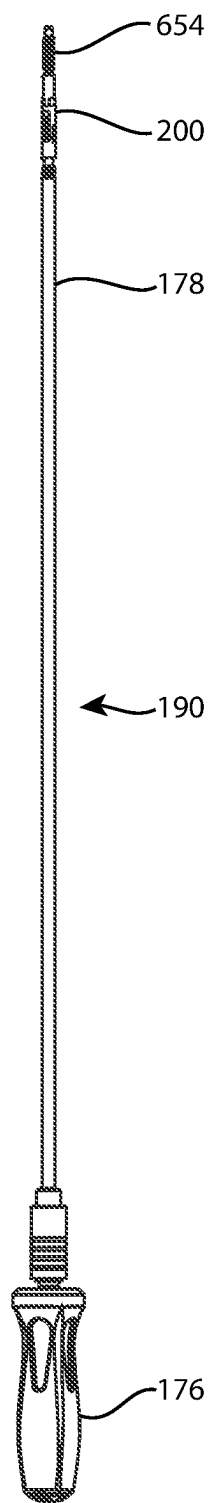


FIG. 29

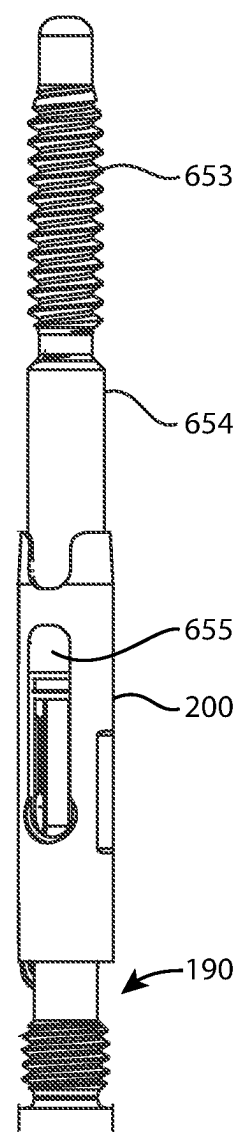
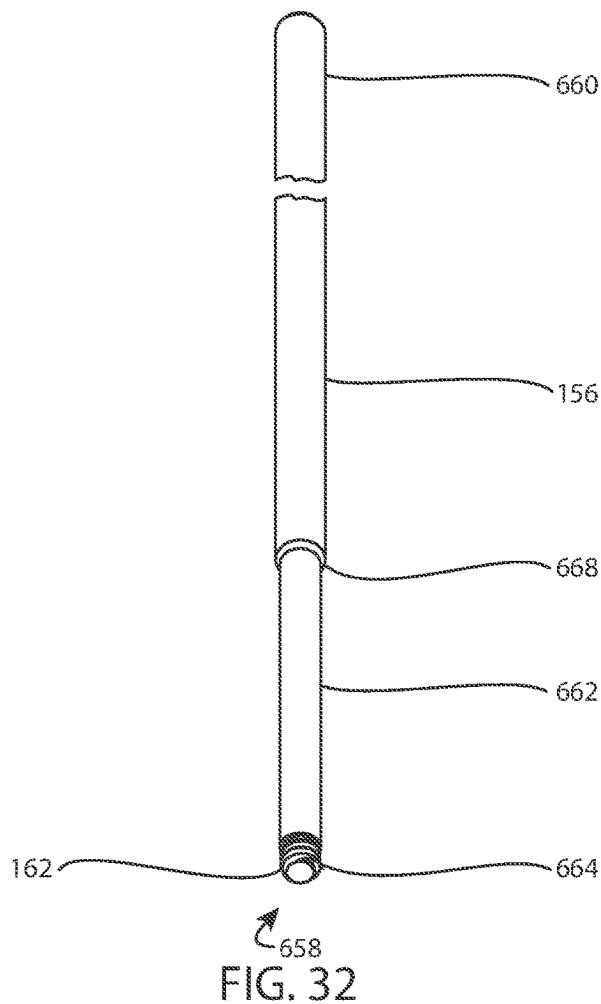
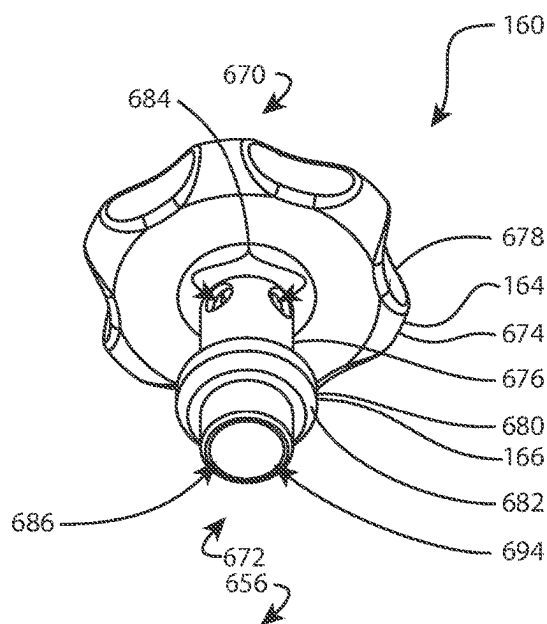
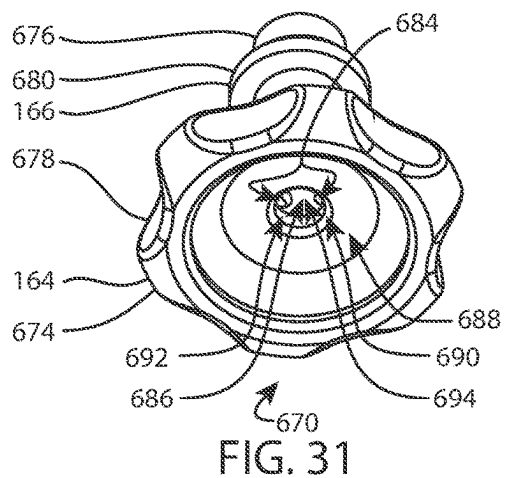
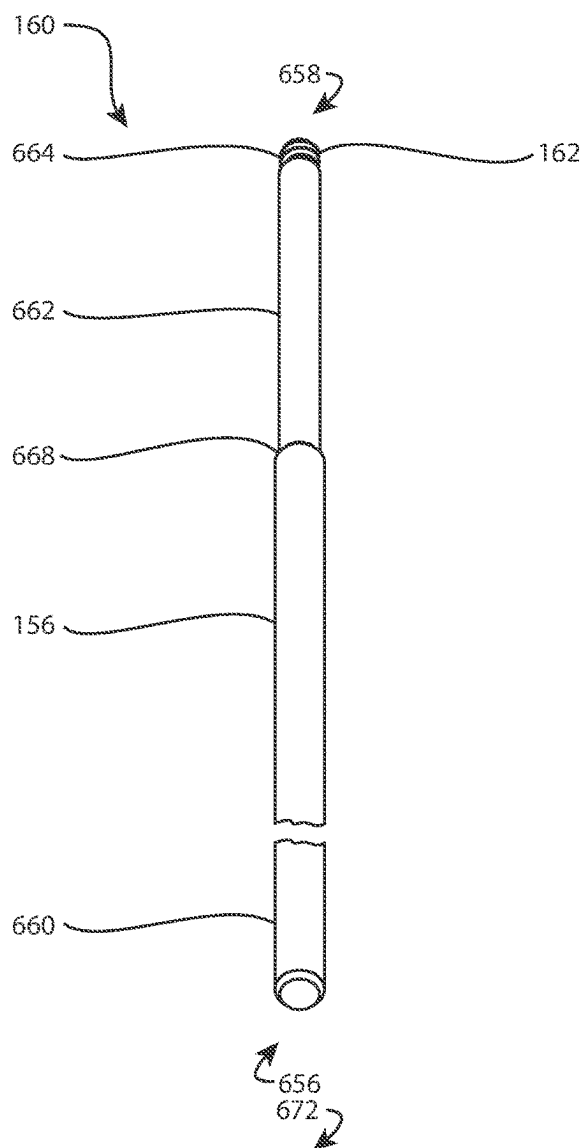


FIG. 30



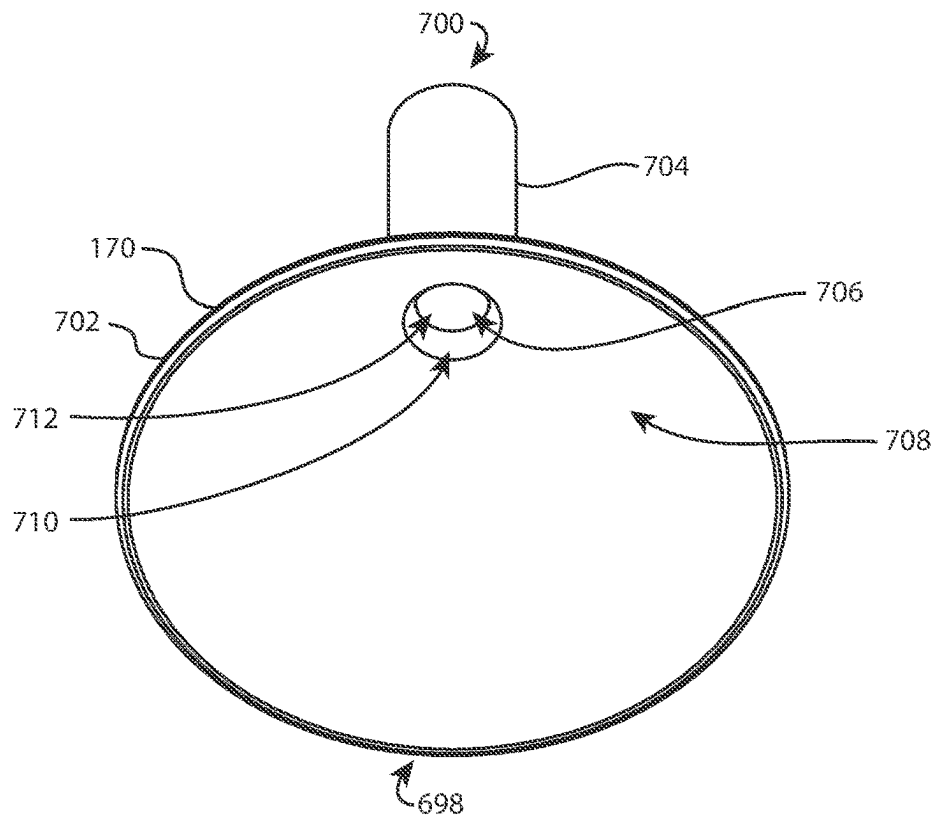


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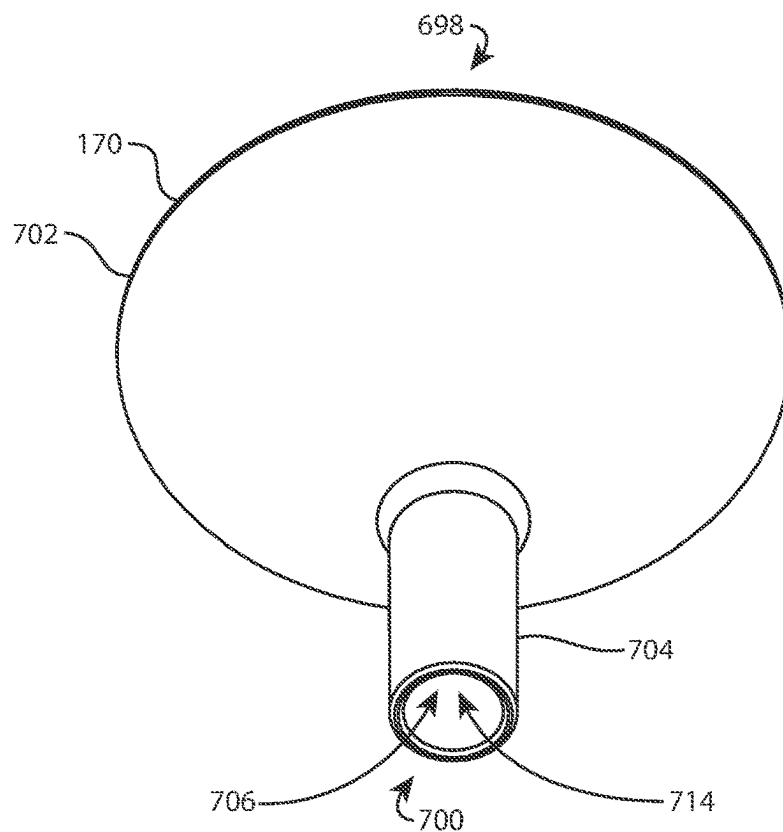
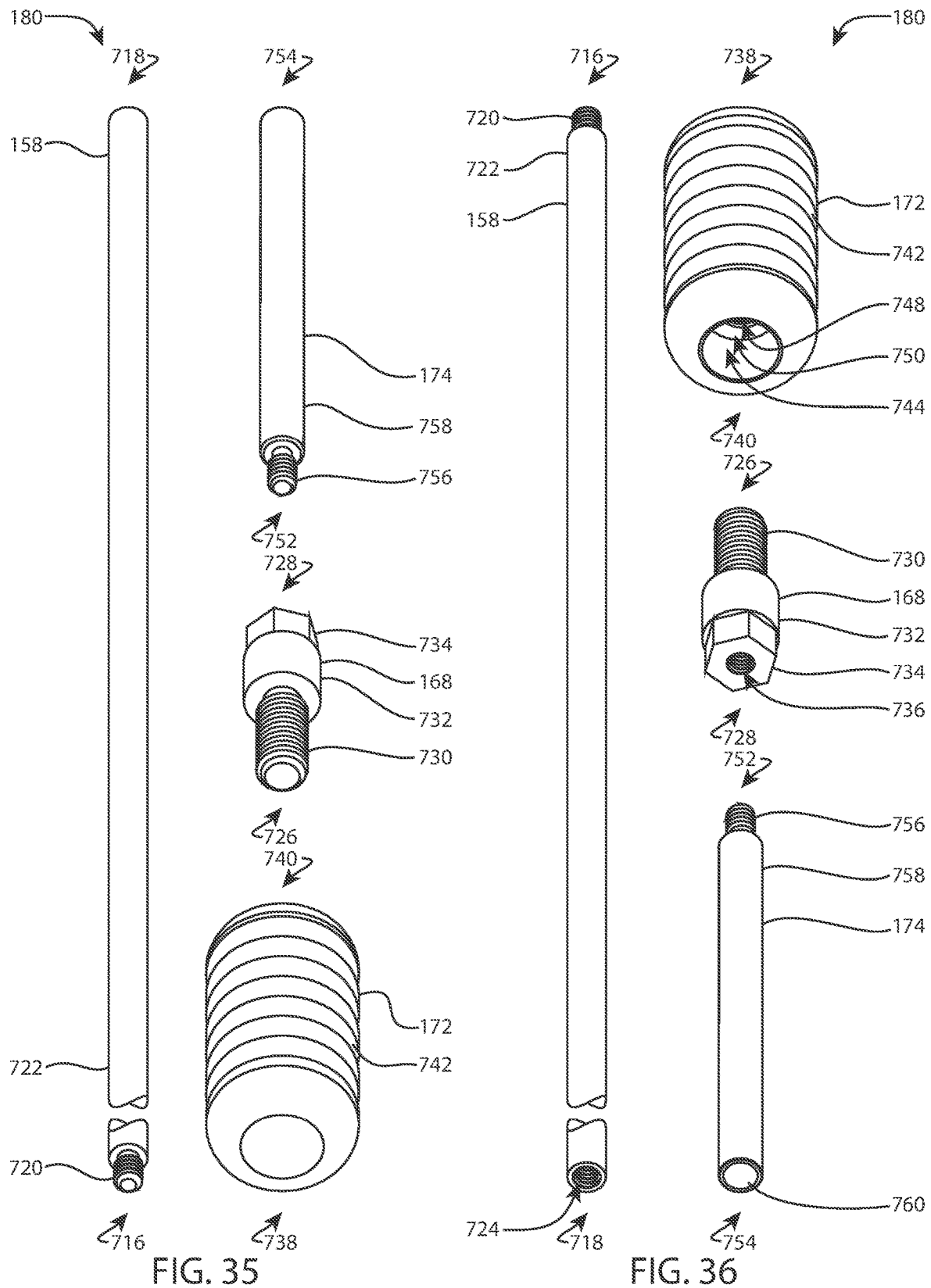


FIG. 34



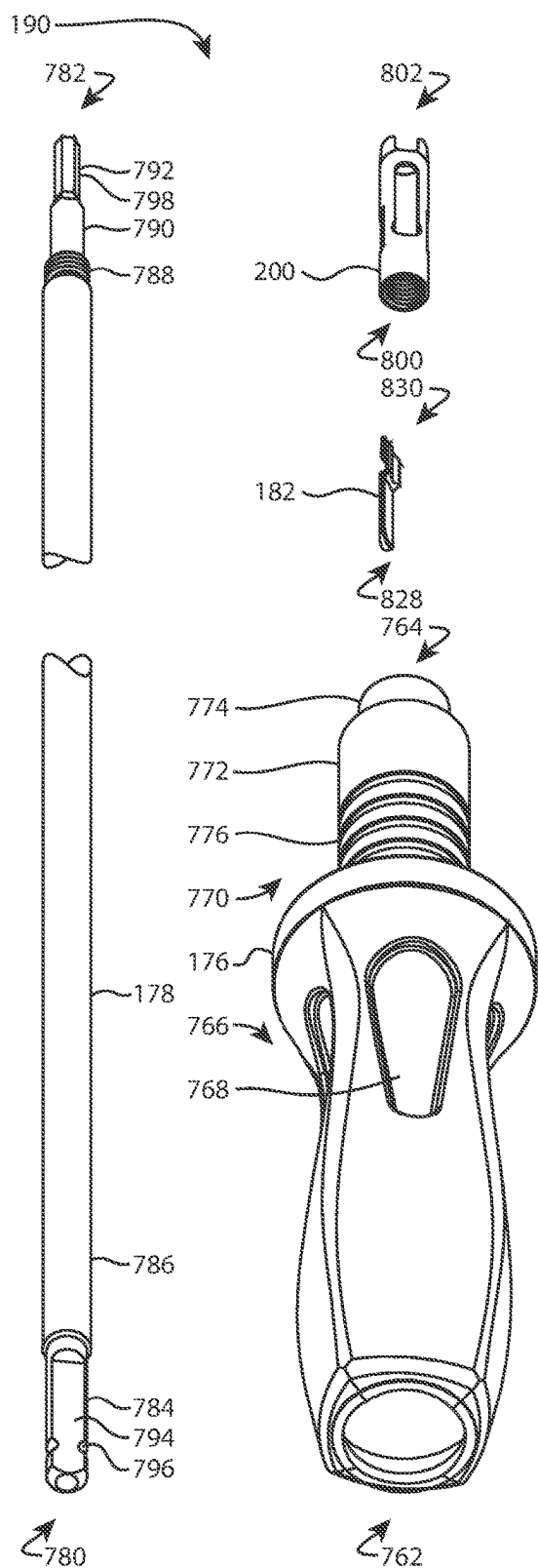


FIG. 37

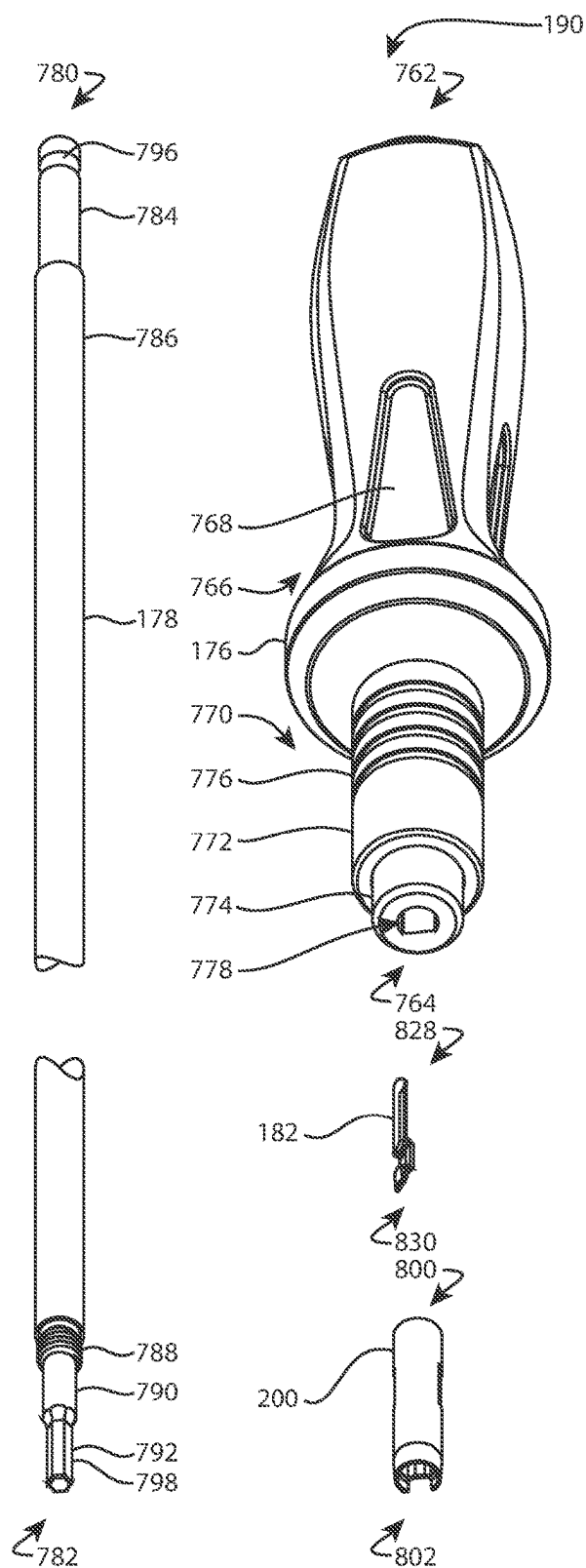
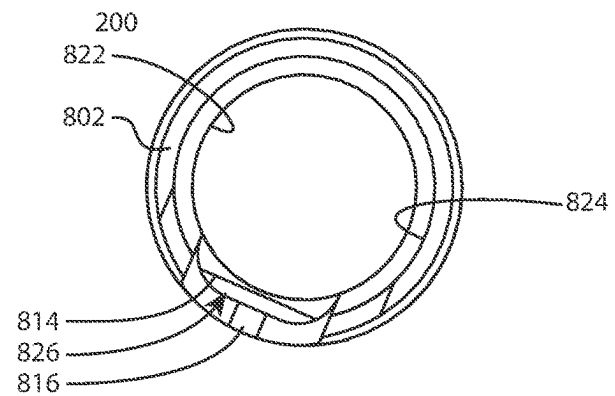
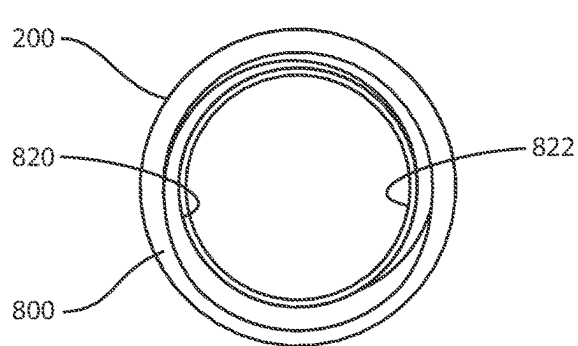
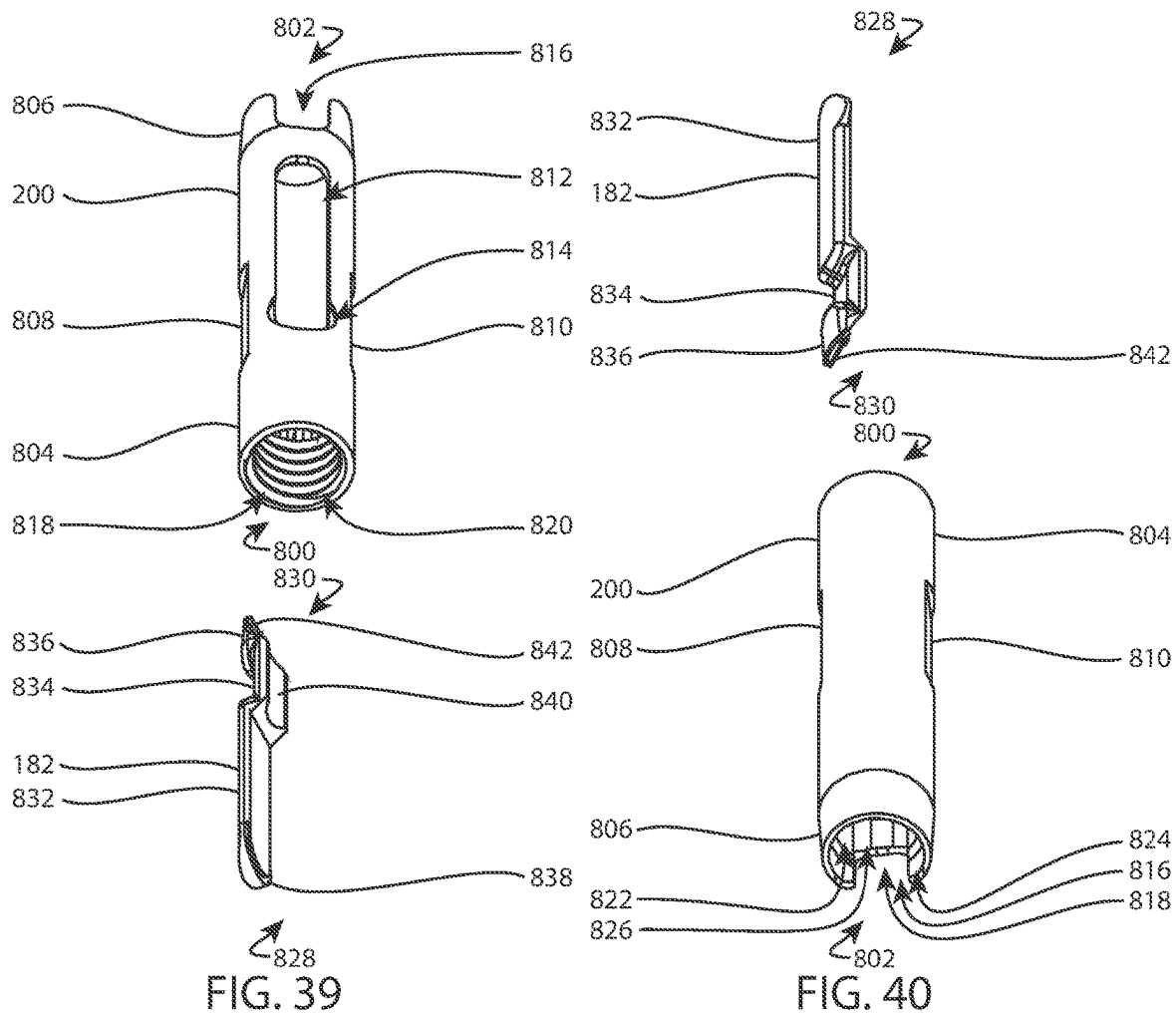


FIG. 38



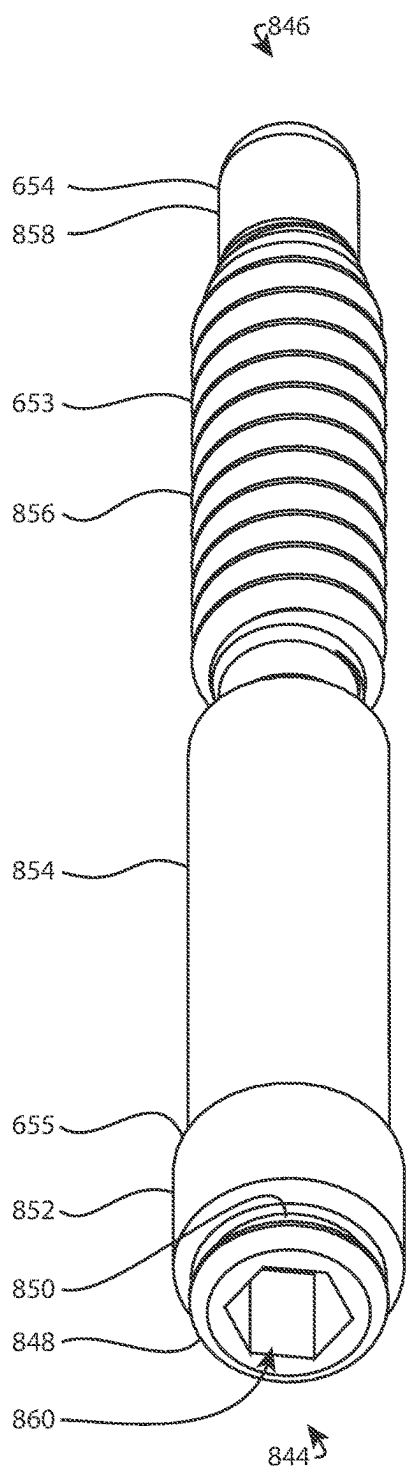


FIG. 43

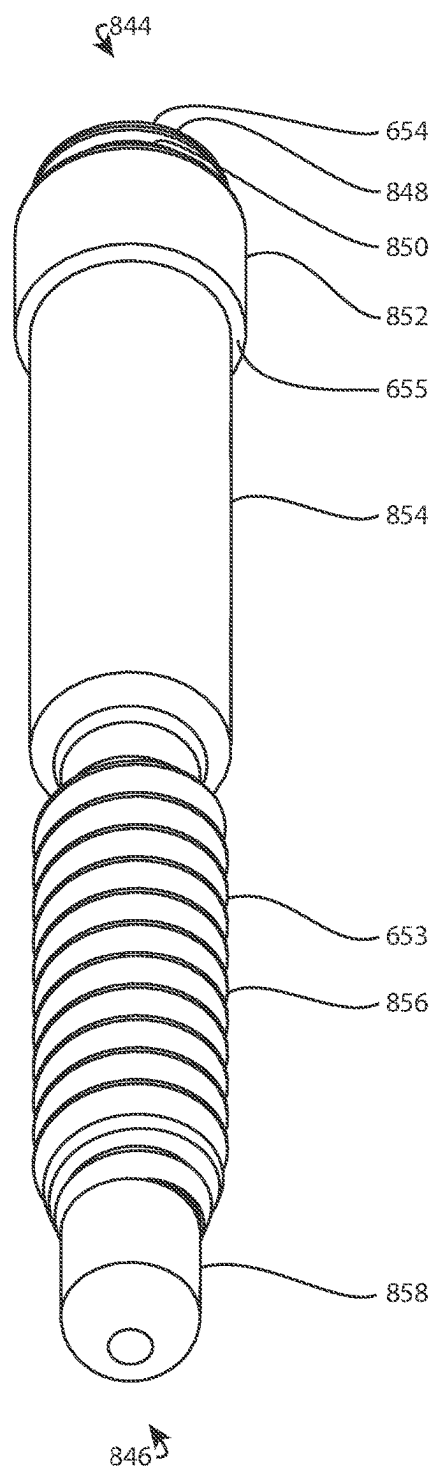
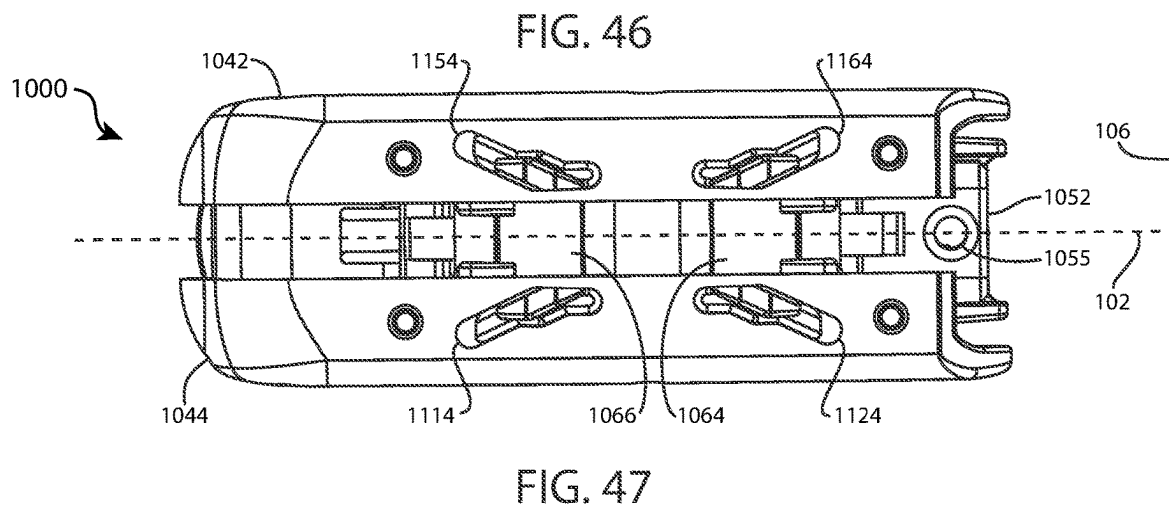
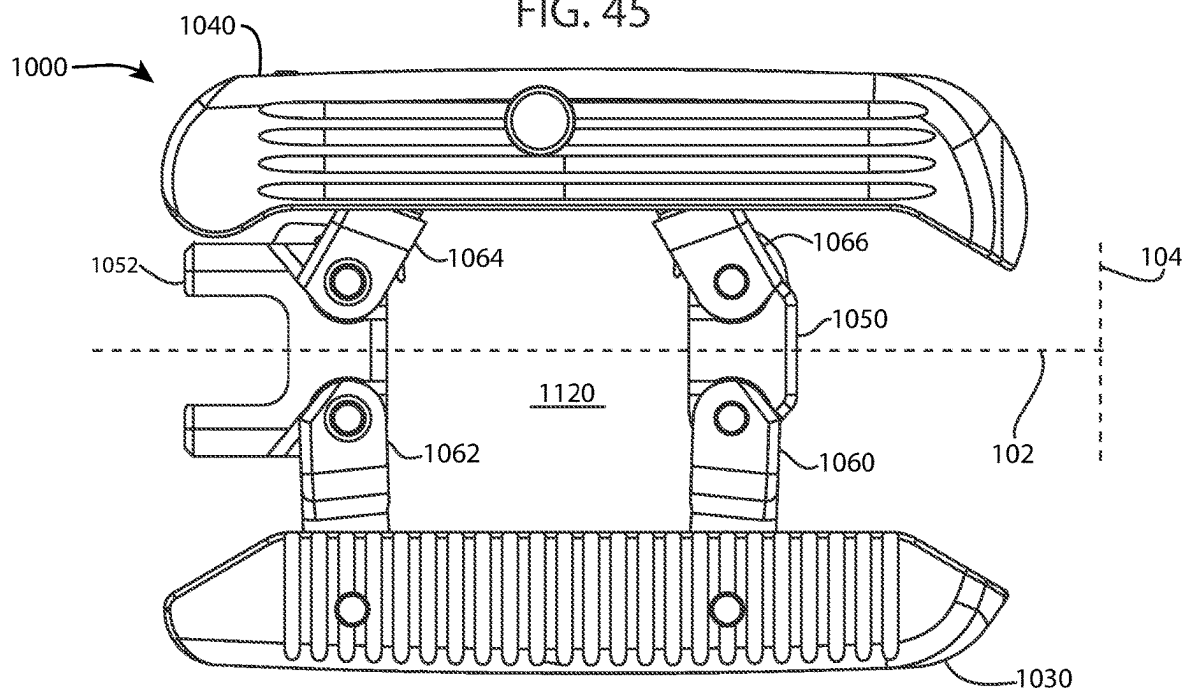
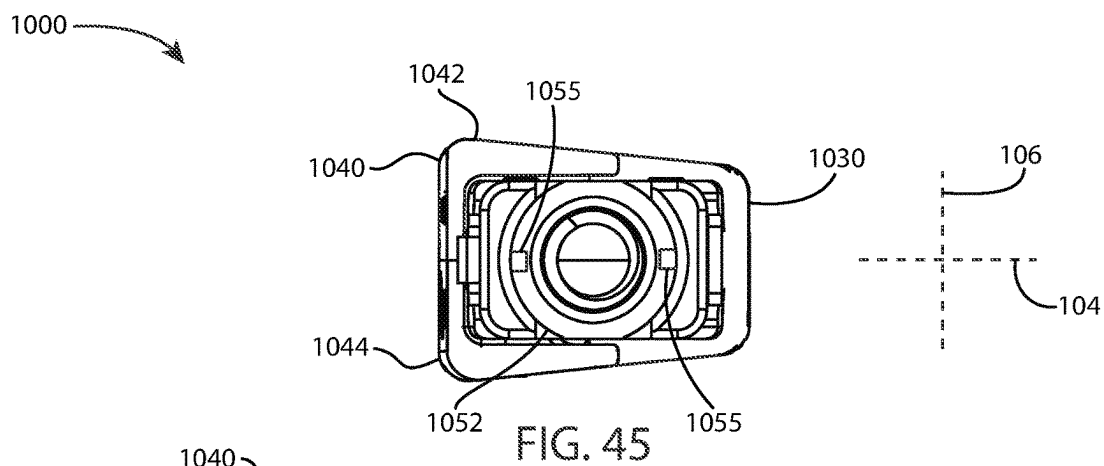


FIG. 44



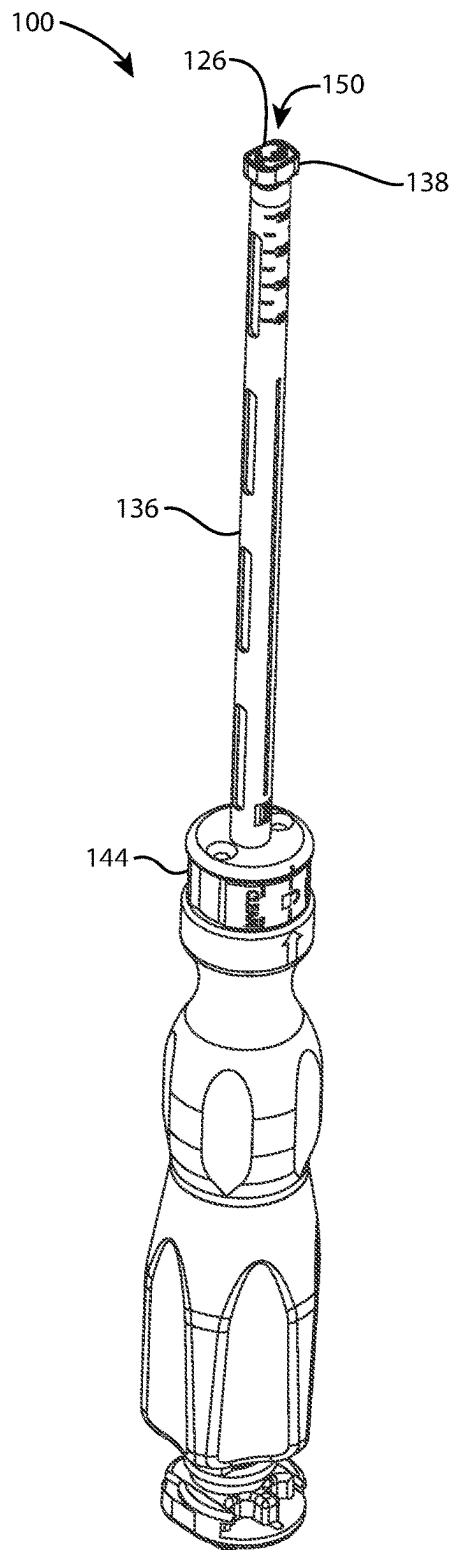


FIG. 48

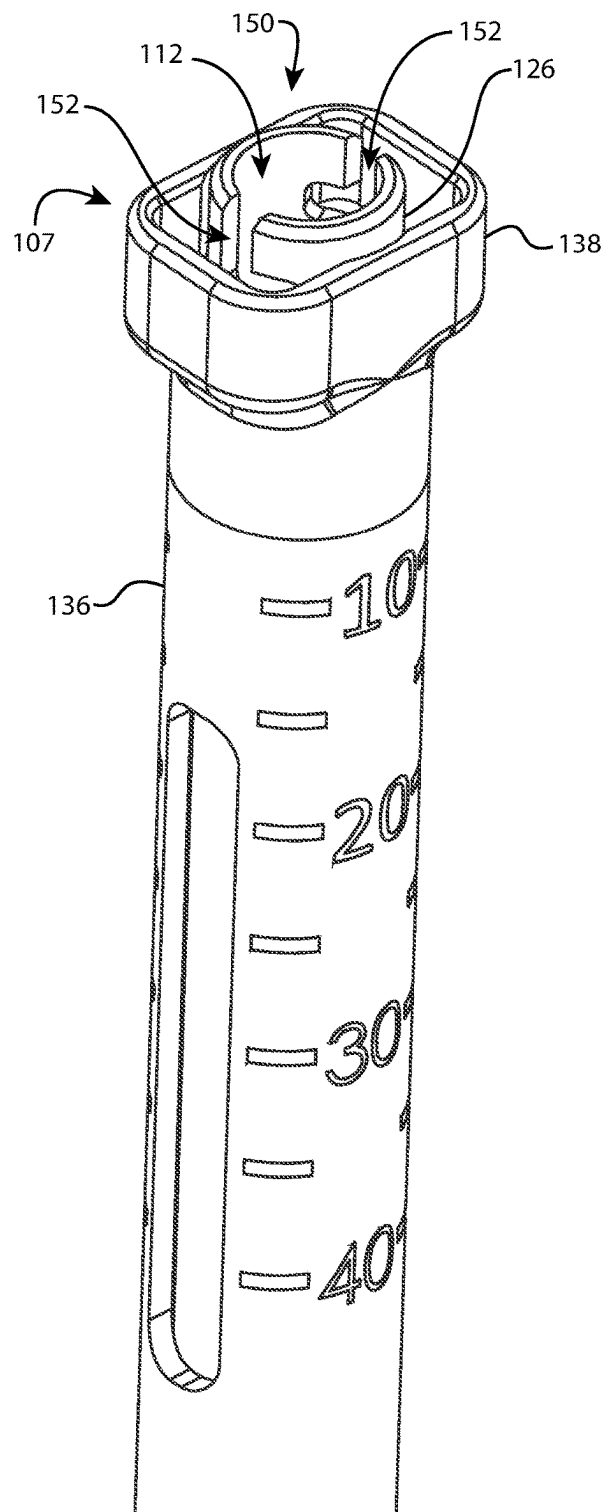


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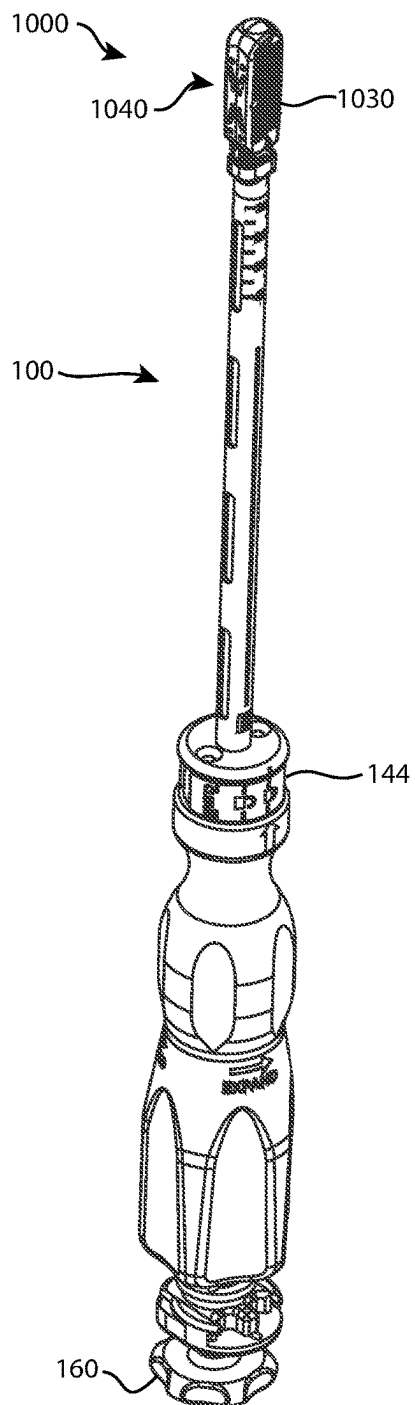


FIG. 50

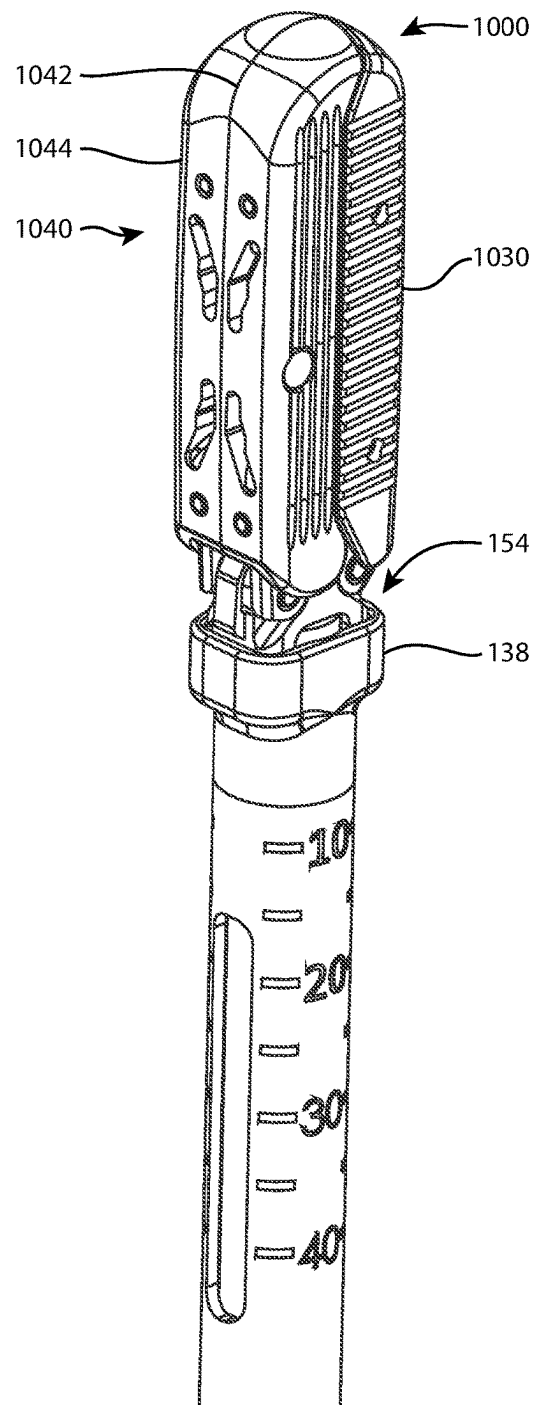


FIG. 51

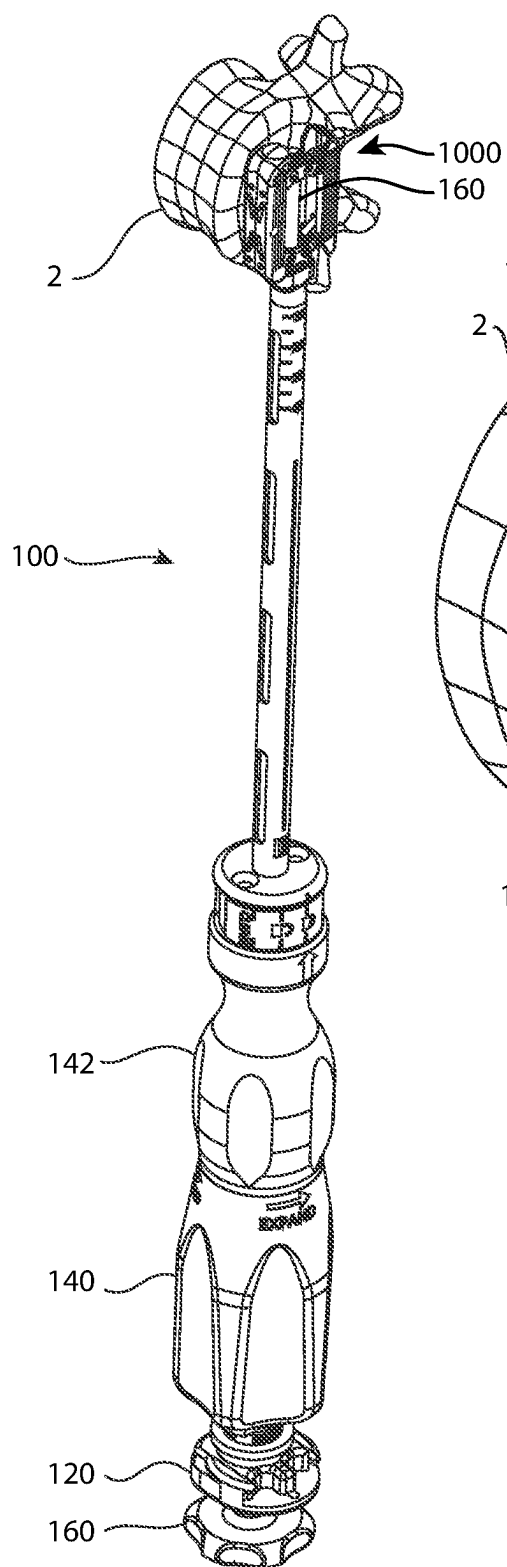


FIG. 52

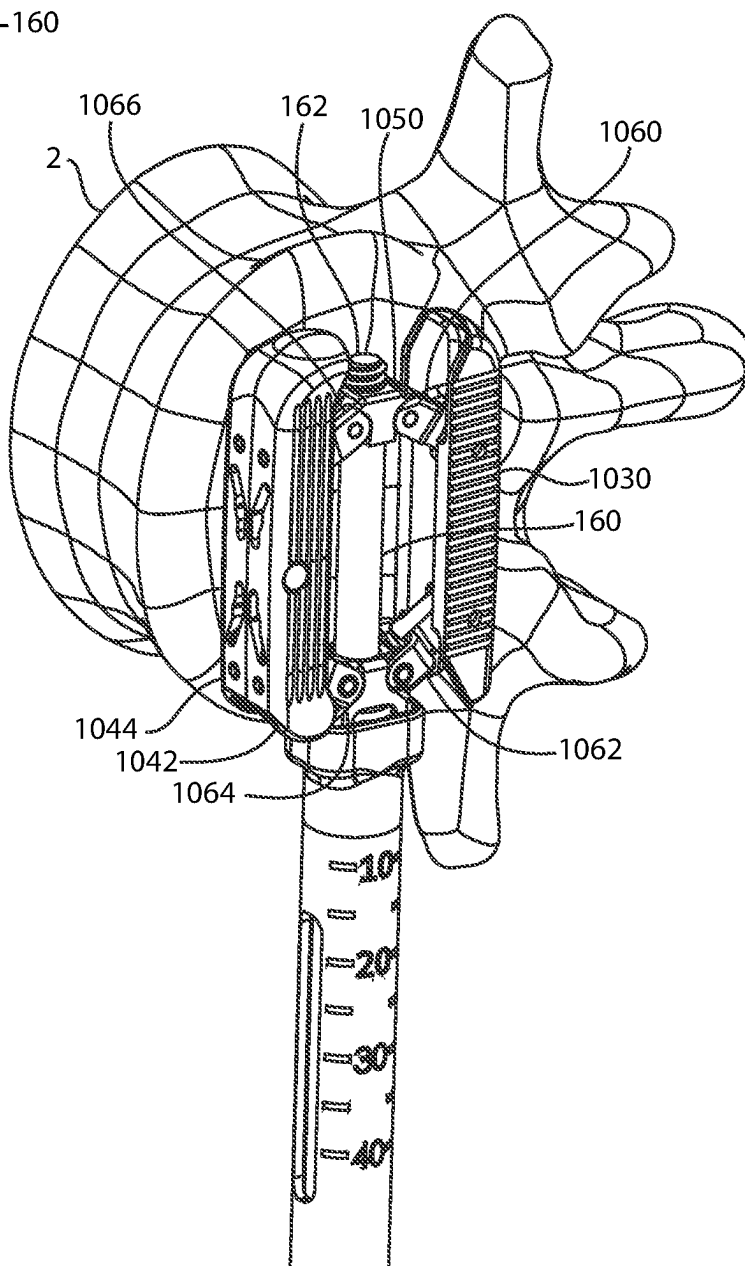


FIG. 53

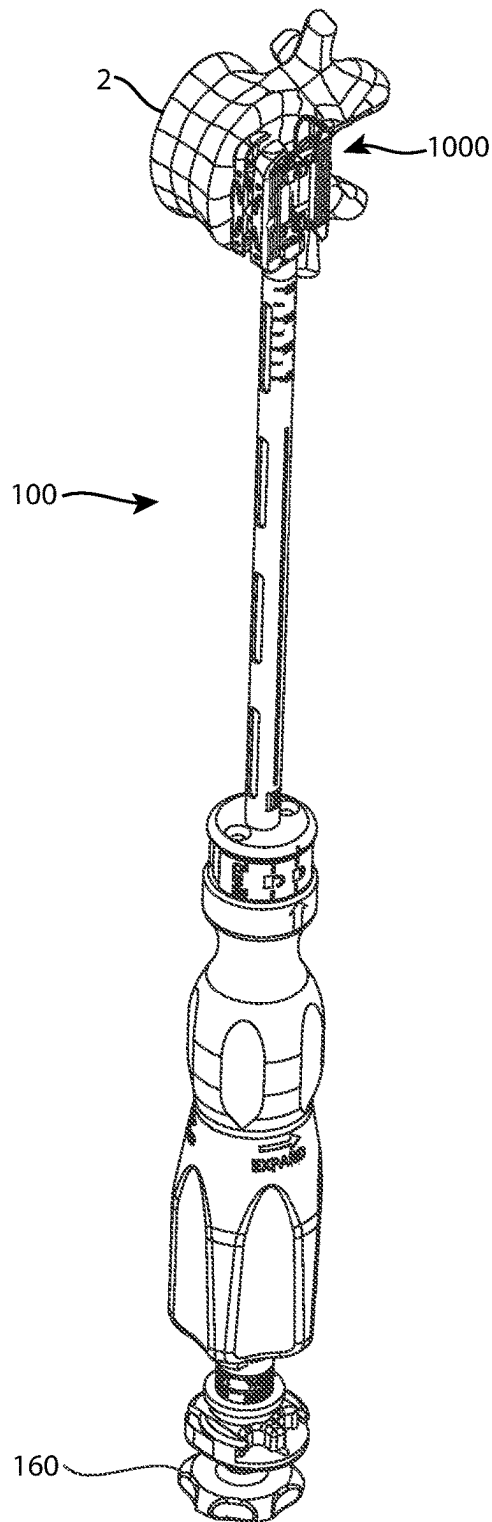


FIG. 54

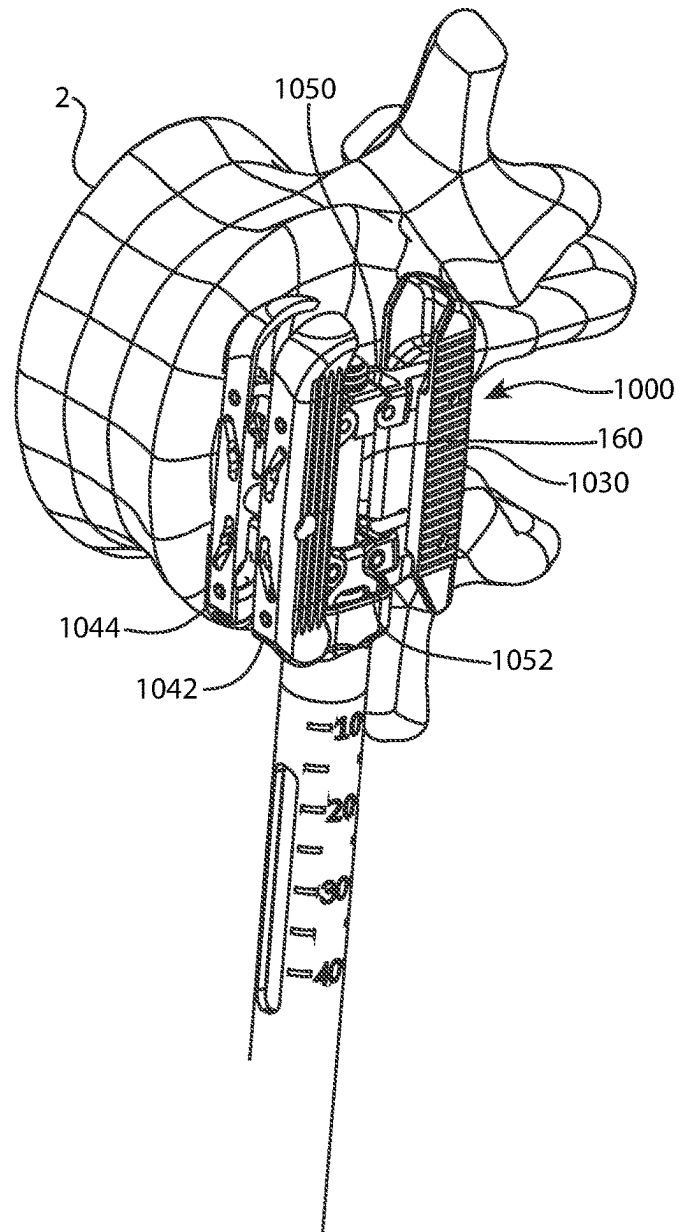


FIG. 55

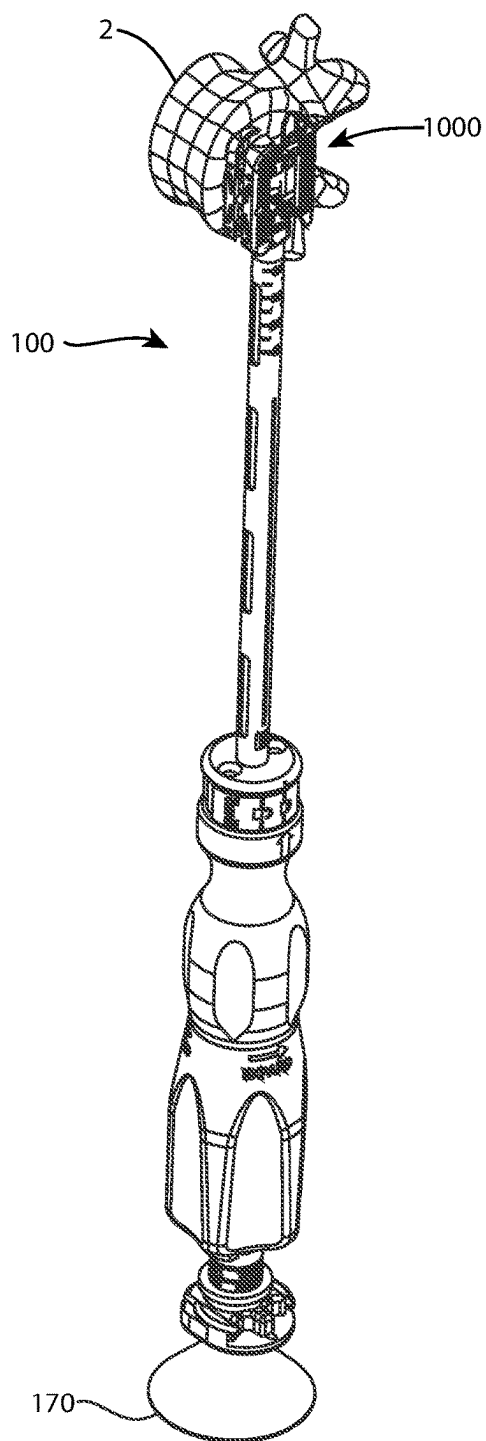


FIG. 56

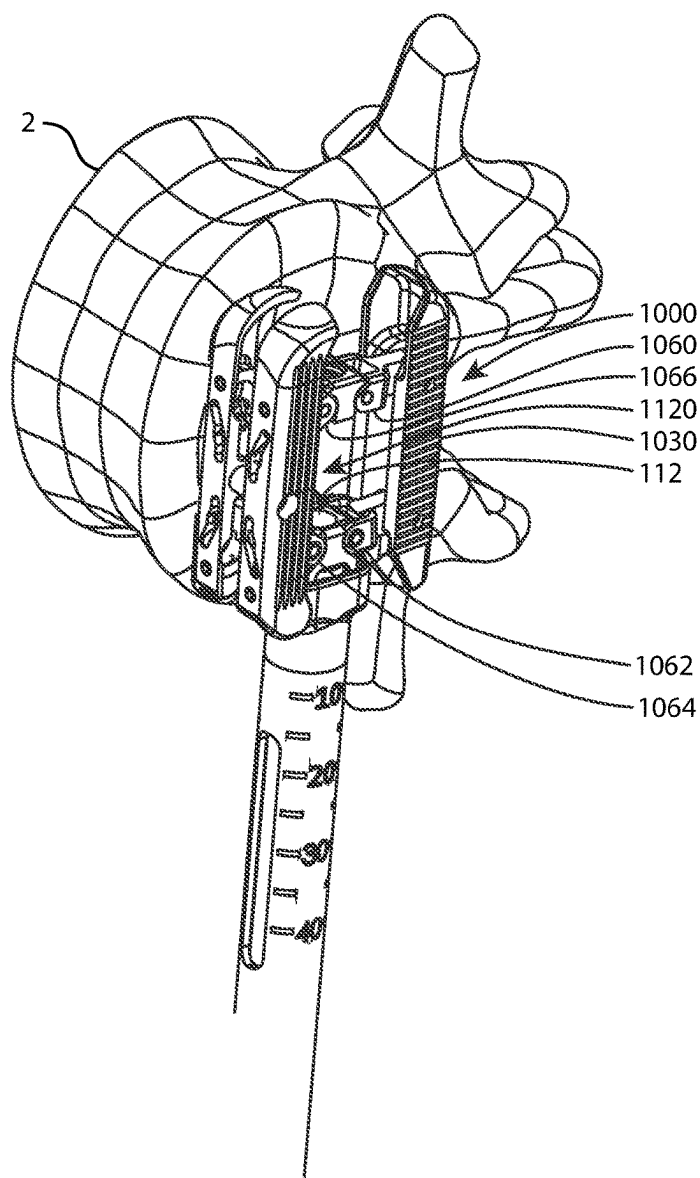


FIG. 57

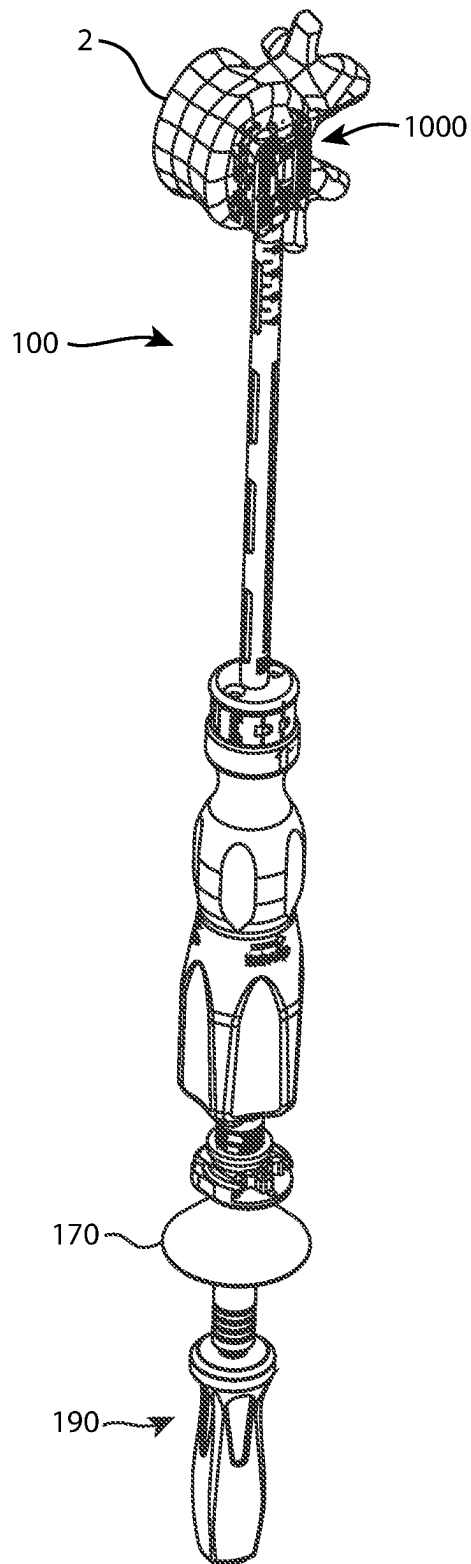


FIG. 58

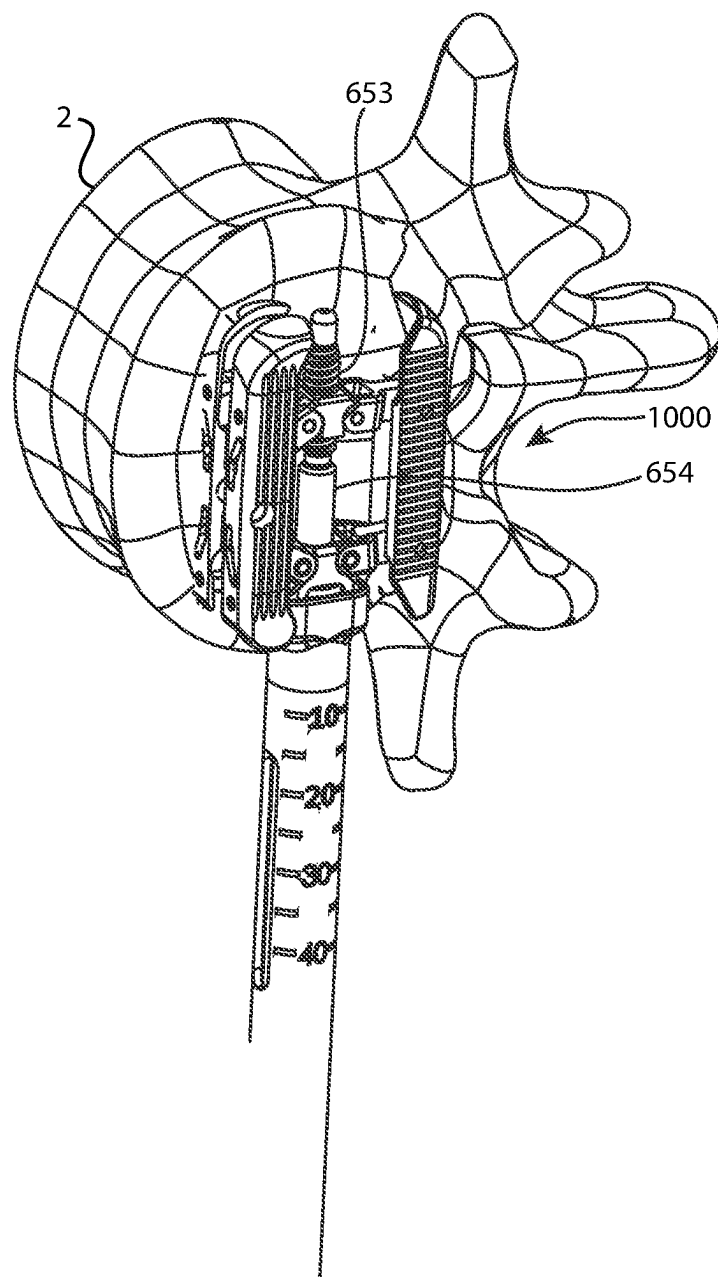


FIG. 59

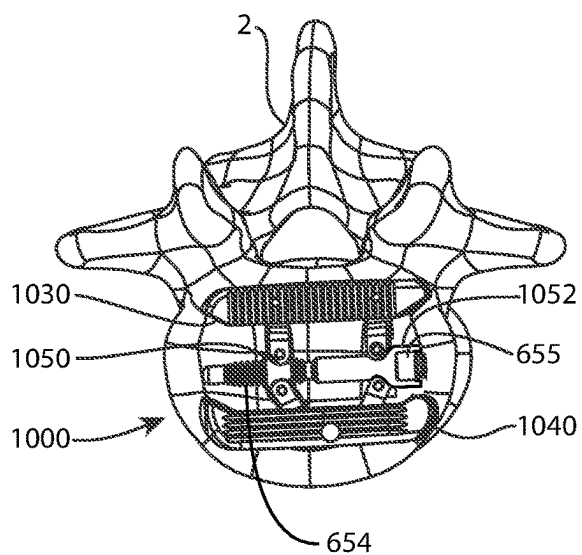


FIG. 60

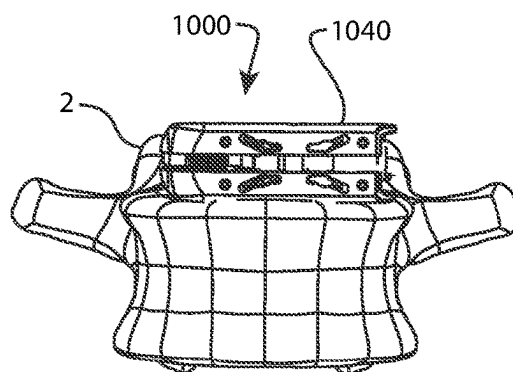


FIG. 61

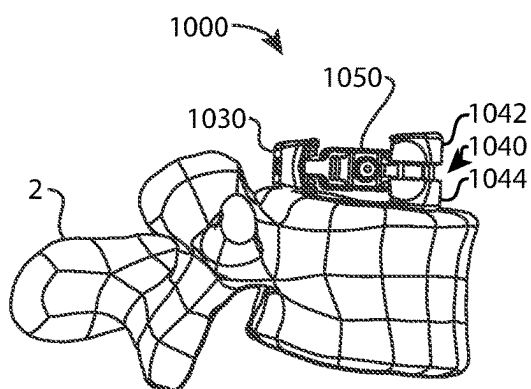


FIG. 62

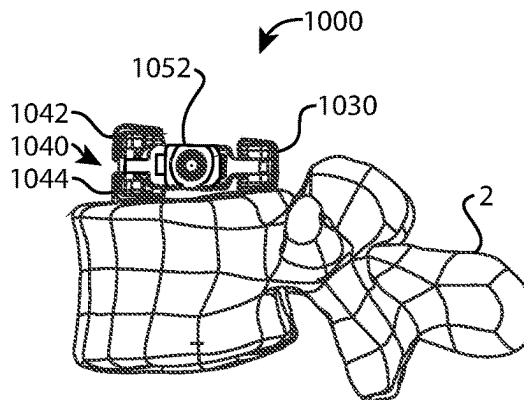


FIG. 63

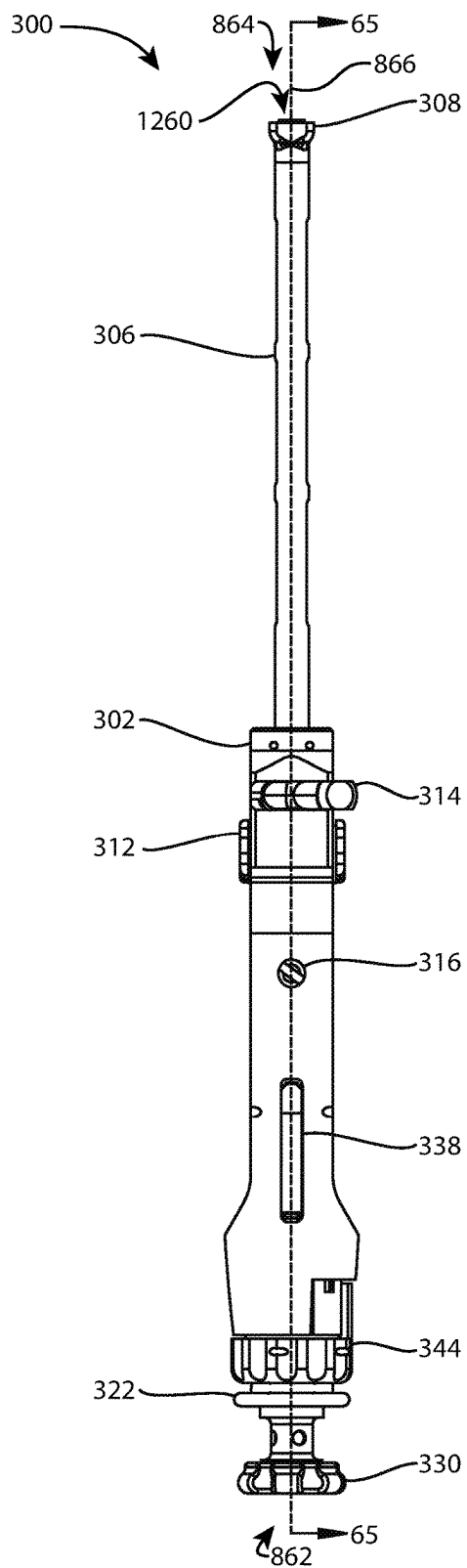


FIG. 64

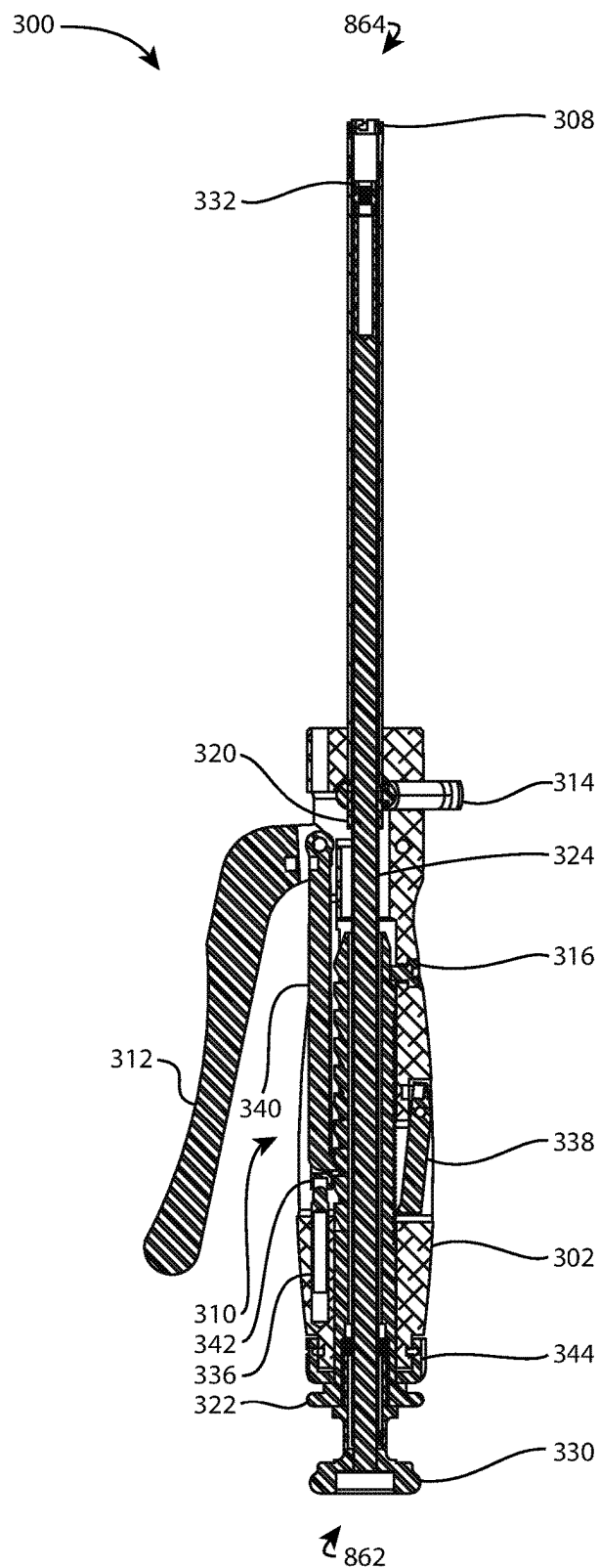
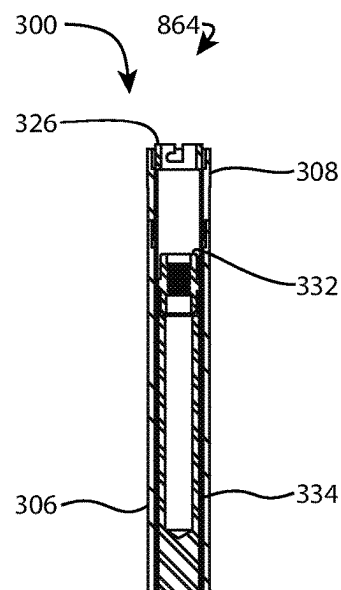
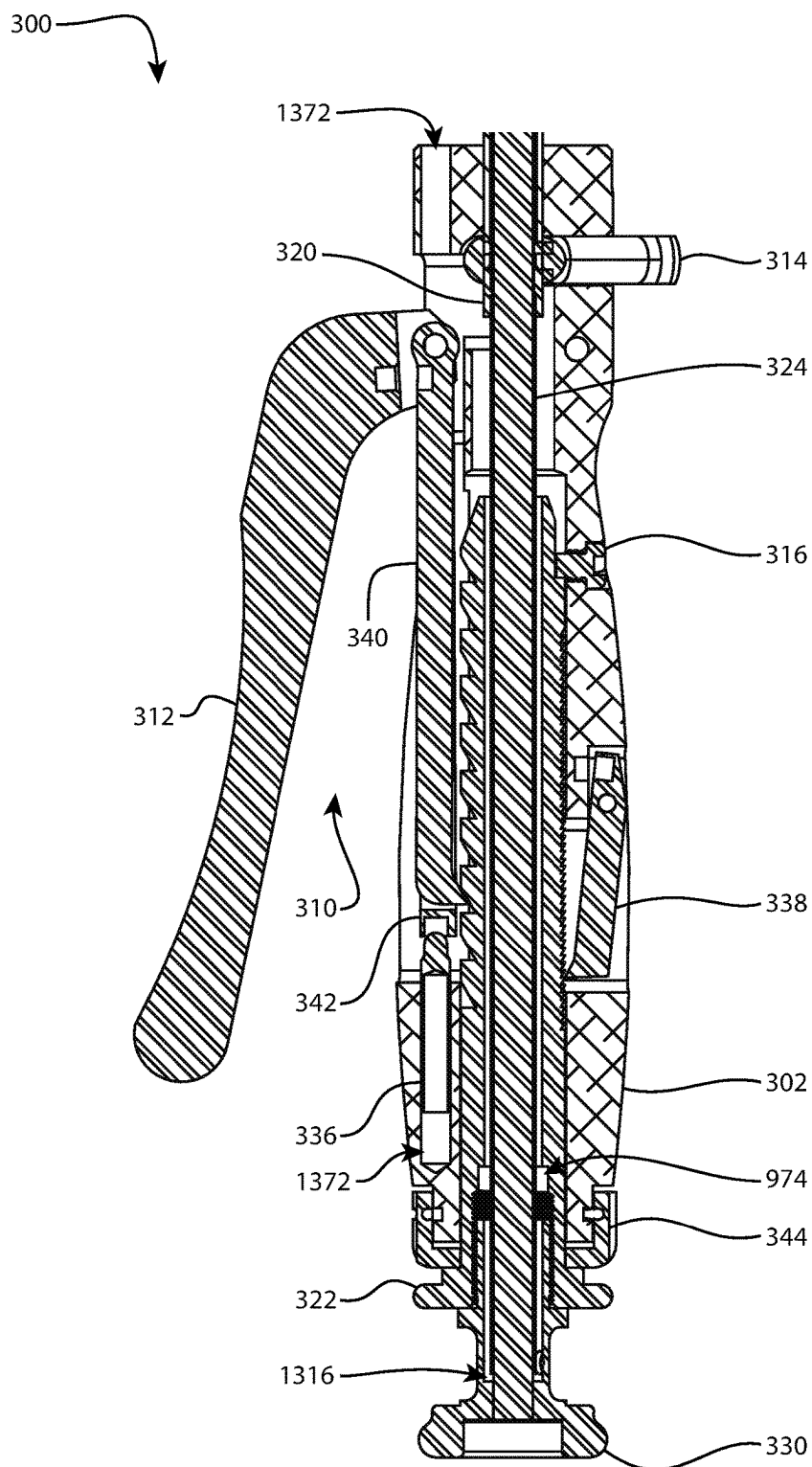
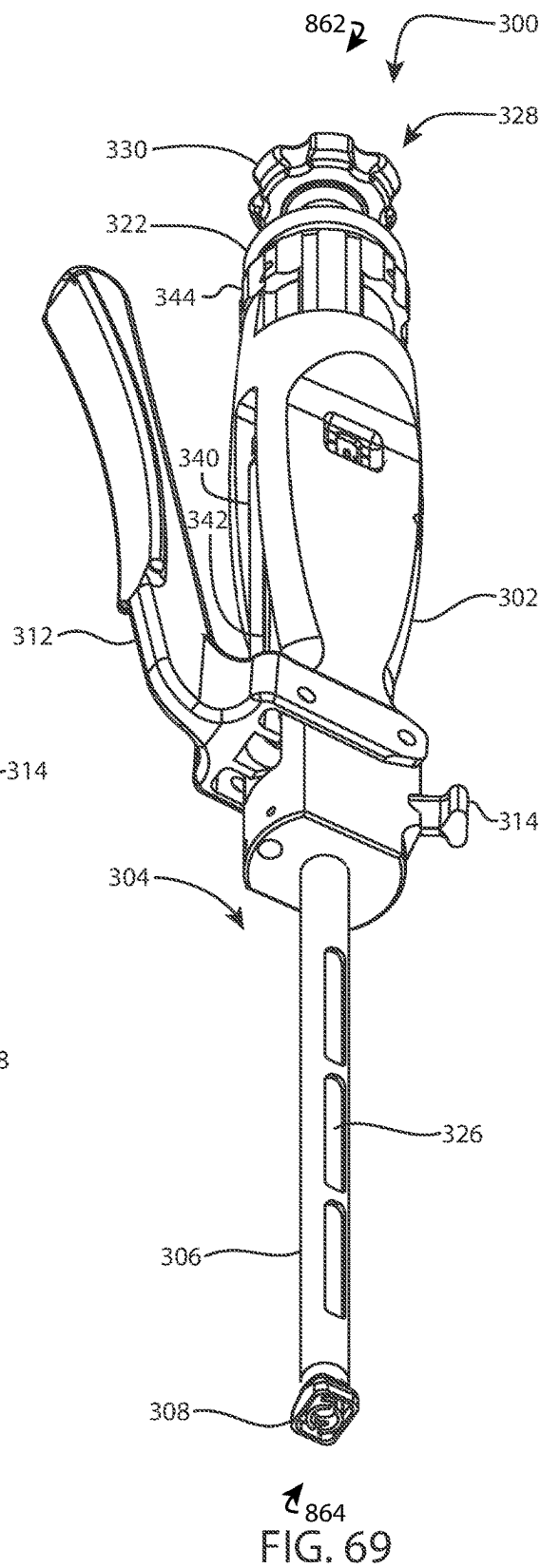
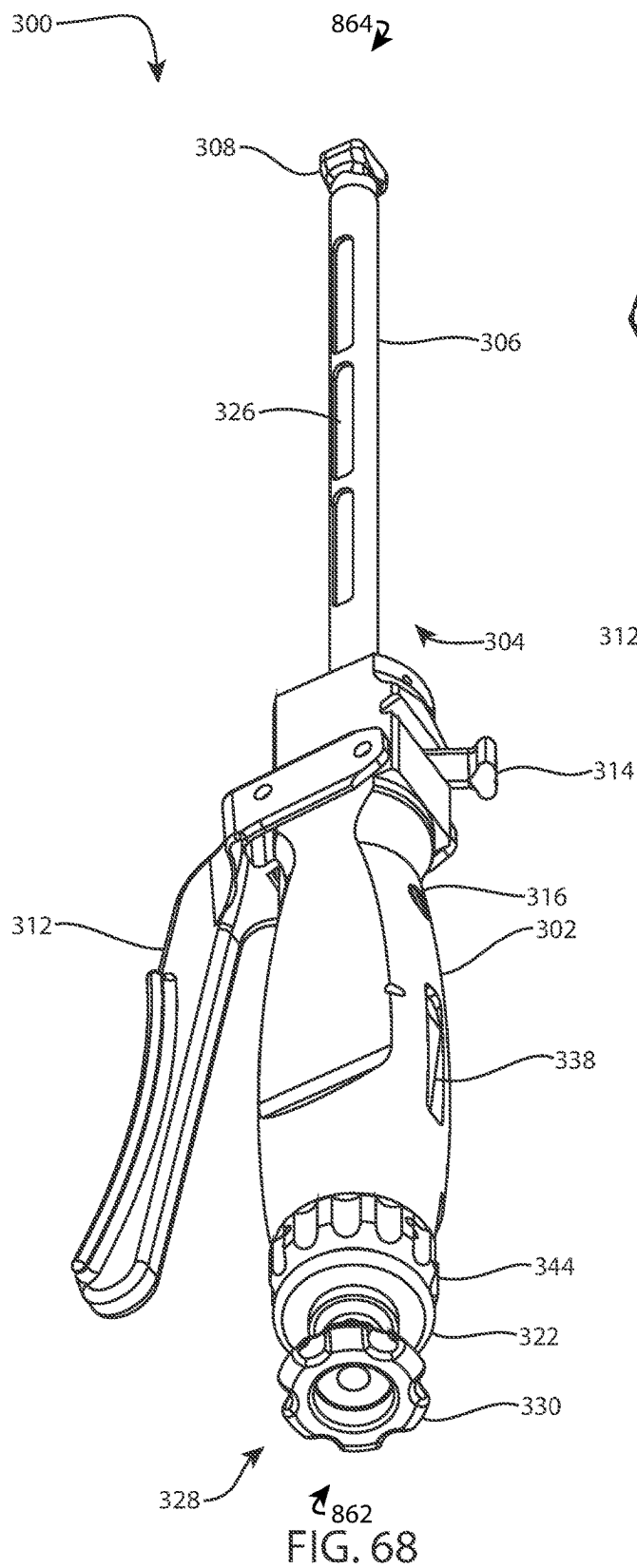
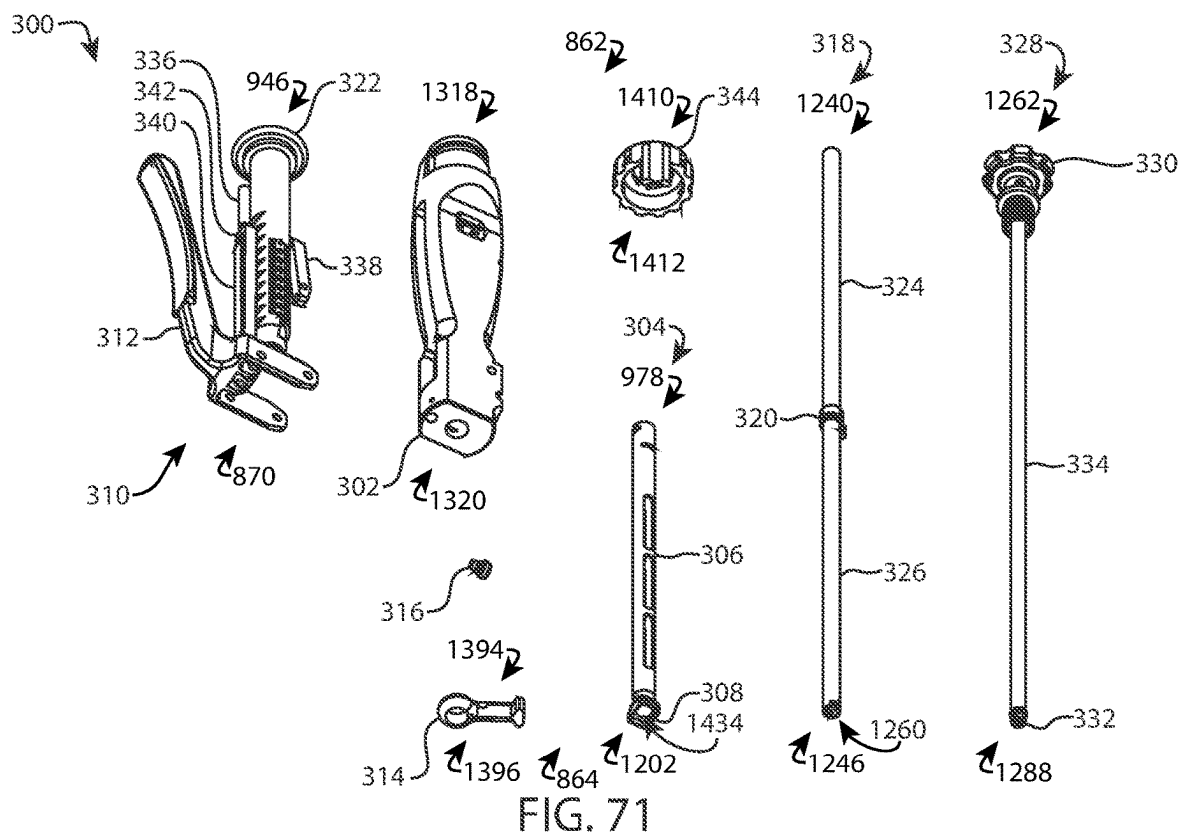
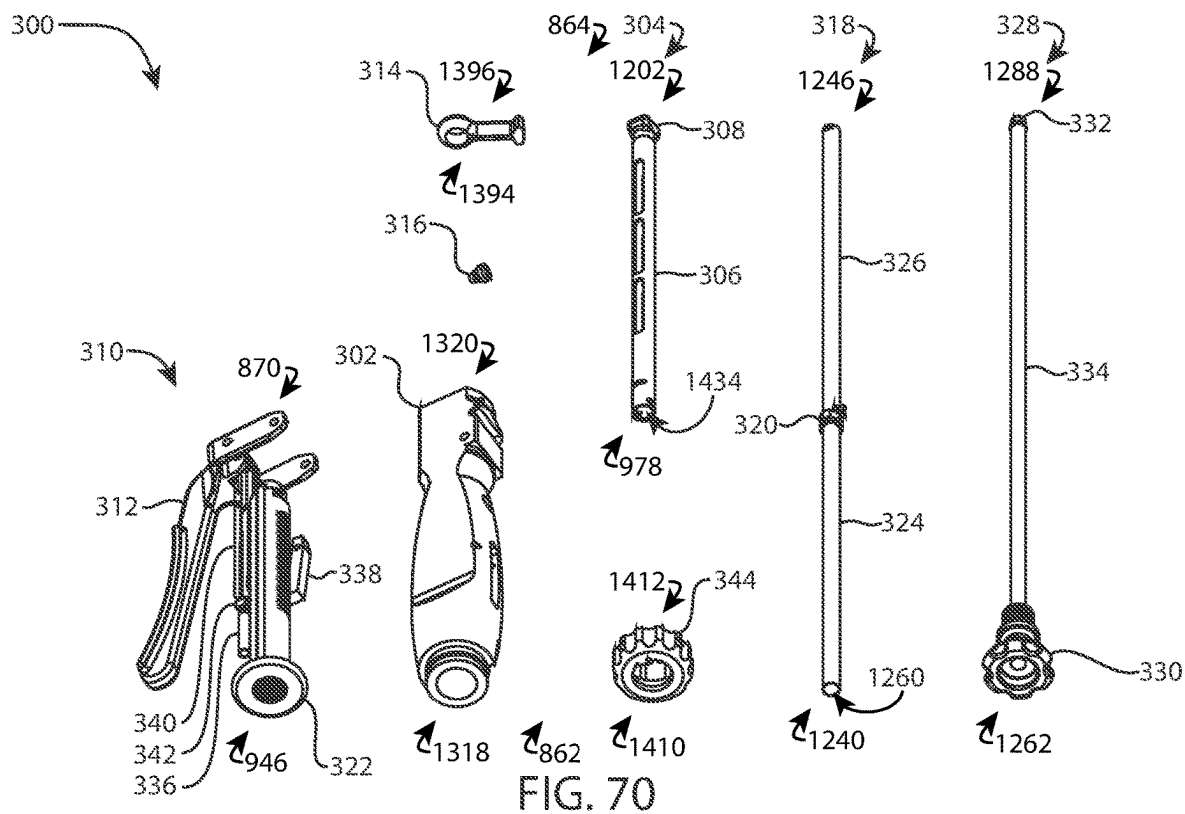


FIG. 65







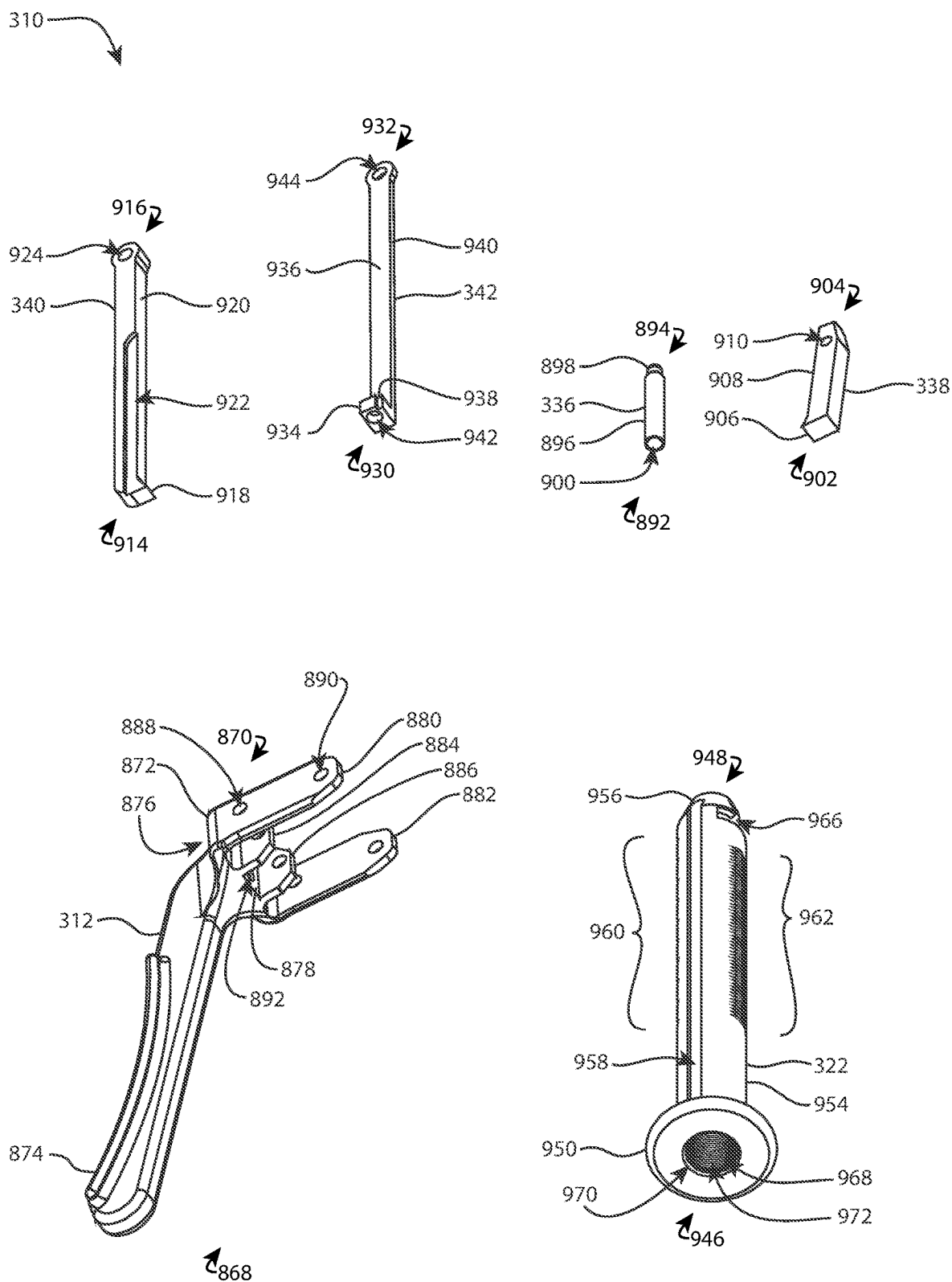


FIG. 72

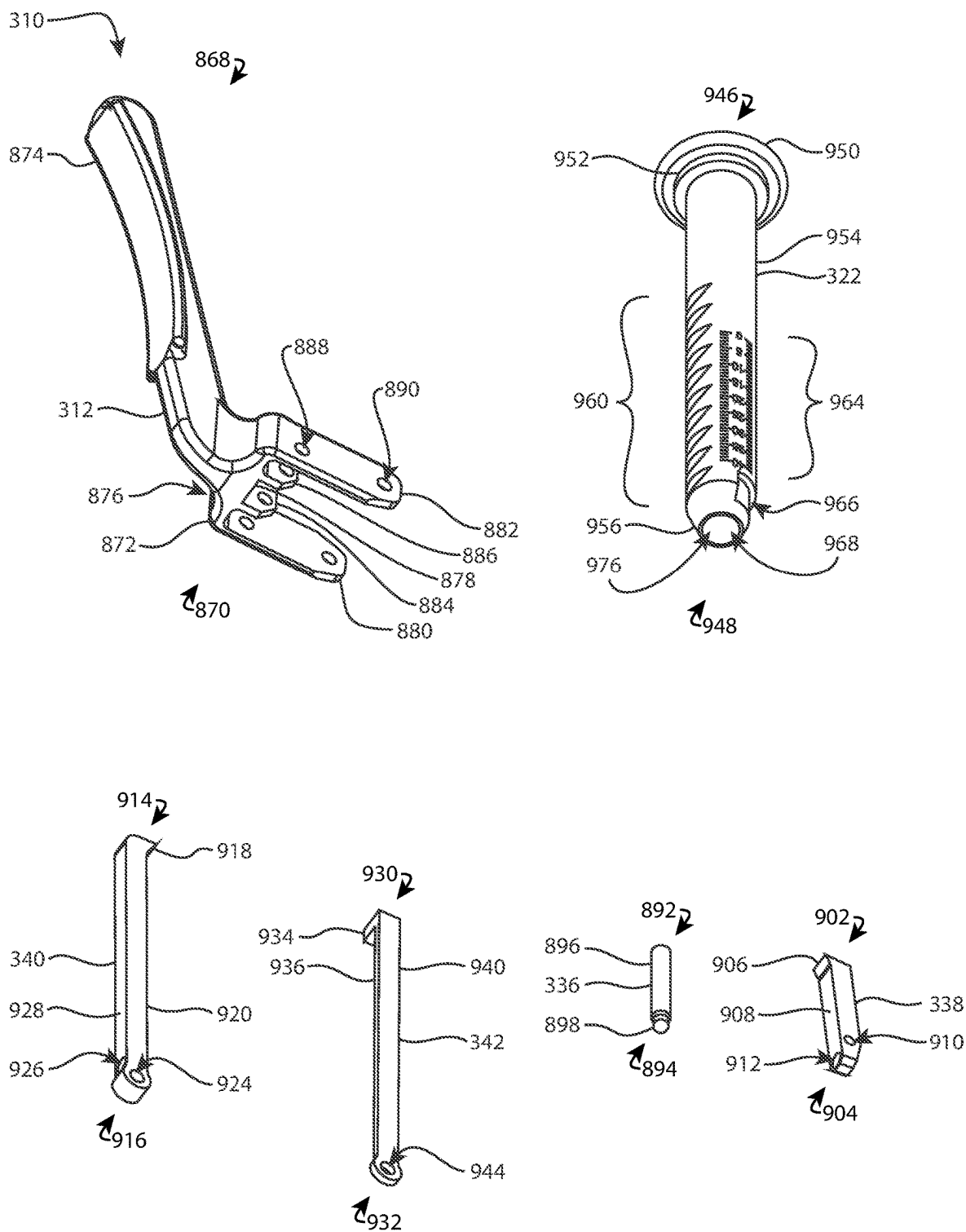
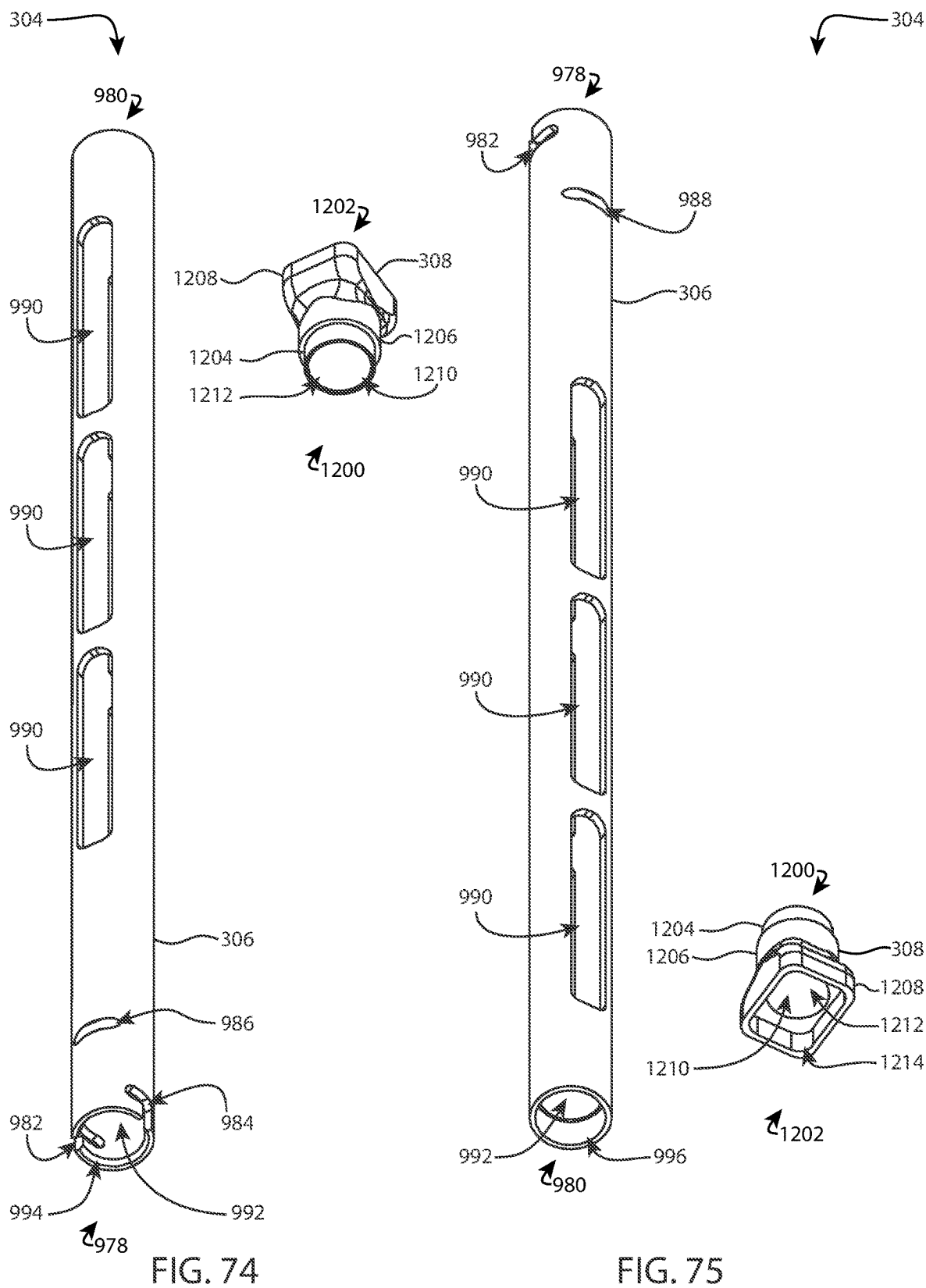


FIG. 73



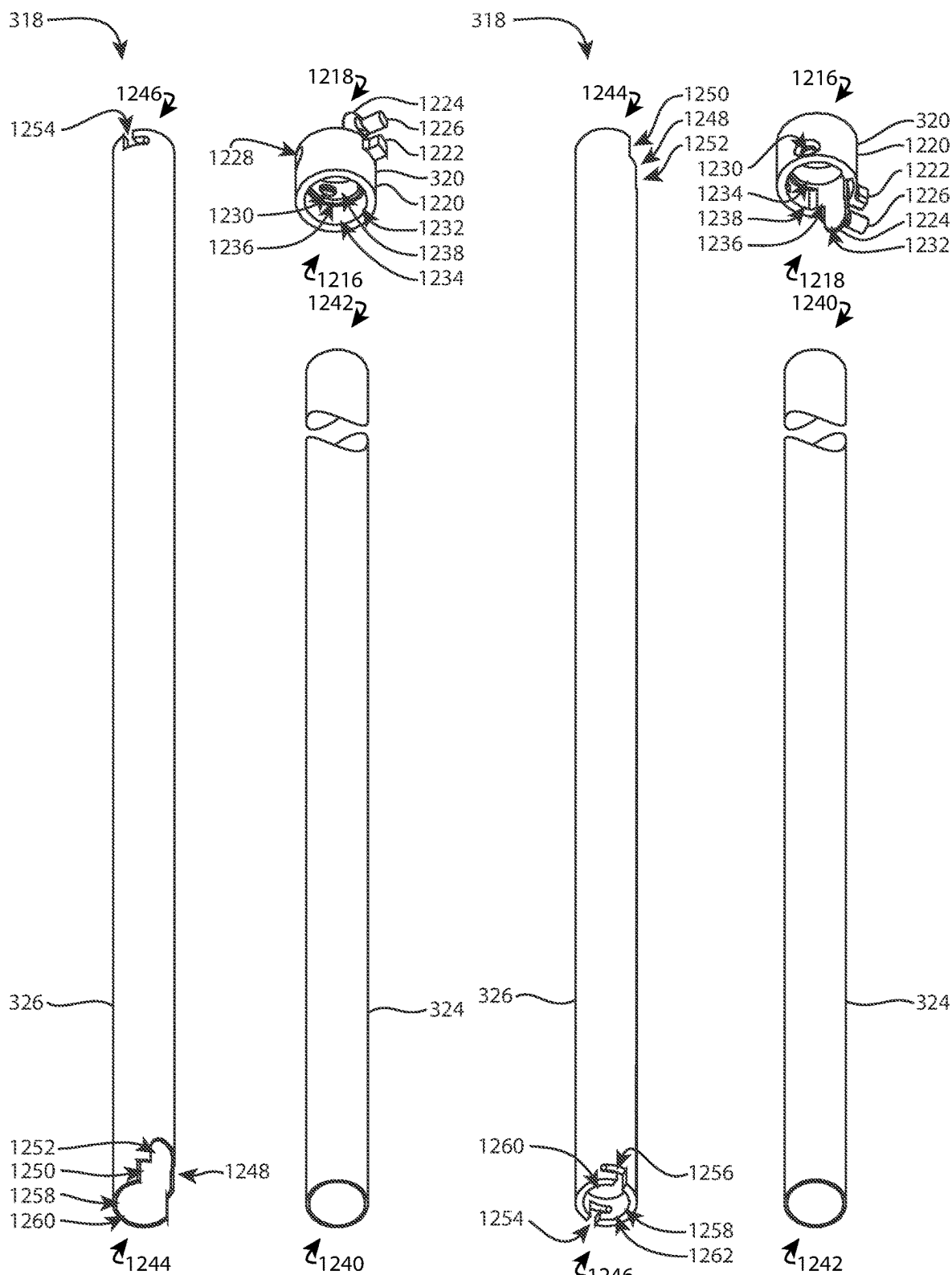
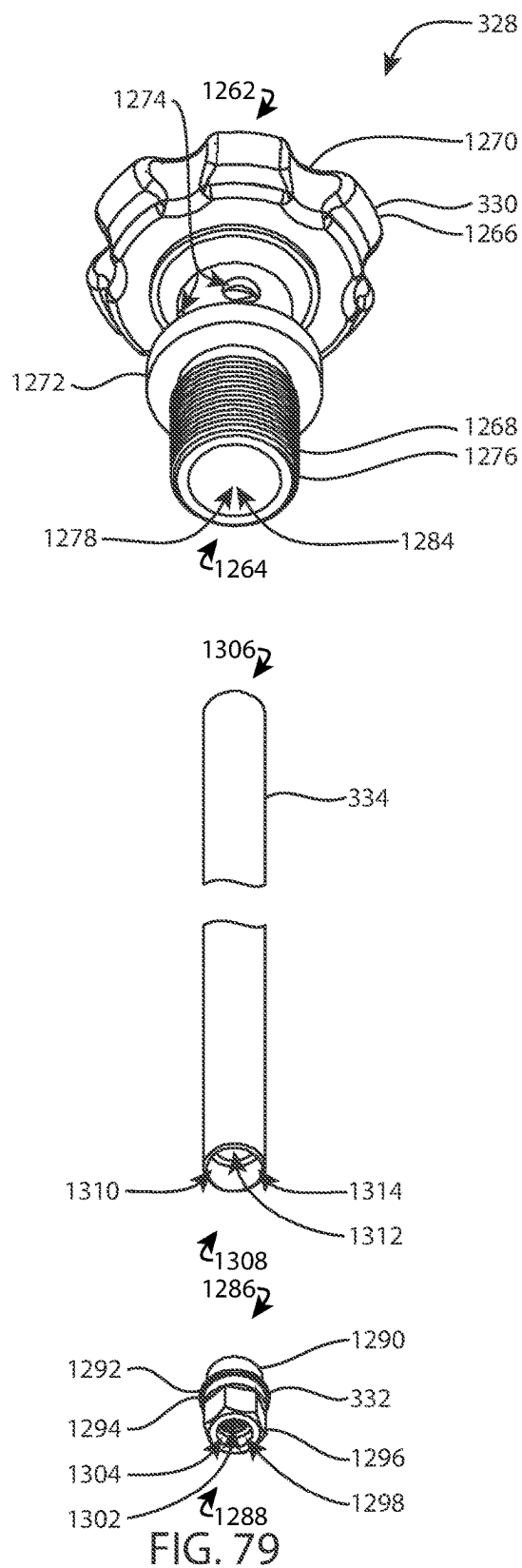
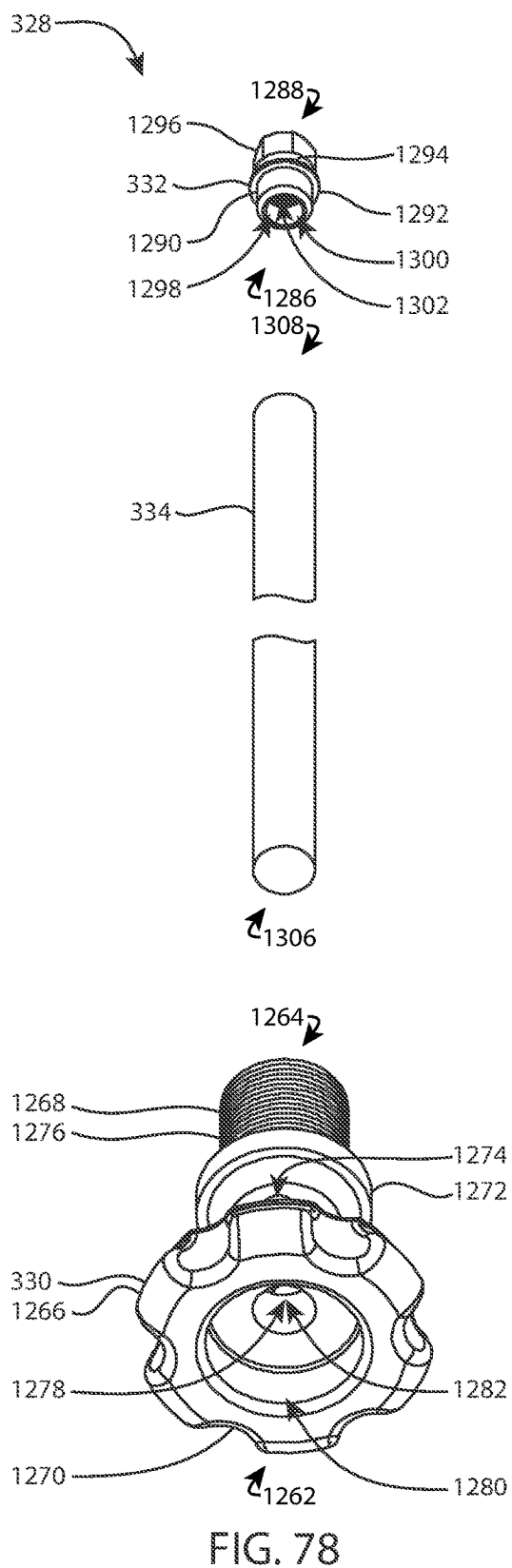
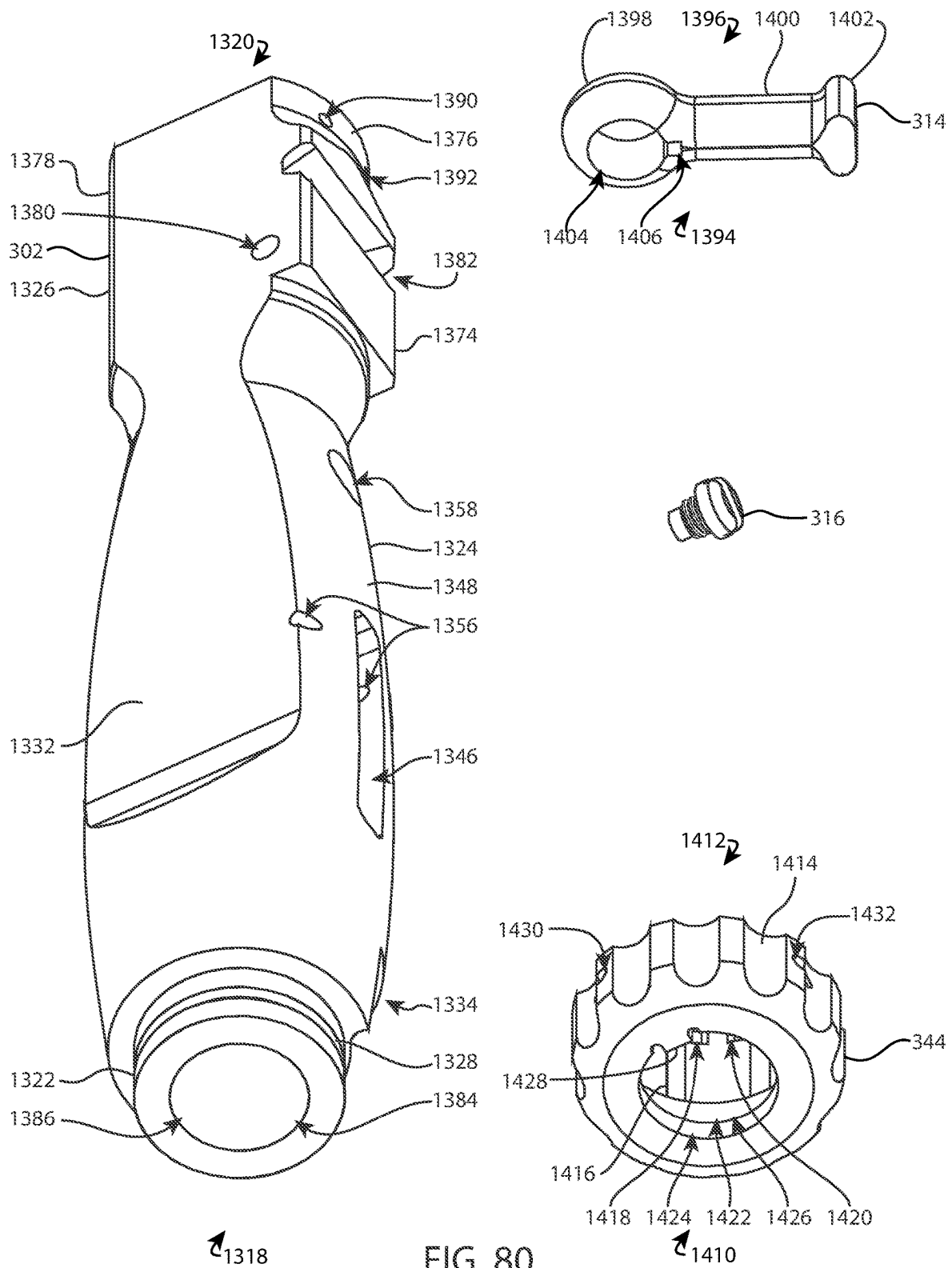
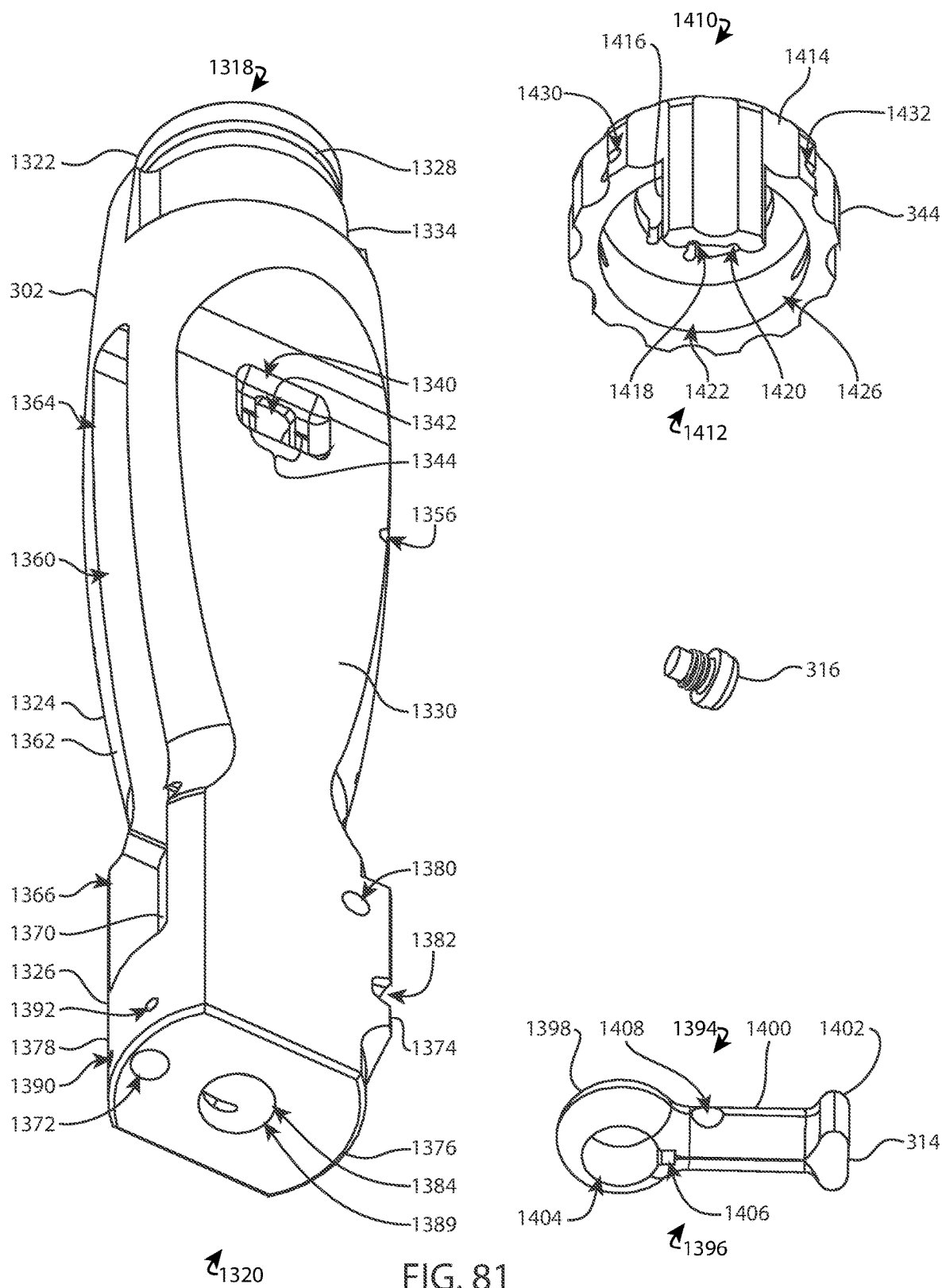


FIG. 76

FIG. 77







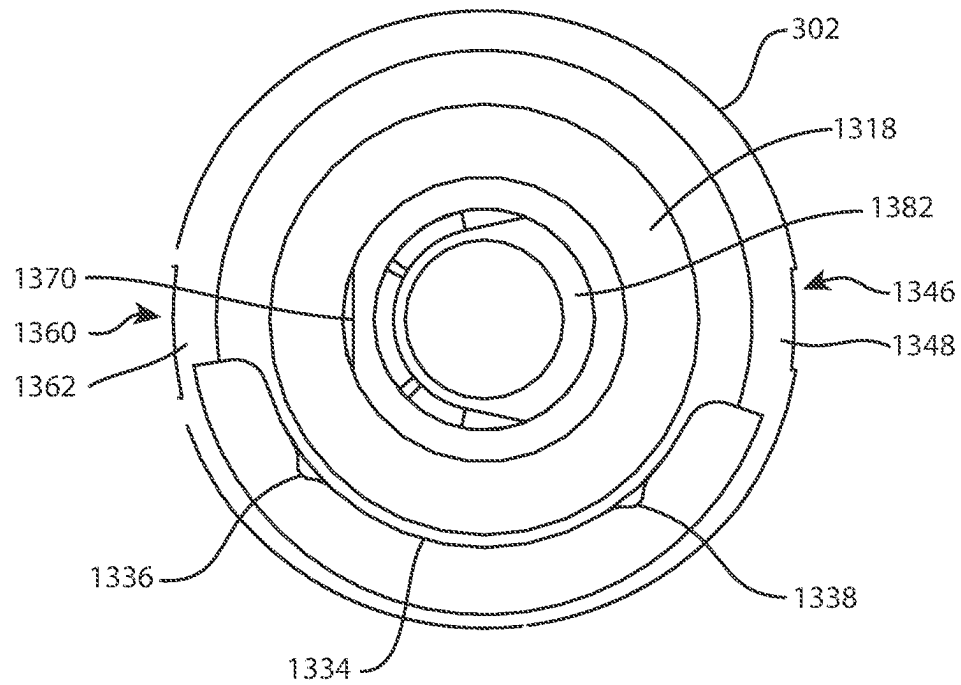


FIG. 82

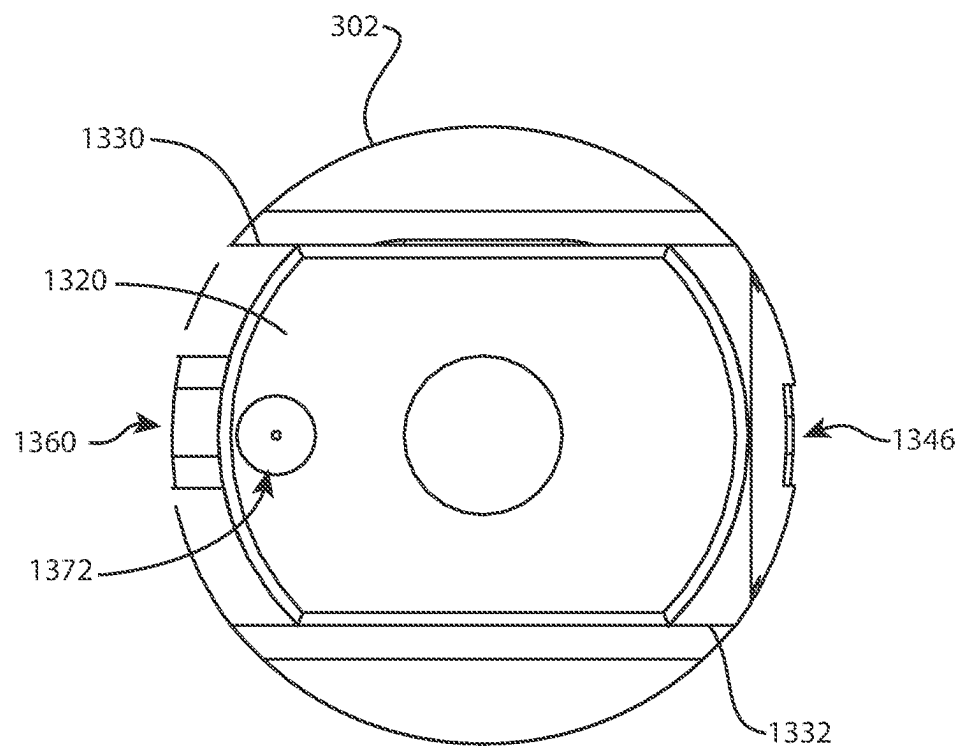


FIG. 83

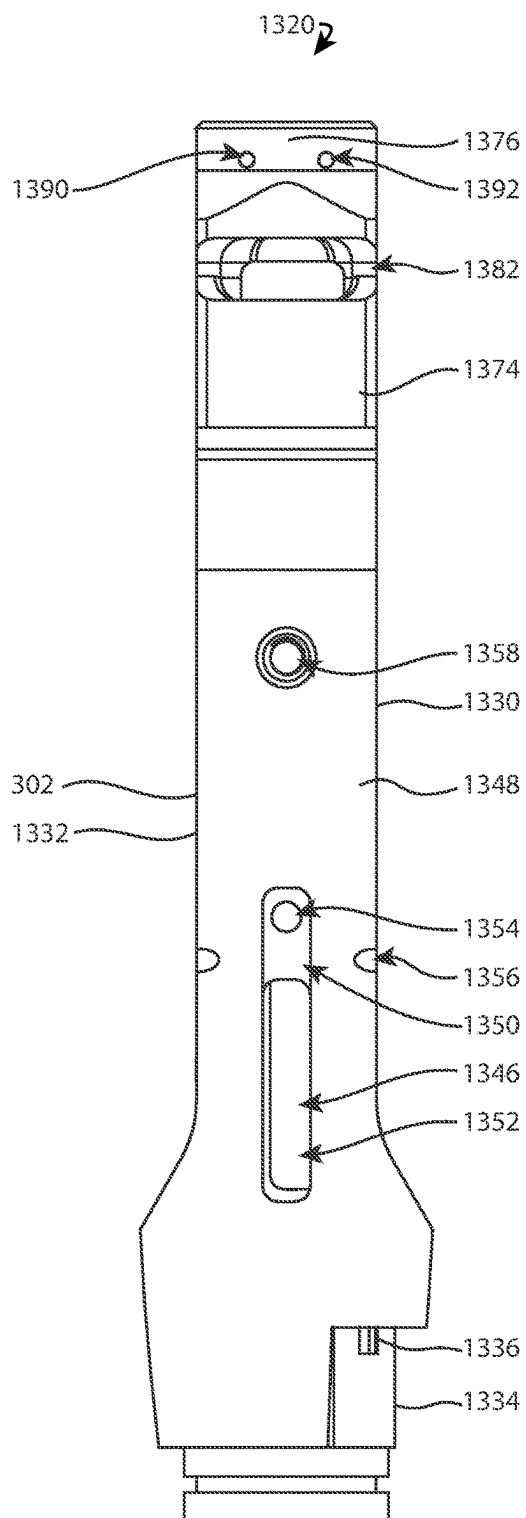


FIG. 84

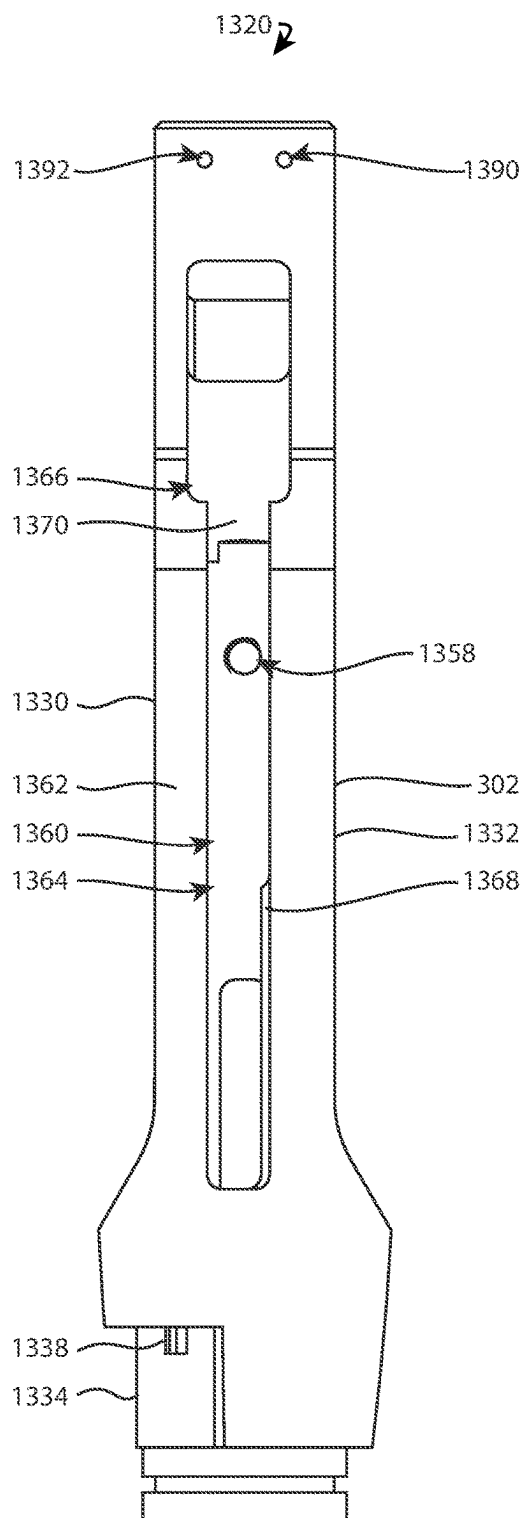
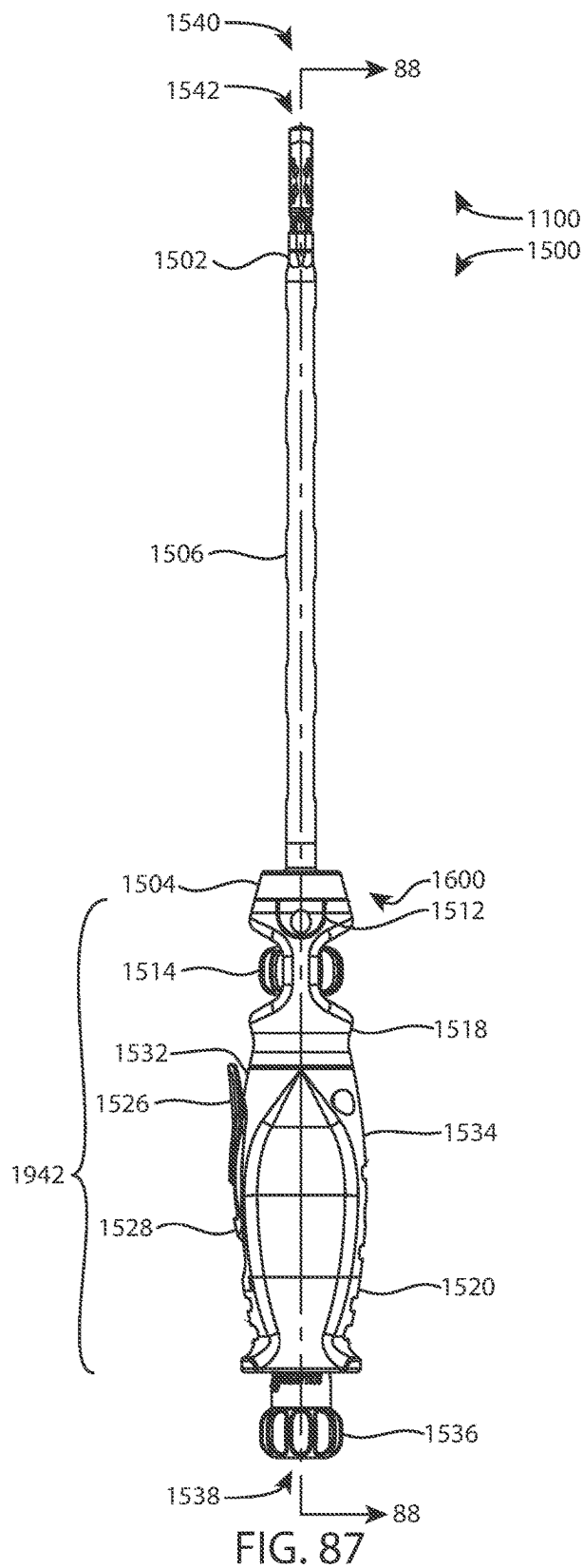
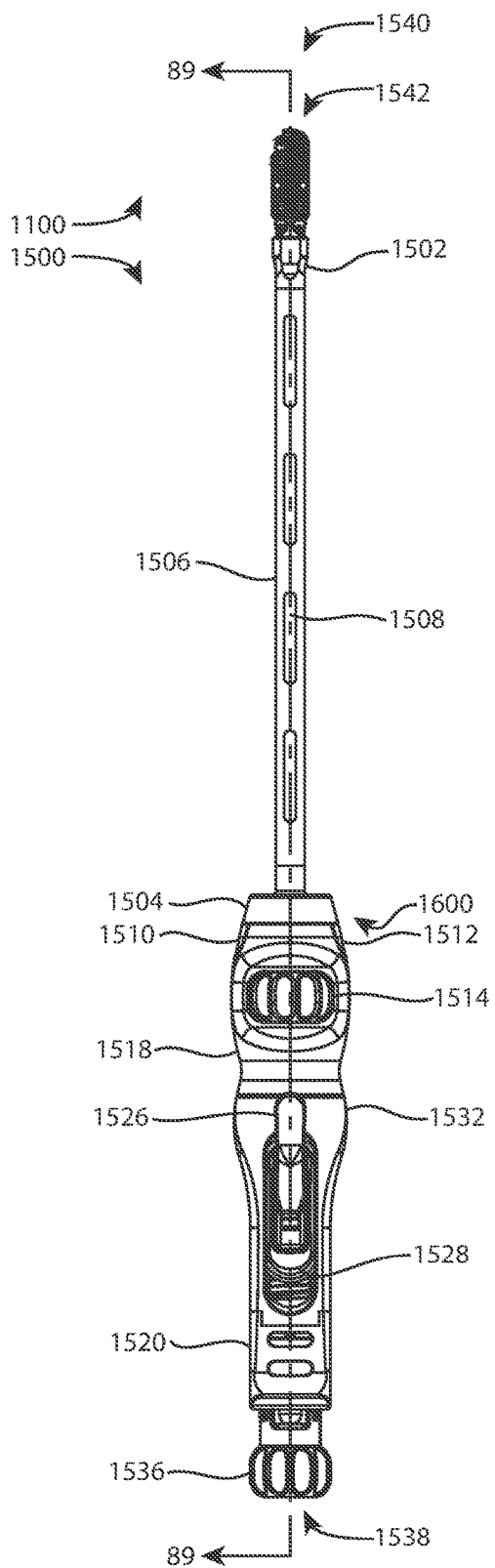


FIG. 85



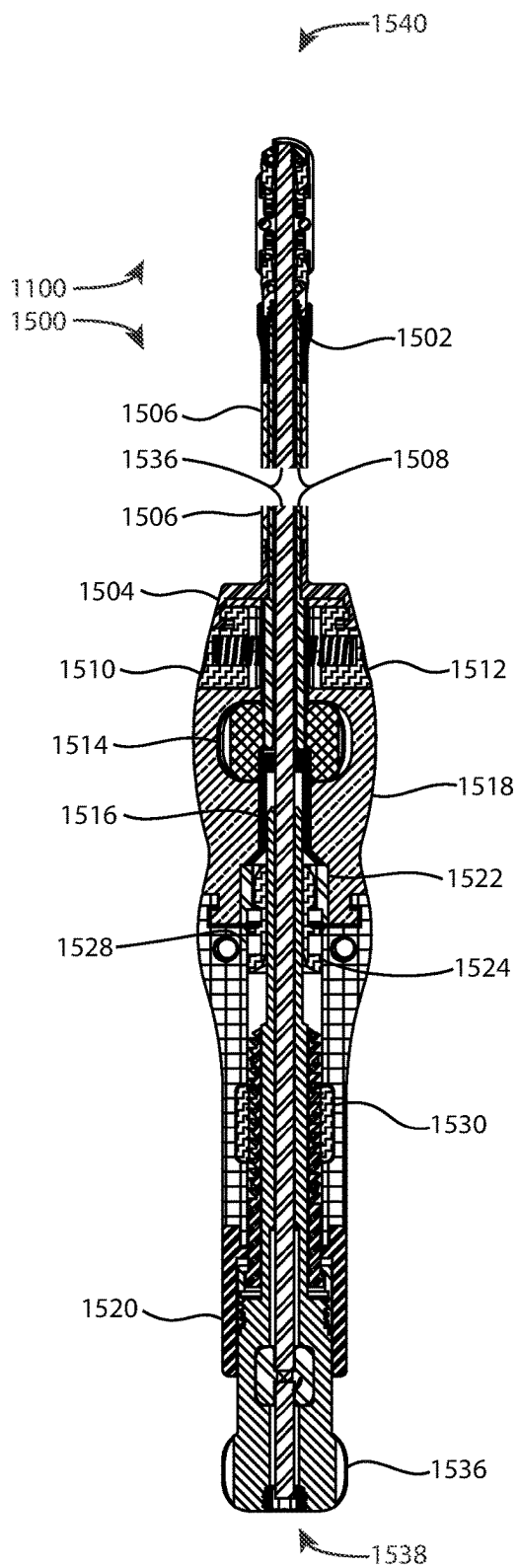


FIG. 88

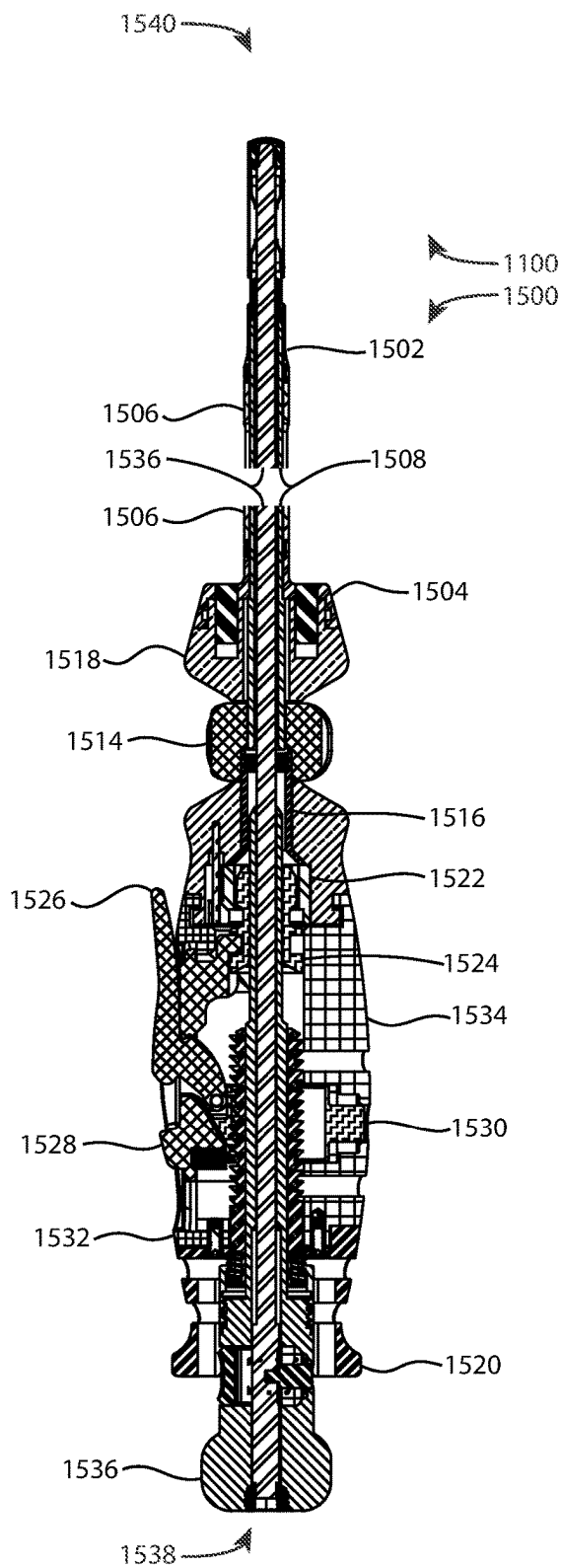


FIG. 89

1500 1540 1540 1500

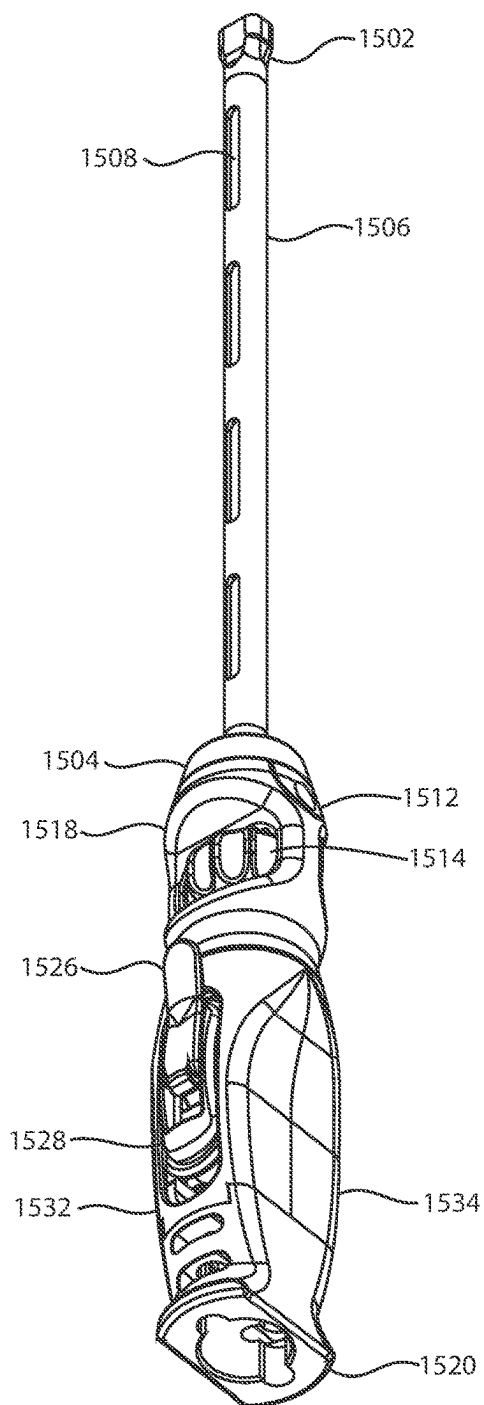


FIG. 90

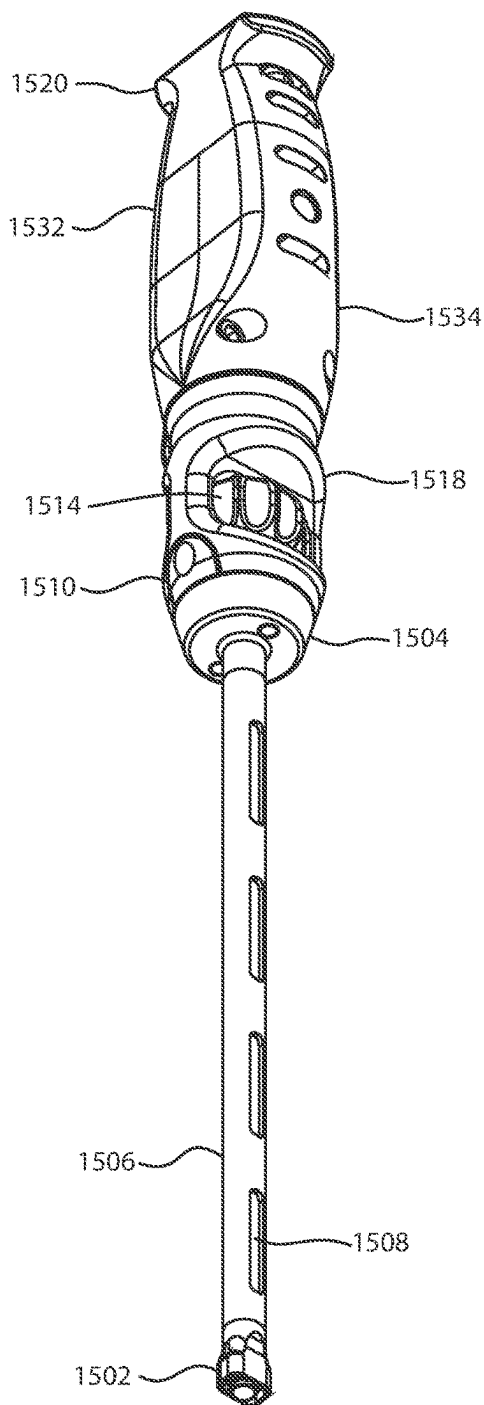
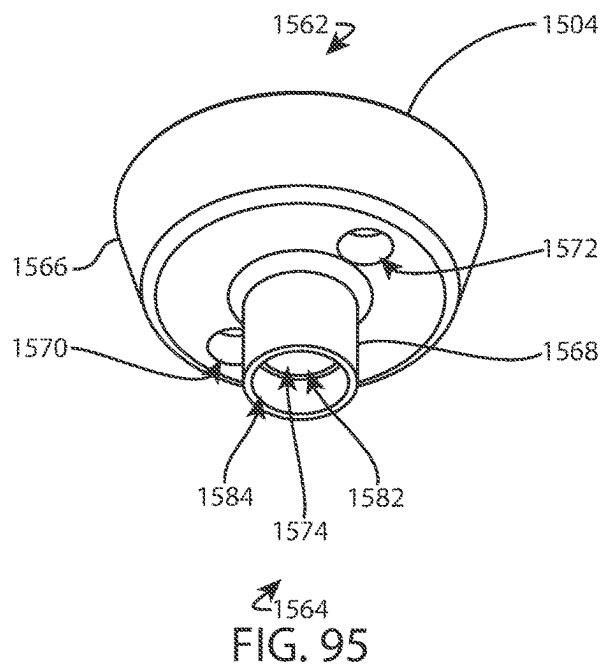
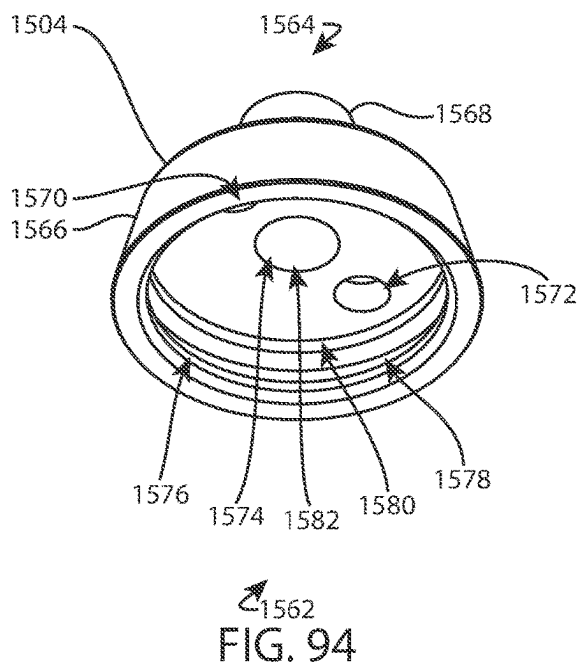
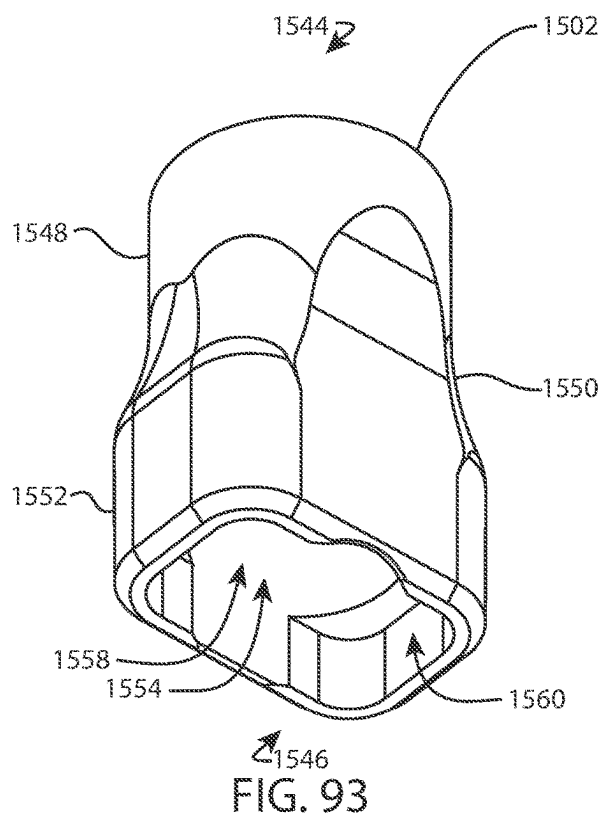
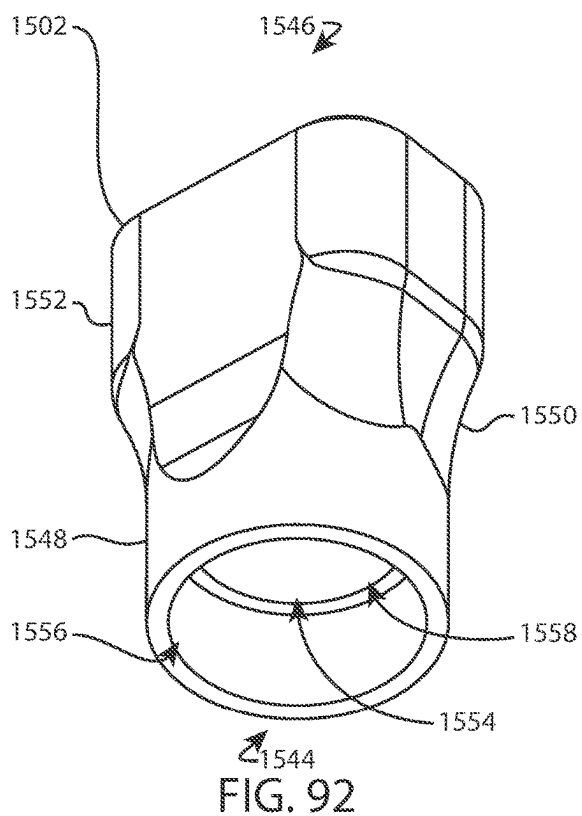
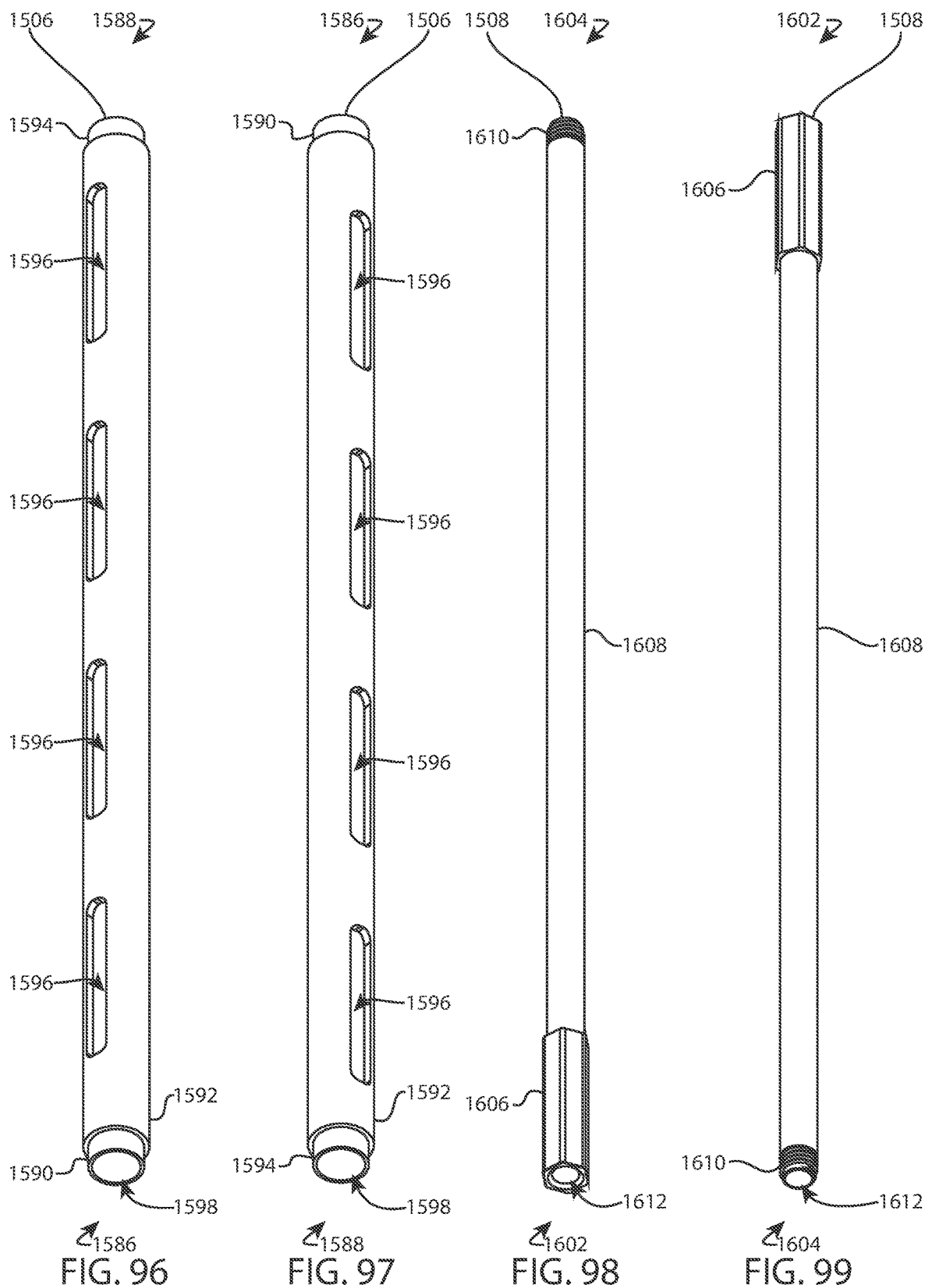


FIG. 91





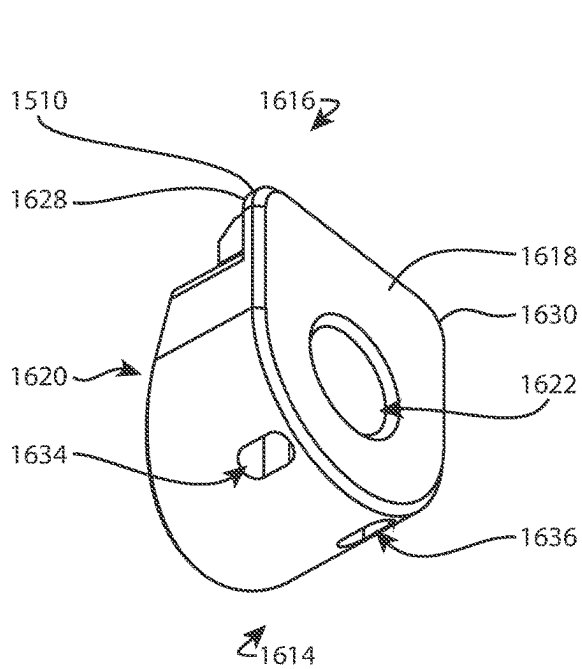


FIG. 100

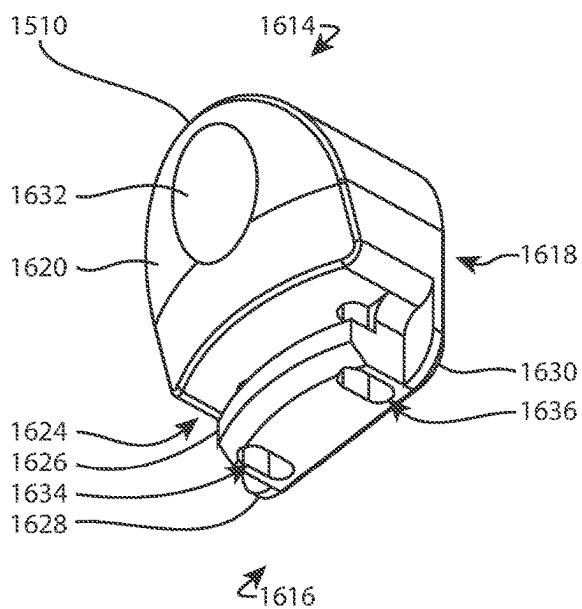
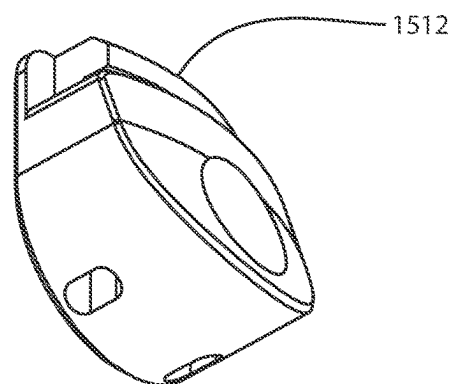
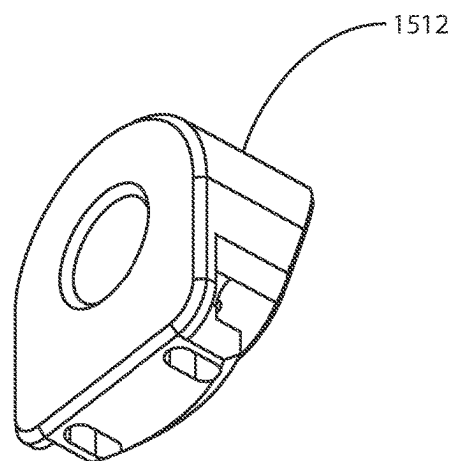


FIG. 101



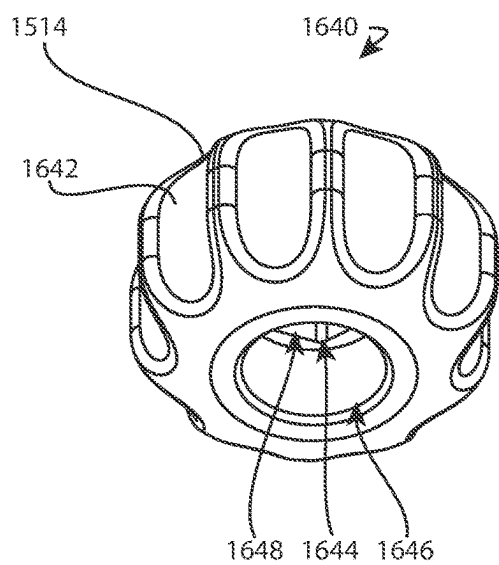


FIG. 102

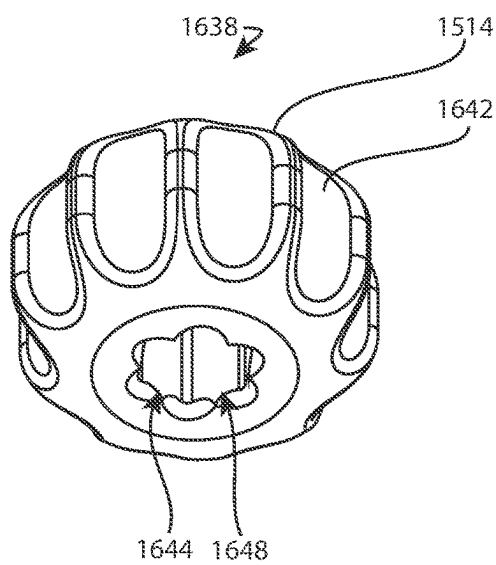


FIG. 103

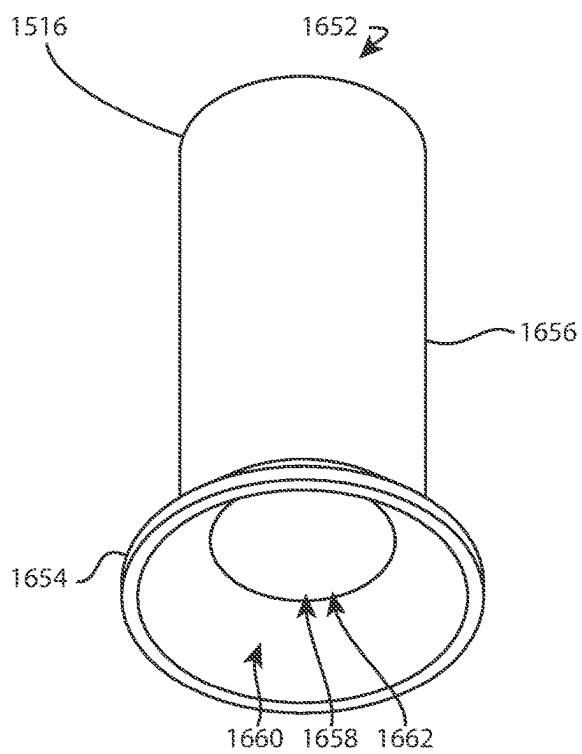


FIG. 104

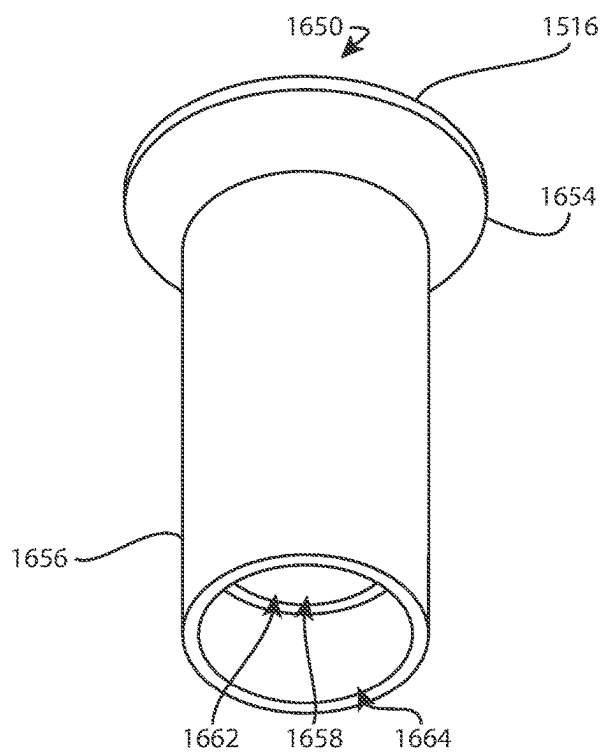
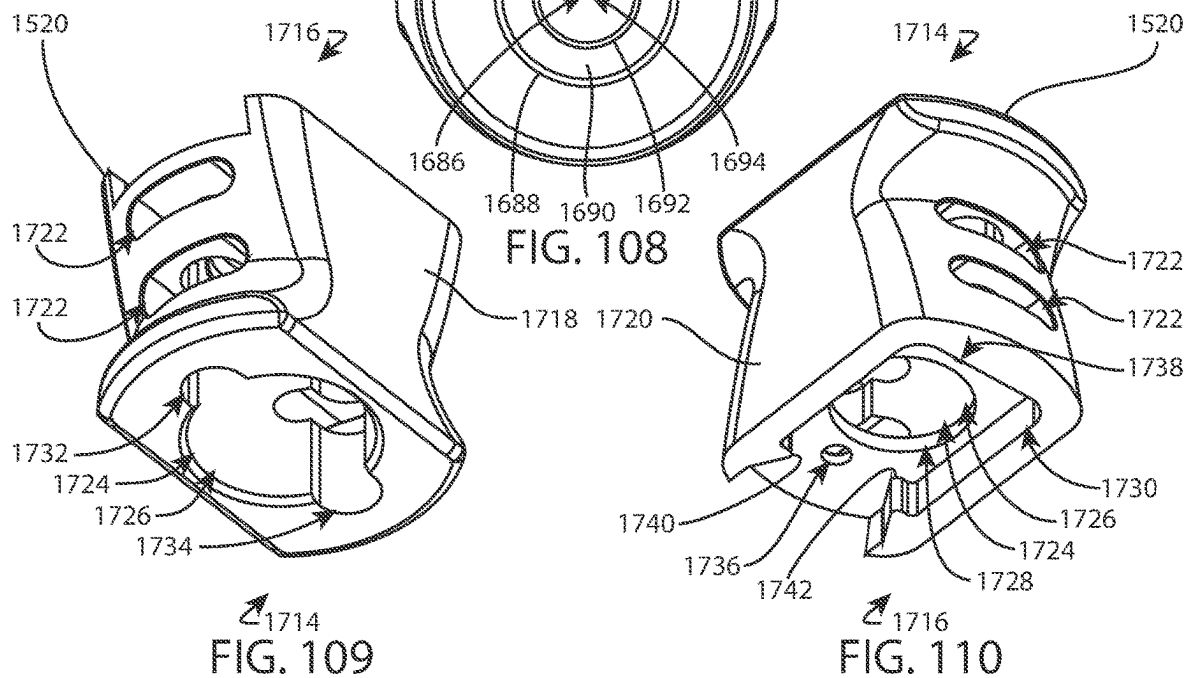
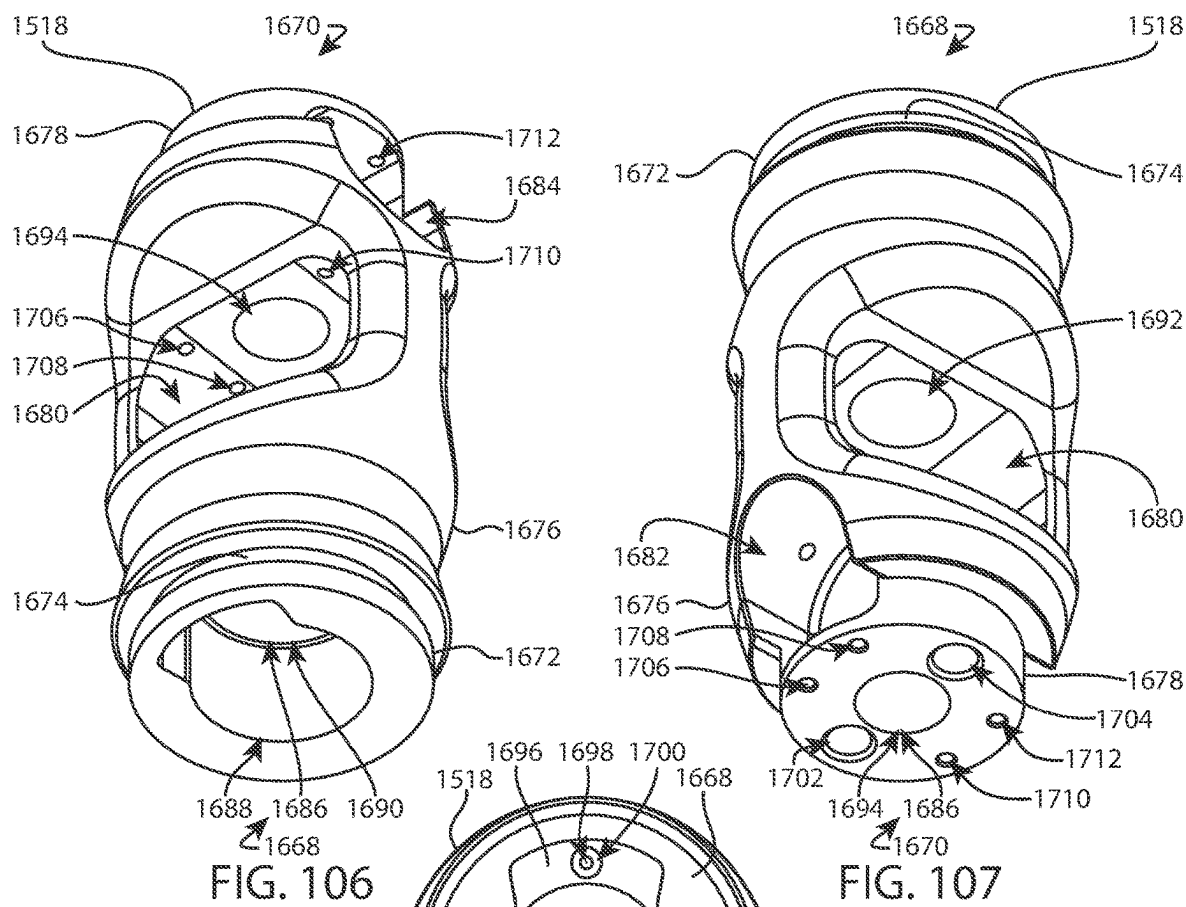


FIG. 105



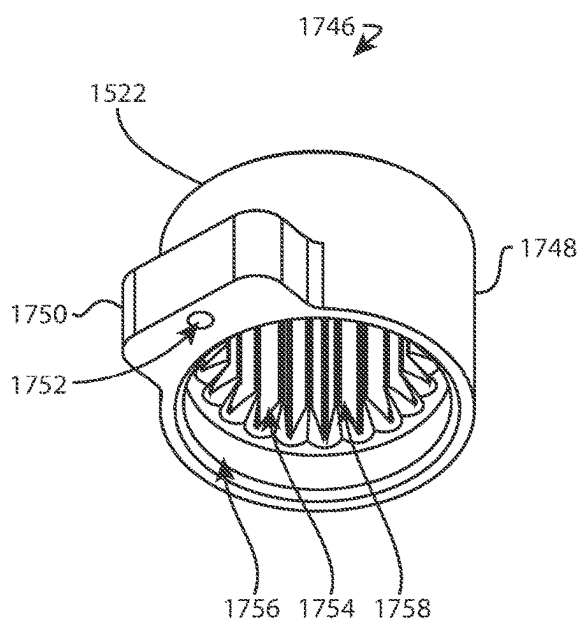


FIG. 111

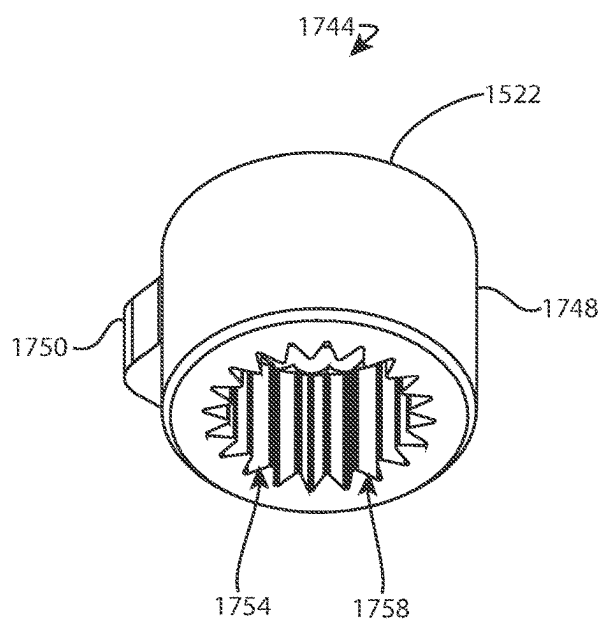


FIG. 112

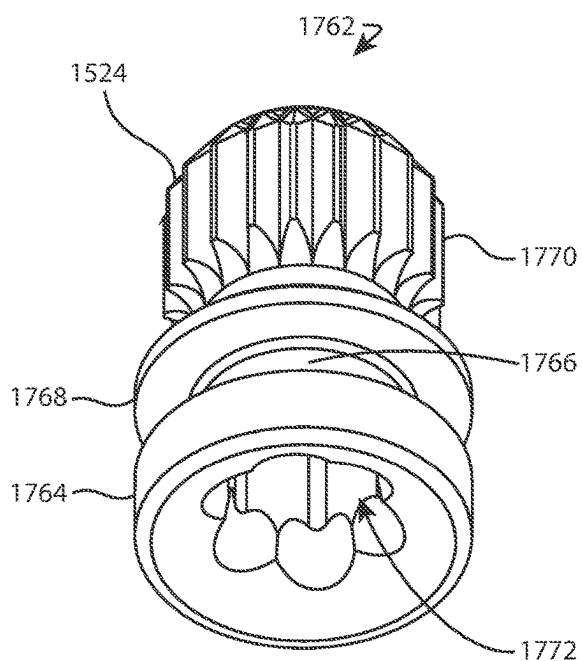


FIG. 113

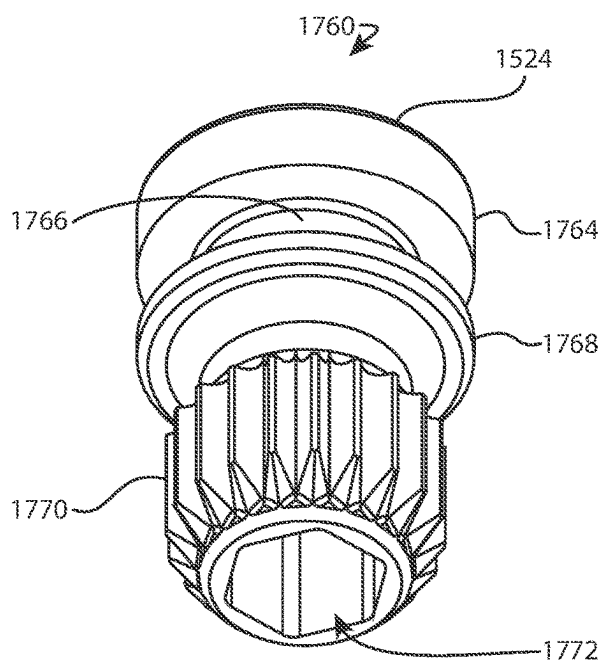


FIG. 114

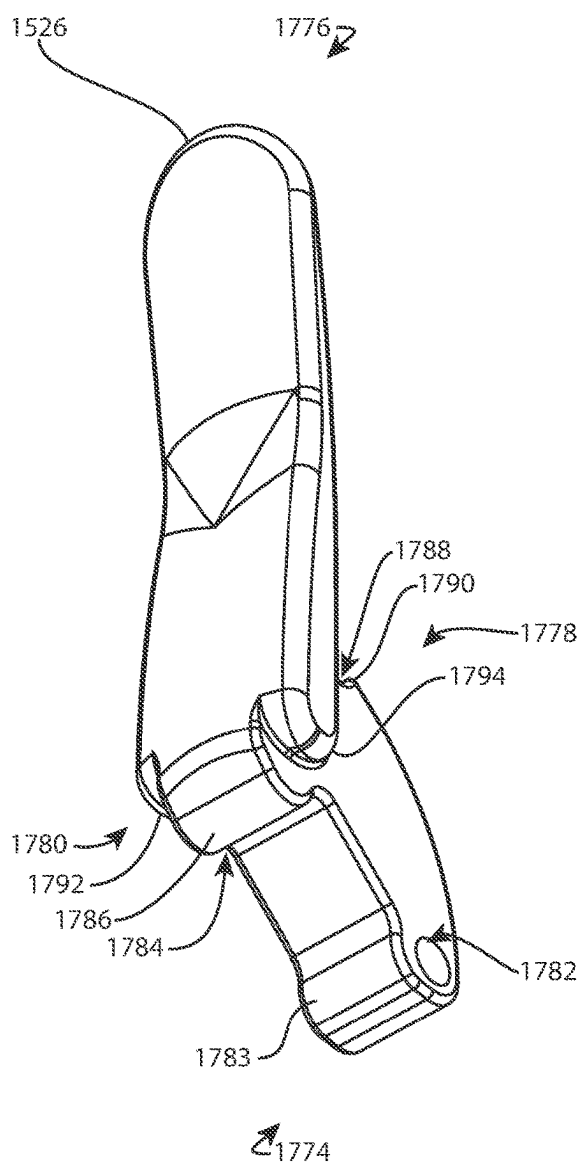


FIG. 115

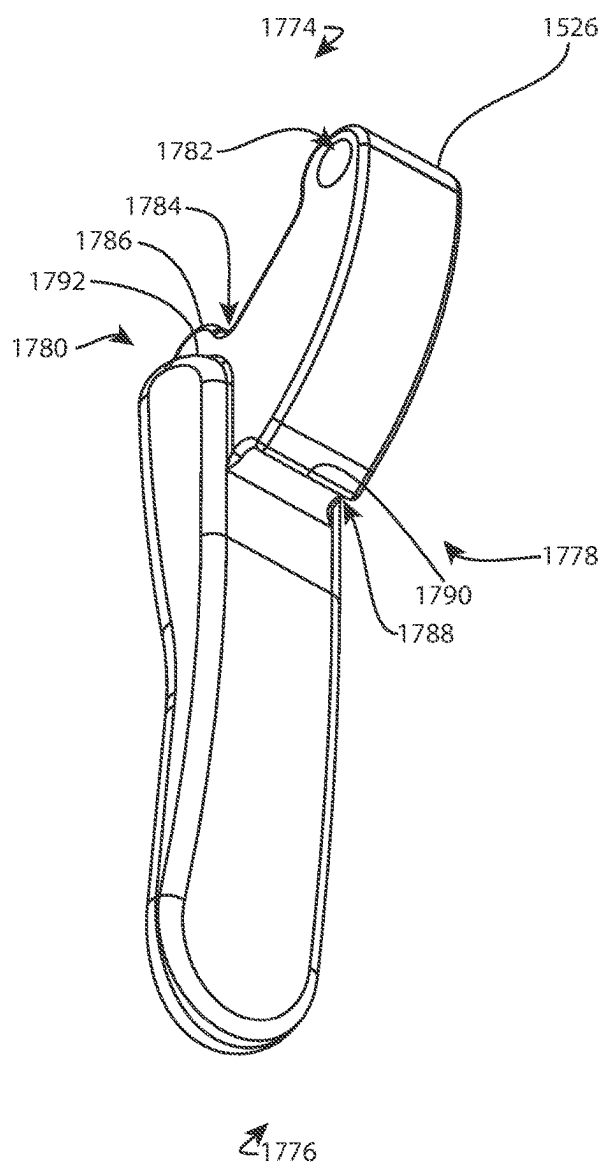


FIG. 116

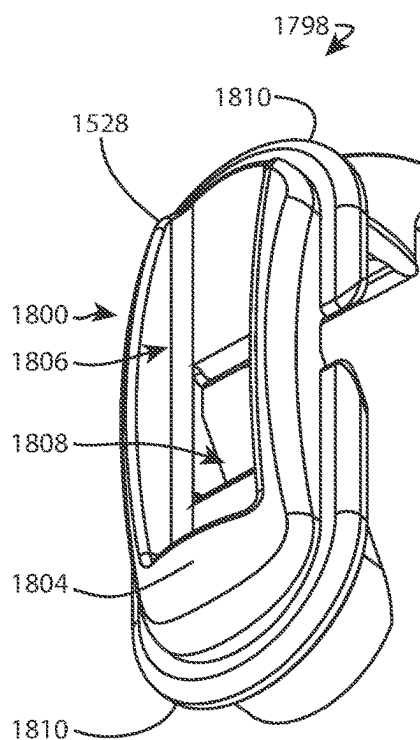


FIG. 117

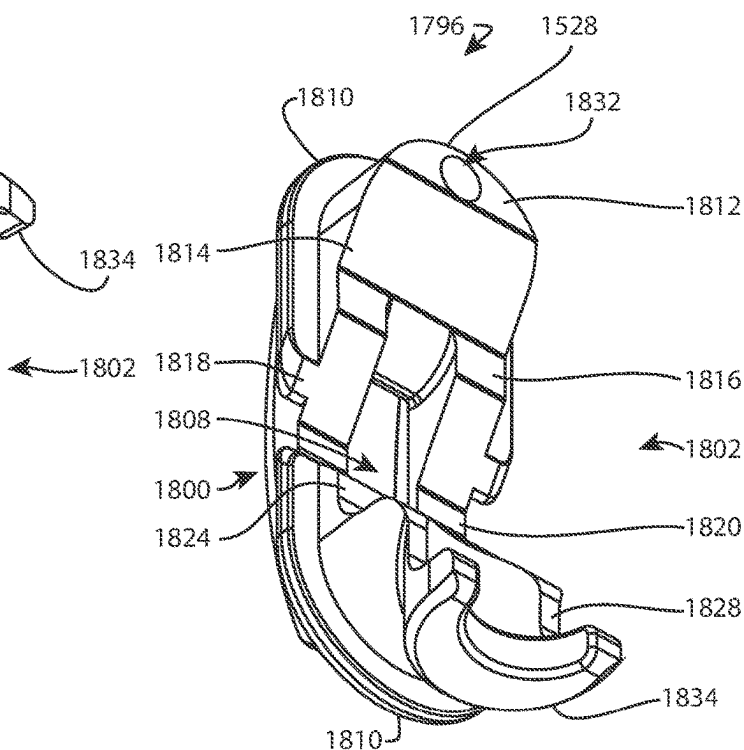


FIG. 118

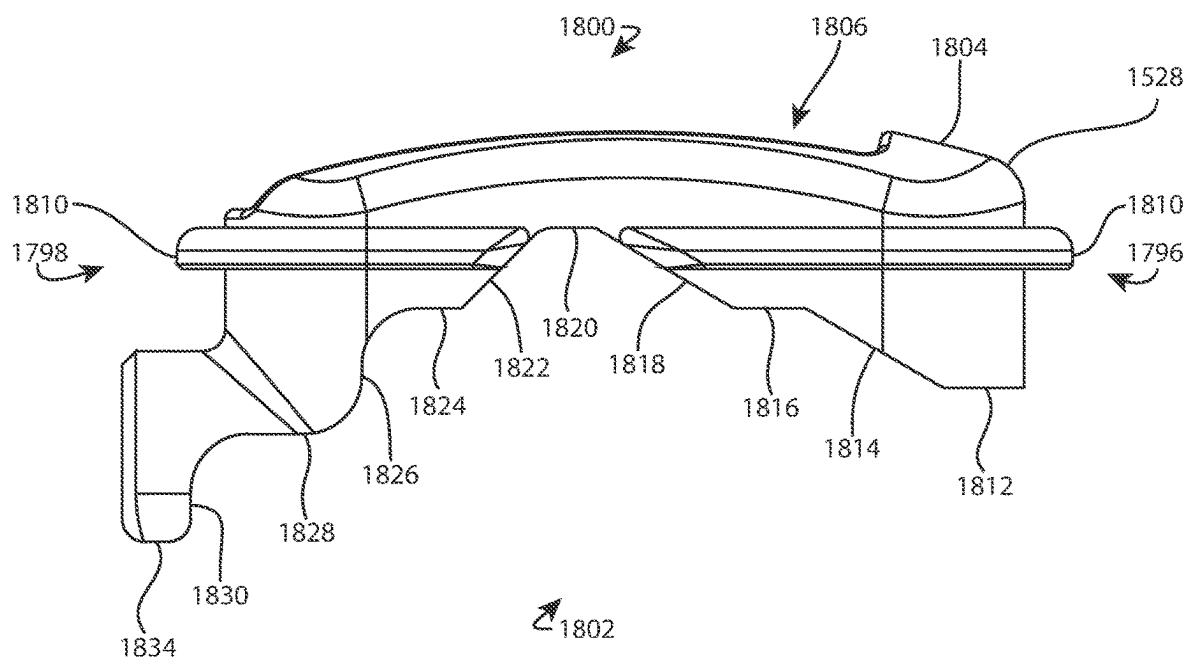


FIG. 119

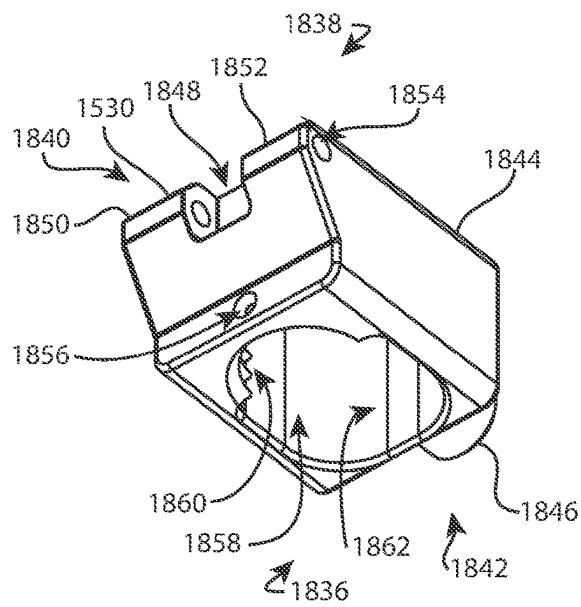


FIG. 120

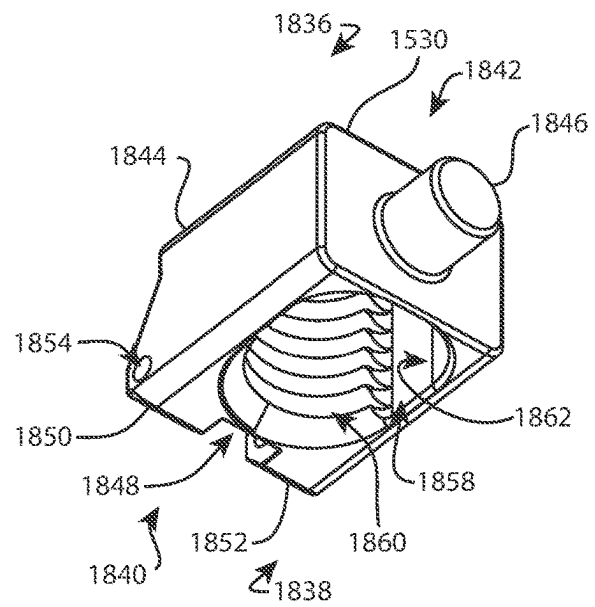


FIG. 121

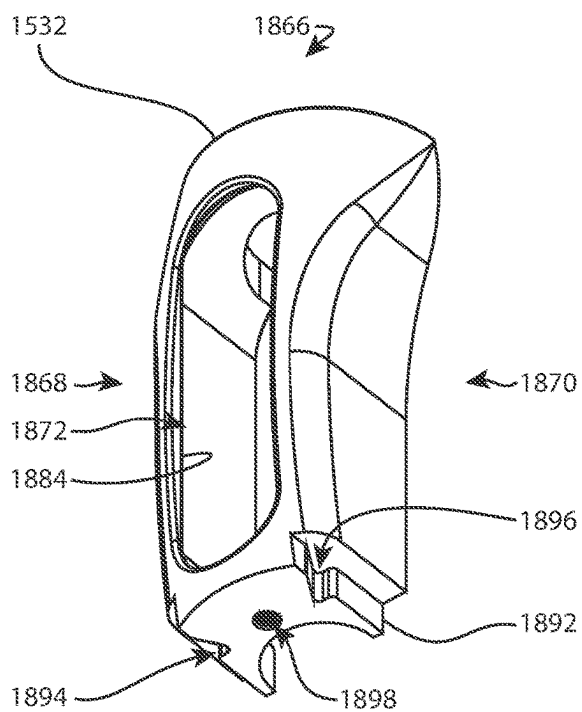


FIG. 122

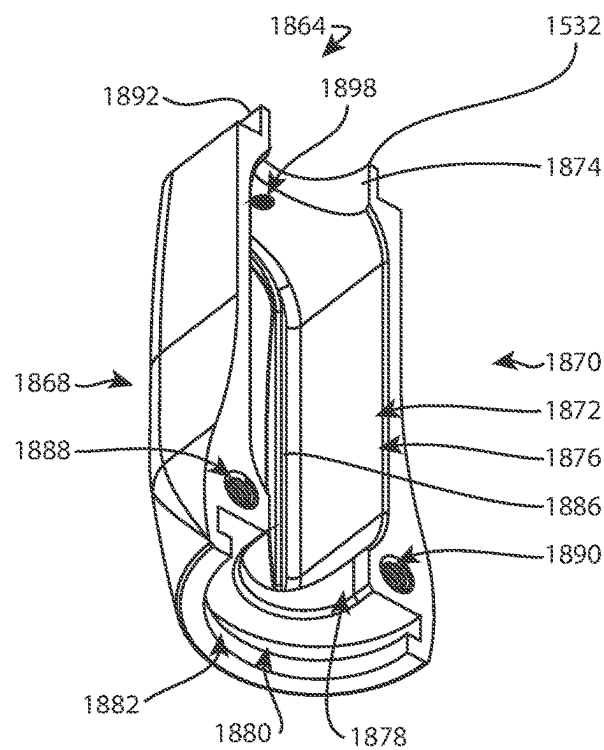


FIG. 123

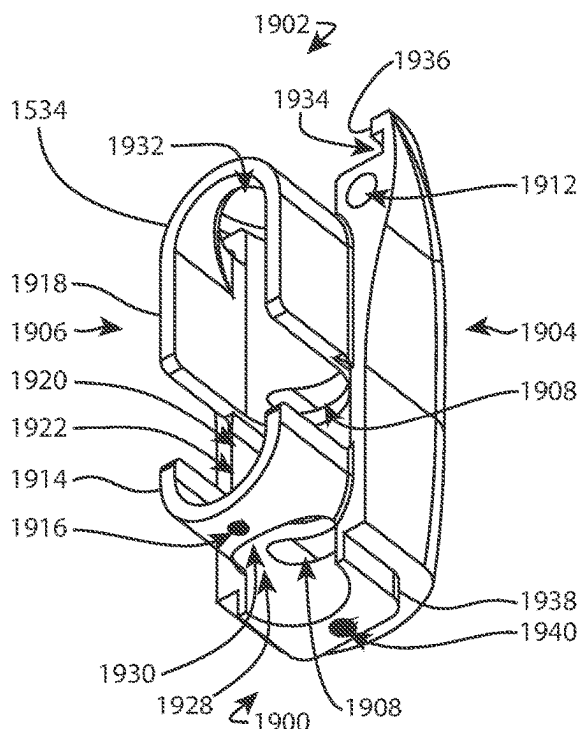


FIG. 124

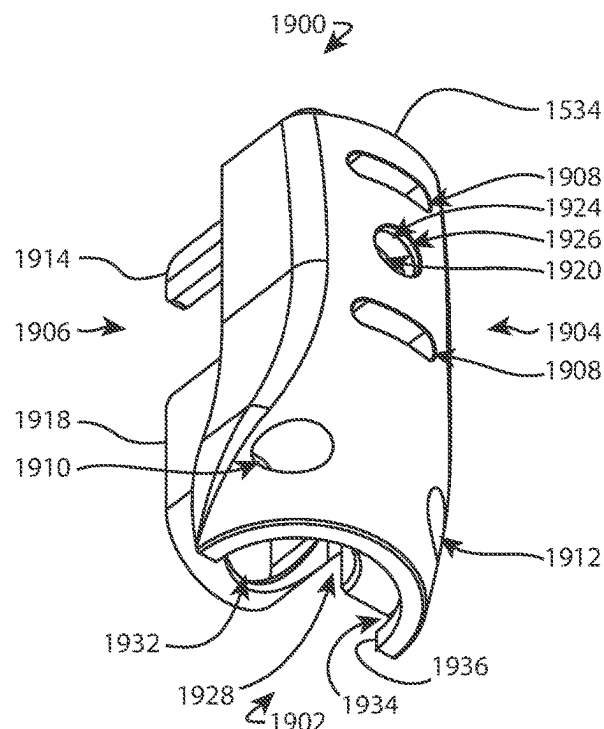


FIG. 125

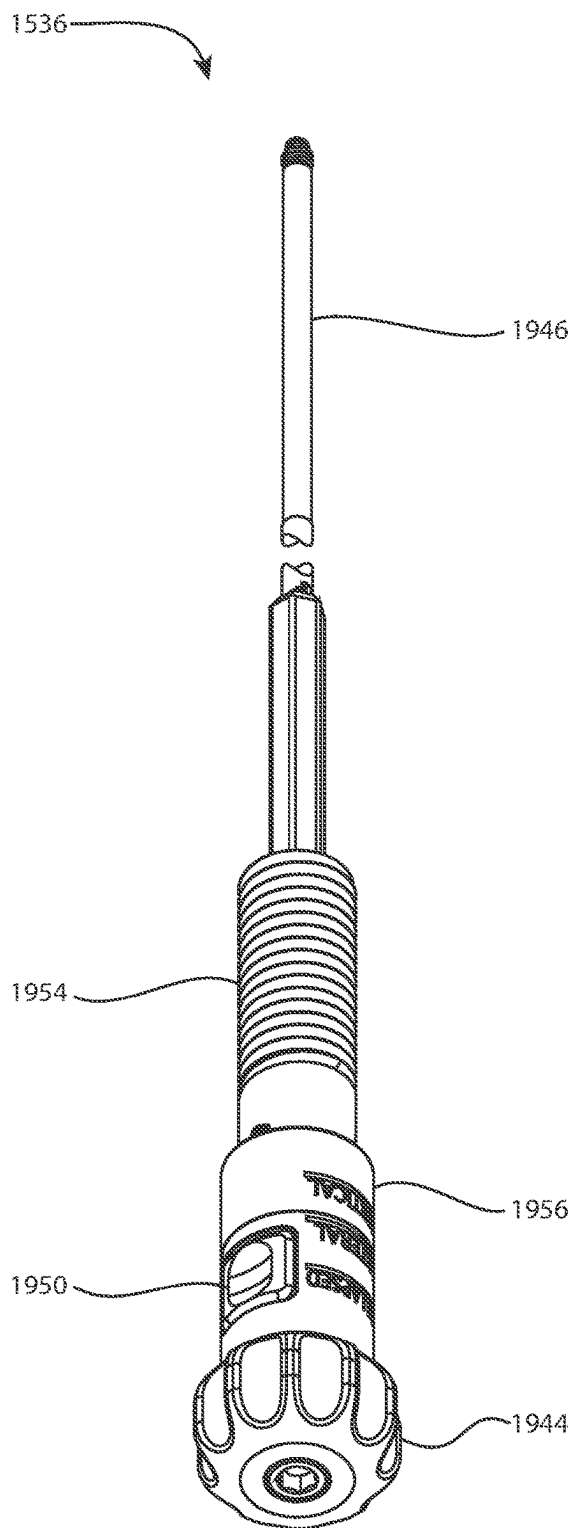


FIG. 126

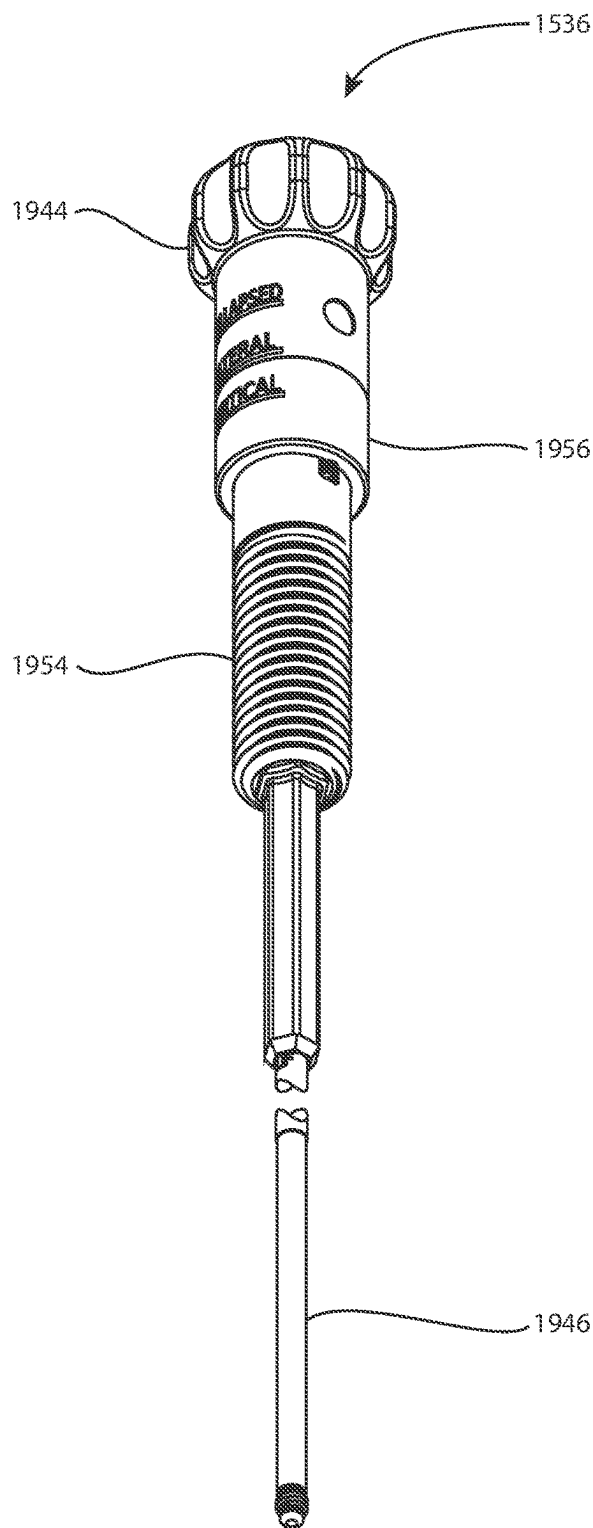


FIG. 127

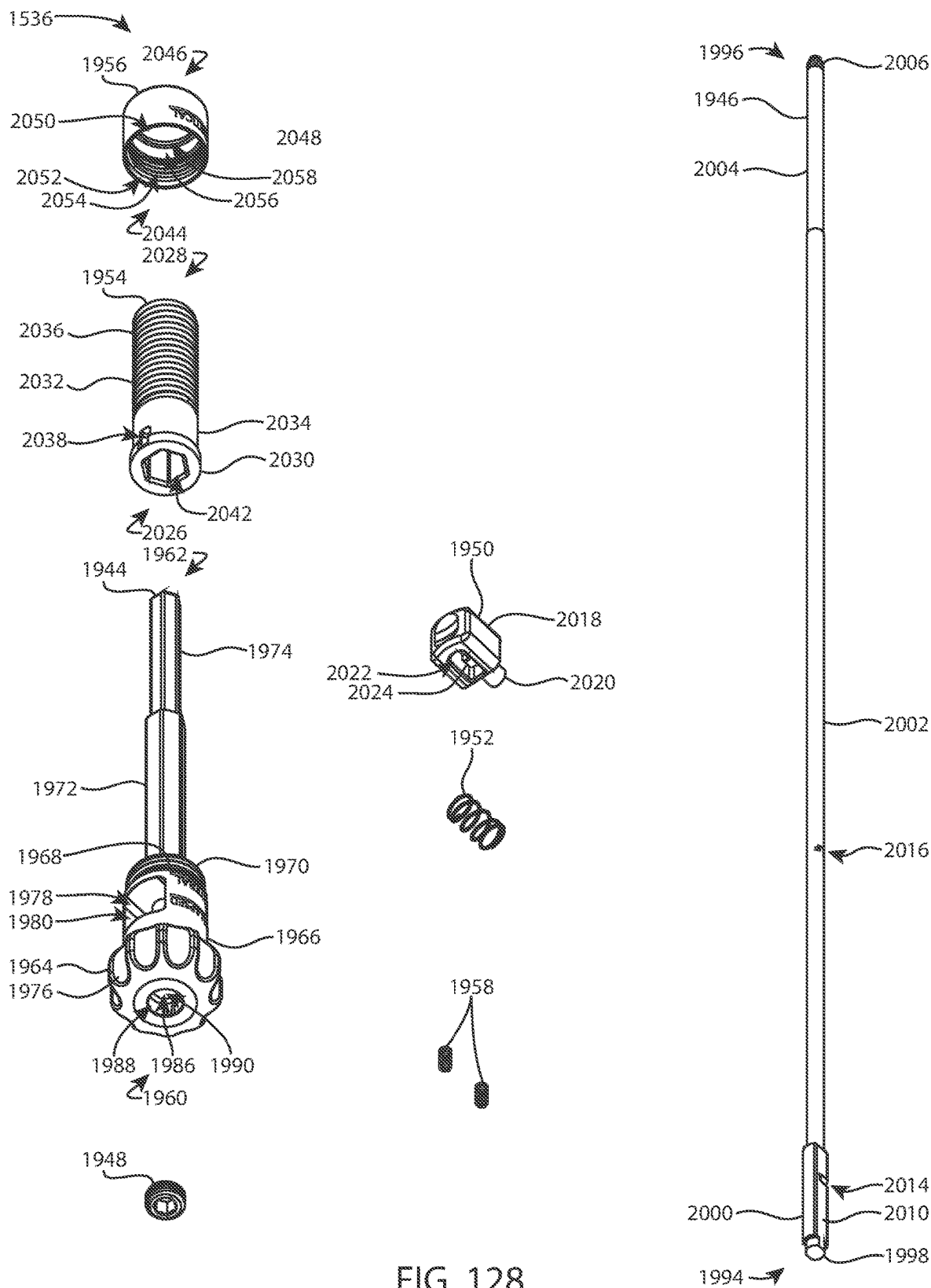
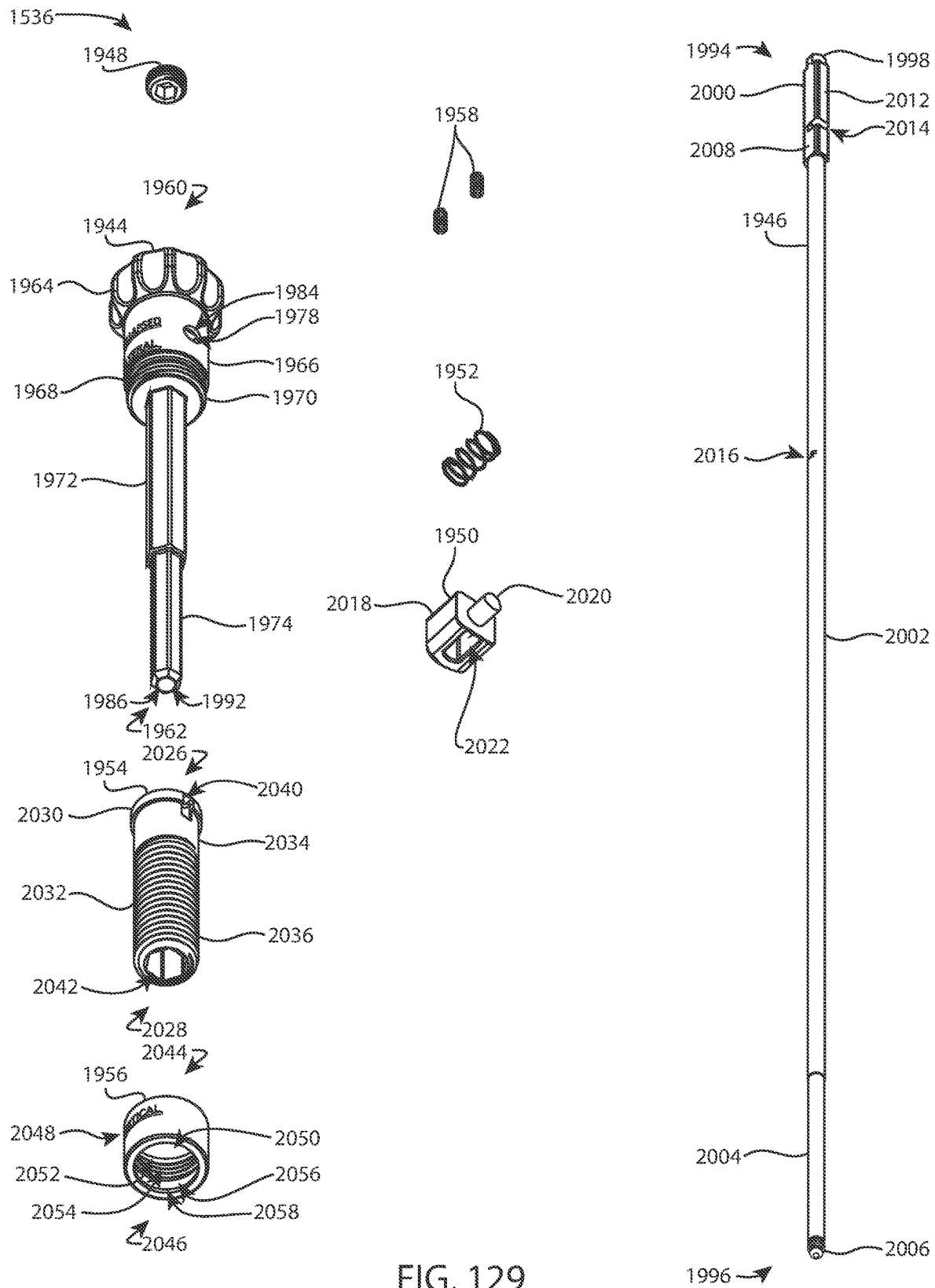


FIG. 128



2060

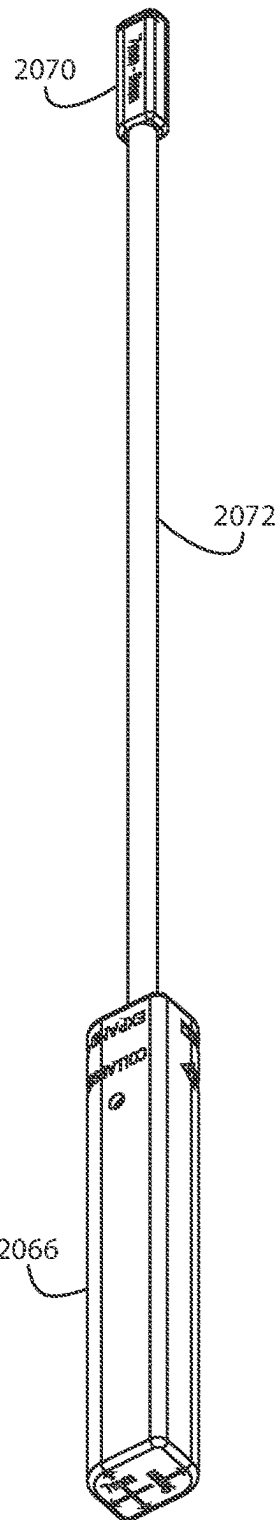


FIG. 130

2060

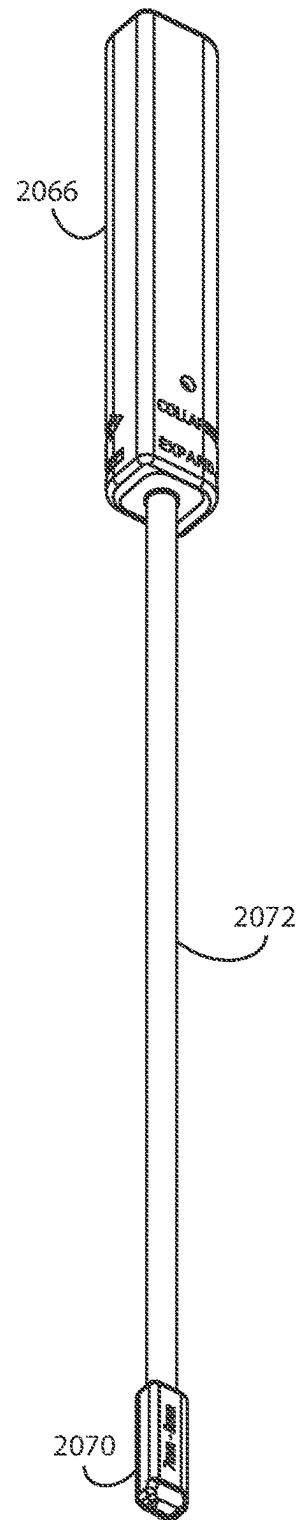
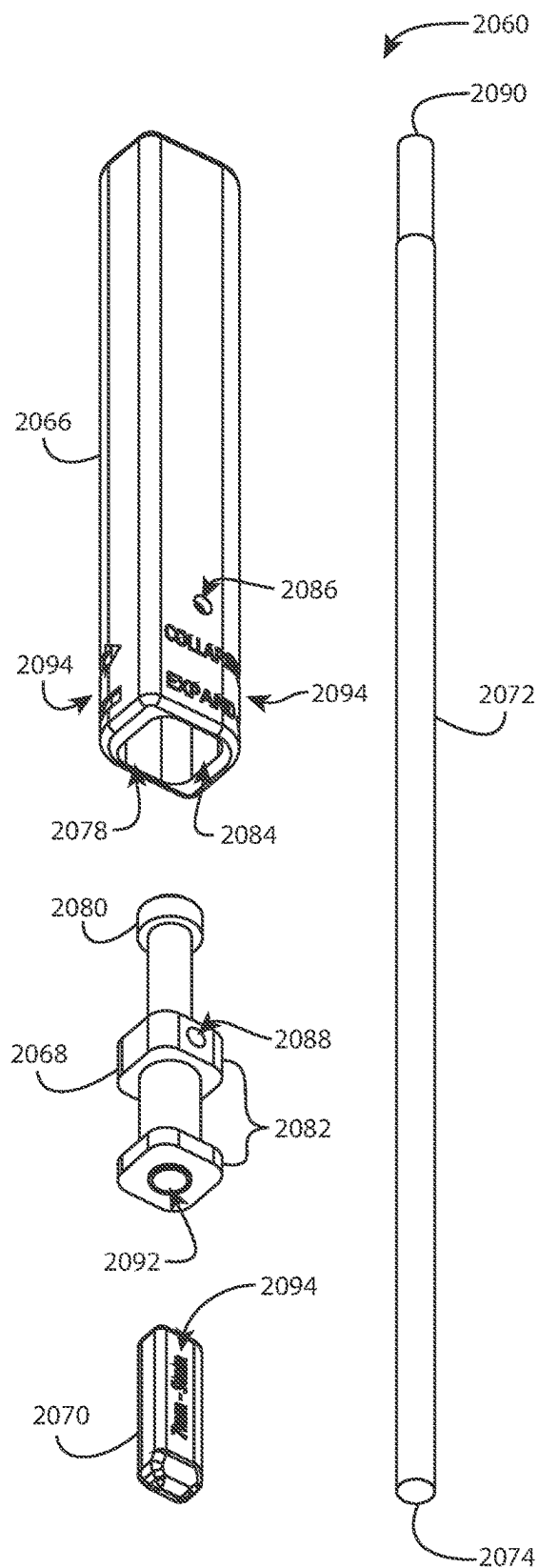
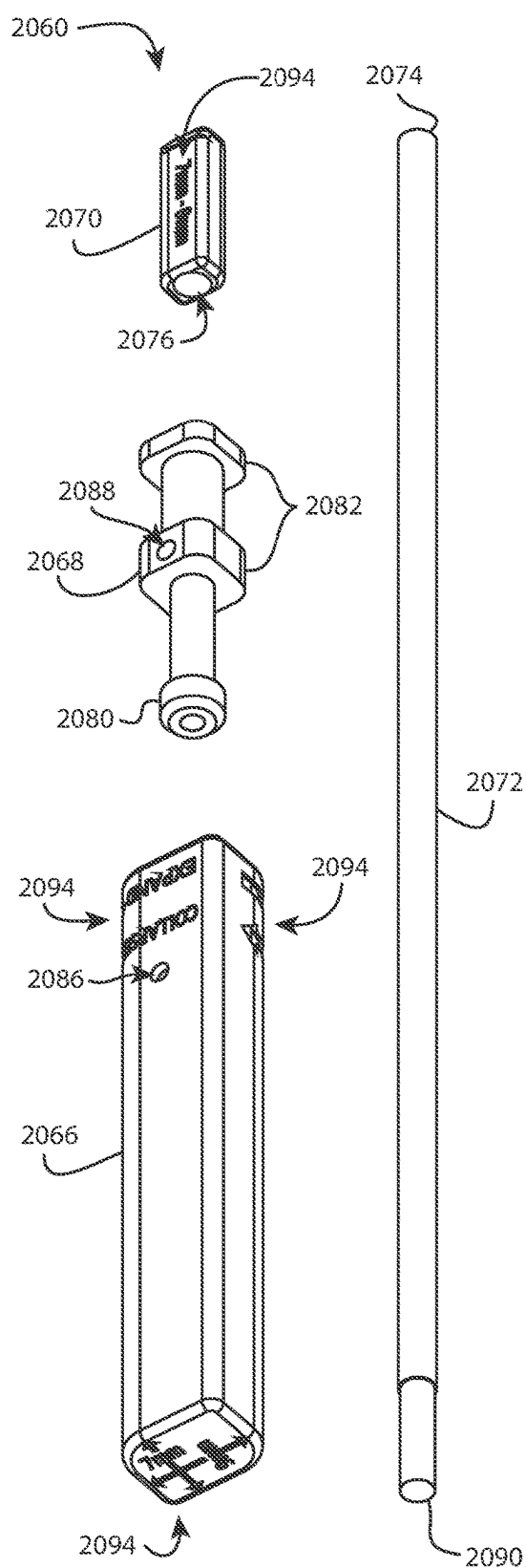


FIG. 131



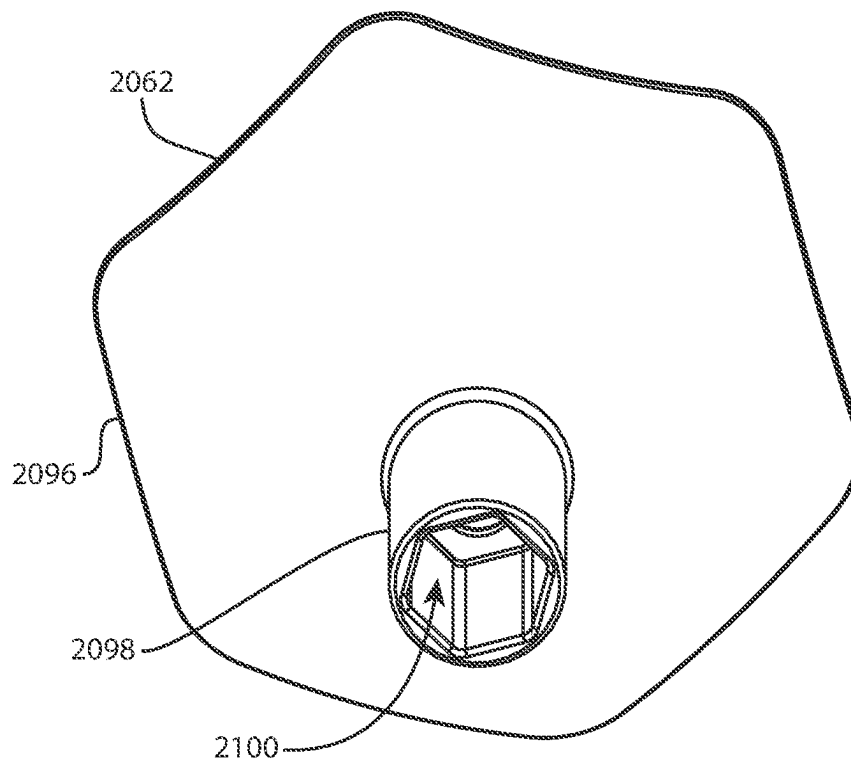


FIG. 134

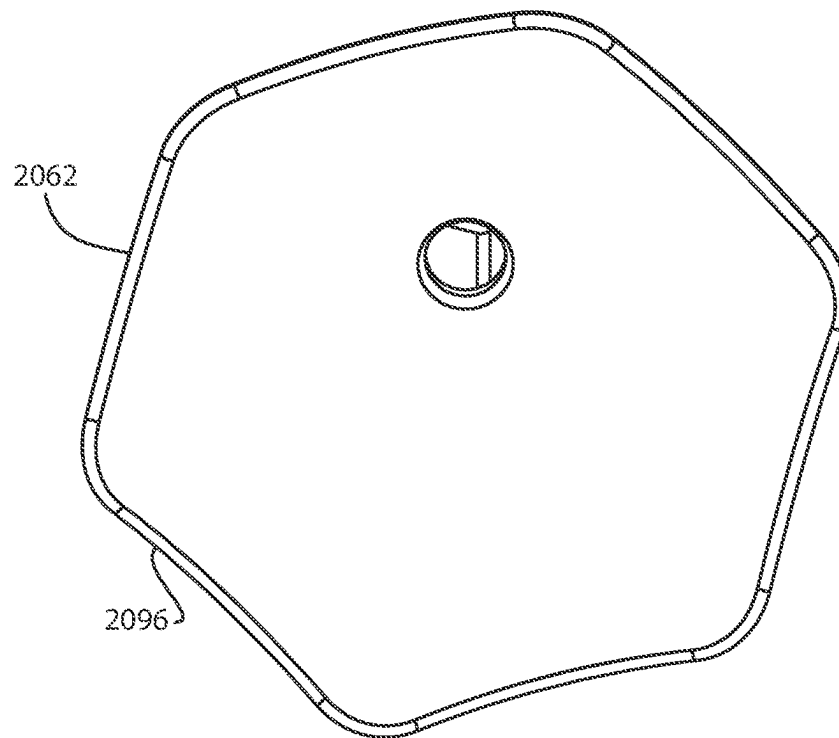


FIG. 135

2064

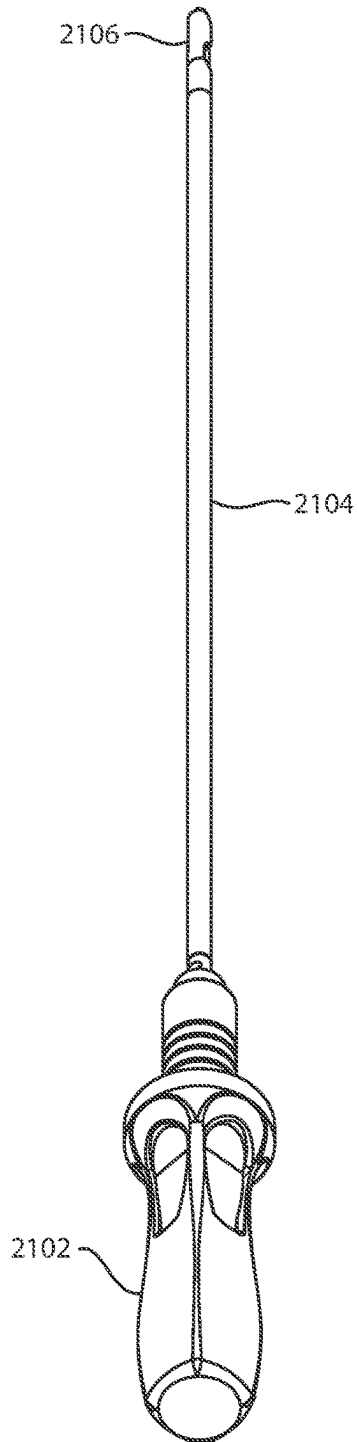


FIG. 136

2064

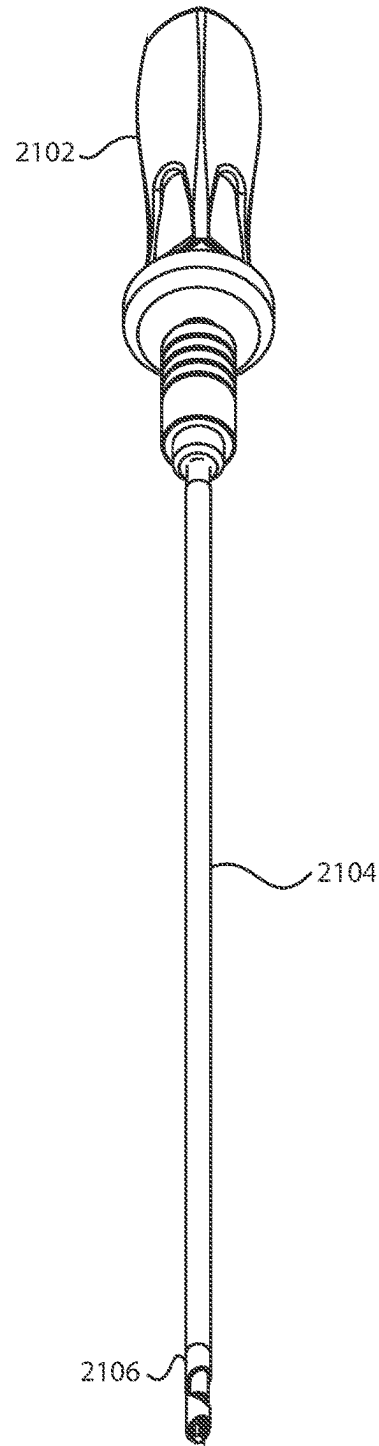


FIG. 137

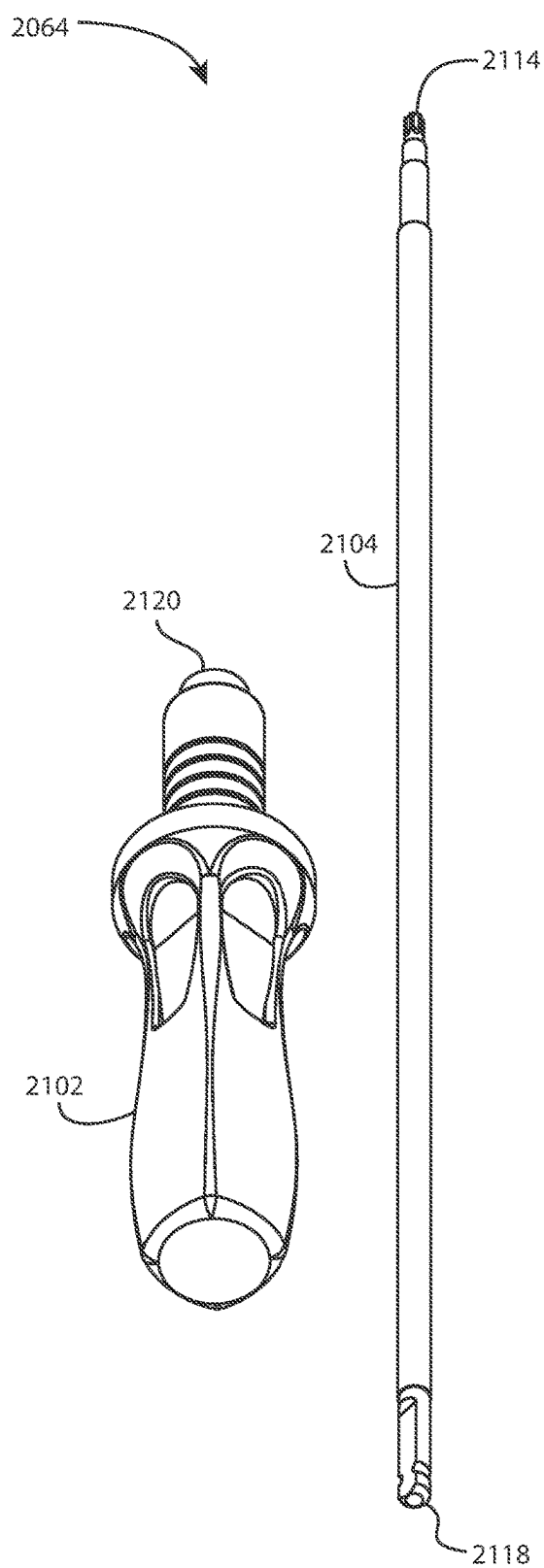


FIG. 138

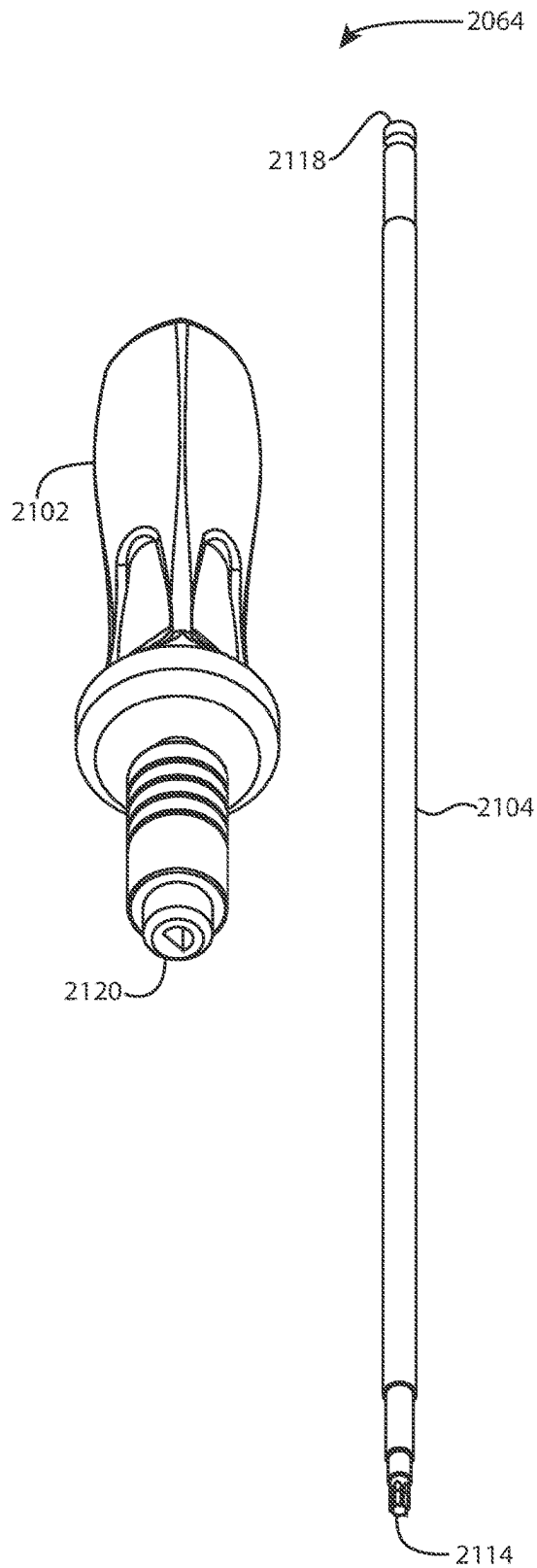


FIG. 139

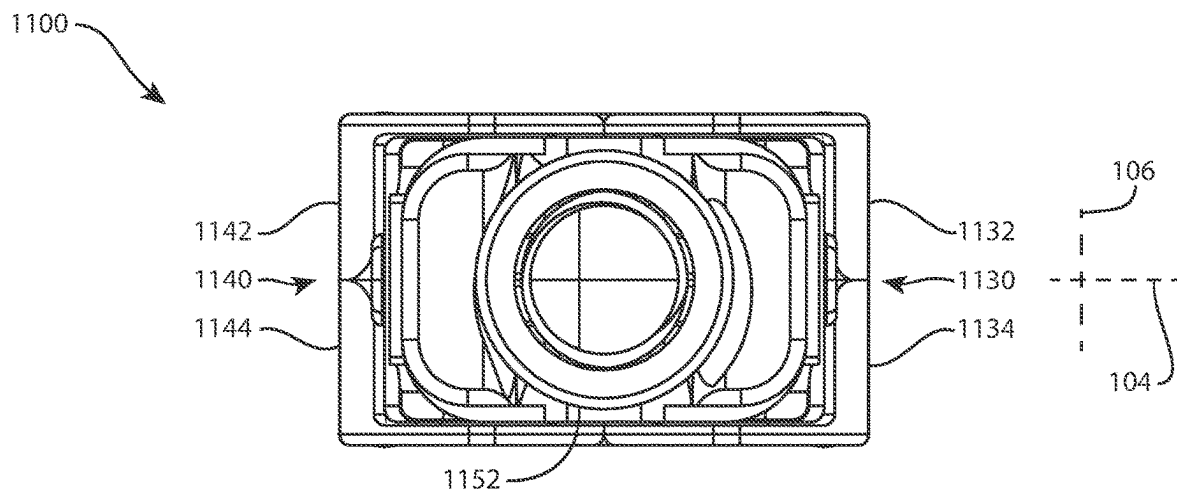


FIG. 140

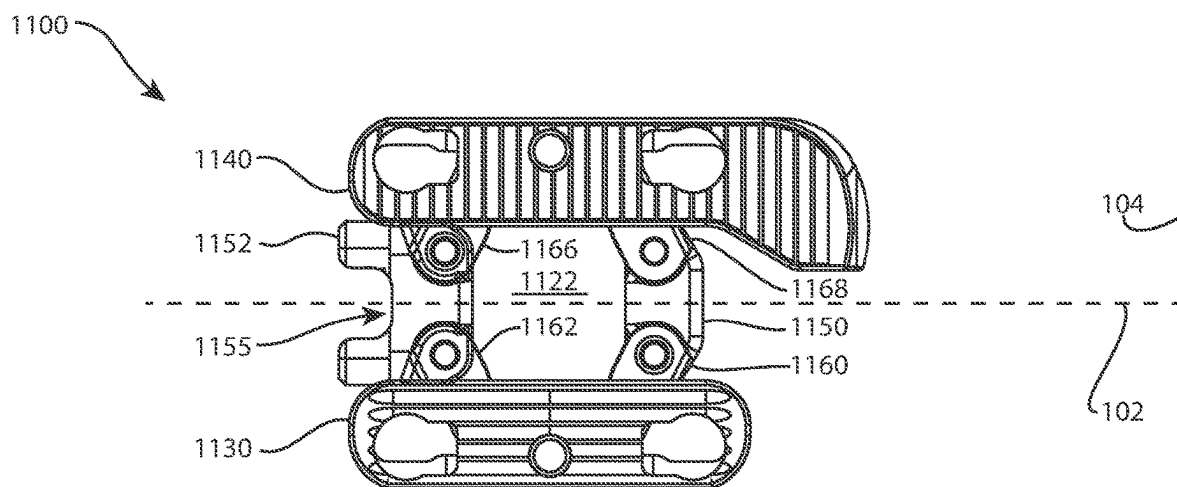


FIG. 141

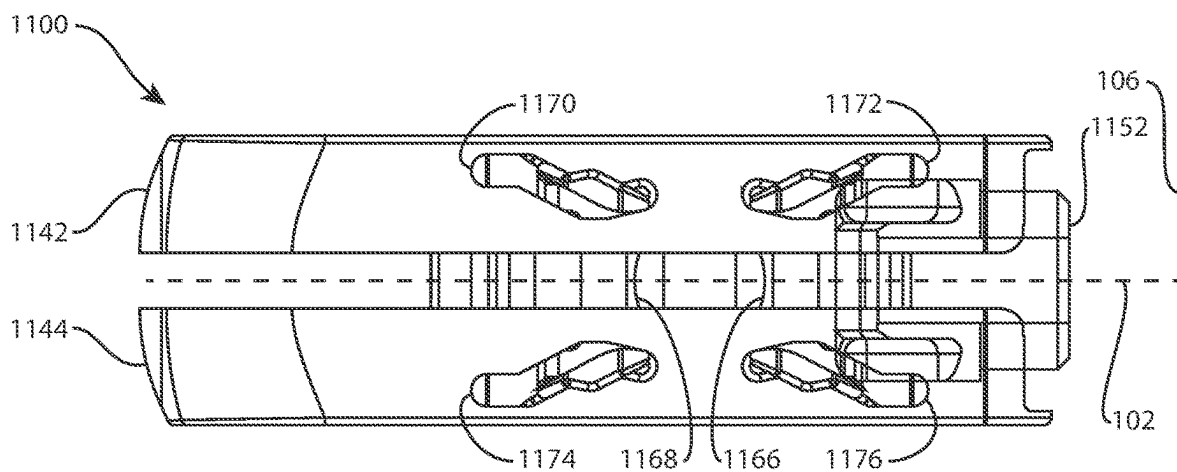


FIG. 142

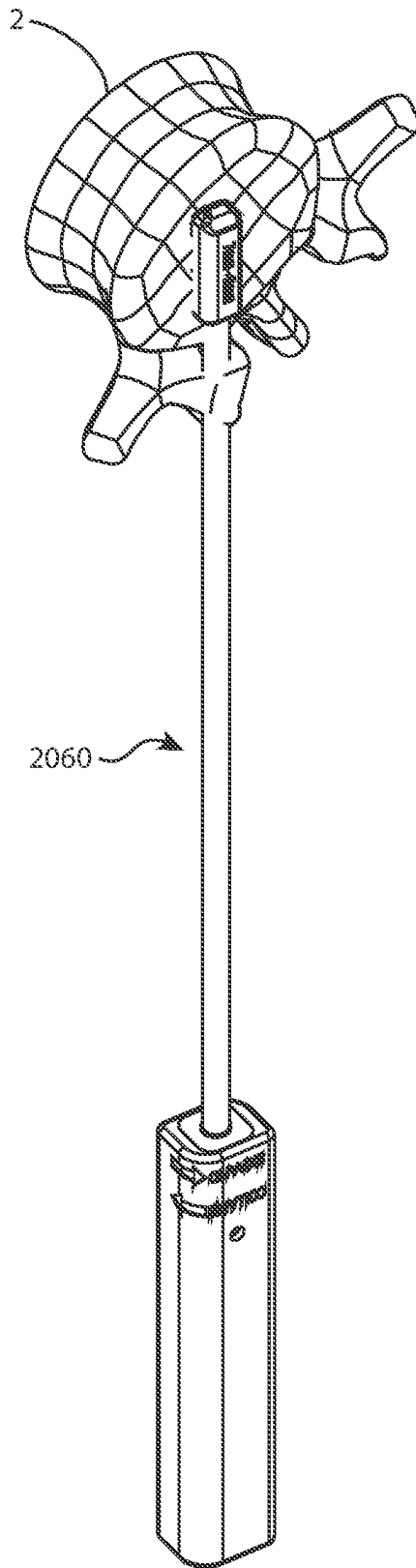


FIG. 143

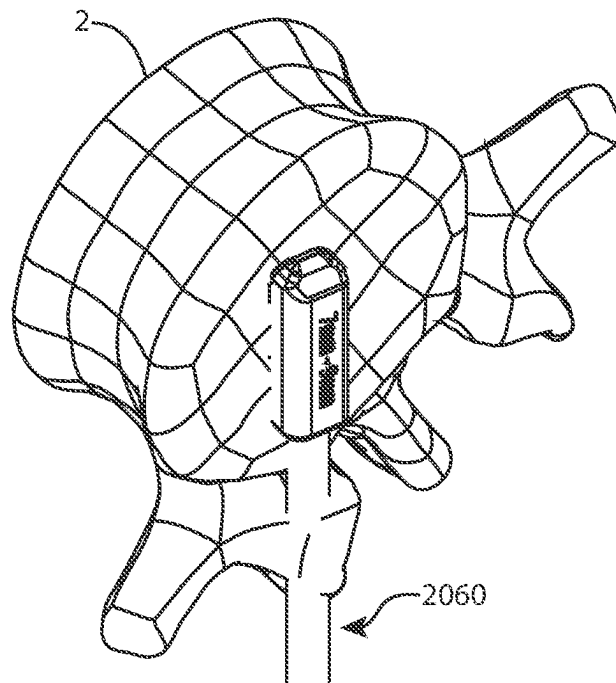


FIG. 144

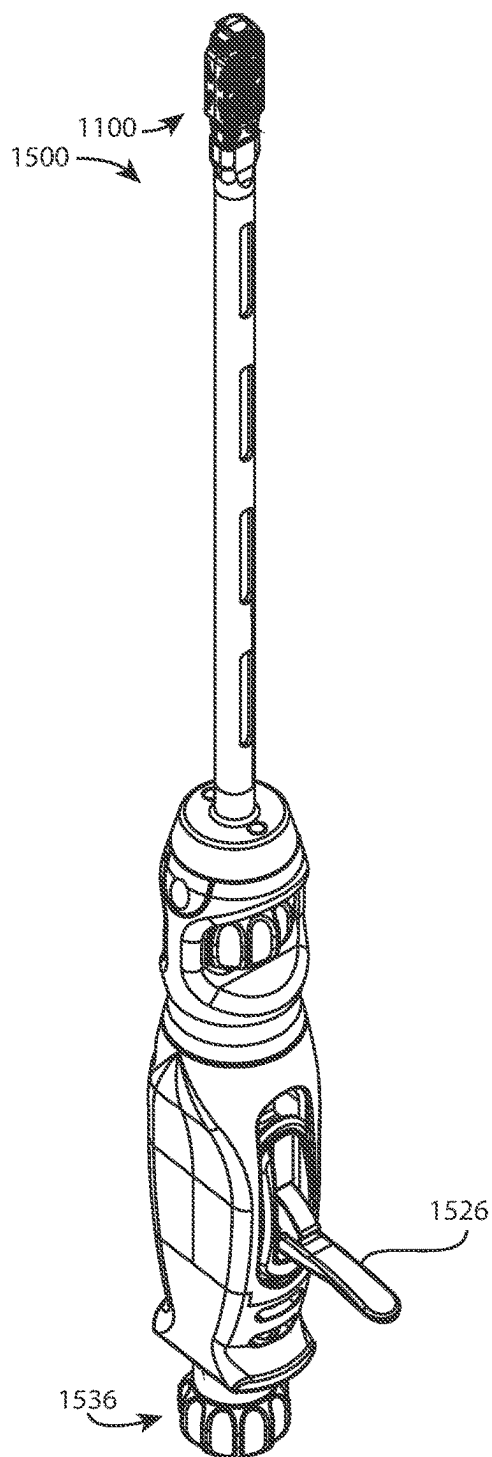


FIG. 145

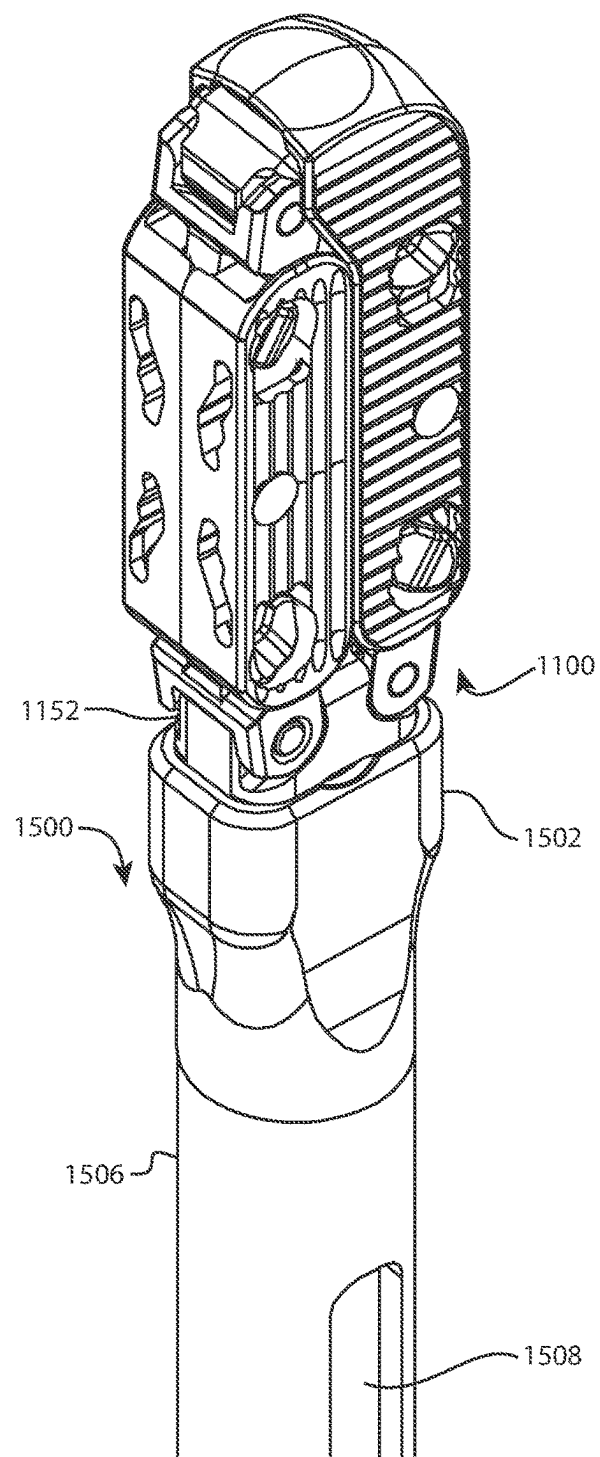


FIG. 146

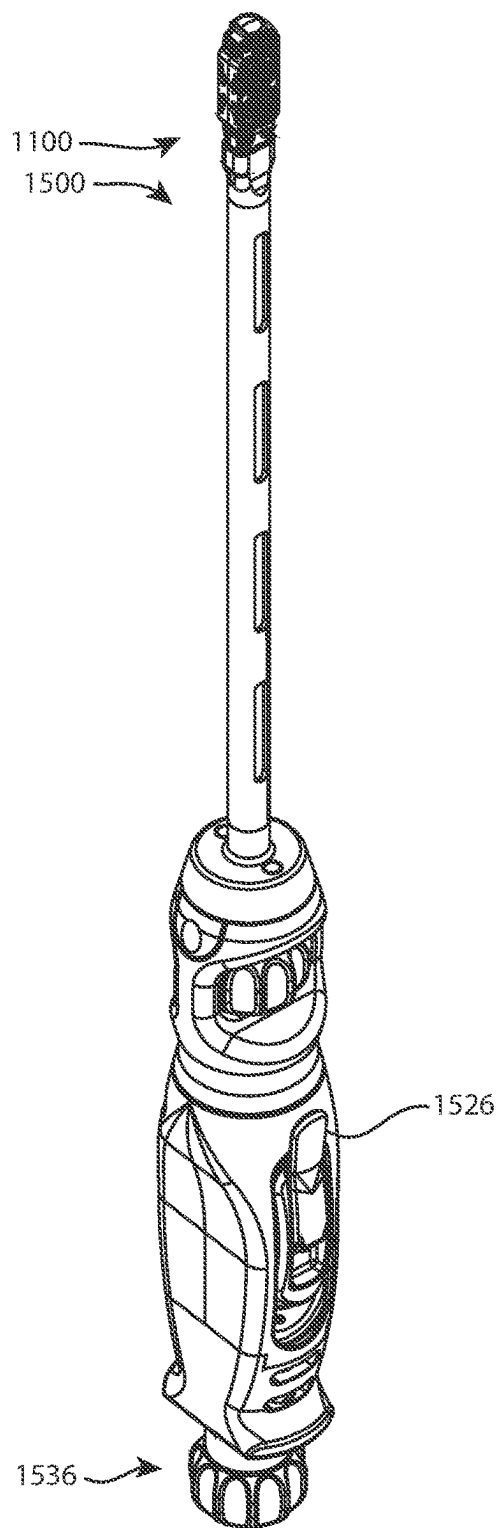


FIG. 147

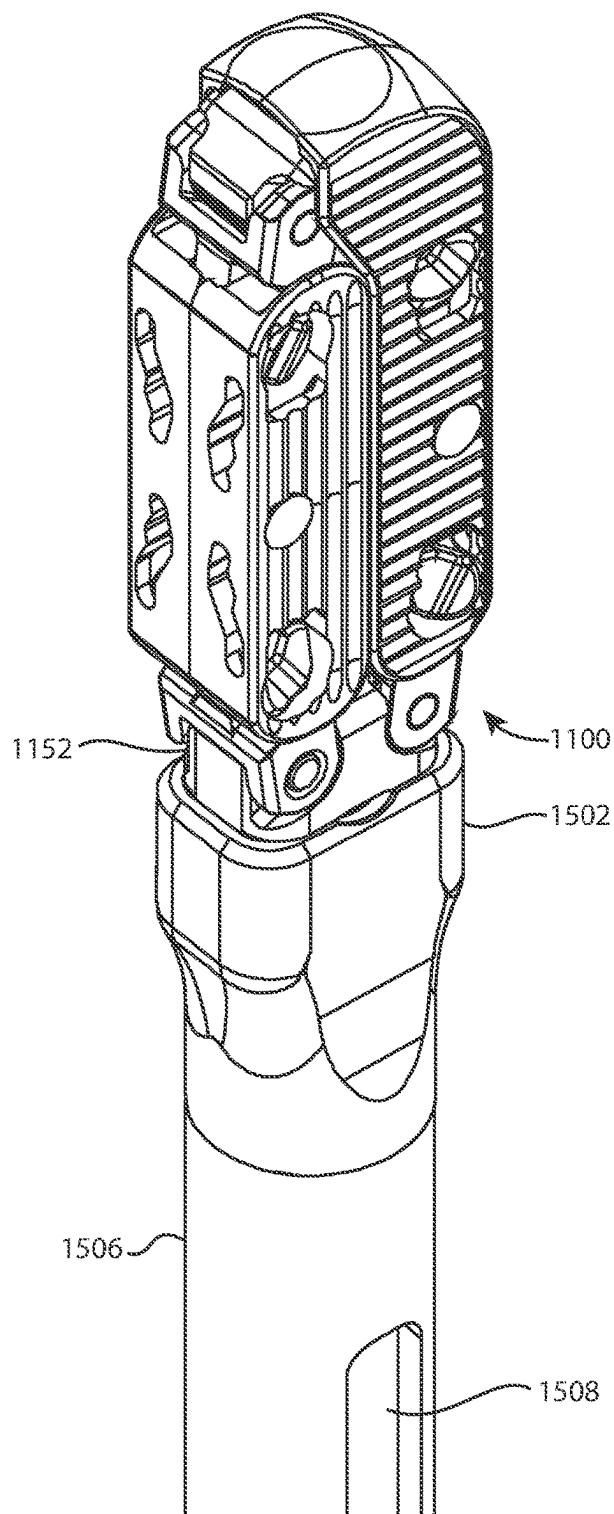


FIG. 148

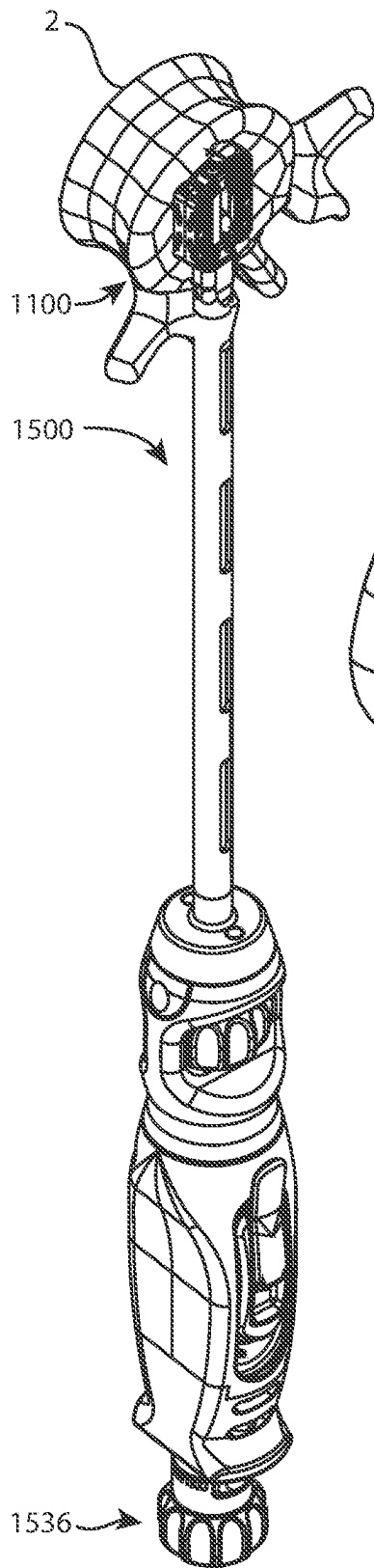


FIG. 149

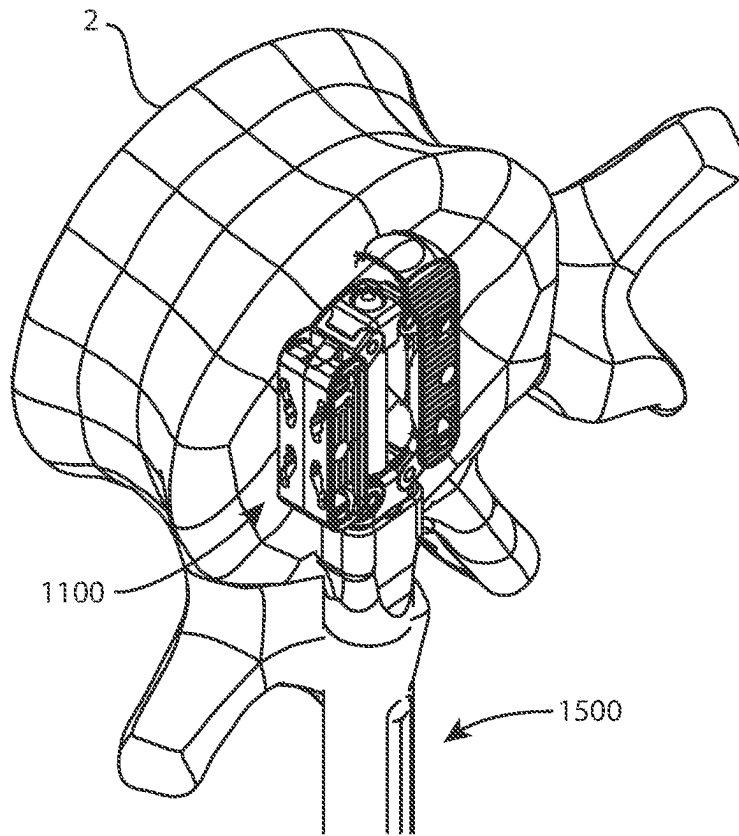


FIG. 150

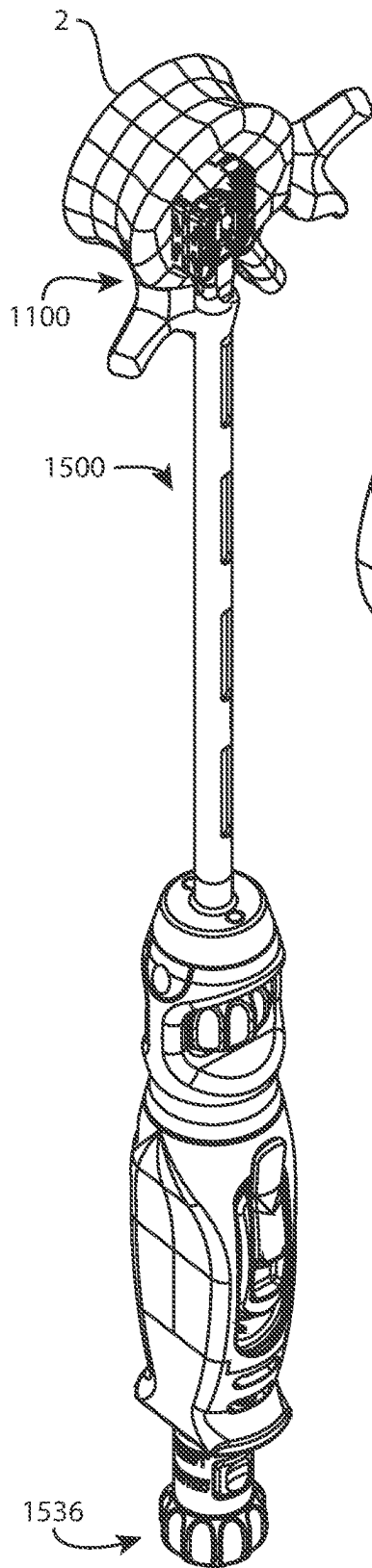


FIG. 151

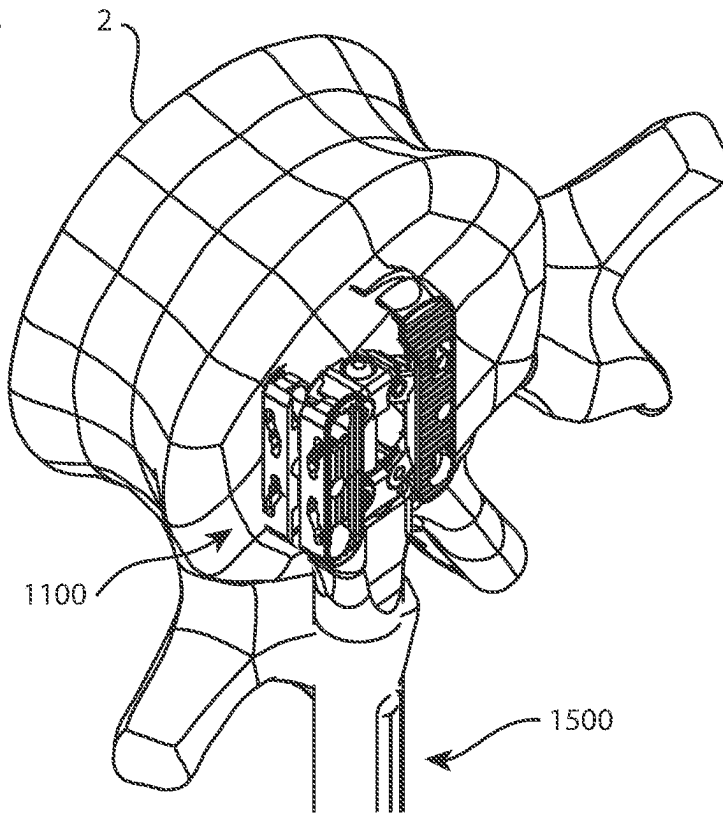


FIG. 152

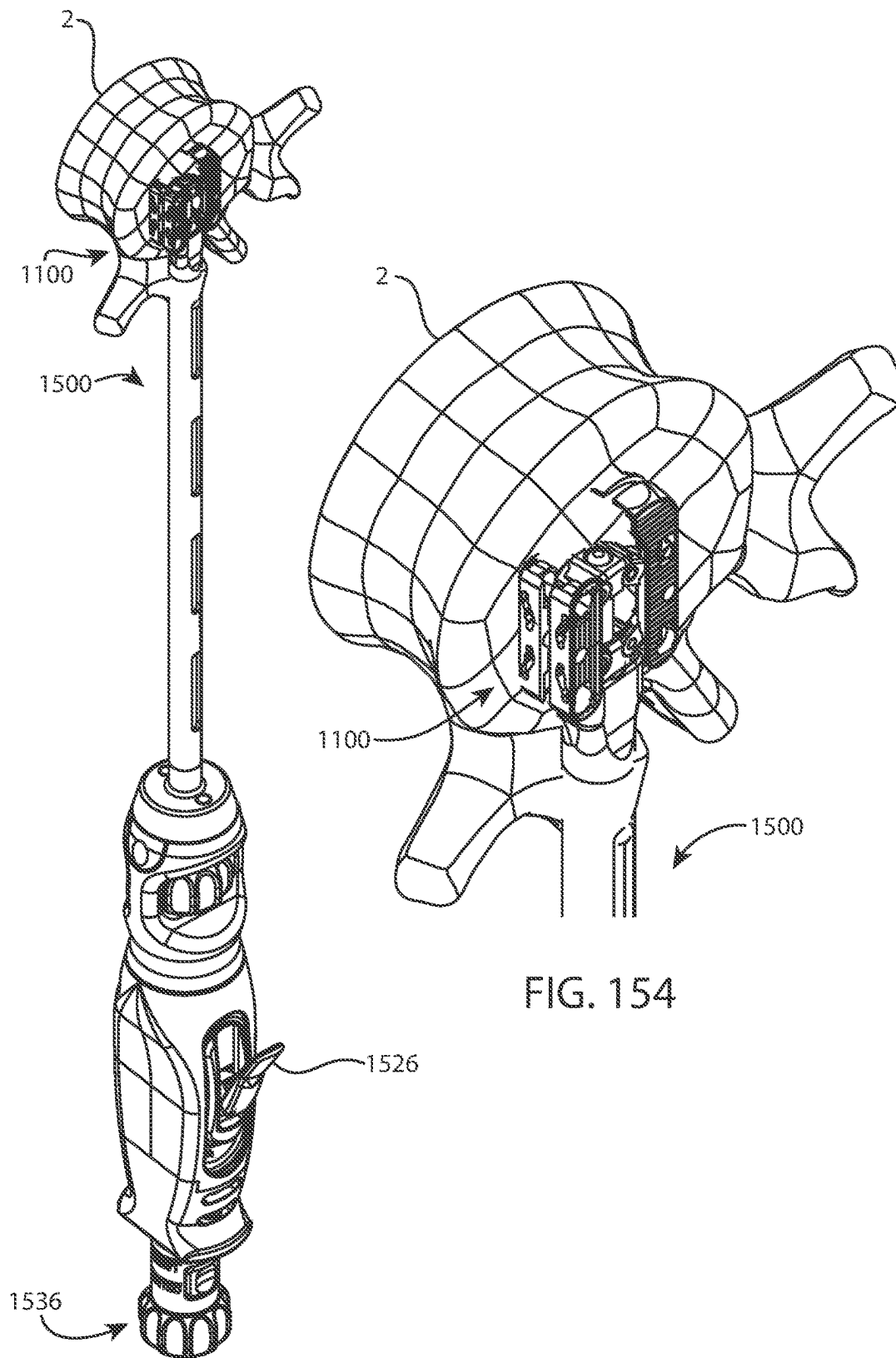


FIG. 154

FIG. 153

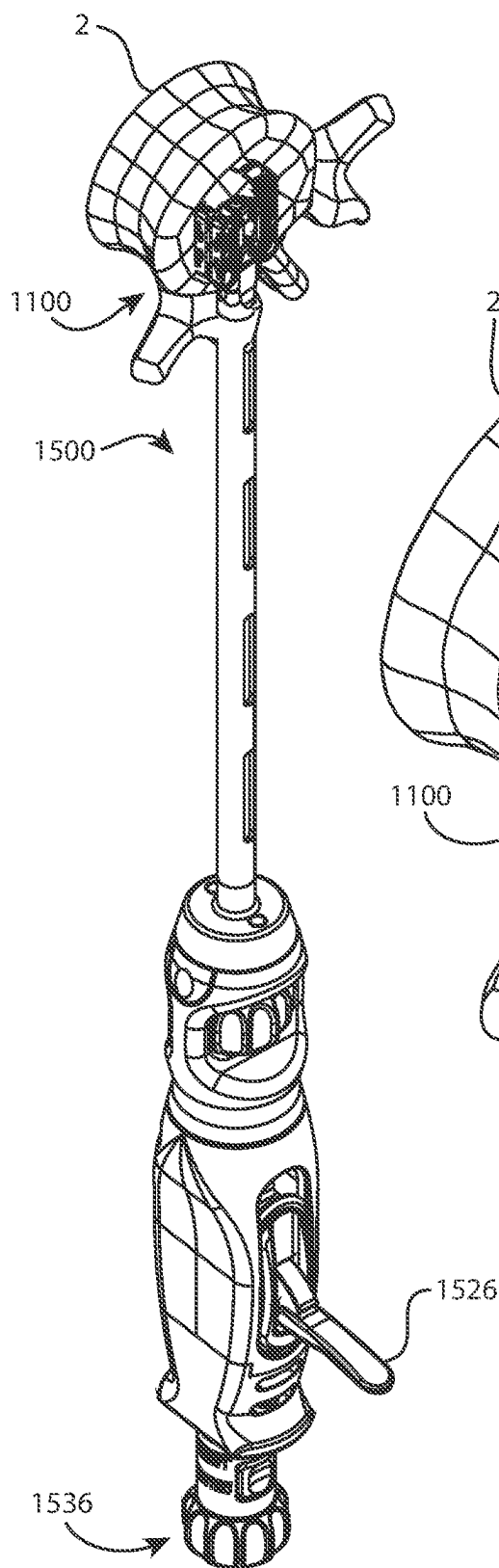


FIG. 155

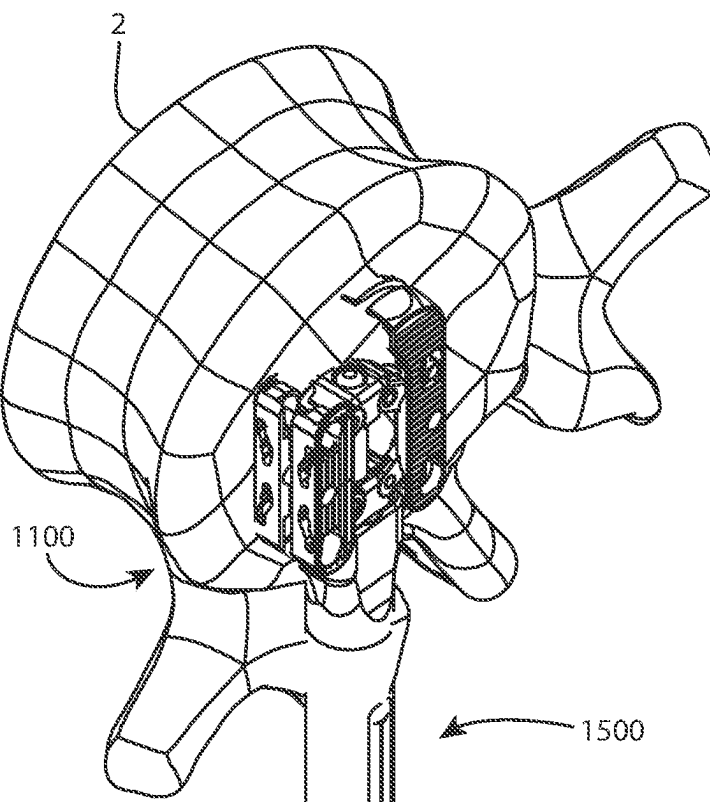


FIG. 156

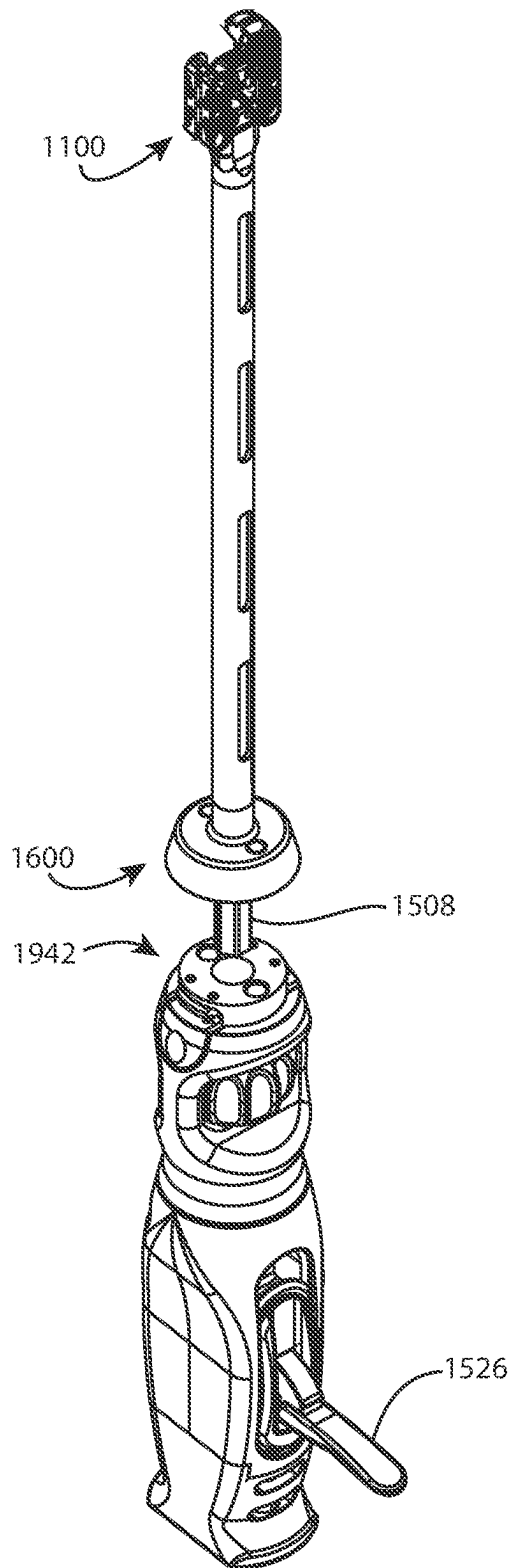


FIG. 157

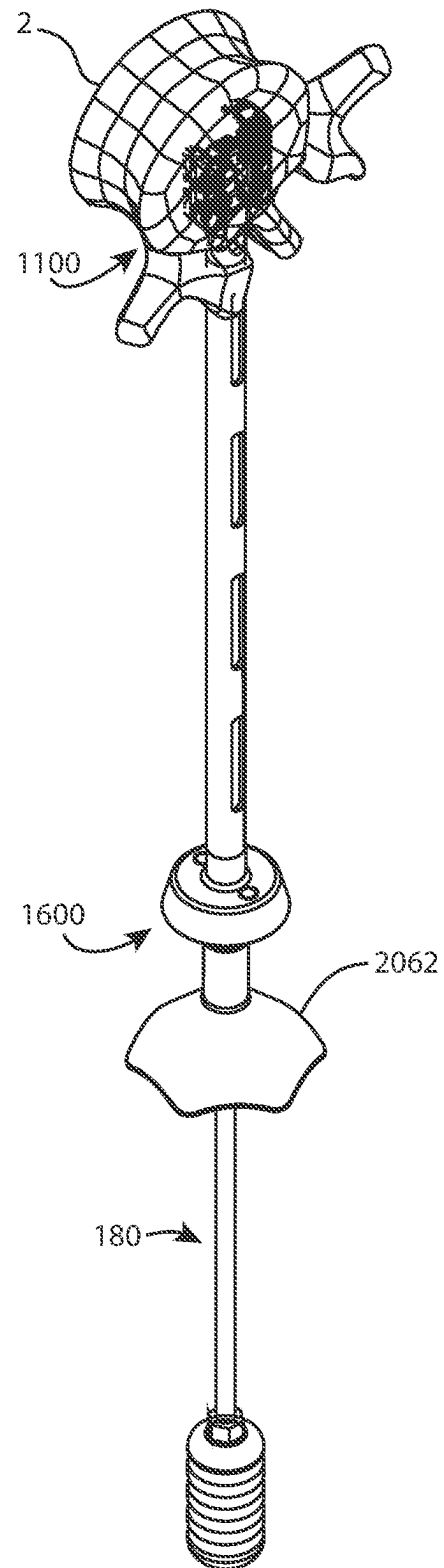


FIG. 158

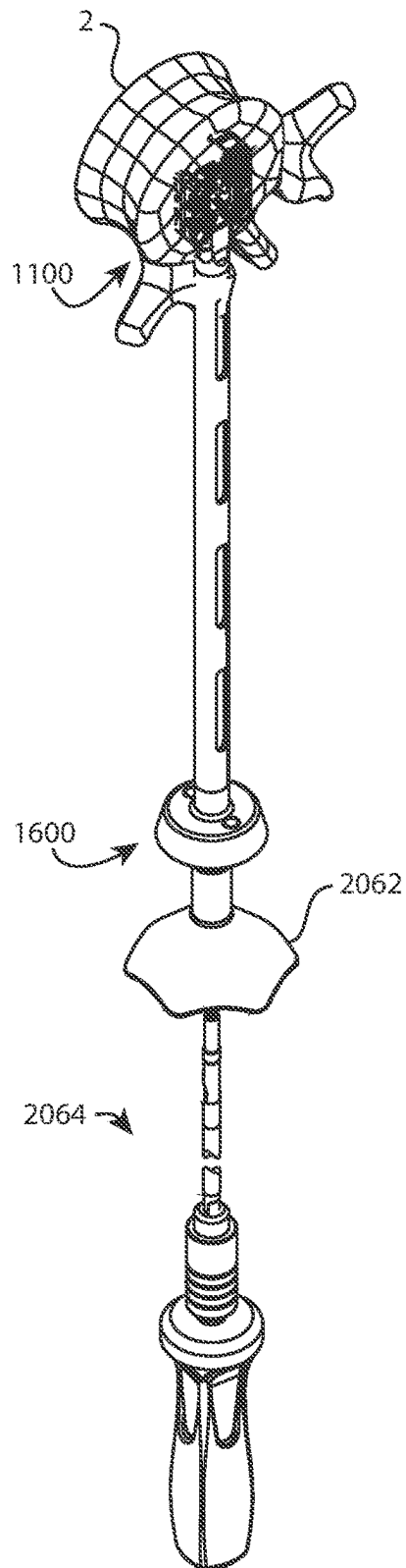


FIG. 159

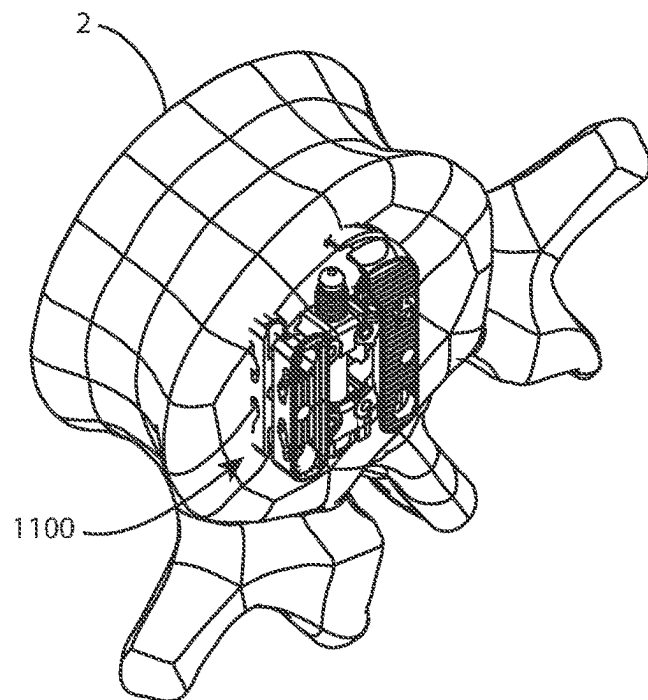


FIG. 160

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METHODS AND INSTRUMENTATION FOR INTERVERTEBRAL CAGE EXPANSION

CROSS-REFERENCE TO RELATED APPLICATIONS

The present application is a continuation of U.S. patent application Ser. No. 15/793,950 filed on Oct. 25, 2017, entitled "Methods and Instrumentation for Intervertebral Cage Expansion", which claims the benefit of U. S. Provisional Patent Application No. 62/412,781, filed on Oct. 25, 2016, entitled "Methods and Instrumentation for Intervertebral Cage Expansion". The foregoing is incorporated by reference as though set forth herein in its entirety.

TECHNICAL FIELD

The present disclosure relates to instruments and methods for implantation and expansion of an intervertebral implant. The instruments and methods described herein may be used to expand an intervertebral implant or spacer which is capable of being expanded in one or more directions: horizontally (laterally) and/or vertically (cephalocaudally). More specifically, an embodiment of the instruments and methods described herein may be used to implant and expand the intervertebral cages described in pending U. S. patent application Ser. No. 15/244,446, entitled EXPANDABLE INTERVERTEBRAL IMPLANTS, filed on Aug. 23, 2016, which is incorporated by reference as though set forth in its entirety.

BACKGROUND

There is a need for an instrument that may be coupled to a multi-axis expandable intervertebral cage so that the cage may be inserted into an intervertebral space, expanded along multiple different directions, filled with bone graft, and/or locked with a fastener. There is also a need for an instrument set that includes the implant insertion instrument and auxiliary instruments to determine implant size, insert bone graft into the cage, and deliver the fastener.

The implant insertion instruments disclosed herein address certain characteristics of the expandable intervertebral cages. The cages expand horizontally and/or vertically as first and second implant end bodies are drawn together along a longitudinal instrument axis. Each of the implant insertion instruments includes a connector that couples to the first implant end body and a draw bar that couples to the second implant end body. The draw bar may be rotatable about the instrument axis to connect with and disconnect from the second implant end body, however the draw bar does not rotate about the axis during implant expansion. In some examples, in order to expand the implant, the implant insertion instrument may first be actuated to lock the draw bar to prevent axial rotation, then actuated to accomplish implant expansion. These actuations may be coupled together so as to occur together.

In some examples, the implant insertion instruments disclosed herein also address certain characteristics of specific approaches to the intervertebral space in the spine. For example, in transforaminal lumbar interbody fusion (TLIF), a patient may be in the prone position (face down) and the approach trajectory may lie in the transverse plane of the operative intervertebral space and inclined about 30 degrees from the sagittal plane (vertical) or 60 degrees from the coronal plane (horizontal). The weight of the implant insertion instrument thus causes a nontrivial force moment on the

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implant which tends to rotate the implant laterally out of position. Disclosed implant insertion instruments have removable handle assemblies to reduce instrument weight and thus the force moment on the implant.

In some examples, the implant insertion instruments disclosed herein also provide a useful and convenient collateral function for a graft funnel, wherein the funnel may be used to unthread an implant retainer shaft after bone graft has been delivered to the operative site.

SUMMARY

The various systems and methods of the present technology have been developed in response to the present state of the art, and in particular, in response to the problems and needs in the art that have not yet been fully solved by currently available expandable implants and related instruments. The systems and methods of the present technology may provide improved instruments and methods for implanting expandable implants, such as spinal interbody fusion spacers.

To achieve the foregoing, and in accordance with the technology as embodied and broadly described herein, in an aspect of the technology, an instrument includes: a first instrument part that is removably connectable to a first implant part of an implant; and a second instrument part that is removably connectable to a second implant part of the implant; wherein, when the first instrument part is connected to the first implant part, the second instrument part is connected to the second implant part, and the instrument is actuated, the first instrument part and the first implant part translate relative to the second instrument part and the second implant part along a central longitudinal first axis of the instrument without rotating about the first axis during instrument actuation for implant expansion.

Embodiments of this aspect may include any or all of the following attributes. The first instrument part includes a draw bar that is removably connectable to the first implant part by rotation of the draw bar about the first axis; wherein the instrument further includes an actuator and a rotation lock; wherein instrument actuation includes movement of the actuator in engagement with the draw bar to translate the draw bar relative to the second instrument part and the second implant part along the first axis; wherein when the rotation lock is engaged, the draw bar cannot rotate about the first axis, wherein when the rotation lock is disengaged, the draw bar is free to rotate about the first axis; wherein the instrument includes a first state in which the actuator is disengaged from the draw bar, the rotation lock is disengaged, and the draw bar is free to translate along and rotate about the first axis, wherein the instrument includes a second state in which the actuator is engaged with the draw bar, the rotation lock is engaged, and the draw bar is prevented from rotating about the first axis. The rotation lock includes a rotation lock ring and a spline insert, wherein the rotation lock ring and the draw bar are fixed together for rotation about the first axis, wherein the spline insert is fixed in translation along and rotation about the first axis, wherein the spline insert is removably connectable to the rotation lock ring; wherein when the rotation lock is engaged, the spline insert is connected to the rotation lock ring so that the rotation lock ring cannot rotate about the first axis, wherein when the rotation lock is disengaged, the spline insert is disconnected from the rotation lock ring so that the rotation lock ring is free to rotate about the first axis; wherein the actuator and the rotation lock are coupled together so that the actuator engages the draw bar when the rotation lock is

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engaged and the actuator disengages from the draw bar when the rotation lock is disengaged. The draw bar includes threads; wherein the actuator includes a threaded slider, wherein the threaded slider includes threads that engage the threads of the draw bar, wherein instrument actuation includes rotation of the threaded slider threads in engagement with the draw bar threads to translate the draw bar relative to the second instrument part and the second implant part along the first axis. The first instrument part includes a draw bar that is removably connectable to the first implant part by rotation of the draw bar about the first axis; wherein the instrument further includes a shuttle and an actuator; wherein the shuttle is fixed in rotation about the first axis, wherein the shuttle and the draw bar are fixed together for translation along the first axis, wherein instrument actuation includes movement of the actuator in engagement with the shuttle to translate the shuttle and the draw bar relative to the second instrument part and the second implant part along the first axis; The instrument of claim 5, wherein the shuttle includes threads, wherein the actuator includes a threaded driver, wherein the threaded driver includes threads that engage the threads of the shuttle, wherein instrument actuation includes rotation of the threaded driver threads in engagement with the shuttle threads to translate the shuttle and the draw bar relative to the second instrument part and the second implant part along the first axis. The shuttle includes a series of transverse grooves, wherein the actuator includes a ratchet mechanism, wherein the ratchet mechanism includes a drive link that engages the series of transverse grooves, wherein instrument actuation includes translation of the drive link in engagement with the series of transverse grooves to translate the shuttle and the draw bar relative to the second instrument part and the second implant part along the first axis.

In another aspect of the technology, an instrument includes a first instrument part that is removably connectable to a first implant part of an implant by rotation of the first instrument part relative to the first implant part about a central longitudinal axis of the instrument; and a second instrument part that is removably connectable to a second implant part of the implant; wherein, when the first instrument part is connected to the first implant part, the second instrument part is connected to the second implant part, and the instrument is actuated, the first instrument part and the first implant part translate relative to the second instrument part and the second implant part along the central longitudinal axis of the instrument so that the first implant part approaches the second implant part, wherein the first instrument part is prevented from rotating about the central longitudinal axis while the first instrument part and the first implant part are translating.

Embodiments of this aspect may include any or all of the following attributes. The instrument further includes an actuator and a rotation lock; wherein instrument actuation includes movement of the actuator in engagement with the first instrument part to translate the first instrument part relative to the second instrument part and the second implant part along the first axis; wherein when the rotation lock is engaged, the first instrument part is prevented from rotating about the central longitudinal axis, wherein when the rotation lock is disengaged, the first instrument part is free to rotate about the first axis; wherein the instrument includes a first state in which the actuator is disengaged from the first instrument part, the rotation lock is disengaged, and the first instrument part is free to translate along and rotate about the first axis, wherein the instrument includes a second state in which the actuator is engaged with the first instrument part,

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the rotation lock is engaged, and the first instrument part is prevented from rotating about the first axis. The rotation lock includes a rotation lock ring and a spline insert, wherein the rotation lock ring and the first instrument part are fixed together for rotation about the first axis, wherein the spline insert is fixed in translation along and rotation about the first axis, wherein the spline insert is removably connectable to the rotation lock ring; wherein when the rotation lock is engaged, the spline insert is connected to the rotation lock ring so that the rotation lock ring cannot rotate about the first axis, wherein when the rotation lock is disengaged, the spline insert is disconnected from the rotation lock ring so that the rotation lock ring is free to rotate about the first axis; wherein the actuator and the rotation lock are coupled together so that the actuator engages the first instrument part when the rotation lock is engaged and the actuator disengages from the first instrument part when the rotation lock is disengaged. The first instrument part includes threads; wherein the actuator includes a threaded slider, wherein the threaded slider includes threads that engage the threads of the first instrument part, wherein instrument actuation includes rotation of the threaded slider threads in engagement with the first instrument part threads to translate the first instrument part relative to the second instrument part and the second implant part along the first axis. The instrument further includes a shuttle and an actuator; wherein the shuttle is fixed in rotation about the first axis, wherein the shuttle and the first instrument part are fixed together for translation along the first axis, wherein instrument actuation includes movement of the actuator in engagement with the shuttle to translate the shuttle and the first instrument part relative to the second instrument part and the second implant part along the first axis; The shuttle includes threads, wherein the actuator includes a threaded driver, wherein the threaded driver includes threads that engage the threads of the shuttle, wherein instrument actuation includes rotation of the threaded driver threads in engagement with the shuttle threads to translate the shuttle and the first instrument part relative to the second instrument part and the second implant part along the first axis. The shuttle includes a series of transverse grooves, wherein the actuator includes a ratchet mechanism, wherein the ratchet mechanism includes a drive link that engages the series of transverse grooves, wherein instrument actuation includes translation of the drive link in engagement with the series of transverse grooves to translate the shuttle and the first instrument part relative to the second instrument part and the second implant part along the first axis.

In yet another aspect of the technology, an instrument includes a draw bar that is removably connectable to a first implant part of an implant by rotation of the draw bar relative to the first implant part about a central longitudinal axis of the instrument, wherein the draw bar is carried at a first end of a first shaft; and a second instrument part that is removably connectable to a second implant part of the implant, wherein the second instrument part is carried at a first end of a second shaft; wherein the instrument is actuatable to translate the draw bar toward the second instrument part along the central longitudinal axis so that the draw bar does not rotate about the central longitudinal axis during instrument actuation.

Embodiments of this aspect may include any or all of the following attributes. The instrument further includes an actuator and a rotation lock; wherein instrument actuation includes movement of the actuator in engagement with the draw bar to translate the draw bar relative to the second instrument part and the second implant part along the first

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axis; wherein when the rotation lock is engaged, the draw bar cannot rotate about the first axis, wherein when the rotation lock is disengaged, the draw bar is free to rotate about the first axis; wherein the instrument includes a first state in which the actuator is disengaged from the draw bar, the rotation lock is disengaged, and the draw bar is free to translate along and rotate about the first axis, wherein the instrument includes a second state in which the actuator is engaged with the draw bar, the rotation lock is engaged, and the draw bar is prevented from rotating about the first axis. The rotation lock includes a rotation lock ring and a spline insert, wherein the rotation lock ring and the draw bar are fixed together for rotation about the first axis, wherein the spline insert is fixed in translation along and rotation about the first axis, wherein the spline insert is removably connectable to the rotation lock ring; wherein when the rotation lock is engaged, the spline insert is connected to the rotation lock ring so that the rotation lock ring cannot rotate about the first axis, wherein when the rotation lock is disengaged, the spline insert is disconnected from the rotation lock ring so that the rotation lock ring is free to rotate about the first axis; wherein the actuator and the rotation lock are coupled together so that the actuator engages the draw bar when the rotation lock is engaged and the actuator disengages from the draw bar when the rotation lock is disengaged. The draw bar includes threads; wherein the actuator includes a threaded slider, wherein the threaded slider includes threads that engage the threads of the draw bar, wherein instrument actuation includes rotation of the threaded slider threads in engagement with the draw bar threads to translate the draw bar relative to the second instrument part and the second implant part along the first axis. The instrument further includes a shuttle and an actuator; wherein the shuttle is fixed in rotation about the first axis, wherein the shuttle and the draw bar are fixed together for translation along the first axis, wherein instrument actuation includes movement of the actuator in engagement with the shuttle to translate the shuttle and the draw bar relative to the second instrument part and the second implant part along the first axis. The shuttle includes threads, wherein the actuator includes a threaded driver, wherein the threaded driver includes threads that engage the threads of the shuttle, wherein instrument actuation includes rotation of the threaded driver threads in engagement with the shuttle threads to translate the shuttle and the draw bar relative to the second instrument part and the second implant part along the first axis. The shuttle includes a series of transverse grooves, wherein the actuator includes a ratchet mechanism, wherein the ratchet mechanism includes a drive link that engages the series of transverse grooves, wherein instrument actuation includes translation of the drive link in engagement with the series of transverse grooves to translate the shuttle and the draw bar relative to the second instrument part and the second implant part along the first axis.

These and other features and advantages of the present technology will become more fully apparent from the following description and appended claims, or may be learned by the practice of the technology as set forth hereinafter.

BRIEF DESCRIPTION OF THE DRAWINGS

Exemplary embodiments of the technology will become more fully apparent from the following description and appended claims, taken in conjunction with the accompanying drawings. Understanding that these drawings depict only exemplary embodiments and are, therefore, not to be considered limiting of the scope of the technology, the

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exemplary embodiments will be described with additional specificity and detail through use of the accompanying drawings in which:

FIG. 1 is a top view of an insertion and expansion instrument;

FIG. 2 is a side view of the instrument of FIG. 1 with a draw bar assembly inserted into the instrument and an expandable intervertebral implant connected to a distal end of the instrument;

FIG. 3 is a cross-sectional view of the instrument and implant of FIG. 2 taken along line 3-3 of FIG. 2;

FIG. 4 is a distal detail cross-sectional view of the instrument and implant of FIG. 3;

FIG. 5 is a proximal detail cross-sectional view of the instrument of FIG. 3;

FIG. 6 is a top view of the instrument of FIG. 1 in a first partially assembled state;

FIG. 7 is a top view of the instrument of FIG. 1 in a second partially assembled state;

FIG. 8 is a top view of the instrument of FIG. 1 in a fully assembled state;

FIG. 9 is an exploded perspective view of the instrument of FIG. 1;

FIG. 10 is another exploded perspective view of the instrument of FIG. 1, from a different direction;

FIG. 11 is an exploded perspective view of a shuttle assembly of the instrument of FIG. 1;

FIG. 12 is another exploded perspective view of the shuttle assembly of FIG. 11, from a different direction;

FIG. 13 is a proximal end view of an inserter handle of the instrument of FIG. 9;

FIG. 14 is a distal end view of the inserter handle of the instrument of FIG. 9;

FIG. 15 is a proximal end view of a selector ring of the instrument of FIG. 9;

FIG. 16 is a proximal end view of an expansion knob of the instrument of FIG. 9;

FIG. 17 is a distal end view of the expansion knob of the instrument of FIG. 9;

FIG. 18 is a proximal end view of a threaded driver of the instrument of FIG. 9;

FIG. 19 is a distal end view of the threaded driver of the instrument of FIG. 9;

FIG. 20 is a proximal end view of a lock tube coupling of the lock tube assembly of FIG. 24;

FIG. 21 is a distal end view of the lock tube coupling of the lock tube assembly of FIG. 24;

FIG. 22 is an exploded perspective view of a shaft assembly of the instrument of FIG. 1;

FIG. 23 is another exploded perspective view of the shaft assembly of FIG. 22, from a different direction;

FIG. 24 is an exploded perspective view of a lock tube assembly of the instrument of FIG. 1;

FIG. 25 is another exploded perspective view of the lock tube assembly of FIG. 24, from a different direction;

FIG. 26 is a top view of the draw bar assembly of FIG. 2;

FIG. 27 is a top view of a tamp assembly;

FIG. 28 is a top view of a funnel;

FIG. 29 is a top view of a screwdriver assembly coupled to a lockout screw;

FIG. 30 is a distal detail view of the screwdriver assembly and lockout screw of FIG. 29;

FIG. 31 is an exploded perspective view of the draw bar assembly of FIG. 26;

FIG. 32 is another exploded perspective view of the draw bar assembly of FIG. 26, from a different direction;

FIG. 33 is a perspective view of the funnel of FIG. 28;

FIG. 34 is another perspective view of the funnel of FIG. 28, from a different direction;

FIG. 35 is an exploded perspective view of the tamp assembly of FIG. 27;

FIG. 36 is another exploded perspective view of the tamp assembly of FIG. 27, from a different direction;

FIG. 37 is an exploded perspective view of the screwdriver assembly of FIG. 29;

FIG. 38 is another exploded perspective view of the screwdriver assembly of FIG. 29, from a different direction;

FIG. 39 is a perspective view of a screw holder and a screw holder catch of the screwdriver assembly of FIG. 29;

FIG. 40 is another perspective view of a screw holder and a screw holder catch of the screwdriver assembly of FIG. 29, from a different direction;

FIG. 41 is a proximal end view of the screw holder of FIG. 40;

FIG. 42 is a distal end view of the screw holder of FIG. 40;

FIG. 43 is a perspective view of the lockout screw of FIG. 29;

FIG. 44 is another perspective view of the lockout screw of FIG. 29, from a different direction;

FIG. 45 is a rear or second end view of the implant of FIG. 2 in a collapsed configuration;

FIG. 46 is a top view of the implant of FIG. 2;

FIG. 47 is a side view of the implant of FIG. 2;

FIG. 48 is an isometric view of the instrument of FIG. 1 in an unlocked configuration;

FIG. 49 is a closeup view of a distal end of the instrument of FIG. 48;

FIG. 50 is an isometric view of the instrument of FIG. 1 with the implant of FIG. 2 connected to the distal end of the instrument;

FIG. 51 is a closeup view of the distal end of the instrument and implant of FIG. 50;

FIG. 52 is an isometric view of the instrument and implant of FIG. 50, the implant adjacent a vertebral body and in a laterally expanded configuration;

FIG. 53 is a closeup view of the distal end of the instrument, implant, and vertebra of FIG. 52;

FIG. 54 is an isometric view of the instrument and implant of FIG. 52, the implant adjacent a vertebral end body and in a laterally and vertically expanded configuration;

FIG. 55 is a closeup view of the distal end of the instrument, implant, and vertebra of FIG. 54;

FIG. 56 is an isometric view of the instrument and implant of FIG. 54, the implant adjacent a vertebral end body and in a laterally and vertically expanded configuration, and the funnel of FIG. 28 connected to a proximal end of the instrument;

FIG. 57 is a closeup view of the distal end of the instrument, implant, and vertebra of FIG. 56;

FIG. 58 is an isometric view of the instrument and implant of FIG. 56, the implant adjacent a vertebral end body and in a laterally and vertically expanded configuration, the lockout screw of FIG. 29 extending through the implant and the screwdriver assembly of FIG. 29 inserted through the instrument;

FIG. 59 is a closeup view of the distal end of the instrument, implant, and vertebra of FIG. 58;

FIG. 60 is a top view of the implant and vertebra of FIG. 58;

FIG. 61 is an anterior view of the implant and vertebra of FIG. 58;

FIG. 62 is a right lateral view of the implant and vertebra of FIG. 58;

FIG. 63 is a left lateral view of the implant and vertebra of FIG. 58;

FIG. 64 is a top view of another insertion and expansion instrument;

FIG. 65 is a cross-sectional view of the instrument of FIG. 64, taken along line 65-65 of FIG. 64;

FIG. 66 is a proximal detail cross-sectional view of the instrument of FIG. 65;

FIG. 67 is a distal detail cross-sectional view of the instrument of FIG. 65;

FIG. 68 is a perspective view of the instrument of FIG. 64;

FIG. 69 is another perspective view of the instrument of FIG. 64, from a different direction;

FIG. 70 is an exploded perspective view of the instrument of FIG. 64;

FIG. 71 is another exploded perspective view of the instrument of FIG. 64, from a different direction;

FIG. 72 is an exploded perspective view of a ratchet mechanism of the instrument of FIG. 64;

FIG. 73 is another exploded perspective view of the ratchet mechanism of FIG. 72, from a different direction;

FIG. 74 is an exploded perspective view of a shaft assembly of the instrument of FIG. 64;

FIG. 75 is another exploded perspective view of the shaft assembly of FIG. 74, from a different direction;

FIG. 76 is an exploded perspective view of a lock tube assembly of the instrument of FIG. 64;

FIG. 77 is another exploded perspective view of the lock tube assembly of FIG. 76, from a different direction;

FIG. 78 is an exploded perspective view of a draw bar assembly of the instrument of FIG. 64;

FIG. 79 is another exploded perspective view of the draw bar assembly of FIG. 78, from a different direction;

FIG. 80 is an exploded perspective view of a handle, a lock lever, a stop screw, and a release ring of the instrument of FIG. 64;

FIG. 81 is another exploded perspective view of the handle, lock lever, stop screw, and release ring of FIG. 80, from a different direction;

FIG. 82 is a proximal end view of an inserter handle of the instrument of FIG. 70;

FIG. 83 is a distal end view of the inserter handle of the instrument of FIG. 70;

FIG. 84 is a right side view of the inserter handle of the instrument of FIG. 70;

FIG. 85 is a left side view of the inserter handle of the instrument of FIG. 70;

FIG. 86 is a top view of another insertion and expansion instrument coupled to another expandable intervertebral implant;

FIG. 87 is a side view of the instrument and implant of FIG. 86;

FIG. 88 is a cross-sectional view of the instrument and implant of FIG. 87, taken along line 88-88 of FIG. 87;

FIG. 89 is a cross-sectional view of the instrument and implant of FIG. 86, taken along line 89-89 of FIG. 86;

FIG. 90 is a perspective view of the instrument of FIG. 86;

FIG. 91 is another perspective view of the instrument of FIG. 86, from a different direction;

FIG. 92 is a perspective view of an inserter connector of the instrument of FIG. 86;

FIG. 93 is another perspective view of the inserter connector of FIG. 92, from a different direction;

FIG. 94 is a perspective view of a connector plate of the instrument of FIG. 86;

FIG. 95 is another perspective view of the connector plate of FIG. 94, from a different direction;

FIG. 96 is a perspective view of an inserter shaft of the instrument of FIG. 86;

FIG. 97 is another perspective view of the inserter shaft of FIG. 96, from a different direction;

FIG. 98 is a perspective view of an implant retainer shaft of the instrument of FIG. 86;

FIG. 99 is another perspective view of the implant retainer shaft of FIG. 98, from a different direction;

FIG. 100 is a perspective view of connector buttons of the instrument of FIG. 86;

FIG. 101 is another perspective view of the connector buttons of FIG. 100, from a different direction;

FIG. 102 is a perspective view of a thumb wheel of the instrument of FIG. 86;

FIG. 103 is another perspective view of the thumb wheel of FIG. 102, from a different direction;

FIG. 104 is a perspective view of a wheel axle of the instrument of FIG. 86;

FIG. 105 is another perspective view of the wheel axle of FIG. 104, from a different direction;

FIG. 106 is a perspective view of an inserter handle of the instrument of FIG. 86;

FIG. 107 is another perspective view of the inserter handle of FIG. 106, from a different direction;

FIG. 108 is a proximal end view of the inserter handle of FIG. 106;

FIG. 109 is a perspective view of a strike plate of the instrument of FIG. 86;

FIG. 110 is another perspective view of the strike plate of FIG. 109, from a different direction;

FIG. 111 is a perspective view of a spline insert of the instrument of FIG. 86;

FIG. 112 is another perspective view of the spline insert of FIG. 111, from a different direction;

FIG. 113 is a perspective view of a rotation lock ring of the instrument of FIG. 86;

FIG. 114 is another perspective view of the rotation lock ring of FIG. 113, from a different direction;

FIG. 115 is a perspective view of a cam lever of the instrument of FIG. 86;

FIG. 116 is another perspective view of the cam lever of FIG. 115, from a different direction;

FIG. 117 is a perspective view of a slider button of the instrument of FIG. 86;

FIG. 118 is another perspective view of the slider button of FIG. 117, from a different direction;

FIG. 119 is a side view of the slider button of FIG. 117;

FIG. 120 is a perspective view of a threaded slider of the instrument of FIG. 86;

FIG. 121 is another perspective view of the threaded slider of FIG. 120, from a different direction;

FIG. 122 is a perspective view of a knob top part of the instrument of FIG. 86;

FIG. 123 is another perspective view of the knob top part of FIG. 122, from a different direction;

FIG. 124 is a perspective view of a knob bottom part of the instrument of FIG. 86;

FIG. 125 is another perspective view of the knob bottom part of FIG. 124, from a different direction;

FIG. 126 is a perspective view of a draw bar assembly for use with the instrument of FIG. 86;

FIG. 127 is another perspective view of the draw bar assembly of FIG. 126, from a different direction;

FIG. 128 is an exploded perspective view of the draw bar assembly of FIG. 126;

FIG. 129 is another exploded perspective view of the draw bar assembly of FIG. 126, from a different direction;

FIG. 130 is a perspective view of a trial assembly for use with the instrument of FIG. 86;

FIG. 131 is another perspective view of the trial assembly of FIG. 130, from a different direction;

FIG. 132 is an exploded perspective view of the trial assembly of FIG. 130;

FIG. 133 is another exploded perspective view of the trial assembly of FIG. 130, from a different direction;

FIG. 134 is a perspective view of a funnel for use with the instrument of FIG. 86;

FIG. 135 is another perspective view of the funnel of FIG. 134, from a different direction;

FIG. 136 is a perspective view of a screwdriver assembly for use with the instrument of FIG. 86;

FIG. 137 is another perspective view of the screwdriver assembly of FIG. 136, from a different direction;

FIG. 138 is an exploded perspective view of the screwdriver assembly of FIG. 136;

FIG. 139 is another exploded perspective view of the screwdriver assembly of FIG. 136, from a different direction;

FIG. 140 is a rear or second end view of the implant of FIG. 86 in a collapsed configuration;

FIG. 141 is a top view of the implant of FIG. 86;

FIG. 142 is a side view of the implant of FIG. 86;

FIG. 143 is a perspective view of the trial assembly of FIG. 130 adjacent a vertebral body;

FIG. 144 is a distal detail perspective view of the trial assembly and vertebral body of FIG. 143;

FIG. 145 is a perspective view of the instrument and implant of FIG. 86 coupled to the draw bar assembly of FIG. 126, the slider button of the instrument in a proximal position;

FIG. 146 is a distal detail perspective view of the instrument and implant of FIG. 145;

FIG. 147 is a perspective view of the instrument, draw bar assembly, and implant of FIG. 145, the slider button of the instrument in a distal position;

FIG. 148 is a distal detail perspective view of the instrument and implant of FIG. 147;

FIG. 149 is a perspective view of the instrument, draw bar assembly, and implant of FIG. 144, the implant adjacent a vertebral body and in a laterally expanded configuration;

FIG. 150 is a distal detail perspective view of the instrument, implant, and vertebra of FIG. 149;

FIG. 151 is a perspective view of the instrument, draw bar assembly, and implant of FIG. 149, the implant adjacent a vertebral body and in a laterally and vertically expanded configuration;

FIG. 152 is a distal detail perspective view of the instrument, implant, and vertebra of FIG. 151;

FIG. 153 is a perspective view of the instrument, draw bar assembly, and implant of FIG. 151, the implant adjacent a vertebral body and in a laterally and vertically expanded configuration, the slider button of the instrument released;

FIG. 154 is a distal detail perspective view of the instrument, implant, and vertebra of FIG. 153;

FIG. 155 is a perspective view of the instrument, draw bar assembly, and implant of FIG. 153, the implant adjacent a vertebral body and in a laterally and vertically expanded configuration, the slider button in the proximal position;

FIG. 156 is a distal detail perspective view of the instrument, implant, and vertebra of FIG. 155;

FIG. 157 is a perspective view of the instrument and implant of FIG. 155, the implant in a laterally and vertically expanded configuration, a handle assembly of the instrument partially removed from a shaft assembly of the instrument;

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FIG. 158 is a perspective view of the shaft assembly and implant of FIG. 157, the implant adjacent a vertebral body and in a laterally and vertically expanded configuration, with the funnel of FIG. 134 and the tamp assembly of FIG. 27 coupled to the shaft assembly;

FIG. 159 is a perspective view of the shaft assembly, implant, and funnel of FIG. 158, the implant adjacent a vertebral body and in a laterally and vertically expanded configuration, with the screwdriver assembly of FIG. 136 coupled to the lockout screw of FIG. 43 and oriented for insertion into the shaft assembly;

FIG. 160 is a perspective view of the implant and lockout screw of FIG. 159 adjacent a vertebral body and in a laterally and vertically expanded configuration;

DETAILED DESCRIPTION

Exemplary embodiments of the technology will be best understood by reference to the drawings, wherein like parts are designated by like numerals throughout. It will be readily understood that the components of the technology, as generally described and illustrated in the figures herein, could be arranged and designed in a wide variety of different configurations. Thus, the following more detailed description of the embodiments of the apparatus, system, and method is not intended to limit the scope of the invention, as claimed, but is merely representative of exemplary embodiments of the technology.

The phrases “connected to,” “coupled to” and “in communication with” refer to any form of interaction between two or more entities, including mechanical, electrical, magnetic, electromagnetic, fluid, and thermal interaction. Two components may be functionally coupled to each other even though they are not in direct contact with each other. The term “abutting” refers to items that are in direct physical contact with each other, although the items may not necessarily be attached together. The phrase “fluid communication” refers to two features that are connected such that a fluid within one feature is able to pass into the other feature.

The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any embodiment described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments. While the various aspects of the embodiments are presented in drawings, the drawings are not necessarily drawn to scale unless specifically indicated.

Standard medical planes of reference and descriptive terminology are employed in this specification. A sagittal plane divides a body into right and left portions. A mid-sagittal plane divides the body into bilaterally symmetric right and left halves. A coronal plane divides a body into anterior and posterior portions. A transverse plane divides a body into superior and inferior portions. Anterior means toward the front of the body. Posterior means toward the back of the body. Superior means toward the head. Inferior means toward the feet. Medial means toward the midline of the body. Lateral means away from the midline of the body. Axial means toward a central axis of the body. Abaxial means away from a central axis of the body. Ipsilateral means on the same side of the body. Contralateral means on the opposite side of the body. Proximal means located toward a center of a body, or toward a user. Distal means located away from the center of the body, or away from a user. These descriptive terms may be applied to an animate or inanimate body.

Standard spinal descriptive terminology is used herein with the ordinary and customary meanings.

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Referring to FIGS. 1 and 2, an instrument 100 for insertion and expansion of an intervertebral implant is disclosed. The instrument 100 extends from a proximal end 105 to a distal end 107 along a longitudinal instrument axis 102. The instrument 100 includes a handle portion 108 and a shaft portion 110. In FIG. 2, an expandable intervertebral implant 1000 is connected to the instrument distal end 107 and a draw bar assembly 160 is received in the instrument 100 for expansion of the implant.

Referring to FIGS. 3-8, components and features of instrument 100 are disclosed. As seen in FIGS. 3 and 6, instrument 100 includes a lock collar 120 attachable to a shuttle 122. A proximal lock tube 124 extends distally from the shuttle 122, and a distal lock tube 126 extends distally from the proximal lock tube. A lock tube coupling 128 couples the proximal and distal lock tubes 124, 126 together. The lock collar 120, shuttle 122, and proximal and distal lock tubes 124, 126 are cannulated, surrounding a longitudinal instrument bore 112 extending along the longitudinal instrument axis 102. As seen in FIGS. 3 and 7, a threaded driver 130, standoff 132, and standoff nut 134 are coupled to the shuttle 122. An inserter shaft 136 extends distally from lock tube coupling 128, and an inserter connector 138 is at the distal end of the inserter shaft 136. The outer inserter shaft 136 may include one or more openings that aid in cleaning the instrument. As seen in FIGS. 3 and 8, the fully assembled instrument further includes an expansion knob 140, inserter handle 142, selector ring 144, and impaction cap 146. Markings or indicia 248 on the shuttle 122 or another component may provide indication when a particular step of implant expansion is attained. In one example the words “collapsed”, “lateral” and “vertical” may be indicated when the implant is in collapsed, laterally expanded, and vertically expanded configurations, respectively. In other embodiments, letters, numbers, symbols, or other markings may be used to indicate stages of insertion and/or expansion.

Referring to FIGS. 9 and 10, the instrument 100 may include a shuttle assembly 114, a standoff nut 134, an inserter handle 142, a selector ring 144, an expansion knob 140, a threaded driver 130, a shaft assembly 116, a standoff 132, and a lock tube assembly 118.

Referring to FIGS. 11 and 12, the shuttle assembly 114 may include a shuttle 122 and a lock collar 120.

The shuttle 122 extends between a proximal end 184 and a distal end 186. The proximal end may include a head 188. A shaft 192 may extend from the head 188 toward the distal end 186. The head 188 may be wider than the shaft 192, as shown.

The head 188 may include, from proximal to distal, first, second, third (FIG. 6), and fourth levels 194, 196, 198, 202. The first level 194 may have a generally rectangular cross section and may overhang the shaft 192 on three sides, as shown. The first level 194 may include a tab 204 that protrudes from a first side of the first level. The tab 204 may be flanked by bilateral teeth 206, 208 that may also protrude from the first side of the first level. The teeth 206, 208 may be smaller and/or shorter than the tab 204. Bilateral indentations 210, 212 may be present between the tab 204 and the teeth 206, 208. The second and third overhanging sides 214, 216 of the first level 194 may be flat. A hole 234 may extend through the tab 204 and may be parallel to the shaft 192. Bilateral holes 236, 238 may extend into the first level 194 opposite the tab 204. Hole 236 is visible in FIG. 12; hole 238 is hidden by the second level 196. The second level 196 may have a generally rectangular cross section and may overhang the shaft 192 all around. The second level 196 may include the tab 204, the teeth 206, 208, and the indentations 210, 212

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arranged on a first overhanging side of the second level. Second and third overhanging sides **218**, **220** may be flat, and may be parallel to the sides **214**, **216**. The side **218** may be recessed relative to the side **214** to form a ledge **222**. The side **220** may be recessed relative to the side **216** to form a ledge **224**. A fourth overhanging side **226** may include a shallow central indentation **228** opposite the tab **204**. The third level **198** (FIG. 6) may have a circular cross section and may include a circumferential groove **230**. The fourth level **202** may have a circular cross section and may include a circumferential flange **232**.

The shaft **192** may include, from proximal to distal, first, second, third, and fourth outer portions **240**, **242**, **244**, **246**. The first outer portion **240** may have a circular cross section and may include indicia **248** (FIG. 6) such as lines and/or text to indicate various implant configurations, such as collapsed, laterally expanded, and/or vertically expanded. The outer diameter of the first outer portion **240** may be equal to the minor diameter in the groove **230**. The second outer portion **242** may include external threads **250**. The major diameter of the threads **250** may be equal to the outer diameter of the first outer portion **240**. The third outer portion **244** may have a circular cross section. The outer diameter of the third outer portion **244** may be smaller than the minor diameter of the threads **250**. The fourth outer portion **246** may have a generally rectangular cross section. The cross section may be square with rounded corners as shown. Two opposite flat sides of the fourth outer portion **246** may be parallel to the sides **214**, **216** and/or the sides **218**, **220**. The width across the flat sides may be equal to the outer diameter of the third outer portion **244**. The width across the rounded corners may be greater than the outer diameter of the third outer portion **244** and may be less than the outer diameter of the first outer portion **240**.

The shuttle **122** may include a central longitudinal hole **252** that extends through the shuttle between the proximal end **184** and the distal end **186**. The hole **252** may include, from proximal to distal, first, second, third, fourth (FIG. 5), and fifth inner portions **254**, **256**, **258**, **260**, **262**. Each of these inner portions may have a circular cross section. The first inner portion **254** may extend into the proximal end **184** through the first level **194** and may end at the second level **196**. The first inner portion **254** may be referred to as a counterbore associated with the second inner portion **256**. The first inner portion **254** may break through the side of the first level **194** opposite the tab **204**. The second inner portion **256** may extend through the second level **196**. The diameter of the second inner portion **256** may be less than the diameter of the first inner portion **254**. The third inner portion **258** may extend through the third and fourth levels **198**, **202** and most of the first outer portion **240**, and may end just proximal to the second outer portion **242**. There may be a short conical tapered transition between the second inner portion **256** and the third inner portion **258**. The diameter of the third inner portion **258** may be less than the diameter of the first inner portion **254** and may be greater than the diameter of the second inner portion **256**. The fourth inner portion **260** (FIG. 5) may extend between the third and fifth inner portions **258**, **262**. The fourth inner portion **260** may begin just proximal to the second outer portion **242** and may end just distal to the first outer portion **240**. There may be a short conical tapered transition between the third inner portion **258** and the fourth inner portion **260**. The diameter of the fourth inner portion **260** may be less than the diameter of the second inner portion **256**. The fifth inner portion **262** may extend into the distal end **186** through the fourth and third outer portions **246**, **244** and most of the second outer

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portion **242**, and may end just distal to the first outer portion **240**. The diameter of the fifth inner portion **262** may be less than the diameter of the first inner portion **254** and may be greater than the diameter of the second inner portion **256**. The diameter of the fifth inner portion **262** may be equal to the diameter of the third inner portion **258**.

The lock collar **120** extends between a proximal side **264** and a distal side **266**. The lock collar **120** may have a generally D-shaped profile in a proximal view, with a curved side **268** and three substantially flat sides **270**, **272**, **274**. The proximal side **264** may be flat. A central hole **276** may extend through the lock collar **120** between the proximal side **264** and the distal side **266**. The hole **276** may be non-circular in a proximal view. The hole **276** may include a first curved side **278**, flat tapered sides **280**, **282**, flat parallel sides **284**, **286**, and a second curved side **288**. Transitional fillet surfaces may be present between some or all of the sides **278**, **280**, **282**, **284**, **286**, **288**. A conical surface **290** may be associated with the first curved side **278**, and may taper into the hole **276** from the proximal side **264**. A slot **292** may extend through the lock collar **120** from the proximal side **264** to the distal side **266**. The slot **292** may be located between the curved side **268** and the first curved side **278** and may be elongated in a direction from the curved side **268** to the flat side **272**. The distal side **266** may include bilateral feet **294**, **296** that protrude distally on either side of the hole **276**. The foot **294** may be associated with the flat side **270**. The foot **296** may be associated with the flat side **274**. In a distal view, the outer profile of the curved side **268** may be concentric with the outer profiles of the feet **294**, **296**. The inner profiles of the feet **294**, **296** may be flat and parallel to each other, and may also be parallel to the flat sides **270**, **274**. The feet **294**, **296** may be undercut. An undercut groove **298** may extend along the inner profile of the foot **294** and an undercut groove **346** may extend along the inner profile of the foot **296**. Taken together, the feet **294**, **296** and undercut grooves **298**, **346** may form a T-slot across the distal side **266**. The distal side **266** may include a first surface **348** that is proximal to the distal aspect of the feet **294**, **296**. The first surface **348** may be at the same or a similar level as the distal-most surfaces of the undercut grooves **298**, **346**. The distal side **266** may include a second surface **350** that is proximal to the first surface **348**. The second surface **350** may be at the same or a similar level as the proximal-most surfaces of the undercut grooves **298**, **346**. Bilateral holes **352**, **354** (not shown) may extend into the lock collar, between the first surface **348** and the second surface **350**, along a direction from the curved side **268** to the flat side **272**. The bilateral holes **352**, **354** may be complementary to the bilateral holes **236**, **238** in the head **188** of the shuttle **122**. The flat side **272** may include friction features **356**, such as alternating ridges and grooves, knurling, grit blast, or the like.

The shuttle assembly **114** may be assembled by orienting the lock collar **120** relative to the head **188** of the shuttle **122** so that the proximal end **184** and the proximal side **264** face the same way, the side **226** and the side **272** face the same way, and the sides **218**, **220** are aligned between the feet **294**, **296**; and sliding the lock collar **120** along a direction from the flat side **272** to the curved side **268** so that the first level **194** is received in the undercut grooves **298**, **346** and the second level **196** is received between the feet **294**, **296**. Bilateral pins or compression springs (not shown) may be inserted into the bilateral holes **236**, **238** and **352**, **354**. A pin (not shown) may be inserted through the slot **292** and into the hole **234**. When compression springs are inserted in the holes **236**, **238** and **352**, **354** and a pin is inserted through the

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slot **292** and into the hole **234**, the springs bias the lock collar **120** toward a position in which the pin contacts the end of the slot **292** that is closest to the outer curved side **268**.

Referring to FIGS. **9** and **10**, the standoff nut **134** extends between a proximal side **358** and a distal side **360**. The standoff nut **134** may be a generally cylindrical part. The proximal side **358** may include a circumferential outer chamfer **362** and/or bilateral notches **364**, **366**. Each notch may include a longitudinal surface **368** and a transverse surface **370**. The distal side **360** may include a circumferential outer chamfer **372**. A central hole **374** may extend through the standoff nut **134** between the proximal side **358** and the distal side **360**. The hole **374** may include, from proximal to distal, first, second, and third inner portions **376**, **378**, **380**. The first inner portion **376** may have a circular cross section. The second inner portion **378** may have a circular cross section, and may have a larger diameter than the first portion **376**. The second inner portion **378** may have a flat proximal end. The third inner portion **380** may have internal threads, which may have a major diameter the same or similar to the diameter of the second portion **378**. The hole **374** may include an inner chamfer **382** at the distal side **360**.

Referring to FIGS. **5**, **9**, **10**, **13**, and **14**, the inserter handle **142** extends between a proximal end **384** and a distal end **386**. The inserter handle **142** may be generally circular in cross section. The inserter handle **142** may include, from proximal to distal, first, second, third, fourth, fifth, and sixth outer portions **388**, **390**, **392**, **394**, **396**, **398**. The first outer portion **388** may be cylindrical with external threads. The second outer portion **390** may be cylindrical with a diameter that may be the same or similar to the minor diameter of the external threads of the first outer portion **388**. The third outer portion **392** may be cylindrical with a diameter that may be greater than the major diameter of the external threads of the first outer portion **388**. The fourth outer portion **394** may be generally hourglass-shaped and may include friction features **400**, such as longitudinal grooves, alternating ridges and grooves, knurling, grit blast, or the like. The fifth outer portion **396** may be cylindrical with a diameter that may be greater than the diameter of the third outer portion **392**. The fifth outer portion **396** may include indicia **402**, such as a distally-pointing arrow as shown, and/or an outer chamfer **404**. The sixth outer portion **398** may be cylindrical with a diameter that may be the same or similar to the diameter of the second outer portion **390**. The sixth outer portion **398** may include a short base portion **406** which may be directly adjacent to the fifth outer portion **396**, and a collar portion **408** which is at the distal end **386**. The collar portion **408** may be horseshoe-shaped or C-shaped in cross section, with an open side that faces the distally-pointing arrow indicium **402**. Bilateral holes **410**, **412** may extend proximally into the ends of the collar portion **408** (the ends of the C-shape).

The inserter handle **142** may include a central longitudinal hole **414** that extends through the inserter handle between the proximal end **384** and the distal end **386**. The hole **414** may include, from proximal to distal, first, second, third, fourth, and fifth inner portions **416**, **418**, **420**, **422**, **424**. The first inner portion **416** may have a circular cross section and may extend distally through the first and second outer portions **388**, **390** and most of the third outer portion **392**, and may end proximal to the fourth outer portion **394**. The second inner portion **418** may have a generally rectangular cross section. The cross section may be square with rounded corners as shown, and the open side of the collar portion **408** may face one of the flat sides. The width across the flat sides and the width across the rounded corners may each be less

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than the diameter of the first inner portion **416**. The second inner portion **418** may extend to a location that is near the middle of the overall proximal-distal length of the inserter handle **142**. The third inner portion **420** may have a circular cross section. The diameter of the third inner portion **420** may be the same or similar to the width across the flat sides of the second inner portion **418**. The third inner portion **420** may end just proximal to the sixth outer portion **398**. The fourth inner portion **422** may have a circular cross section. The diameter of the fourth inner portion **422** may be less than the width across the flat sides of the second inner portion **418** or the diameter of the third inner portion **420**. Bilateral longitudinal grooves **426**, **428** may extend along the fourth inner portion **422**. The width across the grooves **426**, **428** may be the same or similar to the width across the flat sides of the second inner portion **418** or the diameter of the third inner portion **420**. The groove **428** may point into the open side of the collar portion **408**, near the end with the hole **412**. The grooves **426**, **428** may point into the rounded corners of the second inner portion **418**. The fourth inner portion **422** may end just distal to the fifth outer portion **396**, or near the middle of the overall proximal-distal length of the base portion **406**. The fifth inner portion **424** may have a circular cross section and may be continuous with the inner wall of the collar portion **408**. The diameter of the fifth inner portion **424** may be less than the width across the rounded corners of the second inner portion **418**, greater than the width across the flat sides of the second inner portion or the diameter of the third inner portion **420**, and greater than the width across the grooves **426**, **428**. The fifth inner portion **424** may be referred to as a shallow counterbore associated with the fourth inner portion **422**.

Referring to FIGS. **9**, **10**, and **15**, the selector ring **144** extends between a proximal side **430** and a distal side **432**. The selector ring **144** may include an outer ring **434**, an inner ring **436** within the outer ring, and an arm **438** that connects the inner ring to the outer ring. The outer ring **434** may be a generally cylindrical part with flat proximal and distal sides **430**, **432**. The outer surface of the outer ring **434** may include a longitudinal groove **440** and a longitudinal rib **442**. The illustrated example includes four longitudinal grooves **440**, **444**, **446**, **448** and three longitudinal ribs **442**, **450**, **452**. The outer surface of the outer ring **434** may also include indicia **454** such as lines, text, and/or icons as shown to indicate various settings, such as a locked setting, an unlocked setting, and a setting for cleaning. The inner surface of the outer ring **434** may be cylindrical. The inner ring **436** may be a generally cylindrical part with flat proximal and distal sides **456**, **458**. The proximal side **456** may be recessed relative to the proximal side **430** and the distal side **458** may be recessed relative to the distal side **432**. The outer surface of the inner ring **436** may be cylindrical as shown. The inner surface of the inner ring **436** may be cylindrical with bilateral longitudinal rectangular grooves **460**, **462**. The arm **438** may extend between the inner surface of the outer ring **434** and the outer surface of the inner ring **436**. The arm **438** may be aligned with the groove **462**. The arm **438** may have a longitudinal hole **464** that extends between the proximal and distal sides **456**, **458**.

Referring to FIGS. **9**, **10**, **16**, and **17**, the expansion knob **140** extends between a proximal end **466** and a distal end **468**. The expansion knob **140** may be generally circular in cross section. The expansion knob **140** may be wider toward the proximal end **466** and narrower toward the distal end **468**. The expansion knob **140** may include friction features **470**, such as longitudinal grooves, alternating ridges and grooves, knurling, grit blast, or the like. Bilateral tabs **472**,

474 may extend proximally. The tabs 472, 474 may have circular outer and inner profiles. The outer sides of the tabs 472, 474 may include indicia 476, such as transverse lines as shown. Bilateral holes 478, 480 may extend distally into the expansion knob 140 and may be located between the tabs 472, 474 (i.e., rotated 90 degrees away from the tabs). The holes 478, 480 may include proximal counterbores.

The expansion knob 140 may include a central longitudinal hole 482 that extends through the expansion knob between the proximal end 466 and the distal end 468. The hole 482 may include, from proximal to distal, first, second, third, and fourth inner portions 484, 486, 488, 490. The first inner portion 484 may have a circular cross section. The first inner portion 484 may extend distally about one third of the overall proximal-distal length of the expansion knob 140. The second inner portion 486 may have a generally rectangular cross section. The cross section may be square with rounded corners as shown. The width across the flat sides and the width across the rounded corners may each be greater than the diameter of the first inner portion 484. The second inner portion 486 may extend distally about one third of the overall proximal-distal length of the expansion knob 140, in other words, the second inner portion may occupy the middle third of the overall proximal-distal length of the expansion knob. The third inner portion 488 may have a circular cross section. The diameter of the third inner portion 488 may be greater than the width across the rounded corners of the second inner portion 486. The third inner portion 488 may end just proximal to the distal end 468. The fourth inner portion 490 may have a circular cross section. The diameter of the fourth inner portion 490 may be the same, similar to, or greater than the width across the rounded corners of the second inner portion 486 and less than the diameter of the third inner portion 488. The fourth inner portion 490 may form a shallow undercut with the third inner portion 488.

Referring to FIGS. 9, 10, 18, and 19, the threaded driver 130 extends between a proximal end 492 and a distal end 494. The threaded driver 130 may be referred to as an actuator. The threaded driver 130 may be a rectangular solid, and may be a cube. Referring to FIGS. 18 and 19, the threaded driver 130 may have rounded corners in a proximal or distal view. Bilateral holes 496, 498 may extend into the proximal end 492 about a third of the overall proximal-distal length of the threaded driver 130. The holes 496, 498 may be located in opposite corners of the proximal end 492. The proximal ends of the holes 496, 498 may include shallow countersinks. The threaded driver 130 may include a central longitudinal hole 500 that extends through the threaded driver between the proximal end 492 and the distal end 494. The hole 500 may include, from proximal to distal, first, second, third, fourth, and fifth inner portions 502, 504, 506, 508, 510. The first inner portion 502 may be a shallow conical countersink around the hole 500 in the proximal end 492. The second inner portion 504 may be a shallow cylindrical counterbore. The third inner portion 506 may include internal threads 512 that receive the external threads 250 of the shuttle 122. Referring to FIGS. 18 and 19, the cross-sectional shape of the third inner portion 506 may be the superposition of a circle defining the minor diameter of the internal threads 512 and a rectangle. The minor diameter of the internal threads 512 may be smaller than the diameter of the second inner portion 504. The major diameter of the internal threads 512 may be greater than the diameter of the second inner portion 504. The rectangle may be a square, and may have rounded corners. The width across the flat sides may be slightly less than the minor diameter of the

internal threads 512. The width across the corners may be slightly less than the major diameter of the internal threads 512. The fourth inner portion 508 may be a shallow cylindrical counterbore, with a diameter greater than the minor diameter of the internal threads 512 and less than the major diameter of the internal threads 512. The diameter of the fourth inner portion 508 may be equal to, or similar to, the diameter of the second inner portion 504. The fifth inner portion 510 may be a shallow counterbore around the hole 500 in the distal end 494, with a diameter greater than the major diameter of the internal threads 512.

Referring to FIGS. 9, 10, 22, and 23, the shaft assembly 116 may include an impaction cap 146, an inserter connector 138, and an inserter shaft 136.

The impaction cap 146 extends between a proximal side 514 and a distal side 516. The impaction cap 146 may be circular in a proximal or distal view. The impaction cap 146 may include, from proximal to distal, first, second, and third outer portions 518, 520, 522. The first outer portion 518 may be cylindrical. The second outer portion 520 may be conical, and may be referred to as a chamfer. The third outer portion 522 may be cylindrical, and may have a diameter greater than the diameter of the first outer portion 518. An outer distal edge of the third outer portion 522 may be rounded. Indicia 524, such as the mark 524 shown in FIG. 22, may be included on the third outer portion 522. One or more dimples 526 may be included in the proximal side; three evenly spaced dimples 526, 528, 530 are shown. The middle dimple 528 may be aligned with the mark 524. Bilateral holes 532, 534 may extend through the impaction cap 146 between the proximal and distal sides 514, 516. The holes 532, 534 may be symmetrically located relative to the mark 524. Referring to FIG. 23, each hole 532, 534 may include a conical distal countersink. A central hole 536 may extend through the impaction cap 146 between the proximal and distal sides 514, 516.

The inserter connector 138 extends between a proximal end 538 and a distal end 540. The proximal end 538 may be circular in a proximal view. A notch 542 may be present in an outer proximal edge of the proximal end 538. The notch 542 may be rounded or semicircular in a side view. The distal end 540 may be rectangular in a distal view. The distal end 540 may have rounded corners in the distal view. The major dimension (length) of the rectangle may be aligned with the notch 542. The inserter connector 138 may include a central longitudinal hole 544 that extends through the inserter connector between the proximal end 538 and the distal end 540. The hole 544 may include, from proximal to distal, first, second, and third inner portions 546, 548, 550. The first inner portion 546 may have a circular cross section and may extend distally about one third of the overall proximal-distal length of the inserter connector 138. The first inner portion 546 may have a flat distal end. The second inner portion 548 may have a circular cross section with a diameter less than the first inner portion 546. The second inner portion 548 may extend distally about one third of the overall proximal-distal length of the inserter connector 138. Referring to FIG. 23, the cross-sectional shape of the third inner portion 550 may be the superposition of a circle and a rectangle. The diameter of the circle may be less than the diameter of the first inner portion 546 and greater than the diameter of the second inner portion 548. The major dimension (length) of the rectangle may be greater than the diameter of the first inner portion 546, and may be aligned with the major dimension (length) of the rectangle of the distal end 540. The minor dimension (width) of the rectangle may be less than the diameter of the circle, and may be less

than the diameter of the second inner portion 548. The rectangle may have rounded corners. The third inner portion 550 may have a flat proximal end. An inner distal edge of the rectangle may be chamfered.

The inserter shaft 136 extends between a proximal end 552 and a distal end 554. The inserter shaft 136 may be a tubular part with circular outer and inner cross-sectional shapes. The inserter shaft 136 may include, from proximal to distal, first, second, and third outer portions 556, 558, 560. The first outer portion 556 may include a pair of shaped slots 562, 564 which each include an approximately 90° bend. The slots 562, 564 may extend longitudinally into the proximal end 552 before making the approximately 90° bend. Each slot 562, 564 may extend through one wall of the first outer portion 556. The outer diameter of the second outer portion 558 may be greater than the outer diameter of the first outer portion 556. The second outer portion 558 may occupy most of the overall proximal-distal length of the inserter shaft 136. The second outer portion 558 may include one or more openings 566. Four openings 566 are shown evenly spaced along the length of the second outer portion 558. Each opening 566 may extend through one wall of the second outer portion 558, or through both walls as shown. The second outer portion 558 may include one or more indicia or markings. Three indicia are shown, from proximal to distal: a sideways “8” 568, a longitudinal line 570, and a set of numerals and transverse lines 572. The indicia may be repeated on the opposite side of the second outer portion 558 as shown in FIGS. 22 and 23. The outer diameter of the third outer portion 560 may be less than the outer diameter of the first outer portion 556. The inserter shaft 136 may include a central longitudinal hole 574 that extends through the inserter shaft between the proximal end 552 and the distal end 554. The hole 574 may be chamfered around its proximal inner edge. The diameter of the hole 574 may be the same as, or similar to, the diameter of the second inner portion 548 of the inserter connector.

The shaft assembly 116 may be assembled by inserting the first outer portion 556 of the inserter shaft 136 into the central hole 536 of the impaction cap 146 so that the proximal side 514 and the proximal end 552 face the same direction, the distal side 516 abuts the proximal end of the second outer portion 558, and the mark 524 is aligned with the sideways “8” 568 and/or the longitudinal line 570; and by inserting the third outer portion 560 of the inserter shaft 136 into the first inner portion 546 of the inserter connector 138 so that the major dimension of the rectangle of the distal end 540 is transverse to, or perpendicular to, the sideways “8” 568 and/or the longitudinal line 570. The shaft assembly 116 may be a weldment. Alternatively, the impaction cap 146, inserter connector 138, and/or inserter shaft 136 may be combined into a single part.

Referring to FIGS. 9 and 10, the standoff 132 extends between a proximal end 576 and a distal end 578. The standoff 132 may be a tubular part with circular outer and inner cross-sectional shapes. The standoff 132 may include, from proximal to distal, first and second outer portions 580, 582. The first outer portion 580 may extend over most of the overall proximal-distal length of the standoff 132. The outer diameter of the second outer portion 580 may be greater than the outer diameter of the first outer portion 580 so that the second outer portion 580 forms a flange around the distal end 578. The standoff 132 may include a central longitudinal hole 584 that extends through the standoff between the proximal end 576 and the distal end 578.

Referring to FIGS. 9, 10, 24, and 25, the lock tube assembly 118 may include a proximal lock tube 124, a proximal lock tube end part 148, a distal lock tube 126, and a lock tube coupling 128.

The proximal lock tube 124 extends between a proximal end 586 and a distal end 588. The proximal lock tube 124 may be a tubular part with circular outer and inner cross-sectional shapes. The proximal lock tube 124 may include, from proximal to distal, first and second outer portions 590, 592. The first outer portion 590 may extend distally a short distance. The diameter of the second outer portion 592 may be greater than the diameter of the first outer portion 590. The second outer portion 592 may extend over most of the overall proximal-distal length of the proximal lock tube 124. The proximal lock tube 124 may include a central longitudinal hole 594 that extends through the proximal lock tube between the proximal end 586 and the distal end 588. The hole 594 may be chamfered around its proximal inner edge.

The proximal lock tube end part 148 extends between a proximal end 596 and a distal end 598. The proximal lock tube end part 148 may be a tubular part with circular outer and inner cross-sectional shapes. The proximal lock tube end part 148 may include, from proximal to distal, first, second, and third outer portions 600, 602, 604. The first outer portion 600 may extend distally a short distance, and may include a chamfer around its proximal outer edge. The diameter of the second outer portion 602 may be less than the diameter of the first outer portion 600. The second outer portion 602 may also extend distally a short distance. The diameter of the third outer portion 604 may be greater than the diameter of the second outer portion 602, and may be equal to the diameter of the first outer portion 600. Thus, the second outer portion 602 may be referred to as an exterior groove around the proximal end 596. There may be a short tapered transition between the distal end of the second outer portion 602 and the proximal end of the third outer portion 604. The proximal lock tube end part 148 may include a central longitudinal hole 606 that extends through the proximal lock tube end part between the proximal end 596 and the distal end 598. The hole 606 may include a chamfer 608 around its proximal inner edge, and may include a distal counterbore 610. The inner diameter of the hole 606 may be the same as, or similar to, the inner diameter of the hole 594.

The distal lock tube 126 extends between a proximal end 612 and a distal end 614. The distal lock tube 126 may be a tubular part with circular outer and inner cross-sectional shapes. The proximal end 612 may include a transverse notch 616 that extends through both walls of the distal lock tube 126. The notch 616 may include a rounded distal end. The notch 616 may divide the proximal end 612 into a pair of tabs 618, 620. The notch 616 may step up in width proximally and correspondingly, each tab 618, 620 may step down in width proximally to form fingers 622, 624. The distal end 614 may be chamfered around its outer distal edge. With brief reference to FIG. 49, the distal end 614 may include a pair of shaped slots 152 which each include an approximately 90° bend. The slots 152 may extend longitudinally into the distal end 614 before making the approximately 90° bend. Each slot 152 may extend through one wall of the distal lock tube 126. The distal lock tube 126 may include a central longitudinal hole 626 that extends through the distal lock tube between the proximal end 612 and the distal end 614. The inner diameter of the hole 626 may be the same as, or similar to, the inner diameter of the hole 594.

Referring to FIGS. 24, 25, 20, and 21, the lock tube coupling 128 extends between a proximal end 628 and a distal end 630. The lock tube coupling 128 may be a tubular

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part. The lock tube coupling 128 may include a proximal body 632 which may have a circular cross-sectional outer shape, and may extend distally about half or three-fifths of the overall proximal-distal length of the lock tube coupling 128. Bilateral tabs 634, 636 may protrude distally from opposite sides of the body 632. The tabs 634, 636 may be inset relative to the outer diameter of the body 632. The distal portions of the tabs 634, 636 may include posts 638, 640, respectively. The posts 638, 640 may protrude outwardly from the outer sides of the tabs 634, 636, and may protrude beyond the outer diameter of the body 632 as shown in FIGS. 20 and 21. The lock tube coupling 128 may include a central longitudinal hole 642 that extends through the lock tube coupling between the proximal end 628 and the distal end 630. The hole 642 may include, from proximal to distal, first, second, and third inner portions 644, 646, 648. The first inner portion 644 may have a circular cross section and may extend distally about a third of the overall proximal-distal length of the lock tube coupling 128, or about half of the length of the body 632. The second inner portion 646 may have a circular cross section with a diameter less than the diameter of the first inner portion 644. The diameter of the second inner portion 646 may be the same as, or similar to, the inner diameter of the hole 594. The second inner portion 646 may extend distally only a short distance as a complete diameter, but sections of it may extend through the distal end to form the inner sides of the tabs 634, 636. The third inner portion 648 may have a non-circular cross section that includes bilateral alcoves 650, 652 in the distal walls of the body 632 between the tabs 634, 636 (in other words, the alcoves 650, 652 are located 90° away from the tabs 634, 636).

The lock tube assembly 118 may be assembled by inserting the first outer portion 590 of the proximal lock tube 124 into the distal counterbore 610 of the proximal lock tube end part 148; inserting the distal end 588 of the proximal lock tube 124 into the first inner portion 644 of the lock tube coupling 128; and inserting the proximal end 612 of the distal lock tube 126 into the third inner portion 648 of the lock tube coupling 128 so that the fingers 622, 624 are received in the alcoves 650, 652 and the tabs 634, 636 are received in the notch 616. The hole 594 of the proximal lock tube 124, the hole 606 of the proximal lock tube end part 148, the hole 626 of the distal lock tube 126, and the second inner portion 646 of the hole 642 of the lock tube coupling 128 may together form the longitudinal instrument bore 112. The proximal lock tube 124, proximal lock tube end part 148, distal lock tube 126, and lock tube coupling 128 may be permanently connected, for example by welding. Alternatively, the proximal lock tube 124, proximal lock tube end part 148, distal lock tube 126, and/or lock tube coupling 128 may be combined into a single part.

The instrument 100 may be assembled by coupling the shaft assembly 116 to the lock tube assembly 118; coupling the selector ring 144 to the shaft assembly 116 and lock tube assembly 118; coupling the inserter handle 142 to the shaft assembly 116, lock tube assembly 118, and selector ring 144; coupling the standoff 132 and the standoff nut 134 to the shaft assembly 116, lock tube assembly 118, selector ring 144, and inserter handle 142; coupling the threaded driver 130 and the shuttle assembly 114 to the expansion knob 140; coupling the shuttle assembly, threaded driver, and expansion knob 140 to the shaft assembly 116, lock tube assembly 118, selector ring 144, and inserter handle 142.

Coupling the shaft assembly 116 to the lock tube assembly 118 may include inserting the distal end 614 of the distal lock tube 126 into the hole 574 in the proximal end 552 of

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the inserter shaft 136, advancing the lock tube assembly 118 distally until the posts 638, 640 enter the shaped slots 562, 564, and twisting the lock tube assembly 118 counterclockwise so that the posts 638, 640 are captured in the shaped slots 562, 564.

Coupling the selector ring 144 to the shaft assembly 116 and lock tube assembly 118 may include inserting the proximal end 596 of the proximal lock tube end part 148 into the inner ring 436 of the selector ring 144 and advancing the lock tube assembly 118 proximally until the posts 638, 640 enter the grooves 460, 462 and the hole 464 is aligned with the middle dimple 528. A compression spring or a ball detent may be received in the hole 464 to engage the dimples 526, 528, 530.

Coupling the inserter handle 142 to the shaft assembly 116, lock tube assembly 118, and selector ring 144 may include inserting the proximal end 596 of the proximal lock tube end part 148 into the hole 414 in the distal end 386 of the inserter handle 142, advancing the lock tube assembly 118 proximally until the collar portion 408 enters the space between the outer ring 434 and the inner ring 436, and inserting fasteners such as screws through the holes 532, 410 and the holes 534, 412.

Coupling the standoff 132 and the standoff nut 134 to the shaft assembly 116, lock tube assembly 118, selector ring 144, and inserter handle 142 may include inserting the proximal end 596 of the proximal lock tube end part 148 into the hole 584 in the distal end 578 of the standoff 132, advancing the lock tube assembly 118 proximally until the distal end 578 abuts the proximal end 384 of the inserter handle 142, inserting the proximal end 596 into the hole 374 in the distal side 360 of the standoff nut 134, advancing the lock tube assembly 118 proximally until the distal side 360 abuts the proximal end 384, and engaging the internal threads of the third inner portion 380 with the external threads of the first outer portion 388 to make a tight connection. The notches 364, 366 may facilitate tightening.

Coupling the threaded driver 130 and the shuttle assembly 114 to the expansion knob 140 may include inserting the proximal end 492 of the threaded driver 130 into the hole 482 in the distal end 468 of the expansion knob 140, aligning the hole 478 with the hole 496, advancing the threaded driver 130 proximally into the second inner portion until the proximal end 492 abuts the proximal end of the second inner portion 486, inserting the distal end 186 of the shuttle 122 into the hole 482 in the proximal end 466 of the expansion knob 140, advancing the shuttle assembly 114 distally until the distal end 186 abuts the proximal end 492 of the threaded driver 130, aligning the fourth outer portion 246 with the third inner portion 506, advancing the shuttle assembly 114 farther distally until the distal end of the second outer portion 242 abuts the proximal end 492, and engaging the external threads 250 of the second outer portion 242 with the internal threads 512 of the third inner portion 506 until the shuttle assembly 114, expansion knob 140, and threaded driver 130 achieve a snug connection.

Coupling the shuttle assembly 114, threaded driver 130, and expansion knob 140 to the shaft assembly 116, lock tube assembly 118, selector ring 144, and inserter handle 142 may include inserting the proximal end 596 of the proximal lock tube end part 148 into the hole 252 in the distal end 186 of the shuttle; advancing the lock tube assembly 118 proximally until the fifth inner portion 510 of the threaded driver 130 receives the proximal end 576 of the standoff 132, the third inner portion 488 of the expansion knob 140 receives the standoff nut 134, the fourth inner portion 490 of the expansion knob 140 receives the third outer portion 392 of

the inserter handle 142, the distal end 468 of the expansion knob 140 abuts the distal end of the third outer portion 392, and the second inner portion 418 of the hole 414 of the inserter handle 142 receives the fourth outer portion 246 of the 122; and inserting fasteners such as screws through the holes 478, 496 and the holes 480, 498.

The instrument 100 may be operated by sliding the lock collar 120, rotating the selector ring 144, or rotating the expansion knob 140.

Referring to FIGS. 11 and 12, the lock collar 120 is coupled to the shuttle 122 so that the lock collar 120 can slide relative to the shuttle 122 along a direction established by the undercut grooves 298, 346, feet 294, 296, and slot 292. The travel of the lock collar 120 may be limited by the presence of a pin in the slot 292 and the hole 234 of the shuttle 122. In this arrangement, the lock collar 120 may slide between a first position and a second position. In the first position, the pin may contact the end of the slot 292 that is closest to the first curved side 278 of the hole 276. The first position may be referred to as an open or unlocked position. In the second position, the pin may contact the end of the slot 292 that is closest to the outer curved side 268. The second position may be referred to as a closed or locked position. The lock collar 120 may be biased toward the second position.

Referring to FIGS. 9, 10, and 15, the selector ring 144 is coupled to the posts 638, 640 of the lock tube coupling 128 so that rotation of the selector ring 144 causes rotation of the lock tube assembly 118 relative to the rest of the instrument 100. Referring to FIGS. 14 and 15, rotation of the selector ring 144 is limited by the presence of the collar portion 408 in the space between the outer ring 434 and the inner ring 436. In the example shown, the selector ring 144 is rotatable between a first position and a third position. In the first position, the arm 438 may contact the end of the collar portion 408 that includes the hole 412, the hole 464 may align with the dimple 530, the indicia 454 for the cleaning setting may align with the distally-pointing arrow indicium 402, the posts 638, 640 may be positioned in the shaped slots 562, 564 so that the posts are free to slide axially out of the slots, and the posts 638, 640 may be aligned with the grooves 426, 428. The first position may be referred to as a cleaning position because the lock tube assembly 118 and/or other parts of the instrument 100 are positioned for disassembly for cleaning or reassembly after cleaning. In the third position, the arm 438 may contact the end of the collar portion 408 that includes the hole 410, the hole 464 may align with the dimple 526, the indicia 454 for the locked setting may align with the distally-pointing arrow indicium 402, the posts 638, 640 may be positioned in the shaped slots 562, 564 so that the posts are captured in the slots, the posts 638, 640 may be rotated out of alignment with the grooves 426, 428, and the shaped slots 152 may be positioned with their distal openings aimed into the corners of the third inner portion 550 of the hole 544 of the inserter connector 138. The third position may be referred to as a locked position because the parts of the instrument 100 are positioned to lock the implant 1000 to the instrument. The selector ring 144 may be rotatable to a second position which is between the first and third positions. In the second position, the arm 438 may be centered in the open side of the collar portion 408, the hole 464 may align with the dimple 528, the indicia 454 for the unlocked setting may align with the distally-pointing arrow indicium 402, the posts 638, 640 may be positioned in the shaped slots 562, 564 so that the posts are captured in the slots, the posts 638, 640 may be rotated out of alignment with the grooves 426, 428, and the shaped slots

152 may be positioned with their distal openings aimed along the major dimension (length) of the third inner portion 550 of the hole 544 of the inserter connector 138. The second position may be referred to as an unlocked position because the parts of the instrument 100 are positioned so that the implant 1000 may be coupled to or decoupled from the instrument 100.

Referring to FIGS. 9-12, 13, 14, and 16-19, the expansion knob 140 is coupled to the shuttle assembly 114, the threaded driver 130, and the inserter handle 142 so that rotation of the expansion knob 140 causes linear translation of the shuttle assembly 114 along the longitudinal instrument axis 102 without rotation of the shuttle assembly 114 about the axis 102. The threaded driver 130 is received in the second inner portion 486 of the hole 482 of the expansion knob 140 so that rotation of the expansion knob 140 causes rotation of the threaded driver 130. The external threads 250 of the second outer portion 242 of the shuttle 122 are received by the internal threads 512 of the third inner portion 506 of the threaded driver 130 and the fourth outer portion 246 of the shuttle 122 is received in the second inner portion 418 of the hole 414 of the inserter handle 142 so that rotation of the threaded driver 130 causes linear translation of the shuttle assembly 114 along the longitudinal instrument axis 102. In the example shown, the shuttle assembly 114 is movable between a first position and a third position. In the first position, the distal side of the flange 232 of the shuttle 122 may abut the proximal end 466 (tabs 472, 474) of the expansion knob 140, the proximal end of the external threads 250 of the second outer portion of the shuttle 122 may be even with, or close to, the proximal end 492 of the threaded driver 130, and the indicia 248 of the shuttle 122 for the collapsed configuration may be aligned with the indicia 476 of the expansion knob 140. The first position may be referred to as a collapsed position for this reason. In the third position, the distal end of the external threads 250 may be even with, or close to, the distal end 494 of the threaded driver 130, and the indicia 248 of the shuttle 122 for the vertically expanded configuration may be aligned with the indicia 476 of the expansion knob 140. The third position may be referred to as a vertically expanded position for this reason. The shuttle assembly 114 may be movable to a second position which is between the first and third positions. In the second position, the distal end of the external threads 250 may be even with, or close to, the proximal end 358 of the standoff nut 134 and the indicia 248 of the shuttle 122 for the laterally expanded configuration may be aligned with the indicia 476 of the expansion knob 140. The second position may be referred to as a laterally expanded position for this reason. The shuttle assembly 114 may be movable to any number of positions between the first and third positions due to the threaded interconnection between the shuttle 122 and the threaded driver 130.

Various additional tools may be used with instrument 100 to implant and expand an intervertebral implant. The instrument 100 and selected additional tools may be provided in a kit and may be collectively referred to as an instrument system 115. FIGS. 26-30 disclose additional tools of the instrument system 115, including a draw bar assembly 160, a graft funnel 170, a tamp assembly 180, and a screwdriver assembly 190. The draw bar assembly 160 may include a threaded tip 162 and a handle 164. The screwdriver assembly 190 may be assembled with a screw holder 200 to hold and drive a lockout screw 654 into an implant such as implant 1000 to lock the implant in an expanded configuration. The instrument system 115 may be used to implant and/or expand other expandable implants.

Referring to FIGS. 26, 31, and 32, the draw bar assembly 160 may include a draw bar 156 and a draw bar knob or handle 164.

The draw bar 156 extends between a proximal end 656 and a distal end 658. The draw bar 156 may be a cylindrical shaft as shown. The draw bar 156 may include, from proximal to distal, first, second, and third outer portions 660, 662, 664. The first outer portion 660 may extend distally most of the overall proximal-distal length of the draw bar 156, and may be chamfered around its proximal outer edge. The diameter of the second outer portion 662 may be less than the diameter of the first outer portion 660. A tapered shoulder 668 may be included between the first and second outer portions 660, 662. The third outer portion 664 may include the threaded tip 162 with external threads. The major diameter of the threads may be less than the diameter of the second outer portion 662.

The draw bar handle 164 extends between a proximal end 670 and a distal end 672. The draw bar handle 164 may include, from proximal to distal, a knob body 674 and a shaft 676. The knob body 674 may be generally disc-shaped with an outer diameter that is greater than its proximal-distal length. The knob body 674 may include one or more grip features 678 such as the six rim scallops shown. The outer diameter of the shaft 676 may be less than the outer diameter of the knob body 674 and the proximal-distal length of the shaft may be greater than the proximal-distal length of the knob body 674. The shaft 676 may include a flange 680 (annulus 166) that extends circumferentially around a middle portion of the shaft. The flange 680 may include a distal chamfer 682. The shaft 676 may include one or more holes 684 that extend transversely into, or through, the shaft; two through holes 684 are shown just distal to the distal side of the knob body 674. The two through holes 684 are perpendicular to each other. The draw bar handle 164 may include a central longitudinal hole 686 that extends through the draw bar handle between the proximal end 670 and the distal end 672. The hole 686 may include, from proximal to distal, first, second, third, and fourth inner portions 688, 690, 692, 694. The first inner portion 688 may have a circular cross section and may extend distally about half, or most, of the proximal-distal length of the knob body 674. The first inner portion 688 may have a flat distal end. A circumferential fillet radius may be present around the proximal inner edge of the first inner portion 688. A circumferential fillet radius may be present in the distal inner corner of the first inner portion 688. The second inner portion 690 may be conical. The second inner portion 690 may be referred to as a countersink or circumferential chamfer associated with the third inner portion 692. The major diameter of the second inner portion 690 may be less than the diameter of the first inner portion 688. The third inner portion 692 may have a circular cross section and may extend distally to the same or similar level as the distal side of the knob body 674. The diameter of the third inner portion 692 may be less than the diameter of the first inner portion 688. The fourth inner portion 694 may have a circular cross section and may extend distally through the distal end of the 672 of the draw bar handle 164. The diameter of the fourth inner portion 694 may be less than the diameter of the first inner portion 688 and greater than the major diameter of the second inner portion 690 and the diameter of the third inner portion 692. The fourth inner portion 694 may have a flat proximal end. The hole(s) 684 may extend into a proximal end of the fourth inner portion 694.

The draw bar assembly 160 may be assembled by inserting the proximal end 656 of the draw bar 156 into the hole

686 of the draw bar handle 164 so that the first portion 660 is received in the third inner portion 692 and the proximal end 656 is at the same or similar level with the distal end of the first inner portion 688. A circumferential gap 696 may exist between the outer surface of the first portion 660 and the inner surface of the fourth inner portion 694. See FIG. 5. The draw bar 156 and draw bar handle 164 may be permanently connected, for example by welding. Alternatively, the draw bar 156 and draw bar handle 164 may be combined into a single part.

The draw bar 156 may be received in the lock tube assembly 118, more specifically in the hole 606 of the proximal lock tube end part 148, the hole 594 of the proximal lock tube 124, the second inner portion 646 of the hole 642 of the lock tube coupling 128, and the hole 626 of the distal lock tube 126.

Referring to FIGS. 28, 33, and 34, the graft funnel 170 extends between a proximal end 698 and distal end 700. The graft funnel 170 may include a proximal mouth 702 and a distal stem 704. The mouth 702 and the stem 704 may have circular cross sections. The mouth 702 may be conical or otherwise tapered. The diameter of the stem 704 may be much less than the diameter of the mouth 702. The graft funnel 170 may include a central longitudinal hole 706 that extends through the graft funnel between the proximal end 698 and the distal end 700. The hole 706 may include, from proximal to distal, first, second, third, and fourth inner portions 708, 710, 712, 714. Each of these inner portions may have a circular cross section. The first inner portion 708 may be conical or otherwise tapered, and may be shaped like the mouth 702. The second inner portion 710 may be described as a fillet radius transition or throat between the first inner portion 708 and the third inner portion 712. The second inner portion 710 may be at the same or similar level as the transition between the mouth 702 and the stem 704. The diameter of the third inner portion 712 may be less than the minor diameter of the first inner portion 708. The fourth inner portion 714 may extend proximally about half or more of the proximal-distal length of the stem 704. The diameter of the fourth inner portion 714 may be greater than the diameter of the third inner portion 712. The proximal end of the fourth inner portion 714 may be flat, and a circumferential chamfer may be present at the inner edge between the third and fourth inner portions 712, 714.

The stem 704 may be received by the second inner portion 256 of the hole 252 of the shuttle 122 of the shuttle assembly 114. The fourth inner portion 714 may receive the proximal end 596 of the proximal lock tube end part 148 of the lock tube assembly 118. The diameter of the third inner portion 712 may be the same as or similar to the diameter of the hole 606.

Referring to FIGS. 27, 35, and 36, the tamp assembly 180 may include a tamp shaft 158, a tamp shaft top part 168, a tamp handle 172, and a tamp shaft tip part 174.

The tamp shaft 158 extends between a proximal end 716 and a distal end 718. The tamp shaft 158 may be a cylindrical shaft as shown. The tamp shaft 158 may include a proximal externally threaded portion 720 and a distal smooth portion 722. The diameter of the distal smooth portion 722 may be greater than the major diameter of the proximal externally threaded portion 720. The tamp shaft 158 may include a central longitudinal internally threaded hole 724 that extends proximally into the distal end 718.

The tamp shaft top part 168 extends between a proximal end 726 and a distal end 728. The tamp shaft top part 168 may include a proximal externally threaded portion 730, a middle smooth cylindrical portion 732, and a distal torque

fitting 734, such as the hexagonal key shown. The diameter of the middle smooth cylindrical portion 732 may be greater than the major diameter of the proximal externally threaded portion 730. The outer diameter of the distal torque fitting 734 may be less than the diameter of the middle smooth cylindrical portion 732. The tamp shaft top part 168 may include a central longitudinal internally threaded hole 736 that extends proximally into the distal end 728.

The tamp handle 172 extends between a proximal end 738 and a distal end 740. The tamp handle 172 may be a cylindrical part with rounded proximal and distal ends 738, 740 as shown. The tamp handle 172 may include one or more friction features 742, such as the series of shallow circumferential grooves shown. The tamp handle 172 may include a central longitudinal hole 744 that extends proximally into the distal end 740. The hole 744 may extend proximally about half, or most, of the proximal-distal overall length of the tamp handle 172. The hole 744 may include, from proximal to distal, first, second, and third inner portions 746, 748, 750 of approximately equal proximal-distal lengths. Each of the inner portions may have a circular cross section. The first inner portion 746 is not visible in FIG. 36. The first inner portion 746 may be a blind hole included for ease of manufacturing the second inner portion 748, which has internal threads. The minor diameter of the threads may be equal to or similar to the diameter of the first inner portion 750. The diameter of the third inner portion 750 may be greater than the major diameter of the threads. The third inner portion 750 may be referred to as a counterbore around the second inner portion 748. The proximal end of the third inner portion 750 may be flat.

The tamp shaft tip part 174 extends between a proximal end 752 and a distal end 754. The tamp shaft tip part 174 may be a cylindrical shaft as shown. The tamp shaft tip part 174 may include a proximal externally threaded portion 756 and a distal smooth portion 758. The diameter of the distal smooth portion 758 may be greater than the major diameter of the proximal externally threaded portion 756. The distal end 754 may have a concave surface 760.

The tamp assembly 180 may be assembled by threading the proximal externally threaded portion 730 of the tamp shaft top part 168 into the second inner portion 748 of the hole 744 of the tamp handle 172 so that the middle smooth cylindrical portion 732 is received in the third inner portion 750, threading the proximal externally threaded portion 720 of the tamp shaft 158 into the hole 736 of the tamp shaft top part 168, and threading the proximal externally threaded portion 756 of the tamp shaft tip part 174 into the hole 724 of the tamp shaft 158. The tamp assembly 180 may be a weldment. Alternatively, the tamp shaft 158, tamp shaft top part 168, tamp handle 172, and/or tamp shaft tip part 174 may be combined into a single part.

The outer diameters of the distal smooth portion 722 of the tamp shaft 158 and the distal smooth portion 758 of the tamp shaft tip part 174 may be received in the inner diameter of the third inner portion 712 of the hole 706 of the graft funnel 170 and in the lock tube assembly 118, more specifically in the hole 606 of the proximal lock tube end part 148, the hole 594 of the proximal lock tube 124, the second inner portion 646 of the hole 642 of the lock tube coupling 128, and the hole 626 of the distal lock tube 126.

Referring to FIGS. 29, 30, 37, and 38, the screwdriver assembly 190 may include a handle 176, a shaft 178, a screw holder 200, and a screw holder catch 182.

The handle 176 extends between a proximal end 762 and a distal end 764. The handle 176 may include a proximal grip portion 766 with one or more friction features 768 such

as the four generally pear-shaped indentations shown, and a distal shaft 770. The shaft 770 may include a proximal first portion 772 and a distal second portion 774. The first portion 772 may include one or more friction features 776, such as the series of shallow circumferential grooves shown. The first portion 772 may be a sleeve that is longitudinally movable relative to the second portion 774. A central longitudinal D-shaped hole 778 may extend proximally into the distal end 764. One or more balls (not shown) may be captured between the first and second portions 772, 774 and may protrude at least partially into the D-shaped hole 778 when the first portion 772 is in a first position relative to the second portion 774. The balls may be free to move radially away from the hole 778 when the first portion 772 is in a second position relative to the second portion 774. The first portion 772 may be biased towards the first position.

The shaft 178 extends between a proximal end 780 and a distal end 782. The shaft 178 may be generally cylindrical as shown. The shaft 178 may include, from proximal to distal, first, second, third, fourth, and fifth outer portions 784, 786, 788, 790, 792. The first outer portion 784 may have a generally circular cross section, and may include a unilateral longitudinal flat 794 so that the first outer portion 784 is complementary to the D-shaped hole 778 in the shaft 770 of the handle 176. A circumferential groove 796 may extend around the circular side of the first outer portion 784 to engage the balls in the shaft 770 of the handle 176. The diameter of the second outer portion 786 may be greater than the diameter of the first outer portion 784. The second outer portion 786 may extend over most of the proximal-distal length of the shaft 178. The third outer portion 788 may include external threads. The major diameter of the threads may be less than the diameter of the second outer portion 786. The fourth outer portion 790 may have a circular cross section and may be smooth. The diameter of the fourth outer portion 790 may be the same as, or similar to, the minor diameter of the threads of the third outer portion 788. The fifth outer portion 792 may include a torque fitting 798 such as the hexagonal key shown.

Referring to FIGS. 39-42, the screw holder 200 extends between a proximal end 800 and a distal end 802. The screw holder 200 may be a tubular part with generally circular outer and inner cross-sectional shapes. The screw holder 200 may include, from proximal to distal, first and second outer portions 804, 806. The first outer portion 804 may have a generally circular cross section, and may extend distally over most of the proximal-distal length of the screw holder 200. Bilateral flats 808, 810 may be included at or near the middle of the proximal-distal length of the first outer portion. A longitudinal slot 812 may extend through one wall of the first outer portion 804, and may be located between the flats 808, 810. The proximal end of the slot 812 may be at the same or similar level as the proximal ends of the flats 808, 810. The proximal end of the slot 812 may include a circular counterbore 814. The distal end of the slot 812 may extend distally about twice as far as the distal ends of the flats 808, 810. The slot 812 may be rectangular or oval. The second outer portion 806 may taper inwardly toward the distal end 802. The second outer portion 806 may be conical. The second outer portion 806 may include a notch 816 that extends proximally from the distal end 802 through one wall of the second outer portion 806. The notch 816 may be rectangular or square, and may have rounded corners. The notch 816 may be in line with the slot 812. The screw holder 200 may include a central longitudinal hole 818 that extends through the screw holder between the proximal end 800 and the distal end 802. The hole 818 may include, from proximal

to distal, first, second, and third inner portions **820**, **822**, **824**. The first inner portion **820** may include internal threads. The diameter of the second inner portion **822** may be less than, equal to, or similar to the minor diameter of the threads. The proximal end of the second inner portion **822** may be proximal to the proximal ends of the flats **808**, **810**. The distal end of the second inner portion **822** may be distal to the proximal end of the notch **816**, and may be at or near the middle of the proximal-distal length of the notch **816**. A short tapered surface may be included at the transition between the second and third inner portions **822**, **824**. The second inner portion **822** may include a unilateral longitudinal alcove **826** extending into one wall of the second inner portion **822**. The alcove **826** may be in line with the slot **812** and the notch **816**. The diameter of the third inner portion **824** may be less than the major diameter of the threads of the first inner portion **820**, and may be greater than the minor diameter of the threads.

The screw holder catch **182** extends between a proximal end **828** and a distal end **830**. The screw holder catch **182** may be a generally plate-like part, and may include, from proximal to distal, first, second, and third portions **832**, **834**, **836**. The first portion **832** may have a size and shape that is complementary to the slot **812** of the screw holder **200**. The proximal end of the first portion **832** may include a circular shelf **838** that is complementary to the counterbore **814**. The second portion **834** may be offset from the first portion **832** in a direction opposite the shelf **838**, and may be substantially parallel to the first portion **832**. The second portion **834** may include a longitudinal arcuate groove **840** that faces the same direction as the shelf **838**. The third portion **836** may have a size and shape that is complementary to the notch **816** of the screw holder **200**. The third portion **836** may be in line with the first portion **832** (thus offset from the second portion in a direction toward the shelf **838**). The third portion **836** may include a distal tooth **842** that faces the same direction as the shelf **838**.

The screwdriver assembly **190** may be assembled by coupling the screw holder catch **182** to the screw holder **200** so that the shelf **838** is in the counterbore **814**, the second portion **834** is in the alcove **826**, the third portion **836** is in the notch **816**, and the groove **840** faces the interior of the screw holder **200**; threading the first inner portion **820** of the hole **818** of the screw holder **200** onto the third outer portion **788** of the shaft **178**; and inserting the first outer portion **784** of the shaft **178** into the D-shaped hole **778** of the handle **176** so that the balls engage the groove **796**. Alternatively, the handle **176**, shaft **178**, screw holder **200**, and/or screw holder catch **182** may be combined into a single part.

The outer diameters of the first outer portion **804** of the screw holder **200** and the second outer portion **786** of the shaft **178** may be received in the inner diameter of the third inner portion **712** of the hole **706** of the graft funnel **170** and in the lock tube assembly **118**, more specifically in the hole **606** of the proximal lock tube end part **148**, the hole **594** of the proximal lock tube **124**, the second inner portion **646** of the hole **642** of the lock tube coupling **128**, and the hole **626** of the distal lock tube **126**.

Referring to FIGS. **43** and **44**, the lockout screw **654** extends between a proximal end **844** and a distal end **846**. The lockout screw **654** may include, from proximal to distal, first, second, third, fourth, fifth, and sixth portions **848**, **850**, **852**, **854**, **856**, **858**, each with a circular cross section. The first portion **848** may include a rounded outer proximal circumferential edge, and may extend a short distance in the proximal-distal direction. The first portion **848** may form a circumferential lip or flange around the proximal end of the

lockout screw **654**. The diameter of the second portion **850** may be less than the diameter of the first portion **848**. There may be a 90 degree corner between the first portion **848** and the second portion **850**. The diameter of the third portion **852** may be greater than the diameter of the first portion **848**. There may be a 90 degree corner between the second portion **850** and the third portion **852**, so that the second portion **850** forms a circumferential groove between the first and third portions **848**, **852**. The third portion **852** may have a rounded outer proximal circumferential edge. The diameter of the fourth portion **854** may be the same as or similar to the diameter of the first portion **848**, greater than the diameter of the second portion **850**, and less than the diameter of the third portion **852**. There may be a shoulder **655** between the third and fourth portions **852**, **854**. The shoulder **655** may taper, and may be conical. The fourth portion **854** may extend distally to about the middle of the proximal-distal length of the lockout screw **654**. The fifth portion **856** may include an externally threaded portion **653**. The major diameter of the threads may be the same as or similar to the diameter of the fourth portion **854**. There may be a tapered transition between the fourth and fifth portions **854**, **856**. The diameter of the sixth portion **858** may be the same as or similar to the minor diameter of the threaded portion **653**. The sixth portion **858** may have a smooth shaft and may include a rounded outer distal circumferential edge. The lockout screw **654** may include a torque fitting **860** in or on the proximal end **844**, such as the hexagonal socket shown. The lockout screw **654** is described in pending U. S. patent application Ser. No. 15/244,446.

The outer diameter of the third portion **852** of the lockout screw **654** may be received in the inner diameter of the third inner portion **712** of the hole **706** of the graft funnel **170** and in the lock tube assembly **118**, more specifically in the hole **606** of the proximal lock tube end part **148**, the hole **594** of the proximal lock tube **124**, the second inner portion **646** of the hole **642** of the lock tube coupling **128**, and the hole **626** of the distal lock tube **126**.

The lockout screw **654** may be coupled to the screwdriver assembly **190** by coupling the torque fitting **798** of the shaft **178** to the torque fitting **860** of the lockout screw **654**, inserting the first portion **848** of the lockout screw **654** into the third inner portion **824** of the screw holder **200**, and engaging the tooth **842** of the screw holder catch **182** in the second portion **850** of the lockout screw **654**. Alternatively, the lockout screw **654** may be coupled to the screwdriver assembly **190** by inserting the first portion **848** of the lockout screw **654** into the third inner portion **824** of the screw holder **200**, engaging the tooth **842** of the screw holder catch **182** in the second portion **850** of the lockout screw **654**, coupling the torque fitting **798** of the shaft **178** to the torque fitting **860** of the lockout screw **654**, and threading the first inner portion **820** of the hole **818** of the screw holder **200** onto the third outer portion **788** of the shaft **178**. The lockout screw **654** may be decoupled from the screwdriver assembly **190** by reversing these steps.

Referring to FIGS. **45-47**, implant **1000** is expandable along a horizontal or lateral or transverse or medial-lateral axis **104** and a vertical or cephalocaudal axis **106**. Implant **1000** includes a first support member **1040** and a second support member **1030**, the support members connected by a first end body **1050** and a second end body **1052**. First support member **1040** may include an upper support body **1042** and a lower support body **1044**. A pair of connection pins **1055** project from the second end body **1052**. A plurality of link members **1060**, **1062**, **1064**, **1066** pivotably connect each of the first and second end bodies **1050**, **1052**

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to each of the support members **1030**, **1040**. Lower support body **1044** includes ramped expansion slots **1114**, **1124**, and upper support body includes ramped expansion slots **1154**, **1164**. The implant **1000** is representative of the intervertebral cages described in pending U.S. patent application Ser. No. 15/244,446.

Referring to FIGS. **48-59**, methods for inserting and expanding an expandable intervertebral implant with instrument system **115** are shown. The instrument **100**, as seen in FIGS. **48** and **49**, includes a bayonet receiver **150** at the distal end **107** of the distal lock tube **126**. The bayonet receiver **150** includes a pair of shaped slots **152** which each include an approximately 90° bend. The slots are in the distal end **614** of the distal lock tube **126** of the lock tube assembly **118**. The bayonet receiver **150** may also include the inserter connector **138** of the shaft assembly **116**. In one step of the method, the implant **1000** is attached to the distal end **107** of the instrument **100**. The connection pins **1055** are received in the shaped slots **152** of the bayonet receiver **150** so that the implant **1000** extends distally from the instrument **100**. The distal lock tube **126** is rotated so that the pins **1055** traverse the 90° bend of the shaped slots **152**, forming a bayonet connection **154** and locking the implant **1000** to the instrument **100**. The implant second end body **1052** may be stabilized against rotation by its fit within the third inner portion **550** of the inserter connector **138**. Other embodiments within the scope of the disclosure may include a threaded connection between the implant **1000** and instrument **100** instead of the bayonet connection. In an embodiment, the distal lock tube **126** may be rotated via rotation of the selector ring **144**. In an embodiment, the lock tube is rotated 30° to complete the bayonet connection. The lock tube assembly (distal lock tube **126**, lock tube coupling **128**, proximal lock tube **124**) may additionally be unlocked from the inserter shaft **136** and removed from the instrument **100** for cleaning.

Referring to FIGS. **26**, **50**, and **51**, another step of the method includes inserting the draw bar assembly **160** distally through the instrument bore **112**. The distal threaded tip **162** of the draw bar assembly **160** passes through the implant second end body **1052** and is threadably engaged with the implant first end body **1050**. The instrument lock collar **120** may be actuated to engage the instrument shuttle **122** with an annulus **166** near the proximal end of the draw bar assembly **160**. The lock collar **120** may be moved to its first position, the draw bar assembly **160** may be advanced within the instrument bore **112** until the annulus is distal to the lock collar **120**, and the lock collar **120** may be moved to its second position. When the draw bar assembly **160** is inserted into connection with implant **1000** but before expansion of the implant, indicia **248** or markings may indicate the implant is in the collapsed configuration.

Referring to FIGS. **52** and **53**, the instrument **100** may be used to insert the implant **1000** into a prepared intervertebral space between a first vertebra **2** and a second vertebra (not shown for clarity), in another step of the method. The implant **1000** may be positioned in the intervertebral space before or after insertion of the draw bar assembly **160** into the instrument **100**.

When the implant is properly positioned in the intervertebral space and the draw bar assembly **160** is inserted into the instrument **100**, in another step of the method the expansion knob **140** is rotated, turning threaded driver **130**. The knob **140**, threaded driver **130** and shuttle **122** form a motion conversion mechanism to convert rotational motion to linear or axial motion along the longitudinal instrument axis **102**. Interaction of threaded driver **130** with shuttle **122**

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translates shuttle **122** and draw bar assembly **160** proximally, providing axial force in a first direction along axis **102** to draw the implant first end **1050** proximally toward the implant second end **1052** to laterally expand the implant in a second direction perpendicular to the first direction, along axis **104**. As the axial force is applied to the implant, the link members **1060**, **1062**, **1064**, **1066** pivot outward away from the central longitudinal axis **102**, carrying the first and second support members **1040**, **1030** away from one another along axis **104**, laterally expanding the implant. Stop surfaces on the implant end bodies and the support members may interact to limit the lateral expansion. When the lateral expansion is attained, indicia **248** or markings on the instrument may indicate the implant is in the laterally expanded configuration.

In another step of the method, the implant **1000** is expanded vertically. Referring to FIGS. **54** and **55**, the expansion knob **140** is rotated farther to continue to translate draw bar assembly **160** proximally along the longitudinal instrument axis **102**. The continued axial force along the first direction causes the link members **1064**, **1066** within the implant **1000** to travel along the expansion slots **1114**, **1124**, **1154**, **1164** in the upper and lower support bodies **1042**, **1044**, urging the upper and lower support bodies away from one another and vertically expanding the first support member **1040** along a third direction indicated by axis **106**, perpendicular to the first and second directions. When the vertical expansion is attained, indicia **248** or markings on the instrument may indicate the implant is in the vertically expanded configuration.

It is appreciated that the expansion knob may be actuated in discrete steps to first laterally and then vertically expand the implant, or the expansion knob may be actuated in a continuous motion to first laterally and then vertically expand the implant. It is also appreciated that in another embodiment, the vertical expansion of the implant may precede the horizontal expansion. In the embodiment shown, only the first support member **1040** expands vertically; the height of the second support member **1030** is fixed. This may provide a lordotic correction between the first and second vertebrae. In other embodiments, the first support member **1040** may be fixed and the second support member **1030** may be vertically expandable, or the first and second support members **1040**, **1030** may both be vertically expandable. When expansion is complete, the draw bar assembly **160** may be disengaged from the shuttle **122** by actuating the lock collar **120**, unthreaded from the implant **1000** and removed from the instrument **100**.

In another step of the method, graft material may be inserted through the instrument into the expanded implant **1000**. Referring to FIGS. **27**, **28**, **56**, and **57**, funnel **170** may be connected to the proximal end of instrument bore **112**. Graft material may be inserted axially through the funnel **170** and bore **112** and deposited into the implant inner chamber **1120**. The tamp assembly **180** may be deployed through instrument **100** to push and pack the graft through the bore **112** and in the chamber **1120**.

In another step of the method, a locking fastener such as lockout screw **654** may be inserted through the instrument and into fixed engagement with the expanded implant **1000**, to lock the implant in the horizontally and vertically expanded configuration. Referring to FIGS. **29**, **30**, **58**, and **59**, screw **654** may be loaded into screw holder **200** and connected to a distal end of screwdriver assembly **190**. A distal tip of screwdriver assembly **190** may include a tip **792** shaped to complementarily engage the proximal end of screw **654**. Screwdriver assembly **190**, with attached screw

holder 200 and screw 654, may be inserted longitudinally through bore 112 of instrument 100. The screwdriver assembly 190 may be actuated to engage a threaded portion 653 of screw 654 with the threaded first end body 1050 of implant 1000. As the screw is driven distally, a proximal shoulder 655 of screw 654 may abut a surface of the second end body 1052, rigidly locking the cage in the expanded configuration. Screw holder 200 may include a release feature or mechanism to release screw 654 from the holder 200 after the screw is in engagement with implant 1000. For example, the screw holder catch 182 may be biased to disengage the tooth 842 from the second portion 850 of the screw 654 when the screw is engaged with the implant 1000. When deployment and release of screw 654 is complete, the screwdriver assembly 190 and attached screw holder 200 may be withdrawn from the instrument 100.

Referring to FIGS. 60-63, implant 1000 is shown in situ after insertion between the first vertebral body 2 and the second vertebral body. In the embodiment depicted, support members 1040 and 1030 have been expanded horizontally along an anterior-posterior direction to provide support between the first and second vertebral bodies. Support member 1040 has been expanded vertically along a cephalocaudal direction to provide a lordotic correction between the first and second vertebrae. It is appreciated that in other embodiments of the invention, both support members could be expanded vertically, or support member 1030 could be solely expanded vertically to provide a kyphotic correction. One or more implants 1000 may be implanted between first and second vertebrae. Other approaches may be used within the scope of the disclosure, including lateral, transforaminal and anterior. One or more implants may be inserted in any orientation relative to the first and second vertebrae.

Referring to FIGS. 64-71, another insertion and expansion instrument is shown. Instrument 300 may be deployed to insert and expand implant 1000 or another expandable intervertebral implant. Instrument 300 may include a bayonet or other connection to removably connect implant 1000 to the instrument. Instrument 300 includes a ratchet mechanism 310 operable by a lever 312 to axially translate a shuttle 322 and draw bar assembly 328 proximally. The ratchet mechanism 310 may be referred to as an actuator. Similar to instrument 100, translation of draw bar assembly 328 provides axial force to laterally and/or vertically expand the connected implant. Other tools such as the funnel 170, tamp assembly 180, and screwdriver assembly 190 disclosed herein may be used, or adapted for use, with instrument 300 to deposit graft material within the implant and secure a locking fastener such as screw 654.

Instrument 300 extends between a proximal end 862 and a distal end 864 along a longitudinal instrument axis 866. Instrument 300 may include the ratchet mechanism 310, the lever 312, the shuttle 322 which may be similar to shuttle 122, an inserter handle 302 which may be similar to inserter handle 142, a shaft assembly 304 which may be similar to shaft assembly 116, an inserter shaft 306 which may be similar to inserter shaft 136, a distal tip part 308 which may be similar to inserter connector 138, a lock lever 314, a stop screw 316, a lock tube assembly 318 which may be similar to lock tube assembly 118, a lock tube coupling 320 which may be similar to lock tube coupling 128, a proximal lock tube 324 which may be similar to proximal lock tube 124, a distal lock tube 326 which may be similar to distal lock tube 126, a draw bar assembly 328 which may be similar to draw bar assembly 160, a draw bar knob 330 which may be similar to handle 164, a draw bar tip part 332, a draw bar 334

which may be similar to draw bar 156, a spring plunger 336, a ratchet 338, a lever link 340, a return link 342, and/or a release ring 344.

Referring to FIGS. 70-73, the ratchet mechanism 310 may include the lever 312, spring plunger 336, ratchet 338, lever link 340, and return link 342. The ratchet mechanism 310 may include the shuttle 322.

The lever 312 extends between a proximal end 868 and a distal end 870. The lever 312 may include a body 872 and an arm 874. The arm 874 may extend proximally obliquely away from an outer side 876 of the body 872. An opposite inner side 878 of the body may include a pair of outer arms 880, 882 forming an outer clevis, and a pair of inner arms 884, 886 forming an inner clevis. The inner arms 884, 886 may be located between the outer arms 880, 882. The inner arms 884, 886 may be shorter than the outer arms 880, 882, and may extend from the inner side 878 about one third as far as the outer arms. The arms 880, 882, 884, 886 may all be parallel as shown, and may be perpendicular to the inner side 878. A first hole 888 may extend through the arms 880, 882, 884, 886 near the inner side 878. A second hole 890 may extend through the outer arms 880, 882 near the free ends of the outer arms. The first and second holes 888, 890 may be parallel to the inner side 878. A third hole 892 may extend into the inner side 878 between the inner arms 884, 886. The third hole 892 may be parallel to the arms 880, 882, 884, 886.

The spring plunger 336 extends between a proximal end 892 and a distal end 894. The spring plunger 336 may be a generally cylindrical shaft with circular cross sections as shown. The spring plunger 336 may include, from proximal to distal, first and second outer portions 896, 898. The first outer portion 896 may extend distally most of the proximal-distal length of the spring plunger 336. The diameter of the second outer portion 898 may be less than the diameter of the first outer portion 896. There may be a tapered transition between the first and second outer portions 896, 898. The second outer portion 898 may include a rounded distal outer edge. A central longitudinal hole 900 may extend distally into the proximal end of the spring plunger 336. The hole 900 may extend distally almost as far as the first outer portion 896.

The ratchet 338 extends between a proximal end 902 and a distal end 904. The ratchet 338 may have a rectangular or square cross section as shown. A tooth 906 may protrude from an inner side 908 of the ratchet 338. The tooth 906 may be located along an inner proximal edge of the ratchet 338. A first hole 910 may extend through the ratchet 338 near the distal end 904 and may be parallel to the tooth 906. A second hole 912 may extend into the inner side 908 near the distal end 904 and may be perpendicular to the first hole 910. The second hole 912 may be between the first hole 910 and the distal end 904.

The lever link 340 extends between a proximal end 914 and a distal end 916. The lever link 340 may have a rectangular or square cross section as shown. A tooth 918 may protrude from an inner side 920 of the lever link 340. The tooth 918 may be located along an inner proximal edge of the lever link 340. A unilateral notch 922 may extend along the lever link 340 so as to intersect a proximal portion of the inner side 920 and an inner portion of the proximal end 916, so that the tooth 918 is thinner than the distal end 916. A first hole 924 may extend through the lever link 340 near the distal end 916 and may be parallel to the tooth 918. A second hole 926 may extend into an outer side 928 of the lever link 340 opposite the inner side 920 near the distal end

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916 and may be perpendicular to the first hole 924. The first hole 924 may be between the second hole 926 and the distal end 916.

The return link 342 extends between a proximal end 930 and a distal end 932. The return link 342 may have a rectangular or square cross section as shown. A foot 934 may protrude from a first side 936 of the return link 342. The foot 934 may be located at the proximal end 930 of the return link 342. The foot 934 may extend perpendicular to the proximal-distal length of the return link 342. A unilateral notch 938 may extend along the return link 342 so as to intersect a proximal portion of an inner side 940 of the return link and an inner portion of the proximal end 930. A first hole 942 may extend distally into the proximal end 930 and the foot 934. A second hole 944 may extend through the return link 342 near the distal end 932 and may be parallel to the foot 934.

The shuttle 322 extends between a proximal end 946 and a distal end 948. The shuttle 322 may include, from proximal to distal, first, second, third, and fourth outer portions 950, 952, 954, 956. The first outer portion 950 may have a circular cross section and rounded proximal and distal outer circumferential edges. The proximal-distal length of the first outer portion 950 may be less than its diameter. The second outer portion 952 may have a circular cross section with a diameter that is less than the diameter of the first outer portion 950. The second outer portion 952 may form a square (90 degree) inner corner with the first outer portion 950. The proximal-distal length of the second outer portion 952 may be the same as or similar to the length of the first outer portion 950.

The third outer portion 954 may have a circular cross section with a diameter that is less than the diameter of the second outer portion 952. The third outer portion 952 may form a square inner corner with the second outer portion 952. The third outer portion 954 may extend over most of the proximal-distal length of the shuttle 322. The fourth outer portion 956 may have a circular cross section that tapers down in diameter toward the distal end 948. The fourth outer portion 956 may be conical. A unilateral longitudinal groove 958 may extend along the second and third outer portions 952, 954 and may run out in the fourth outer portion 956. The groove 958 may have a rectangular or square cross section with a rounded proximal end at the proximal end of the second outer portion 952. A first series of transverse grooves 960 may extend across the third outer portion 954 and may be located 90 degrees from the longitudinal groove 958. There may be ten grooves 960 as shown, or another number. The size and shape of the grooves 960 may be complementary to the tooth 918 of the lever link 340. A second series of transverse grooves 962 may extend across the third outer portion 954 and may be located 90 degrees from the longitudinal groove 958 and 180 degrees from the first series of transverse grooves 960. There may be 54 grooves 962 as shown, or another number. The size and shape of the grooves 960 may be complementary to the tooth 906 of the ratchet 338. Indicia 964 may be included along the third outer portion 954 opposite the longitudinal groove 958. The indicia 964 may include transverse lines and numerals as shown, or other markings or engravings. A unilateral shaped slot 966 may extend proximally into the distal end of the third outer portion 954 and may run out in the fourth outer portion 956. The proximal end of the slot 966 may make a 90 degree bend toward the longitudinal groove 958. The slot 966 may have a rectangular or square cross section. The slot 966 and other similarly shaped slots disclosed herein may be referred to as a bayonet slot.

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The shuttle 322 may include a central longitudinal hole 968 that extends through the shuttle between the proximal end 946 and the distal end 948. The hole 968 may include, from proximal to distal, first, second, third, and fourth inner portions 970, 972, 974, 976, each with a circular cross section. The first inner portion 970 may be a shallow countersink around the proximal end of the hole 968. The second inner portion 972 may have internal threads. The minor diameter of the threads may be the same as or similar to the minor diameter of the first inner portion 970. The third inner portion 974 may be smooth, with a diameter that is less than the minor diameter of the threads of the second inner portion 972. See FIG. 66. The distal end of the third inner portion 974 may have a square (90 degree) inner corner. The fourth inner portion 976 may be smooth, with a diameter that is less than the diameter of the third inner portion 974.

Referring to FIGS. 70, 71, 74, and 75, the shaft assembly 304 may include the inserter shaft 306 and distal tip part 308.

The inserter shaft 306 extends between a proximal end 978 and a distal end 980. The inserter shaft 306 may be a tubular part with circular outer and inner cross-sectional shapes. The inserter shaft 306 may include a pair of shaped slots 982, 984 which each include an approximately 90 degree bend. The slots 982, 984 may extend longitudinally into the proximal end 978 before making the bend. Each slot 982, 984 may extend through one wall of the inserter shaft 306. The inserter shaft 306 may include bilateral transverse grooves 986, 988 which may be located 90 degrees from, and distal to, the slots 982, 984. The inserter shaft 306 may include one or more openings 990. Three openings 990 are shown spaced along the inserter shaft 306 distal to the transverse grooves 986, 988. Each opening 990 may extend through one wall of the inserter shaft 306, or through both walls as shown. The inserter shaft 306 may include a central longitudinal hole 992 that extends through the inserter shaft between the proximal end 978 and the distal end 980. The hole 992 may include a proximal countersink 994 and/or a distal counterbore 996.

The distal tip part 308 extends between a proximal end 1200 and a distal end 1202. The proximal end 1200 may be circular in a proximal view. The distal end 1202 may be rectangular in a distal view. The distal end 1202 may have rounded corners in the distal view. The distal tip part 308 may include, from proximal to distal, first, second, and third outer portions 1204, 1206, 1208. The first outer portion 1204 may have a circular cross section and may extend distally about one fourth of the proximal-distal length of the distal tip part 308. The second outer portion 1206 may have a circular cross section and may have a diameter that is greater than the diameter of the first outer portion 1204. The second outer portion 1206 may form a square (90 degree) inner corner with the first outer portion 1204. The third outer portion 1208 may include the rectangular distal end 1202, and may also include a curvaceous transition surface that blends with the second outer portion 1206. The distal tip part 308 may include a central longitudinal hole 1210 that extends through the distal tip part between the proximal end 1200 and the distal end 1202. The hole 1210 may include, from proximal to distal, first and second inner portions 1212, 1214. The first inner portion 1212 may have a circular cross section and may extend distally about two thirds to three fourths of the overall proximal-distal length of the distal tip part 308. The cross-sectional shape of the second inner portion 1214 may be rectangular or square. The rectangle or square may have rounded corners. The major dimension (length) of the rectangle may be greater than the diameter of the first inner portion 1212, and may be aligned with the

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major dimension (length) of the rectangle of the distal end **1202**. The minor dimension (width) of the rectangle may be greater than the diameter of the first inner portion **1212**. The second inner portion **1214** may have a flat proximal end.

The shaft assembly **304** may be assembled by inserting the first outer portion **1204** of the distal tip part **308** into the counterbore **996** of the inserter shaft **306** and aligning the length of the rectangle of the third outer portion **1208** to extend over the transverse grooves **986**, **988**. The inserter shaft **306** and distal tip part **308** may be permanently connected, for example by welding. Alternatively, the inserter shaft **306** and distal tip part **308** may be combined into a single part. The inner diameters of the hole **992** of the inserter shaft **306** and the first inner portion **1212** of the hole **1210** of the distal tip part **308** may be the same or similar, and together they may form a shaft assembly bore **1434**. See FIGS. **70** and **71**.

Referring to FIGS. **70**, **71**, **76**, and **77**, the lock tube assembly **318** may include the lock tube coupling **320**, proximal lock tube **324**, and distal lock tube **326**.

The lock tube coupling **320** extends between a proximal end **1216** and a distal end **1218**. The lock tube coupling **320** may be a tubular part. The lock tube coupling **320** may include a proximal body **1220** which may have a circular cross-sectional outer shape, and may extend distally about half or three-fifth of the overall proximal-distal length of the lock tube coupling **320**. A first lateral post **1222** may extend radially outwardly from the distal end of the body **1220**. A unilateral tab **1224** may protrude distally from the body **1220**, and may be located in line with the first lateral post **1222**. The outer surface of the tab **1224** may be radially inset relative to the outer diameter of the body **1220**. A second lateral post **1226** may protrude radially outwardly from the tab **1224**. The second lateral post **1226** may protrude radially beyond the outer diameter of the body **1220** as shown. The second lateral post **1226** may be distal to and in line with the first lateral post **1222**. First and second holes **1228**, **1230** may extend through one wall of the body **1220** on either side of a location opposite the first lateral post **1222**, or in other words, each hole may be located about 135 degrees away from the first lateral post **1222**. The lock tube coupling **320** may include a central longitudinal hole **1232** that extends through the lock tube coupling between the proximal end **1216** and the distal end **1218**. The hole **1232** may include, from proximal to distal, first, second, and third inner portions **1234**, **1236**, **1238**. The first inner portion **1234** may have a circular cross section and may extend distally about a third of the overall proximal-distal length of the lock tube coupling **320**, or about half of the length of the body **1220**. The second inner portion **1236** may have a circular cross section with a diameter less than the diameter of the first inner portion **1234**. The second inner portion **1236** may extend distally only a short distance as a complete diameter, but a section of it may extend through the distal end **1218** to form the inner side of the tab **1224**. The third inner portion **1238** may have a non-circular cross section that is a C-shaped alcove in the distal wall of the body **1220** opposite the tab **1224**. The alcove may extend around more than 180 degrees of arc.

The proximal lock tube **324** extends between a proximal end **1240** and a distal end **1242**. The proximal lock tube **324** may be a tubular part with circular outer and inner cross-sectional shapes and constant outer and inner diameters.

The distal lock tube **326** extends between a proximal end **1244** and a distal end **1246**. The distal lock tube **326** may be a tubular part with circular outer and inner cross-sectional shapes. The proximal end **1244** may include a unilateral

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notch **1248** that extends distally into one side of the distal lock tube **326**. The notch **1248** may extend around less than 180 degrees of arc. The notch **1248** may include, from proximal to distal, first and second portions **1250**, **1252**. The first portion **1250** may have a flat distal end with square corners. The second portion **1252** may be narrower than the first portion **1250**. The second portion **1252** may have a flat distal end with round corners. The distal end **1246** may include a pair of shaped slots **1254**, **1256** which each include an approximately 90° bend. The slots **1254**, **1256** may extend longitudinally into the distal end **1246** before making the approximately 90° bend. Each slot **1254**, **1256** may extend through one wall of the distal lock tube **326**. The distal lock tube **326** may include a central longitudinal hole **1258** that extends through the distal lock tube between the proximal end **612** and the distal end **614**. The hole **1258** may include, from proximal to distal, first and second inner portions **1260**, **1262**. The first inner portion **1260** may extend distally for most of the proximal-distal length of the distal lock tube **326**, and may have a flat distal end. The diameter of the second inner portion **1262** may be less than the diameter of the first inner portion **1260**.

The lock tube assembly **318** may be assembled by inserting the distal end **1242** of the proximal lock tube **324** in the first inner portion **1234** of the lock tube coupling **320**, and inserting the proximal end **1244** of the distal lock tube **326** in the third inner portion **1238** of the lock tube coupling **320** so that the tab **1224** is received in the second portion **1252** of the notch **1248**. The lock tube coupling **320**, proximal lock tube **324**, and distal lock tube **326** may be permanently connected, for example by welding. Alternatively, the lock tube coupling **320**, proximal lock tube **324**, and/or distal lock tube **326** may be combined into a single part. The inner diameters of the second inner portion **1262** of the hole **1232** of the lock tube coupling **320**, proximal lock tube **324**, and first inner portion **1260** of the hole **1258** of the distal lock tube **326** may all be the same or similar, and together they may form a longitudinal instrument bore **1260**. See FIGS. **64**, **70**, and **71**. The distal lock tube **326** and the body **1220** of the lock tube coupling **320** may fit within the bore **1434** of the shaft assembly **304**. The post **1226** of the lock tube coupling **320** may fit within either slot **982**, **984** of the inserter shaft **306** of the shaft assembly **304**.

Referring to FIGS. **70**, **71**, **78**, and **79**, the draw bar assembly **328** may include the draw bar knob **330**, draw bar tip part **332**, and draw bar **334**.

The draw bar knob **330** extends between a proximal end **1262** and a distal end **1264**.

The draw bar handle **330** extends between a proximal end **1262** and a distal end **1264**. The draw bar handle **330** may include, from proximal to distal, a knob body **1266** and a shaft **1268**. The knob body **1266** may be generally disc-shaped with an outer diameter that is greater than its proximal-distal length. The knob body **1266** may include one or more grip features **1270** such as the six rim scallops shown. The outer diameter of the shaft **1268** may be less than the outer diameter of the knob body **1266** and the proximal-distal length of the shaft may be greater than the proximal-distal length of the knob body **1266**. The shaft **1268** may include a flange **1272** or annulus that extends circumferentially around a middle portion of the shaft. The flange **1272** may have square proximal and distal outer circumferential edges. The shaft **1268** may include one or more holes **1274** that extend transversely into, or through, the shaft; two through holes **1274** are shown between the distal side of the knob body **1266** and the proximal side of the flange **1272**. The two through holes **1274** are perpen-

dicular to each other. A distal portion of the shaft **1268** may include external threads **1276**. The outer diameter of the flange **1272** may be greater than the major diameter of the threads **1276**. The minor diameter of the threads **1276** may be greater than the outer diameter of the remaining smooth portions of the shaft **1268**. The draw bar handle **330** may include a central longitudinal hole **1278** that extends through the draw bar handle between the proximal end **1262** and the distal end **1264**. The hole **1278** may include, from proximal to distal, first, second, and third inner portions **1280**, **1282**, **1284**. The first inner portion **1280** may have a circular cross section and may extend distally about half, or most, of the proximal-distal length of the knob body **1266**. The first inner portion **1280** may have a flat distal end. A circumferential fillet radius may be present around the proximal inner edge of the first inner portion **1280**. A circumferential fillet radius may be present in the distal inner corner of the first inner portion **1280**. The second inner portion **1282** may have a circular cross section and may extend distally between the distal side of the knob body **1266** and the holes **1274**. The diameter of the second inner portion **1282** may be less than the diameter of the first inner portion **1280**. The third inner portion **1284** may have a circular cross section and may extend distally through the distal end of the **1264** of the draw bar handle **330**. The diameter of the third inner portion **1284** may be less than the diameter of the first inner portion **1280** and greater than the diameter of the second inner portion **1282**. The third inner portion **1284** may have a flat proximal end. The hole(s) **1274** may extend into a proximal end of the third inner portion **1284**.

The draw bar tip part **332** extends between a proximal end **1286** and a distal end **1288**. The draw bar tip part **332** may include, from proximal to distal, first, second, third, and fourth outer portions **1290**, **1292**, **1294**, **1296**. The first outer portion **1290** may have a circular cross section and may extend distally about one third to half of the overall proximal-distal length of the draw bar tip part **332**. An outer proximal edge of the first outer portion **1290** may include a circumferential chamfer. The second outer portion **1292** may have a circular cross section and may extend distally only a short distance. The diameter of the second outer portion **1292** may be greater than the diameter of the first outer portion **1290**. The second outer portion **1292** may be referred to as a flange or annulus around a middle portion of the draw bar tip part **332**. The third outer portion **1294** may have a circular cross section. The diameter of the third outer portion **1294** may be greater than the diameter of the first outer portion **1290** and less than the diameter of the second outer portion **1292**. The fourth outer portion **1296** may be a torque fitting, such as the hexagonal key shown. An outer distal edge of the fourth outer portion **1296** may include a circumferential chamfer. The draw bar tip part **332** may include a central longitudinal hole **1298** that extends through the draw bar tip part between the proximal end **1286** and the distal end **1288**. The hole **1298** may include, from proximal to distal, first, second, and third inner portions **1300**, **1302**, **1304**. The first inner portion **1300** may have a circular cross section. The distal end of the first inner portion **1300** may be proximal to the second outer portion **1292**. The second inner portion **1302** may include internal threads. The major diameter of the threads may be less than the diameter of the first inner portion **1300**. The third inner portion **1304** may have a circular cross section. The diameter of the third inner portion **1304** may be less than the diameter of the first inner portion **1300** and greater than the major diameter of the threads of the second inner portion **1302**.

The draw bar **334** extends between a proximal end **1306** and a distal end **1308**. The draw bar **334** may be a cylindrical shaft as shown. The draw bar **334** may include a central longitudinal hole **1310** that extends proximally into the distal end **1308** of the draw bar **334**. The hole **1310** may include, from proximal to distal, first and second inner portions **1312**, **1314**. The first inner portion **1312** may have a circular cross section. The second inner portion **1314** may have a circular cross section with a diameter greater than the diameter of the first inner portion **1312**.

The draw bar assembly **328** may be assembled by inserting the proximal end **1306** of the draw bar **334** in the second inner portion **1282** of the hole **1278** of the draw bar handle **330** so that the proximal end **1306** is at the same or similar level with the distal end of the first inner portion **1280**, and inserting the first outer portion **1290** of the draw bar tip part **332** in the second inner portion **1314** of the hole **1310** of the draw bar **334**. A circumferential gap **1316** may exist between the outer surface of the draw bar **334** and the inner surface of the third inner portion **1284**. See FIG. 66. The draw bar knob **330**, draw bar tip part **332**, and draw bar **334** may be permanently connected, for example by welding. Alternatively, the draw bar knob **330**, draw bar tip part **332**, and/or draw bar **334** may be combined into a single part. The draw bar tip part **332** and draw bar **334** may fit within the bore **1260** of the lock tube assembly **318**. The proximal end **1240** of the proximal lock tube **324** may fit in the gap **1316**.

Referring to FIGS. 66, 70, 71, and 80-85, the inserter handle **302** extends between a proximal end **1318** and a distal end **1320**. The inserter handle **302** may include, from proximal to distal, first, second, and third outer portions **1322**, **1324**, **1326**. The first outer portion **1322** may be cylindrical with a circumferential groove **1328** centrally located along the proximal-distal length of the first outer portion **1322**. The groove **1328** may have square inner corners. The second outer portion **1324** may extend distally over most of the overall proximal-distal length of the inserter handle **302**. The second outer portion **1324** may be generally hourglass shaped or otherwise contoured to be comfortably and ergonomically gripped by a user's hand. The third outer portion **1326** may extend over about one fourth of the overall proximal-distal length of the inserter handle **302**. The third outer portion **1326** may include flat front and back surfaces **1330**, **1332** which may extend proximally from the distal end **1320** over the third outer portion **1326** and most of the proximal-distal length of the second outer portion **1324**. The third outer portion **1326** may include a rectangular or square boss **1374** which protrudes from a right side **1376** of the third outer portion. The boss **1374** may extend from a middle portion of the proximal-distal length of the third outer portion **1326**. A left side **1378** of the third outer portion **1326** may be a portion of a cylinder.

The inserter handle **302** may include a unilateral recess **1334** that extends distally into the second outer portion. The recess **1334** may be on the same side as the front surface **1330**. The recess **1334** may be arcuate, and may extend around less than 180 degrees of arc. The recess **1334** may include bilateral ridges **1336**, **1338** as seen best in FIG. 82. The ridges **1336**, **1338** may extend proximally a short distance from the distal end of the recess **1334**. A pocket **1340** may be located in the proximal region of the front surface **1330**. A window **1342** may be located within the pocket **1340**. The window **1342** may extend through the front side of the inserter handle **302**. The pocket **1340** may include indicia **1344** such as the transverse lines shown bracketing the window **1342**. A hole **1380** may extend

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through the inserter handle 302 between the front and back sides and may be located in the proximal right portion of the third outer portion 1326.

A longitudinal slot 1346 may extend into a right side 1348 of the second outer portion 1324 of the inserter handle 302. The slot 1346 may be rectangular with rounded corners. The proximal end of the slot 1346 may be proximal to the pocket 1340 and the distal end of the slot may be distal to the pocket. A distal portion 1350 of the slot 1346 may be shallow, with a flat bottom, and a proximal portion 1352 of the slot may be deep. The distal portion 1350 may extend over the distal one third of the overall proximal-distal length of the slot 1346. The distal portion 1350 may include a hole 1354 in the bottom near the distal end. A transverse hole 1356 may extend through the inserter handle 302 between the front and back sides so as to intersect the side walls of the distal portion 1350 proximal to the hole 1354. A hole 1358 may extend into the right side 1348 of the second outer portion 1324 of the inserter handle 302 distal to the slot 1346. The hole 1358 may include internal threads and a counterbore to complement the stop screw 316. A transverse slot 1382 may extend into the right side 1376 of the third outer portion 1326. The slot 1382 may extend into a distal portion of the boss 1374. The slot 1382 may taper in the transverse direction, becoming narrower as the slot extends into the third outer portion 1326. See FIG. 82.

A longitudinal slot 1360 may extend into a left side 1362 of the second and third outer portions 1324, 1326 of the inserter handle 302. The proximal end of the slot 1360 may be at the same or similar level as the proximal end of the slot 1346 and the distal end of the slot may extend about half of the overall proximal-distal length of the third outer portion 1326. A proximal portion 1364 of the slot 1360 may be narrower than a distal portion 1366 of the slot. The proximal and distal portions 1364, 1366 may each be rectangular with rounded corners. The distal end of the proximal portion 1364 may be distal to the hole 1358 and proximal to the third outer portion 1326. A unilateral narrow ledge 1368 may extend along a proximal back side wall within the proximal portion 1364. A platform 1370 may extend between the side walls of the proximal and distal portions 1364, 1366. The proximal end of the platform 1370 may be distal to the hole 1358 and proximal to the distal end of the proximal portion 1364, and the distal end of the platform may be located about one third of the overall proximal-distal length of the third outer portion 1326, or about one half of the overall proximal-distal length of the distal portion 1366. The platform 1370 may be deeper within the inserter handle 302 than the ledge 1368. A longitudinal hole 1372 may extend proximally into the distal end 1320 of the inserter handle 302 and into the proximal end wall of the proximal portion 1364 of the slot 1360. See FIG. 66.

The inserter handle 302 may include a central longitudinal hole 1384 that extends through the inserter handle between the proximal end 1318 and the distal end 1320. The hole 1384 may include, from proximal to distal, first, second, and third inner portions 1386, 1388, 1389. Each of these inner portions may have a circular cross section. The first inner portion 1386 may extend distally to the proximal end of the platform 1370. The diameter of the second inner portion 1388 may be less than the diameter of the first inner portion 1386. The second inner portion 1388 may extend distally to intersect the slot 1382. The diameter of the third inner portion 1389 may be less than the diameter of the second inner portion 1388. The window 1342, proximal portion 1352 of the slot 1346, hole 1358, and proximal portion 1364 of the slot 1360 may open into the first inner portion 1386.

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The distal portion 1366 of the slot 1360, distal to the platform 1370, and the slot 1382 may open into the second inner portion 1388. Bilateral holes 1390, 1392 may extend through the inserter handle 302 between the right and left sides near the distal end 1320. The holes 1390, 1392 may intersect the third inner portion 1389.

The lock lever 314 extends between a proximal side 1394 and a distal side 1395. The lock lever 314 may include a ring 1398, a stem 1400, and a cap 1402. The stem 1400 may extend from one side of the ring 1398. The cap 1402 may be an enlargement at the free end of the stem 1400. A central longitudinal hole 1404 may extend through the lock lever 314 between the proximal and distal sides 1394, 1396. A keyway 1406 may also extend through the lock lever 314 between the proximal and distal sides 1394, 1396 along an inner wall of the hole 1404 at the base of the stem 1400. A hole 1408 may extend into the stem 1400 near the ring 1398.

The stop screw 316 may be a dog point screw that is complementary to the hole 1358. The stop screw 316 may have a pan head or cheese head.

The release ring 344 extends between a proximal end 1410 and a distal end 1412. The release ring 344 may include one or more friction features 1414, such as the twelve rim scallops shown. The release ring 344 may include a unilateral tab 1416 that may extend distally. An inner side of the tab 1416 may include two longitudinal grooves 1418, 1420 that extend through the distal end 1412. A central longitudinal hole 1422 may extend through the release ring 344 between the proximal and distal sides 1410, 1412. The hole 1422 may include, from proximal to distal, first and second inner portions 1424, 1426. The first inner portion 1424 may have a generally circular cross section with a unilateral key 1428 extending radially into the hole 1422 at the proximal end 1410. The key 1428 may be generally opposite the tab 1416, but may be offset several degrees from being exactly opposite the tab. For example, the key 1428 may be 155 degrees away from the tab 1416. The second inner portion 1426 may have a circular cross section with a diameter that is greater than the diameter of the first inner portion 1424. The proximal end of the second inner portion 1426 may be flat. The inner side of the tab 1416 may be radially offset outwardly from the second inner portion 1426. Bilateral transverse holes 1430, 1432 may extend through the release ring 344. The holes 1430, 1432 may intersect the second inner portion 1426.

The instrument 300 may be assembled by coupling the release ring 344, ratchet 338, stop screw 316, and spring plunger 336 to the inserter handle 302, coupling the lever link 340 and return link 342 to the lever 312, coupling the sub-assembly of the lever link 340, the return link 342, and the lever 312 to the inserter handle 302 (with spring plunger 336), coupling the shaft assembly 304 and the lock lever 314 to the inserter handle 302, inserting the lock tube assembly 318 into the shaft assembly 304, coupling the shuttle 322 to the inserter handle 302 (with stop screw 316 and release ring 344), inserting the draw bar assembly 328 into the lock tube assembly 318, and coupling the draw bar assembly 328 to the shuttle 322. These steps may be performed in any order.

Coupling the release ring 344 to the inserter handle 302 may include inserting the first outer portion 1322 of the inserter handle 302 into the second inner portion 1426 of the hole 1422 of the release ring 344 so that the tab 1416 is in the recess 1334 between the ridges 1336, 1338, aligning the holes 1430, 1432 with the groove 1328, and inserting pins into the holes 1430, 1432 and the groove 1328.

Coupling the ratchet 338 to the inserter handle 302 may include inserting a compression spring (not shown) into the

holes **912**, **1354**, inserting the ratchet **338** into the slot **1346** with the tooth **906** facing into the first inner portion **1386** of the hole **1384**, aligning the holes **1356**, **910**, and inserting a pin (not shown) through the holes **1356**, **910**. The spring may bias the ratchet **338** to protrude into the first inner portion **1386**.

Coupling the stop screw **316** to the inserter handle **302** may include inserting the dog point (tip) of the screw into the hole **1358**, engaging the complementary threads, and tightening the screw in the hole **1358**.

Coupling the spring plunger **336** to the inserter handle **302** may include inserting a compression spring (not shown) into the hole **900** of the spring plunger **336** and inserting the proximal end **892** of the spring plunger **336** into the proximal portion of the hole **1372** of the inserter handle **302** so that the second outer portion **898** protrudes distally into the proximal portion **1364** of the slot **1360**.

Coupling the lever link **340** and the return link **342** to the lever **312** may include positioning the distal end **916** of the lever link **340** between the inner arms **884**, **886** of the lever **312**, next to the arm **884**, with the hole **924** aligned with the hole **888** and the tooth **918** facing away from the arm **874**; positioning the distal end **932** of the return link **342** between the inner arms **884**, **886**, next to the arm **886**, with the hole **944** aligned with the hole **888** and the foot **934** facing towards the proximal end **914** of the lever link; and inserting a pin (not shown) through the holes **888**, **924**, **944** to hingedly connect the lever link **340** and the return link **342** to the lever **312**. The pin may only extend through the inner arms **884**, **886**, the lever link **340**, and the return link **342**. A compression spring (not shown) may be inserted in the holes **892**, **926** to bias the lever link **340** away from the lever **312**.

Coupling the sub-assembly of the lever link **340**, the return link **342**, and the lever **312** to the inserter handle **302** (with spring plunger **336**) may include inserting the inner arms **884**, **886** of the lever **312** into the distal portion **1366** of the slot **1360** of the inserter handle **302** so that the inserter handle **302** is between the outer arms **880**, **882** and the lever link **340** and the return link **342** extend into the proximal portion **1364** of the slot **1360**, inserting the second outer portion **898** of the spring plunger **336** into the hole **942** of the return link **342**, and inserting a pin (not shown) into the holes **890**, **1380**. The spring plunger **336** and associated compression spring may bias the lever **312** away from the inserter handle **302**. The ledge **1368** fits in the notch **922** to control the depth to which the tooth **918** extends into the first inner portion **1386** of the hole **1384**.

Coupling the shaft assembly **304** and the lock lever **314** to the inserter handle **302** may include inserting the ring **1398** of the lock lever **314** into the slot **1382** of the inserter handle **302** so that the stem **1400** and cap **1402** extend laterally outside the slot **1382** and the hole **1408** faces the front surface **1330**, inserting the proximal end **978** of the inserter shaft **306** into the third inner portion **1389** of the hole **1384** in the distal end **1320** of the inserter handle **302** and into the hole **1404** of the lock lever **314**, aligning the grooves **986**, **988** with the holes **1390**, **1392**, and inserting pins (not shown) into the holes **1390**, **1392** and grooves **986**, **988**. A compression spring (not shown) may be inserted in the hole **1408** to bear against a front wall of the slot **1382**. The spring may bias the lock lever **314** toward the back surface **1332**.

Inserting the lock tube assembly **318** into the shaft assembly **304** may include inserting the distal end **1246** of the distal lock tube **326** into the hole **992** in the proximal end **978** of the inserter shaft **306**, advancing the lock tube assembly **318** into the shaft assembly **304** until the body **1220** of the lock tube coupling **320** enters the hole **992** and

the post **1226** enters the slot **982** or the slot **984**, and twisting the lock tube assembly **318** relative to the shaft assembly **304** to move the post **1226** past the slot bend. When the shaft assembly **304**, lock lever **314**, inserter handle **302**, and lock tube assembly **318** are operatively assembled, the body **1220** of the lock tube coupling **320** may fit in the hole **1404** of the lock lever **314** and the posts **1222**, **1226** of the lock tube coupling **320** may fit in the keyway **1406** of the lock lever **314**.

Coupling the shuttle **322** to the inserter handle **302** (with stop screw **316** and release ring **344**) may include inserting the proximal end **1240** of the proximal lock tube **324** in the fourth inner portion **976** of the hole **968** at the distal end **948** of the shuttle **322**, inserting the key **1428** of the release ring **344** in the groove **958** at the distal end **948** of the shuttle **322**, sliding the shuttle **322** into the first inner portion **1386** of the hole **1384** until the dog point of the stop screw **316** slides into the slot **966**, and twisting the release ring **344** and shuttle **322** relative to the inserter handle **302** to move the dog point past the slot bend.

When the instrument **300** is operatively assembled as described in the preceding steps, the proximal lock tube **324** is in the hole **968** of the shuttle **322**, the tooth **918** of the lever link may engage the grooves **960** of the shuttle **322**, the tooth **906** of the ratchet **338** may engage the grooves **962** of the shuttle **322**, and a portion of the indicia **964** of the shuttle **322** may be exposed through the window **1342** of the inserter handle **302**.

Inserting the draw bar assembly **328** into the lock tube assembly **318** may include inserting the distal end **1288** of the draw bar tip **332** into the inner diameter of the proximal end **1240** of the proximal lock tube **324** and advancing the draw bar assembly **328** into the lock tube assembly **318** until the proximal end **1240** enters the gap **1316**.

Coupling the draw bar assembly **328** to the shuttle **322** may include engaging the internal threads of the second inner portion **972** of the hole **968** of the shuttle **322** with the external threads **1276** of the draw bar handle **330** and tightening the threads until the flange **1272** abuts the proximal end **946** of the shuttle **322**.

The instrument **300** may be operated by rotating the release ring **344**, pivoting the lock lever **314**, and pivoting the lever **312**.

The release ring **344** may be coupled to the shuttle **322** by the key **1428** in the groove **958** so that the shuttle is free to slide axially relative to the release ring, and the shuttle is rotationally coupled to the release ring so that rotation of the release ring causes rotation of the shuttle. When the release ring **344** is rotated so that the groove **1420** engages the ridge **1338**, the slot **966** of the shuttle **322** may be free to slide axially relative to the stop screw **316** and the grooves **960**, **962** may rotate out of engagement with the teeth **918**, **906**, respectively. When the release ring **344** is rotated so that the groove **1418** engages the ridge **1336**, the slot **966** may capture the stop screw **316** past the slot bend. Thus the release ring **344** and the stop screw **316** cooperate to connect or disconnect the shuttle **322** and the inserter handle **302**.

The lock lever **314** may be coupled to the lock tube coupling **320** by the posts **1222**, **1226** in the keyway **1406** so that the lock tube coupling is free to slide axially relative to the lock lever, and the lock tube coupling is rotationally coupled to the lock lever so that rotation of the lock lever **314** causes rotation of the lock tube coupling. When the lock lever **314** is rotated toward the front surface **1330** of the inserter handle **302**, the posts **1222**, **1226** may be free to slide axially relative to the keyway **1406** and the slot **982** or **984** of the inserter shaft **306**. When the lock lever **314** is

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rotated toward the back surface 1332, the slot 982 or 984 may capture the post 1226 past the slot bend. The lock lever 314 may be biased toward the back surface 1332. The lock lever 314 thus connects or disconnects the lock tube assembly 318 and the shaft assembly 304.

The lever 312 and associated ratchet mechanism 310 may be coupled to the other parts of the instrument 300 so that actuation of the lever causes proximal axial translation of the shuttle 322 relative to the inserter handle 302. Pivoting the lever 312 away from the inserter handle 302 may move the lever link 340 distally relative to the shuttle 322. Pivoting the lever 312 toward the inserter handle 302 may move the lever link 340 proximally. When the tooth 918 is engaged in a groove 960, pivoting the lever 312 toward the inserter handle 302 also may cause linear translation of the shuttle 322 proximally along the longitudinal instrument axis 866 without rotation of the shuttle 322 about the axis 866. Thus the lever link 340 may be referred to as a drive link of the ratchet mechanism 310. The tooth 906 of the ratchet 338 may engage the grooves 962 to prevent the shuttle 322 from moving distally.

Referring to FIGS. 86-91, another insertion and expansion instrument is shown. Instrument 1500 may be deployed to insert and expand implant 1000, implant 1100 as shown, or another expandable intervertebral implant. Other tools such as the draw bar assembly 160, funnel 170, tamp assembly 180, and screwdriver assembly 190 disclosed herein may be used, or adapted for use, with instrument 1500 to deploy an implant, deposit graft material within the implant, and secure a locking fastener such as screw 654.

The instrument 1500 may differ from instruments 100 and 300 in at least the following attributes. The draw bar may be engaged by a slider button/lever that may prevent the draw bar from rotating, and may engage the rotating portion of the inserter handle to enable expansion. The slider button/lever may reduce the potential for incorrect user inputs, i.e., poka-yoke. The draw bar length may be adjustable so that one draw bar may be compatible with multiple implant lengths. The handle may be removable from the tube portion of the instrument 1500 to minimize the mass attached to the implant during bone graft insertion or other steps in the methods of use. Minimizing the instrument mass attached to the implant, and the associated moment arm through the center of gravity of the instrument mass, may minimize unintended motion or misalignment of the implant during steps in the methods of use. The funnel may be used to unthread the inserter shaft (implant retainer shaft) from the implant, for example, after installation of the lockout screw 654.

The instrument 1500 may include an inserter connector 1502 which may be similar to inserter connector 138 or distal tip part 308, a connector plate 1504 which may be similar to impaction cap 146, an inserter shaft 1506 which may be similar to inserter shaft 136 or 306, an implant retainer shaft 1508 which may function similar to lock tube assembly 118 or 318, a connector button 1510, a thumb wheel 1514, a wheel axle 1516, an inserter handle 1518, a knob strike plate 1520, a spline insert 1522, a rotation lock ring 1524, a cam lever 1526, a slider button 1528, a threaded slider 1530, a top knob part 1532, a bottom knob part 1534, and a draw bar assembly 1536 which may be similar to draw bar assembly 160 or 328. The instrument 1500 may also include various pins, screws and/or bolts, springs, ball detents, and the like. These parts may be basic off-the-shelf components, and they are generally not shown in this

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disclosure. The instrument 1500 extends from a proximal end 1538 to a distal end 1540 along a longitudinal instrument axis 1542.

Referring to FIGS. 92 and 93, the inserter connector 1502 extends between a proximal end 1544 and a distal end 1546. The proximal end 1544 may be circular in a proximal view. The distal end 1546 may be rectangular in a distal view. The distal end 1546 may have rounded corners in the distal view. The inserter connector 1502 may include, from proximal to distal, first, second, and third outer portions 1548, 1550, 1552. The first outer portion 1548 may have a circular cross section and may extend distally about one fourth to one third of the proximal-distal length of the inserter connector 1502. The second outer portion 1550 may be a curvaceous transition surface that blends between the first and third outer portions 1548, 1552. The third outer portion 1552 may include the rectangular distal end 1546. An outer distal edge of the rectangle may be chamfered. The inserter connector 1502 may include a central longitudinal hole 1554 that extends through the distal tip part between the proximal end 1544 and the distal end 1546. The hole 1554 may include, from proximal to distal, first, second, and third inner portions 1556, 1558, 1560. The first inner portion 1556 may have a circular cross section and may extend distally about one fourth to one third of the overall proximal-distal length of the inserter connector 1502. The second inner portion 1558 may have a circular cross section with a diameter that is less than the diameter of the first inner portion 1556. The cross sectional shape of the third inner portion 1560 may be the superposition of a circle and a rectangle. The diameter of the circle may be less than the diameter of the first inner portion 1556 and the same as or similar to the diameter of the second inner portion 1558. The major dimension (length) of the rectangle may be greater than the diameter of the first inner portion 1556, and may be aligned with the major dimension (length) of the rectangle of the third outer portion 1552. The minor dimension (width) of the rectangle may be less than the diameter of the circle, and may be less than the diameter of the second inner portion 1558. The rectangle may have rounded corners. The third inner portion 1560 may have a flat proximal end.

Referring to FIGS. 94 and 95, the connector plate 1504 extends between a proximal side 1562 and a distal side 1564. The connector plate 1504 may be circular in a proximal or distal view. The connector plate 1504 may include, from proximal to distal, first and second outer portions 1566, 1568. The first outer portion 1566 may taper inwardly from proximal to distal, and may be conical.

The first outer portion 1566 may have a flat distal end. The second outer portion 1568 may be cylindrical, with a diameter that is less than the minor diameter of the first outer portion 1566. Bilateral holes 1570, 1572 may extend through the connector plate 1504 between the proximal and distal sides 1562, 1564. A central longitudinal hole 1574 may extend through the connector plate 1504 between the proximal and distal sides 1562, 1564. The hole may include, from proximal to distal, first, second, third, fourth, and fifth inner portions 1576, 1578, 1580, 1582, 1584. Each inner portion may have a circular cross section. The first inner portion 1576 may have a short proximal-distal length and a circumferential chamfer around an inner proximal edge. The second inner portion 1578 may have a diameter that is greater than the diameter of the first inner portion 1576. The third inner portion 1580 may have a diameter that is the same as or similar to the diameter of the first inner portion 1576, and less than the diameter of the second inner portion 1578. There may be a tapered transition between the second

and third inner portions **1578**, **1580**. The distal end of the third inner portion **1580** may be flat and may be proximal to the distal end of the first outer portion **1566**. The fourth inner portion **1582** may have a diameter that is less than the diameter of the first inner portion **1576**. The fifth inner portion **1584** may have a diameter that is less than the diameter of the first inner portion **1576** and greater than the diameter of the fourth inner portion **1582**.

Referring to FIGS. **96** and **97**, the inserter shaft **1506** extends between a proximal end **1586** and a distal end **1588**. The inserter shaft **1506** may be a tubular part with circular outer and inner cross-sectional shapes. The inserter shaft **1506** may include, from proximal to distal, first, second, and third outer portions **1590**, **1592**, **1594**. The first outer portion **1590** may have a short proximal-distal length. The outer diameter of the second outer portion **1592** may be greater than the outer diameter of the first outer portion **1590**. The second outer portion **1592** may occupy most of the overall proximal-distal length of the inserter shaft **1506**. The second outer portion **1592** may include one or more openings **1596**. Four openings **1596** are shown evenly spaced along the length of the second outer portion **1592**. Each opening **1596** may extend through one wall of the second outer portion **1592**, or through both walls as shown. The outer diameter of the third outer portion **1594** may be the same as or similar to the outer diameter of the first outer portion **1590** and less than the outer diameter of the second outer portion **1592**. The inserter shaft **1506** may include a central longitudinal hole **1598** that extends through the inserter shaft between the proximal end **1586** and the distal end **1588**.

The inserter connector **1502**, connector plate **1504**, and inserter shaft **1506** may be coupled together in a shaft assembly **1600** like shaft assembly **116** or **304**. See FIGS. **86** and **87**. The shaft assembly **1600** may also include two pins (not shown) that protrude proximally from the holes **1570**, **1572** in the connector plate **1504**. The shaft assembly **1600** may be assembled by inserting the first outer portion **1590** of the inserter shaft **1506** in the fourth inner portion **1582** of the connector plate **1504** and inserting the third outer portion **1594** in the first inner portion **1556** of the inserter connector **1502**. The holes **1570**, **1572** may be aligned adjacent to the long sides of the rectangle of the third outer portion **1552**. The shaft assembly **1600** may be a weldment. The diameters of the second inner portion **1558** of the inserter connector **1502**, the fourth inner portion **1582** of the connector plate **1504**, and the hole **1598** of the inserter shaft **1506** may be the same or similar, and together, they may form a shaft assembly bore.

Referring to FIGS. **98** and **99**, the implant retainer shaft **1508** extends between a proximal end **1602** and a distal end **1604**. The implant retainer shaft **1508** may include, from proximal to distal, first, second, and third outer portions **1606**, **1608**, **1610**. The first outer portion **1606** may be an elongated torque fitting, such as the hexagonal key shown. The second outer portion **1608** may extend over most of the proximal-distal length of the implant retainer shaft **1508**, and may have a circular cross section with a diameter that is less than the minor diameter of the first outer portion **1606**. The third outer portion **1610** may include external threads. The major diameter of the threads may be the same as or similar to the diameter of the second outer portion **1608**. A central longitudinal hole **1612** may extend through the implant retainer shaft **1508** between the proximal and distal ends **1602**, **1604**.

Referring to FIGS. **100** and **101**, the connector button **1510** extends between a proximal end **1614** and a distal end **1616**. The connector button **1510** may have a flat inner side

1618 and an opposite outer side **1620**. A hole **1622** may extend into the inner side **1618**. A transverse groove **1624** may extend across the outer side **1620** near the proximal end **1614** so as to form a corresponding ledge **1626** along the outer proximal edge of the connector button **1510**. Bilateral thin ears **1628**, **1630** may be located at either end of the ledge **1626**, forming the inner proximal corners of the connector button **1510**. A dimple **1632** may be present on the outer side **1620**. Bilateral slots **1634**, **1636** may extend through the connector button **1510** parallel to the inner side **1618** and through the ledge **1626**. The slots **1634**, **1636** may intersect the groove **1624**. A second connector button **1512** is shown. The connector button **1512** may be a mirror image of the connector button **1510**.

Referring to FIGS. **102** and **103**, the thumb wheel **1514** extends between a proximal side **1638** and a distal side **1640**. The thumb wheel **1514** may be a generally disc- or ring-shaped part with one or more friction features **1642** arranged around its outer perimeter, such as the ten scallops shown. A central longitudinal hole **1644** may extend through the thumb wheel **1514** between the proximal and distal sides **1638**, **1640**. The hole **1644** may include, from proximal to distal, first and second inner portions **1646**, **1648**. The first inner portion **1646** may have a circular cross section and may extend distally about two fifths to half of the overall proximal-distal length of the thumb wheel **1514**. The second inner portion **1648** may be a torque fitting such as the hexagonal hole shown. The major dimension of the second inner portion **1648** may be less than the diameter of the first inner portion **1646**.

Referring to FIGS. **104** and **105**, the wheel axle **1516** extends between a proximal end **1650** and a distal end **1652**. The wheel axle **1516** may be a funnel shaped part with a wide proximal mouth **1654** and a narrow distal stem **1656** or shaft. The mouth **1654** may taper towards the stem **1656**. A central longitudinal hole **1658** may extend through the wheel axle **1516** between the proximal and distal ends **1650**, **1652**. The hole **1658** may include, from proximal to distal, first, second, and third inner portions **1660**, **1662**, **1664**. The first inner portion **1660** may have a round cross sectional shape and may correspond to the outer shape of the mouth **1654**. The second inner portion **1662** may have a circular cross sectional shape and may extend distally into the stem **1656**. The third inner portion **1664** may have a circular cross section with a diameter that is greater than the diameter of the second inner portion **1662**.

Referring to FIGS. **106-108**, the inserter handle **1518** extends between a proximal end **1668** and a distal end **1670**. The inserter handle **1518** may include, from proximal to distal, first, second, third, and fourth outer portions **1672**, **1674**, **1676**, **1678**. The first outer portion **1672** may have a circular cross section and a short proximal-distal length. The second outer portion **1674** may have a circular cross section with a diameter that is less than the diameter of the first outer portion **1672**. The third outer portion **1676** may be generally hourglass shaped or otherwise contoured to be comfortably and ergonomically gripped by a user's hand. The proximal diameter of the third outer portion **1676** may be greater than the diameter of the first outer portion **1672**, so that the second outer portion **1674** forms a circumferential groove between the first and third outer portions **1672**, **1676**. The fourth outer portion **1678** may have a circular cross section with a diameter that is less than the diameter of the second outer portion **1674** and the distal diameter of the third outer portion **1676**. A transverse hole **1680** may extend through the inserter handle **1518** in the third outer portion **1676** along a first direction, which will be referred to as a front-back

direction. The hole **1680** may be rectangular or square and may have rounded corners. A pocket **1682** may extend transversely into the inserter handle **1518** along a second direction, which will be referred to as a right-left direction, which is 90 degrees from the first direction. The pocket **1682** may extend proximally into the third outer portion **1674** and distally into the fourth outer portion **1678**. The pocket may be D-shaped, with a flat distal side and a curved proximal side. A second pocket **1684** may extend into the inserter handle **1518** opposite the pocket **1682**, and may be a mirror image of the pocket **1682**. A central longitudinal hole **1686** may extend through the inserter handle **1518** between the proximal and distal ends **1668**, **1670**. The hole **1686** may include, from proximal to distal, first, second, third, and fourth inner portions **1688**, **1690**, **1692**, **1694**. The first inner portion **1688** may have a cross section that includes a circular portion that is concentric with the second, third, and fourth inner portions **1690**, **1692**, **1694**, and an alcove portion **1696** that extends radially away from the circular portion toward a first side of the inserter handle **1518**, which will be referred to as a front side. The distal end of the first inner portion **1688** may be flat. A hole **1698** may extend distally within the alcove portion **1696**, and may be centered within the alcove portion. The hole **1698** may include a counterbore **1700**. The second inner portion **1690** may taper inwardly as it extends distally. The second inner portion **1690** may be conical. The major diameter of the second inner portion may be less than the diameter of the circular portion of the first inner portion **1688**. The third inner portion **1692** may have a circular cross section and may extend distally to the hole **1680**. The fourth inner portion **1694** may have a circular cross section with a diameter that is the same as or similar to the diameter of the third inner portion **1692**. Bilateral holes **1702**, **1704** may extend proximally into the distal end **1670** of the inserter handle **1518**. The holes **1702**, **1704** may be located toward the front and back. Bilateral holes **1706**, **1708** may extend proximally into the distal end **1670**, and into the proximal wall of the pocket **1682**. Bilateral holes **1710**, **1712** may extend proximally into the distal end **1670**, and into the proximal wall of the pocket **1684**.

Referring to FIGS. **109** and **110**, the knob strike plate **1520** extends between a proximal end **1714** and a distal end **1716**. The knob strike plate **1520** may have flat left and right sides **1718**, **1720** and an hourglass shape in a left or right view. The knob strike plate **1520** may include one or more openings **1722**. Two openings **1722** are shown extending along a front to back direction. Each opening **1722** may extend through one wall of the knob strike plate **1520**, or through both walls as shown. A central longitudinal hole **1724** may extend through the knob strike plate **1520** between the proximal and distal ends **1716**. The hole **1724** may include, from proximal to distal, first, second, and third inner portions **1726**, **1728**, **1730**. The first inner portion **1726** may include a generally circular central portion with bilateral longitudinal grooves **1732**, **1734** extending along the front and back, so that the first inner portion **1726**, taken as a whole, has a noncircular cross section. The first inner portion **1726** may extend distally over most of the proximal-distal length of the knob strike plate **1520**, and may have a flat distal end. The second inner portion **1728** may have a circular cross section with a diameter that is less than the diameter of the circular central portion of the first inner portion **1726**. The second inner portion **1728** may have a short proximal-distal length. Bilateral holes **1736**, **1738** may extend distally through the second inner portion **1728**, and may be aligned with the grooves **1732**, **1734**. The holes **176**,

1738 may be countersunk. The third inner portion **1730** may have a generally rectangular cross section, and may have rounded corners. The major dimension (length) of the rectangle may be greater than the diameter of the circular central portion of the first inner portion **1726**, and may extend along a front to back direction. The third inner portion **1730** may extend through the front side of the knob strike plate **1520**, so that the front side has a distal notch. The minor dimension (width) of the rectangle may be less than the diameter of the circular central portion of the first inner portion **1726** and greater than the diameter of the second inner portion **1728**. Bilateral teeth **1740**, **1742** may protrude toward each other from the left and right walls of the third inner portion **1730** near the front side.

Referring to FIGS. **111** and **112**, the spline insert **1522** extends between a proximal end **1744** and a distal end **1746**. The spline insert **1522** may have a cylindrical body **1748** with a unilateral tab **1750** extending radially outward from the proximal end **1744**. A longitudinal hole **1752** may extend through the tab **1750**. The hole **1752** may be countersunk on the distal side of the tab **1750**. A central longitudinal hole **1754** may include, from proximal to distal, first and second inner portions **1756**, **1758**. The first inner portion **1756** may have a circular cross section and may include a shallow countersink or chamfer around an inner proximal edge. The second inner portion **1758** may include a torque fitting such as the internal spline as shown.

Referring to FIGS. **113** and **114**, the rotation lock ring **1524** extends between a proximal end **1760** and a distal end **1762**. The rotation lock ring **1524** may include, from proximal to distal, first, second, third, and fourth outer portions **1764**, **1766**, **1768**, **1770**. The first outer portion **1764** may have a circular cross section and may include a rounded outer proximal edge. The second outer portion **1766** may have a circular cross section with a diameter that is less than the diameter of the first outer portion **1764**. The third outer portion **1768** may have a circular cross section with a diameter that is the same as or similar to the diameter of the first outer portion **1764**. A distal outer edge of the third outer portion may include a circumferential chamfer. The fourth outer portion **1770** may include a torque fitting such as the external spline shown. The fourth outer portion **1770** may be complementary to the second inner portion **1758** of the spline insert **1522**. A central longitudinal torque fitting **1772** may extend through the rotation lock ring **1524** between the proximal and distal ends **1760**, **1762**. The torque fitting **1772** may be a hexagonal hole as shown. The proximal end **1760** of the rotation lock ring may be concave. The spline insert **1522** and the rotation lock ring **1524** together may be referred to as a rotation locking mechanism.

Referring to FIGS. **115** and **116**, the cam lever **1526** extends between a proximal end **1774** and a distal end **1776**. The cam lever **1526** may include, from proximal to distal, first and second portions **1778**, **1780**. The first portion **1778** may include a hole **1782**, a first notch **1784**, a protrusion **1786**, a second notch **1788**, and a tooth **1790**. The hole **1782** may extend transversely through the cam lever **1526**. The hole **1782** may be near the proximal end **1774**. A proximal/outer bearing surface **1783** may be located next to the hole **1782**; the surface **1783** may be referred to as a first fulcrum of the cam lever **1526**. The first notch **1784** and the protrusion **1786** may extend transversely across an outer side of the cam lever **1526** distal to the hole **1782**. The protrusion **1786** may be directly adjacent to a distal or outer side of the first notch **1784**. The protrusion **1786** may be referred to as a second fulcrum of the cam lever **1526**. The second notch **1788** and the tooth **1790** may extend transversely across an

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inner side of the cam lever **1526** distal to the hole **1782**. The tooth **1790** may be directly adjacent to a proximal or inner side of the second notch **1788**. The second notch **1788** and the tooth **1790** may be distal to the first notch **1784** and the protrusion **1786**. The second portion **1780** may be a distally extending arm for actuation by a user. The second portion **1780** may be wider than the first portion **1778** so that rounded shoulders **1792**, **1794** are formed at the border between the first and second portions **1778**, **1780**. The shoulders **1792**, **1794** may face proximally and inwardly.

Referring to FIGS. **117-119**, the slider button **1528** extends between a proximal end **1796** and a distal end **1798**, and an outer side **1800** and an inner side **1802**. The outer side **1800** may include an oval domed surface **1804** with a longitudinal central trough **1806**. The trough **1806** may be generally rectangular and may extend through the distal end **1798**. A window **1808** may extend through the slider button **1528** between the outer and inner sides **1800**, **1802**. The window **1808** may be rectangular, with its major dimension (length) extending along a proximal-distal direction. A flange **1810** may encircle the surface **1804** and may serve as a boundary between the outer and inner sides **1800**, **1802**. The flange **1810** may be discontinuous in the vicinity of the window **1808**. The inner side **1802** may include, from proximal to distal, first, second, third, fourth, fifth, sixth, seventh, eighth, ninth, and tenth portions **1812**, **1814**, **1816**, **1818**, **1820**, **1822**, **1824**, **1826**, **1828**, **1830**. The first portion **1812** may be parallel to the flange **1810** at a first distance from the inner side of the flange **1810**. A hole **1832** may extend outwardly into the first portion **1812** near the proximal end **1796**. The second portion **1814** may slope between the first and third portions **1812**, **1816**. The third portion **1816** may be parallel to the flange **1810** at a second distance from the inner side of the flange, wherein the second distance is less than the first distance. The fourth portion **1818** may slope between the third and fifth portions **1816**, **1820**. The fourth portion **1818** may be parallel to the second portion **1814**. The fifth portion **1820** may be parallel to the flange **1810** at the same or similar level with the outer side of the flange. The sixth portion **1822** may slope between the fifth and seventh portions **1820**, **1824**. The seventh portion **1824** may be parallel to the flange **1810** and at the same or similar level with the third portion **1816**. The eighth portion **1826** may curve between the seventh and ninth portions **1824**, **1828**. The ninth portion **1828** may be parallel to the flange **1810** at a third distance from the inner side of the flange **1810**, wherein the third distance is greater than the first distance. The tenth portion **1830** may protrude a fourth distance from the inner side of the flange **1810**, and may extend distally past the flange. The tenth portion **1830** may include a semicircular yoke **1834** that opens away from the flange **1810**.

Referring to FIGS. **119** and **120**, the threaded slider **1530** extends between a proximal end **1836** and a distal end **1838**, and an outer side **1840** and an inner side **1842**. The threaded slider **1530** may be referred to as an actuator. The threaded slider **1530** may include a rectangular body **1844** with a post **1846** extending from the inner side **1842**. The outer side **1840** may ramp outwardly from proximal to distal. A notch **1848** may extend into the distal outer edge of the body **1844** between bilateral protrusions **1850**, **1852**. A transverse hole **1854** may extend through the threaded slider **1530**, through the protrusions **1850**, **1852** and intersecting the notch **1848**. A shallow hole **1856** or dimple may extend inwardly into the outer side **1840** near the proximal end **1836** in line with the notch **1848**. A longitudinal hole **1858** extends through the threaded slider **1530** between the proximal and distal ends

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1836, **1838**. The hole **1858** may include a first side portion **1860** toward the outer side **1840** and a second side portion **1862** toward the inner side **1842**. The first side portion **1860** may include internal threads around about 180 degrees of arc, i.e., along a half cylinder. The second side portion **1862** may be smooth, D-shaped, and larger than the major diameter of the internal threads of the first portion **1860**.

Referring to FIGS. **122-123**, the top knob part **1532** extends between a proximal end **1864** and a distal end **1866**, and an outer side **1868** and an inner side **1870**. The outer side **1868** may be domed. A window **1872** may extend through the top knob part **1532** between the outer and inner sides **1868**, **1870**. The window **1872** may be oval or rectangular, with its major dimension (length) extending along a proximal-distal direction. The inner side **1868** may be generally concave, and may include, from proximal to distal, first, second, third, fourth, and fifth portions **1874**, **1876**, **1878**, **1880**, **1882**. The first portion **1874** may have a semicircular cross section and a short proximal-distal length. The second portion **1876** may be oval or rectangular, with its major dimension extending along a proximal-distal direction. The second portion **1876** may be aligned with the window **1872**, and may extend over most of the proximal-distal length of the top knob part **1532**. Bilateral longitudinal ledges **1884**, **1886** may extend along the inner edges of the window **1872**. The third portion **1878** may have a semicircular or half-oval cross section and a short proximal-distal length. The first and third portions **1878** may have the same or similar diameters (or radii), with the third portion **1878** offset toward the outer side **1868**. The fourth portion **1880** may have a semicircular cross section and a short proximal-distal length. The diameter of the fourth portion **1880** may be greater than the diameter of the first portion **1874**. The fourth portion **1880** may be concentric with the first portion **1874**. The fifth portion **1882** may have a semicircular cross section and a short proximal-distal length. The diameter of the fifth portion **1882** may be greater than the diameter of the first portion **1874** and less than the diameter of the fourth portion **1880**. The fifth portion **1882** may be concentric with the first portion **1874**. When taken together, the fourth portion **1880** may be referred to as a groove or undercut and the fifth portion **1882** may be referred to as a lip or overhang. Bilateral holes **1888**, **1890** may extend into the inner side **1870** on either side of the third portion **1878**. The holes **1888**, **1890** may include internal threads. The proximal end **1864** may include a protrusion **1892** that is complementary to the third inner portion **1730** of the knob strike plate **1520**. The protrusion **1892** may be generally rectangular and may include bilateral notches **1894**, **1896** to receive the teeth **1740**, **1742**. An internally threaded hole **1898** may extend distally into the protrusion **1892** between the notches **1894**, **1896** and beside the first portion **1874**.

Referring to FIGS. **124** and **125**, the bottom knob part **1534** extends between a proximal end **1900** and a distal end **1902**, and an outer side **1904** and an inner side **1906**. The outer side **1904** may be domed. One or more openings **1908** may extend through the bottom knob part **1534** between the outer and inner sides **1902**, **1904**. Two openings **1908** are shown. Bilateral holes **1910**, **1912** may extend into the outer side **1904** near the distal end **1902**. The holes **1910**, **1912** may include counterbores. A semicircular wall **1914** or shell may protrude from the inner side **1906** near the proximal end **1900** so that a convex side of the wall **1914** faces proximally. The wall **1914** may include a distally extending internally threaded through hole **1916**. A U-shaped or part-oval wall **1918** may protrude from the inner side **1906** near the distal end **1902** so that a convex side of the wall **1918** faces

distally. A hole 1920 may extend through the bottom knob part 1534 between the outer and inner sides 1904, 1906. The hole 1920 may be located between the distal end of the wall 1914 and the proximal end of the wall 1918. The hole 1920 may include, from inner to outer, first, second, and third inner portions 1922, 1924, 1926. The first inner portion 1922 may have a rectangular or square cross section and may extend outwardly over most of the inner-outer thickness of the bottom knob part 1534. The first inner portion 1922 may have a flat outer end. The second inner portion 1924 may have a circular cross section with a diameter that is less than the minor dimension of the first inner portion 1922. The second inner portion 1924 may have a flat outer end. The third inner portion 1926 may have a circular cross section with a diameter that is less than the diameter of the second inner portion 1924. A longitudinal hole 1928 may extend through the bottom knob part 1534 between the proximal and distal ends 1900, 1902. The hole 1928 may include, from proximal to distal, first, second, third, and fourth inner portions 1930, 1932, 1934, 1936. The first inner portion 1930 may have a circular cross section and may extend distally into the proximal region of the wall 1918. The second inner portion 1932 may have the same or similar diameter (radius) as the first inner portion 1930, but may be elongated toward the outer side 1904 to form an oval cross section. The second inner portion 1932 may extend through the distal end of the wall 1918. The third inner portion 1934 may have a semicircular cross section with a diameter that is greater than the diameter of the first inner portion 1930, and a short proximal-distal length. The fourth inner portion 1936 may have a semicircular cross section with a diameter that is greater than the diameter of the first inner portion 1930 and less than the diameter of the third inner portion 1934, and a short proximal-distal length. When taken together, the third inner portion 1934 may be referred to as a groove or undercut and the fourth inner portion 1936 may be referred to as a lip or overhang.

The proximal end 1900 may include a protrusion 1938 that is complementary to the third inner portion 1730 of the knob strike plate 1520 when taken together with the protrusion 1892 of the top knob part 1532. The outer profile of the protrusion 1938 may be rectangular or square. An internally threaded hole 1940 may extend distally into the protrusion 1938 beside the first inner portion 1930.

The connector buttons 1510, 1512, thumb wheel 1514, wheel axle 1516, inserter handle 1518, knob strike plate 1520, spline insert 1522, rotation lock ring 1524, cam lever 1526, slider button 1528, threaded slider 1530, top knob part 1532, and bottom knob part 1534 may be coupled together in a handle assembly 1942. See FIG. 87.

The handle assembly 1942 may be assembled by coupling the connector button 1510 to the inserter handle 1518; coupling the thumb wheel 1514, wheel axle 1516, and spline insert 1522 to the inserter handle 1518; coupling the cam lever 1526, slider button 1528, threaded slider 1530, rotation lock ring 1524, and the top knob part 1532 together; coupling the top and bottom knob parts 1532, 1534 to the inserter handle 1518; and coupling the knob strike plate 1520 to the top and bottom knob parts 1532, 1534.

Coupling the connector button 1510 to the inserter handle 1518 may include inserting a compression spring (not shown) in the hole 1622, orienting the connector button 1510 and inserter handle 1518 with the proximal ends 1614, 1668 facing the same direction and the inner side 1618 facing into the pocket 1682, inserting the connector button 1510 into the pocket 1682, and inserting pins (not shown) into holes 1706, 1708 and slots 1634, 1636. Coupling the

connector button 1512 to the inserter handle 1518 may include similar steps with pocket 1684 and holes 1710, 1712.

Coupling the thumb wheel 1514, wheel axle 1516, and spline insert 1522 to the inserter handle 1518 may include orienting the thumb wheel 1514 and inserter handle 1518 with the proximal ends 1638, 1668 facing the same direction, inserting the thumb wheel 1514 into the hole 1680, inserting a compression spring (not shown) into the third inner portion 1664 of the wheel axle 1516, inserting the distal end 1652 of the wheel axle 1516 through the third inner portion 1692 of the inserter handle 1518 and into the first inner portion 1646 of the thumb wheel 1516, inserting the distal end 1746 of the spline insert 1522 into the first inner portion 1688 of the inserter handle 1518, inserting the tab 1750 into the alcove portion 1696, and inserting a pin (not shown) into holes 1752, 1698.

Coupling the cam lever 1526, slider button 1528, threaded slider 1530, rotation lock ring 1524, and the top knob part 1532 together may include inserting a ball detent (not shown) in the hole 1832 of the slider button 1528; inserting the first portion 1778 of the cam lever 1526 through the window 1808 of the slider button from the outer side 1800 so that the second portion 1780 extends outwardly, the protrusion 1786 faces proximally, and the tooth 1790 faces distally; inserting the first portion 1778 into the notch 1848 of the threaded slider 1530 between the protrusions 1850, 1852 so that the outwardly ramped outer side 1840 abuts the fourth portion 1818; inserting a pin (not shown) into holes 1854, 1782; engaging the yoke 1834 with the first and second outer portions 1764, 1766 of the rotation lock ring 1524 so that the proximal ends 1760, 1796 face the same direction; and inserting the second portion 1780 and slider button 1528 into the window 1872 of the top knob part 1532 from the inner side 1870 so that the flange 1810 rests against the ledges 1884, 1886 and the proximal ends 1864, 1796 face the same direction.

Coupling the top and bottom knob parts 1532, 1534 to the inserter handle 1518 may include engaging the first outer portion 1672, the fourth portion 1880, and the third inner portion 1934; engaging the second outer portion 1674, the fifth portion 1882, and the fourth inner portion 1936 so that the inner sides 1870, 1904 are facing; and inserting screws (not shown) in the holes 1910, 1888 and the holes 1912, 1890. When this step is performed after the preceding steps, the post 1846 of the threaded slider 1530 is in the hole 1920 of the bottom knob part 1534.

Coupling the knob strike plate 1520 to the top and bottom knob parts 1532, 1534 may include abutting the inner sides 1870, 1904 so that the proximal ends 1864, 1900 face the same direction, inserting the protrusions 1892, 1938 into the third inner portion 1730 so that the notches 1894, 1896 receive the teeth 1740, 1742, and inserting screws (not shown) in the holes 1736, 1898, 1916 and the holes 1738, 1940.

The handle assembly 1942 may be operated by pressing the connector buttons 1510, 1512, rotating the thumb wheel 1514, pivoting the cam lever 1526, and rotating the knob strike plate 1520, top knob part 1532, and bottom knob part 1534.

Pressing the connector buttons 1510, 1512 of the handle assembly 1942 may enable the connector plate 1504 of the shaft assembly 1600 to be disconnected from the handle assembly, and may also enable or facilitate connecting the shaft assembly to the handle assembly. The connector buttons 1510, 1512 are biased outwardly so that the handle assembly will retain the shaft assembly.

Turning the thumb wheel **1514** may rotate the implant retainer shaft **1508** to connect and disconnect the instrument **1500** and implant **1100**. More specifically, turning the thumb wheel **1514** may engage and disengage the external threads of the third outer portion **1610** of the implant retainer shaft **1508** and the internal threads **1155** of the second end body **1152** of the implant **1100**.

Pivoting the cam lever **1526** may engage and disengage the internal threads of the first side portion **1860** of the hole **1858** of the threaded slider **1530** and the externally threaded portion **2036** of the shaft **2032** of the threaded shaft **1954**, discussed below. Pivoting the cam lever **1526** inwardly (distally) may also lock the slider button **1528** in its distal position, due to the interaction of the tooth **1790** with the distal edge of the window **1808**. Pivoting the cam lever **1526** outwardly may provide sufficient mechanical advantage to overcome residual friction and/or potential energy within the instrument **1500** after expanding the implant **1100**, to disengage the threads of the threaded slider **1530** from the threaded shaft **1954**. The mechanical advantage may be due first to the interaction of the first fulcrum, surface **1783**, with the proximal edge of the window **1808**, and second to the interaction of the second fulcrum, protrusion **1786**, with the proximal portion of the trough **1806** next to the proximal edge of the window **1808**.

Rotating the knob strike plate **1520**, top knob part **1532**, and bottom knob part **1534** relative to the inserter handle **1518** may cause linear translation of the draw bar assembly **1536** along the longitudinal instrument axis **1542** without rotation of the draw bar assembly **1536** about the axis **1542**, due to the engaged threads of the threaded slider **1530** and the threaded shaft **1954**, and the sixth outer portion **1974** of the handle **1944** in the torque fitting **1772** of the rotation lock ring **1524**.

Referring to FIGS. **126-129**, the draw bar assembly **1536** may include a handle **1944**, a pull rod **1946**, a set screw **1948**, a button **1950**, a button spring **1952**, a threaded shaft **1954**, a retainer **1956**, and a retainer spring **1958**.

The handle **1944** extends between a proximal end **1960** and a distal end **1962**. The handle **1944** may include, from proximal to distal, first, second, third, fourth, fifth, and sixth outer portions **1964**, **1966**, **1968**, **1970**, **1972**, **1974**. The first outer portion **1964** may have the largest outer diameter of the handle **1944**. Friction features **1976** may be included around the first outer portion **1964**, such as the ten scallops shown. The second outer portion **1966** may have a circular cross section with a diameter that is less than the diameter of the first outer portion **1964**. A transverse hole **1978** may extend through the second outer portion **1966**. The hole **1978** may include, in sequence, first, second, and third inner portions **1980**, **1982**, **1984**. The first inner portion **1980** may be rectangular or square, and may have a flat bottom. The second inner portion **1982** may have a circular cross section with a diameter that is less than the dimension of the first inner portion **1980**. The third inner portion **1984** may also have a circular cross section, with a diameter that is less than the diameter of the second inner portion **1982**. The third outer portion **1968** may have a circular cross section with a diameter less than the diameter of the second outer portion **1966**, and a short proximal-distal length. The fourth outer portion **1970** may include external threads. The fifth outer portion **1972** may be a torque fitting, such as the hex key shown. The dimensions of the torque fitting may be less than the diameters of the first, second, and third outer portions **1964**, **1966**, **1968**. The sixth outer portion **1974** may also be a torque fitting, such as the hex key shown. The dimensions of the sixth outer portion **1974** may be less than the

dimensions of the fifth outer portion **1972**. A central longitudinal hole **1986** may extend through the handle **1944** between the proximal and distal ends **1960**, **1962**. The hole **1986** may include, from proximal to distal, first, second, and third inner portions **1988**, **1990**, **1992**. The first inner portion **1988** may have a short proximal-distal length and may include internal threads. The second inner portion **1990** may be smooth with a circular cross section with a diameter that is less than the minor diameter of the threads of the first inner portion **1988**. The third inner portion **1992** may have a circular cross section with a diameter that is less than the diameter of the second inner portion **1990**.

The pull rod **1946** extends between a proximal end **1994** and a distal end **1996**. The pull rod **1946** may include, from proximal to distal, first, second, third, fourth, and fifth portions **1998**, **2000**, **2002**, **2004**, **2006**. The first portion **1998** may have a generally circular cross section and a short proximal-distal length. The second portion **2000** may have the largest cross sectional dimension of the pull rod **1946**. Bilateral longitudinal flats **2008**, **2010** may extend along the first and second portions **1998**, **2000**. The dimension between the flats **2008**, **2010** may be less than the diameter of the first portion **1998**. A unilateral flat **2012** may extend along the second portion **2000** without affecting the first portion **1998**. A transverse notch **2014** may extend into the second portion **2000** through the side with the flat **2012**. Multiple notches may be present, for example a linear array of notches along the side with the flat **2012**. The third portion **2002** may have a circular cross section with a diameter that is less than the diameter of the first portion **1998**, and may be the same as or similar to the dimension between the flats **2008**, **2010**. The third portion **2002** may extend over most of the proximal-distal length of the pull rod **1946**. The fourth portion **2004** may have a circular cross section with a diameter that is less than the diameter of the third portion **2002**. The fifth portion **2006** may include external threads, with the major diameter of the threads the same as or similar to the diameter of the fourth portion **2004**. The pull rod **1946** may include indicia **2016**, such as text, icons, lines, or other markings.

The button **1950** may include a generally rectangular or square body **2018**. A post **2020** may extend from the body **2018**. A hole **2022** may extend through the body, transverse to the post **2020**. The hole **2022** may be rectangular or oval in cross section, with the major dimension of the rectangle or oval extending in line with the post **2020**. A rib **2024** may protrude from an inner wall of the hole adjacent to the post **2020**. The rib **2024** may be oriented transverse to the hole **2022**.

The threaded shaft **1954** extends between a proximal end **2026** and a distal end **2028**. The threaded shaft **1954** may include a proximal head **2030** and a distal shaft **2032** extending from the head. The shaft **2032** may include a proximal smooth portion **2034** and a distal externally threaded portion **2036**. Bilateral longitudinal troughs **2038**, **2040** may extend across the junction between the head **2030** and the smooth portion **2034**. The thread form of the threaded portion **2036** may be a buttress thread for high axial thrust in one direction. The load-bearing thread face may be substantially perpendicular to the thread axis. The load-bearing thread face of the threaded shaft **1954** may face toward the distal end **2028**. A central longitudinal hole **2042** may extend through the threaded shaft **1954** between the proximal and distal ends **2026**, **2028**. The hole **2042** may be a torque fitting such as the hexagonal hole shown.

The retainer **1956** extends between a proximal end **2044** and a distal end **2046**. The retainer **1956** may be a tubular

part. Indicia **2048** may be present on the outer surface of the retainer, such as text, icons, graphics, lines, and the like. A central longitudinal hole **2050** may extend through the retainer **1956** between the proximal and distal ends **2044**, **2046**. The hole **2050** may include, from proximal to distal, first, second, third, and fourth inner portions **2052**, **2054**, **2056**, **2058**. The first inner portion **2052** may have a circular cross section. The second inner portion **2054** may include internal threads. The major diameter of the threads may be the same as or similar to the diameter of the first inner portion **2052**. The third inner portion **2056** may have a circular cross section with a diameter that is the same as or similar to the minor diameter of the threads of the second inner portion **2054**. The third inner portion **2056** may have a flat distal end. The fourth inner portion **2058** may have a circular cross section with a diameter that is less than the diameter of the third inner portion **2056**.

The draw bar assembly **1536** may be assembled by inserting the spring **1952** into the second inner portion **1982** of the hole **1978** of the handle **1944**; inserting the button **1950** into the hole **1978** of the handle **1944** so that the post **2020** is received in the third inner portion **1984** and the body **2018** is received in the first inner portion **1980**; depressing the button **1950** while inserting the distal end **1996** of the pull rod **1946** into the hole **1986** in the proximal end **1960** of the handle **1944** and advancing the pull rod **1946** until the notch **2014** receives the rib **2024**; tightening the set screw **1948** in the first inner portion **1988**; inserting the distal end **1962** of the handle **1944** into the hole **2042** in the proximal end **2026** of the threaded shaft **1954** and advancing the handle **1944** until the hole **2042** receives the fifth outer portion **1972**; inserting a retainer spring **1958** in each trough **2038**, **2040**; and inserting the distal end **2028** of the threaded shaft **1954** into the hole **2050** in the proximal end **2044** of the retainer **1956** and advancing the threaded shaft **1954** until the first inner portion **2052** of the retainer **1956** receives the third outer portion **1968** of the handle **1944**, the second inner portion **2054** threads onto the fourth outer portion **1970**, and the third inner portion **2056** receives the head **2030** of the threaded shaft **1954**.

The draw bar assembly **1536** may be operated by depressing the button **1950** while sliding the pull rod **1946** proximally or distally to engage different notches like notch **2014**. Different pull rods may be substituted in the draw bar assembly **1536** as well. Different draw bar lengths and/or configurations may be compatible with different implant lengths (in the collapsed configuration) or configurations (i.e., implants for anterior, posterior, lateral, or transforaminal interbody fusion and the like). The draw bar assembly **1536** may be an improvement in the art because it may be adjustable to provide the draw bar length that corresponds to any given implant length. Even if an incorrect draw bar length is initially selected, the draw bar assembly **1536** may be re-adjusted to the correct draw bar length quickly and easily, and with positive confirmation that the correct length has been chosen. The draw bar assembly **1536** also reduces instrument inventory versus a set of draw bars of different lengths.

The instrument **1500** may be assembled by coupling the implant retainer shaft **1508** to the handle assembly **1942**, coupling the shaft assembly **1600** to the handle assembly **1942**, and coupling the draw bar assembly **1536** to the handle assembly **1942**.

Coupling the implant retainer shaft **1508** to the handle assembly **1942** may include inserting the proximal end **1602** of the implant retainer shaft **1508** into the fourth inner portion **1694** of the hole **1686** at the distal end **1670** of the

inserter handle **1518** and advancing the implant retainer shaft **1508** until the first outer portion **1606** is received in the second inner portion **1648** of the hole **1644** of the thumb wheel **1514**.

Coupling the shaft assembly **1600** to the handle assembly **1942** may include inserting the pins protruding proximally from the holes **1570**, **1572** of the connector plate **1504** into the holes **1702**, **1704** of the inserter handle **1518**, inserting the proximal end **1668** of the inserter handle into the hole **1574** in the proximal side **1562** of the connector plate so that the first outer portion **1672** is received in the third inner portion **1580**, the ledge **1626** of the connector button **1510** is received in the second inner portion **1578**, and the first inner portion **1576** is received in the groove **1624** (likewise for connector button **1512**). Optionally, the connector buttons **1510**, **1512** may be depressed during this step.

Coupling the draw bar assembly **1536** to the handle assembly **1942** may include rotating the cam lever **1526** outwardly, inserting the distal end **1996** of the pull rod **1946** into the hole **1724** in the proximal end **1714** of the knob strike plate **1520** and advancing the draw bar assembly **1536** so that the fourth and fifth portions **2004**, **2002** are received in the hole **1612** of the implant retainer shaft **1606**, the sixth outer portion **1974** of the handle **1944** is received in the torque fitting **1772** of the rotation lock ring **1524** and the second inner portion **1662** of the wheel axle **1516**, the threaded portion **2036** of the threaded shaft **1954** is received in the hole **1858** of the threaded slider **1530** and the hole **1928** of the bottom knob part **1534**, and the retainer **1956** is received in the hole **1724** of the knob strike plate **1520**. The draw bar assembly **1536** may slide into the handle assembly **1942** without rotation about the axis **1542**. After the draw bar assembly **1536** is fully inserted into the handle assembly **1942**, the cam lever **1526** may be rotated inwardly to engage the threads of the threaded slider **1530** and the threaded shaft **1954**.

The instrument **1500** may be operated by pressing the connector buttons **1510**, **1512**, rotating the thumb wheel **1514**, pivoting the cam lever **1526**, and rotating the knob strike plate **1520**, top knob part **1532**, and bottom knob part **1534**.

Various additional tools may be used with instrument **1500** to implant and expand an intervertebral implant. The instrument **1500** and selected additional tools may be provided in a kit and together they may be referred to as an instrument set or system. FIGS. **130-139** show additional tools for use with instrument **1500**, including a trial assembly **2060**, a funnel **2062**, and a screwdriver assembly **2064**.

Referring to FIGS. **130-133**, the trial assembly **2060** may include a handle **2066**, a block **2068**, a head **2070**, and a shaft **2072**. A distal end **2074** of the shaft **2072** may be received in a proximal hole **2076** in the head **2070**. The block **2068** may be received in a distal hole **2078** in the handle **2066**. The block **2068** and hole **2078** may have complementary round proximal portions **2080** and complementary rectangular or square distal portions **2082**, **2084**. The block **2068** may be retained in the handle **2066** by a pin (not shown) through holes **2086**, **2088**. A proximal end **2090** of the shaft **2072** may be received in a distal hole **2092** in the block **2068**. The head **2070**, shaft **2072**, and block **2068** may be a weldment. Alternatively, the handle **2066**, block **2068**, head **2070**, and shaft **2072** may be combined in a single part. The trial assembly **2060** may include various markings or other indicia **2094** to convey the dimensions of the trial and the corresponding implant state (for example, expanded or collapsed).

Referring to FIGS. 134 and 135, the funnel 2062 may include a wide mouth 2096 that tapers inwardly to a narrow stem 2098 or spout. The stem 2098 may include a torque fitting 2100, which in this example is a hexagonal hole in the distal end of the stem. The torque fitting 2100 may be sized and shaped to receive the first outer portion 1606 at the proximal end 1602 of the implant retainer shaft 1508. This enables the funnel 2062 to be coupled to the implant retainer shaft 1508 for delivery of graft material, and it enables the funnel 2062 to be used to unthread the implant retainer shaft 1508, for example after the lockout screw 654 has been inserted. See FIGS. 157 and 158.

Referring to FIGS. 27, 35, 36, the tamp assembly 180 may be used with the instrument 1500 and implant 1100 as described above for the instrument 100 and implant 1000.

Referring to FIGS. 136-139, the screwdriver assembly 2064 may include a handle 2102 and a shaft 2104. The handle 2102 may be the same as or similar to the handle 176 of FIGS. 29, 37, and 38. The shaft 2104 may be the same as or similar to the shaft 178. A proximal end 2118 of the shaft 2104 may be received in a distal end 2120 of the handle 2102.

Referring to FIGS. 43 and 44, the lockout screw 654 may be used with the instrument 1500 and implant 1100 as described above for the instrument 100 and implant 1000.

Referring to FIGS. 140-142, implant 1100 is expandable along a horizontal or lateral or transverse or medial-lateral axis 104 and a vertical or cephalocaudal axis 106. Axis 102 is an anterior-posterior axis. Implant 1100 includes a first support member 1140 and a second support member 1130, the support members connected by a first end body 1150 and a second end body 1152. First support member 1140 may include an upper support body 1142 and a lower support body 1144. Second support member 1130 may include an upper support body 1132 and a lower support body 1134. Internal threads 1155 are included in the second end body 1152 for connection to the implant retainer shaft 1508. A plurality of link members 1160, 1162, 1166, 1168 pivotably connect each of the first and second end bodies 1150, 1152 to each of the support members 1130, 1140. Lower support body 1144 includes ramped expansion slots 1174, 1176, and upper support body 1142 includes ramped expansion slots 1170, 1172. A chamber 1122 may exist in the midst of the support bodies 1142, 1144, 1132, 1134, at least when the implant 1100 is expanded. The implant 1100 is representative of the intervertebral cages described in pending U.S. patent application Ser. No. 15/244,446.

Referring to FIGS. 143-160, methods are shown for inserting and expanding an expandable intervertebral implant with instrument 1500 and related instruments.

FIGS. 143 and 144 show a step of using the trial assembly 2060 in an intervertebral space to determine a suitable size for the implant 1100. The trial assembly 2060 may be inserted into the intervertebral space as shown and then rotated 90 degrees about its longitudinal axis to simulate implant vertical expansion.

FIGS. 145 and 146 show a step of coupling the instrument 1500 and implant 1100 together. Note that the cam lever 1526 has been rotated outwardly so that the draw bar assembly 1536 may be inserted into the instrument 1500 and rotated to engage the threads of the fifth portion 2006 of the pull rod 1946 of the draw bar assembly 1536 with internal threads of the first end body 1150 of the implant 1100.

FIGS. 147 and 148 show a step of locking the instrument 1500 and implant 1100 together. Note that the cam lever 1526 has been rotated inwardly (distally). With the cam lever in this position, the external spline of the fourth outer

portion 1770 of the rotation lock ring 1524 is engaged with the internal spline of the second inner portion 1758 of the spline insert 1522 to prevent the draw bar assembly 1536 from rotating about the axis 1542, and the internal threads of the first side portion 1860 of the hole 1858 of the threaded slider 1530 engage the externally threaded portion 2036 of the threaded shaft 1954 of the draw bar assembly 1536 to prepare the instrument 1500 to be actuated to expand the implant 1100.

FIGS. 149 and 150 show a step of expanding the implant 1100 laterally. The knob strike plate 1520, top knob part 1532, and bottom knob part 1534 may be rotated together in a clockwise direction relative to the inserter handle 1518 to rotate the internal threads of the first side portion 1860 of the hole 1858 of the threaded slider 1530 relative to the externally threaded portion 2036 of the threaded shaft 1954 of the draw bar assembly 1536 to urge the draw bar assembly proximally to expand the implant 1100.

FIGS. 151 and 152 show a step of expanding the implant 1100 vertically. This step may be functionally continuous with the step of expanding the implant laterally. Note that implant 1100 features bilateral vertical expansion of the first and second support members 1140, 1130.

FIGS. 153 and 154 show a step of releasing the cam lever 1526 after implant expansion is complete. Releasing the cam lever 1526 disengages the external spline of the fourth outer portion 1770 of the rotation lock ring 1524 from the internal spline of the second inner portion 1758 of the spline insert 1522 so that the draw bar assembly 1536 can rotate about the axis 1542, and disengages the internal threads of the first side portion 1860 of the hole 1858 of the threaded slider 1530 from the externally threaded portion 2036 of the threaded shaft 1954 of the draw bar assembly 1536 by moving the threaded slider 1530 transverse to the axis 1542 so that the smooth second side portion 1862 of the hole 1858 approaches the externally threaded portion 2036.

FIGS. 155 and 156 show a step of unlocking the instrument 1500 and implant 1100. This step may be functionally continuous with the preceding step. At this point, the draw bar assembly 1536 may be removed from the instrument 1500.

FIG. 157 shows a step of removing the handle assembly 1942 from the shaft assembly 1600 and the implant retainer shaft 1508. The connector buttons 1510, 1512 are depressed to disconnect the shaft assembly 1600 from the handle assembly 1942, which also allows the implant retainer shaft 1508 to slide out. Note that the handle assembly 1942 may weigh much more than the shaft assembly 1600 and implant retainer shaft 1508 together, and the handle assembly 1942 is located at the opposite end of the shaft assembly 1600 from the implant 1100. Thus removing the handle assembly 1942 significantly reduces the force moment on the implant 1100 due to the weight of attached instrument apparatus. Force moment is the force times the moment arm. In the current step, the force is the weight of the handle assembly 1942 due at least to gravity. The moment arm is approximately the length of the shaft assembly 1600 modified with trigonometry to take into account the angle between the shaft assembly 1600 and true vertical, for example for TLIF, the length of the shaft assembly 1600 times the cosine of 30 degrees. The true moment arm is the horizontal distance between the centers of gravity of the implant 1100 and the handle assembly 1942. Additionally, the removable handle assembly 1942 shortens the length of the screwdriver 2064 and its associated force moment on the implant 1100.

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FIG. 158 shows a step of coupling the funnel 2062 to the implant retainer shaft 1508 (not visible in this figure) and inserting the tamp assembly 180 through the funnel and into the shaft assembly 1600.

FIG. 159 shows a step of inserting the locking screw 654 using the screw driver assembly 2064 through the funnel 2062 and shaft assembly 1600. A step of using the funnel 2062 to unscrew the implant retainer shaft 1508 from the implant 1100 is not shown.

FIG. 160 shows a view of the implant 1100 in its final implanted state.

The terms “upper” and “lower”, and “top” and “bottom”, “front” and “rear” are used as relative terms herein for ease of description and understanding. It is understood that in embodiments of the disclosure, upper and lower entities may be reversed, as may top and bottom, front and rear.

Any methods disclosed herein includes one or more steps or actions for performing the described method. The method steps and/or actions may be interchanged with one another. In other words, unless a specific order of steps or actions is required for proper operation of the embodiment, the order and/or use of specific steps and/or actions may be modified.

Reference throughout this specification to “an embodiment” or “the embodiment” means that a particular feature, structure or characteristic described in connection with that embodiment is included in at least one embodiment. Thus, the quoted phrases, or variations thereof, as recited throughout this specification are not necessarily all referring to the same embodiment.

Similarly, it should be appreciated that in the above description of embodiments, various features are sometimes grouped together in a single embodiment, figure, or description thereof for the purpose of streamlining the disclosure. This method of disclosure, however, is not to be interpreted as reflecting an intention that any claim require more features than those expressly recited in that claim. Rather, as the following claims reflect, inventive aspects lie in a combination of fewer than all features of any single foregoing disclosed embodiment. Thus, the claims following this Detailed Description are hereby expressly incorporated into this Detailed Description, with each claim standing on its own as a separate embodiment. This disclosure includes all permutations of the independent claims with their dependent claims.

Recitation in the claims of the term “first” with respect to a feature or element does not necessarily imply the existence of a second or additional such feature or element. Elements recited in means-plus-function format are intended to be construed in accordance with 35 U.S.C. § 112 Para. 6. It will be apparent to those having skill in the art that changes may be made to the details of the above-described embodiments without departing from the underlying principles of the technology.

While specific embodiments and applications of the present technology have been illustrated and described, it is to be understood that the technology is not limited to the precise configuration and components disclosed herein. Various modifications, changes, and variations which will be apparent to those skilled in the art may be made in the arrangement, operation, and details of the methods and systems of the present technology disclosed herein without departing from the spirit and scope of the technology.

The invention claimed is:

1. An instrument comprising:

a handle comprising a distal end comprising a distal thumb wheel and a proximal end comprising a proximal thumb wheel;

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a first instrument part that is removably connectable to a first implant part of a distal end of an implant via actuation of the proximal thumb wheel; and

a second instrument part that is removably connectable to a second implant part of a proximal end of the implant via actuation of the distal thumb wheel;

wherein, when the first instrument part is connected to the first implant part, the second instrument part is connected to the second implant part, and the instrument is actuated, the first instrument part and the first implant part translate relative to the second instrument part and the second implant part along a central longitudinal axis of the instrument without rotating about the central longitudinal axis during instrument actuation.

2. The instrument of claim 1, wherein the first instrument part comprises a draw bar that is removably connectable to the first implant part by rotation of the draw bar about the central longitudinal axis;

wherein the instrument further comprises an actuator and a rotation lock;

wherein instrument actuation comprises movement of the actuator in engagement with the draw bar to translate the draw bar relative to the second instrument part and the second implant part along the central longitudinal axis;

wherein when the rotation lock is engaged, the draw bar cannot rotate about the central longitudinal axis, wherein when the rotation lock is disengaged, the draw bar is free to rotate about the central longitudinal axis; wherein the instrument comprises a first state in which the actuator is disengaged from the draw bar, the rotation lock is disengaged, and the draw bar is free to translate along and rotate about the central longitudinal axis;

wherein the instrument comprises a second state in which the actuator is engaged with the draw bar, the rotation lock is engaged, and the draw bar is prevented from rotating about the central longitudinal axis.

3. The instrument of claim 2, wherein the rotation lock comprises a rotation lock ring and a spline insert, wherein the rotation lock ring and the draw bar are fixed together for rotation about the central longitudinal axis, wherein the spline insert is fixed in translation along and rotation about the central longitudinal axis, wherein the spline insert is removably connectable to the rotation lock ring;

wherein when the rotation lock is engaged, the spline insert is connected to the rotation lock ring so that the rotation lock ring cannot rotate about the central longitudinal axis, wherein when the rotation lock is disengaged, the spline insert is disconnected from the rotation lock ring so that the rotation lock ring is free to rotate about the central longitudinal axis;

wherein the actuator and the rotation lock are coupled together so that the actuator engages the draw bar when the rotation lock is engaged and the actuator disengages from the draw bar when the rotation lock is disengaged.

4. The instrument of claim 3, wherein the draw bar comprises threads;

wherein the actuator comprises a threaded slider, wherein the threaded slider comprises threads that engage the threads of the draw bar, wherein instrument actuation comprises rotation of the threaded slider threads in engagement with the draw bar threads to translate the draw bar relative to the second instrument part and the second implant part along the central longitudinal axis.

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5. The instrument of claim 1, wherein:
the handle further comprises a distal portion and a proximal portion rotatable with respect to the distal portion;
and
rotation of the proximal portion actuates the instrument. 5

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